

MASTER

Understanding the Effect of Post-Training Quantization on Large Language Models

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Eindhoven University of Technology



Master Thesis

Understanding the Effect of Post-Training Quantization on Large Language Models

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“There is no knowledge that is not power”

Ralph Waldo Emerson

Abstract

Transformers are currently widely adopted and utilized in the artificial intelligence research space. They do however face scalability and sustainability challenges due to resource-intensive training and inference. Post-training quantization (PTQ), a method for reducing computational and environmental costs, involves using lower bit precision representations of model parameters. The effect, and reasoning behind, the trade-off that PTQ makes in performance is not widely studied. This research investigates the impact of PTQ on transformer models, focusing on large language models (LLMs). The study explores how PTQ affects the performance and behavior of transformer models quantized on a range of bit precisions. The performance of a range of LLMs, quantized on a range of PTQ methods, is measured on a range of benchmark tasks. To this end, a controlled experimental environment is developed. Metrics such as model size, inference time, and benchmark scores are evaluated. The effect of PTQ is quantified on a large scale, showcasing the performance degradation on lower bit precisions across different axes such as training, task and architecture.

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1

Introduction

1.1 Motivation

The transformer architecture [1], with its specific applications in large language models (LLMs) and large vision models (LVMs), has become well-adopted in- and outside the scientific community. The advent of transformers, LLMs, and LVMs has marked a paradigm shift in the field of artificial intelligence. As an example, ChatGPT, a chat-based LLM created by OpenAI, has millions of active monthly users worldwide, showcasing the widespread acknowledgment, adoption, and application of LLMs. Models like BERT [4], GPT [5], and Vision Transformers [6] show their remarkable capabilities in natural language processing and computer vision tasks such as natural language understanding, image classification, and machine translation.

The transformer architecture is based on the attention mechanism. This mechanism allows the model to capture intricate contextual dependencies in the data, making them capable of outperforming traditional models in tasks that require understanding and generation of complex patterns. There are three types of transformer models: encoder, decoder, and sequence-to-sequence models. Encoder models only use the encoder part of the transformer architecture. These models are best suited for understanding entire sentences, such as for sentiment analysis and named-entity recognition. Decoder models only use the decoder part of the transformer architecture. These models are best suited for text generation tasks. Sequence-to-sequences models use both parts of the transformer architecture. These models are best suited for generating new text based on an input, such as machine translation and summarization.

This increase in performance does however come at a cost. Transformers require substantial resources for training and inference [7], which poses challenges for scalability, accessi-

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bility, and sustainability. Furthermore, due to the large nature of the model architecture, expressed in storage size, energy consumption, and memory cost, it is infeasible to deploy these models locally on consumer machines and edge devices. With the demand for these large models growing, it is of great value to investigate methods that improve the efficiency of these models, with as little degradation in performance as possible. Post-training quantization (PTQ) is a promising avenue of research to alleviate the computational and environmental costs that transformers bring along with them.

The challenge to balance the trade-off between efficiency and performance remains. There are several promising PTQ methods available, such as LLM.int8() [8] for 8-bit precision quantization and GPTQ [9] for lower-bit precision quantization. Lower-bit quantization faces more challenges, such as a lack of model expressivity and difficulty with optimizing. The effect of quantization on generative language models, such as GPT-2 [10] and BART [11], has been researched in the past. Homogeneous word embeddings caused by reduced model capacity and varied distribution of weights might be responsible for the degradation of performance [12].

There has not been much research done into the effect that present post-training quantization methods have on a range of encoder, decoder, and sequence-to-sequence models. To effectively use PTQ methods, an understanding of the effect that it has on these models is essential. The resulting knowledge could lead us to mitigate the negative effects that quantization might have on performance, making it more accessible, scalable, and sustainable. This research aims to create a broader understanding of this effect.

1.2 Research questions

There are several research questions to address in this thesis. First of all, we research:

- 1. What kind of post-training quantization methods are available for large language models?**

The research to be conducted will thoroughly include the use of, and investigate, PTQ methods, thus it is essential to know the subtle differences that the PTQ methods used might have. This thesis will provide an overview of post-training quantization methods currently present.

There is a distinction to be found in encoder, decoder, and sequence-to-sequence models. The behavior and inner workings of these models differ, and thus PTQ might affect these models differently as well. The research question is asked:

2. What is the effect of post-training quantization on encoder, decoder, and encoder/decoder models?

This question can be asked in a more general sense. The **main research question** is:

3. What is the effect of post-training quantization across different bit precisions on different large language models?

To answer this main question, We want to quantify the effect that PTQ has on large language models. We quantify the effect of PTQ on LLMs by observing the change in performance metrics on benchmark tasks. We ask the following subquestions in correspondence to the main research question:

- **4. How do different large language models and post-training quantization methods compare in performance across different bit precisions?**
- **5. How big is the effect that post-training quantization has on large language models?**

Furthermore, apart from observing and quantifying this effect, it is useful to know *why* the effect occurs. Regarding this, we ask the following research question:

6. What is the reasoning behind the effect that post-training quantization has on large language models?

Apart from furthering our understanding of PTQ, it might also be interesting to know the next step: what actions to undertake or changes to implement to be able to make the quantized models both as efficient and as performant as possible. As an optional last research question:

7. What are possible solutions for mitigating negative effects that post-training quantization has on large language models?

1.3 Our methodology

To, in a broad sense, investigate the effect that post-training quantization has on the performance of large language models, we conduct a comparative study across a range of large language models, quantization methods, and benchmark tasks. Furthermore, we investigate the effect that post-training quantization has on the weight distribution of the model.

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To this end, we introduce an environment for quantizing large language models, and universally benchmarking (quantized) large language models on widely-used and widely-accepted benchmark tasks in a controlled fashion. This environment is developed to be model implementation-agnostic, such that any model can be benchmarked. The quantization methods, benchmark tasks, and datasets however require implementation.

1.4 Contributions

The following contributions were made:

- We provide a large result set of benchmarks, which provide insight into the performance degradation across bit-precisions.
- We quantify the average effect of post-training quantization on model performance.

Furthermore, open code to individually perform benchmarks on models is provided. The code repository for this project can be found on GitHub: <https://github.com/Generative-AI-TUe/msc-project-1293885>

2

Literature review

2.1 Basic definitions

2.1.1 Transformers

The transformer architecture [1] is a deep learning architecture, based on the multi-head attention mechanism. A diagram of the transformer architecture is displayed in Figure 2.1. This multi-head attention mechanism allows the architecture to capture dependencies between input and output sequences. The multi-head attention mechanism is parallelizable, which makes the transformer architecture highly efficient and scalable. The key components of the transformer architecture are self-attention, multi-head attention, feed-forward networks, encoder-decoder architecture, and positional encoding.

2.1.1.1 Self-attention

Self-attention computes a weighted sum of the input embeddings. The weights are determined by the similarity between tokens in the input sentence. Self-attention allows the architecture to focus on relevant parts of the input sequence. Given a query vector, a set of key vectors, and a set of value vectors, the attention function computes their weighted sum, where the weight assigned to each value is determined by the dot-product of the query and the corresponding key.

Mathematically, given a query Q , keys K , and values V , the output of the self-attention layer is:

$$\text{attention}(Q, K, V) = \text{softmax}((QK^T)/\sqrt{d_k})V$$

where $Q, K, V \in \mathbb{R}^{n \times d_{model}}$ are matrices resulting from multiplying the input by weight matrices $W_q, W_k, W_v \in \mathbb{R}^{d_{model} \times d_k}$, respectively, d_{model} is the dimension of the model, and

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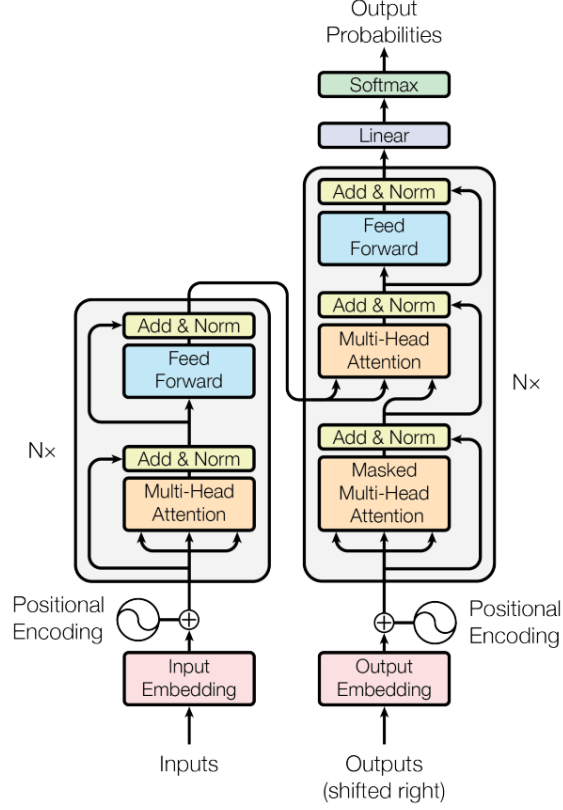


Figure 2.1: The diagram of the transformer architecture [1]. The left column displays the encoder part, and the right column displays the decoder part.

d_k is the dimension of the key vectors. The softmax function ensures that the weights sum up to 1, providing a probability distribution over the values. The output is a weighted sum of the values, where the weight of each value is proportional to the similarity between the query and the corresponding key.

2.1.1.2 Multi-head attention

Transformers utilize multi-head attention mechanisms to capture different aspects of the input sequence simultaneously. This is done by performing the multi-head attention mechanism several times in parallel. The output of multi-head attention is diverse, separated representations of the input.

Mathematically, multi-head attention is defined as follows:

$$\text{multi-head-attention}(Q, K, V) = \text{concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{head}_i = \text{attention}(QW_i^Q, KW_i^K, VW_i^V)$$

where W_i^Q , W_i^K , W_i^V are the weight matrices for the i -th attention head, and $W^O \in \mathbb{R}^{n \times d_{model}}$ is the output weight matrix.

2.1.1.3 Feed-forward networks

In essence, the transformer architecture is a deep learning architecture. This means it utilizes a neural network to model the input and output of the provided data. Feed-forward networks consist of fully connected layers with activation functions, which enables the model to learn complex non-linear transformations of the input data. Transformers include feed-forward networks in their architecture, which process the information extracted by the self-attention mechanism.

2.1.1.4 Encoder-decoder architecture

Transformers typically consist of an encoder component, a decoder component, or both (sequence to sequence). The encoder processes the input sequence and outputs an embedding. The decoder generates the output sequence. Each encoder and decoder layer in a transformer consists of multiple self-attention layers followed by position-wise fully connected feed-forward networks.

2.1.1.5 Positional encoding

Transformers lack recurrence and convolutional operations. The architecture relies on positional encodings to inject information about the order of the tokens into the input embeddings. Positional encodings are added to the input embeddings before inputting them into the model. This allows the model to take into account the sequential order of tokens within the input sequence.

2.1.2 Numeric datatypes

Computers can store numbers in different ways, using a different amount of bits. Depending on the data type, different amounts of bits play different roles in the representation of the number.

With integer (INT) numbers, the number can be represented in a signed and unsigned fashion. Signed n -bit integers have a range from $-(2^{n-1})$ to $(2^{n-1} - 1)$. Unsigned n -bit integers have a range from 0 to $(2^n - 1)$. The number is equal to the binary value.

With signed floating point (FP) numbers, the number is represented by 1 sign bit representing if the number is positive or negative. Range bits (or exponent bits) represent

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the integer number before the radix point. Precision bits (or mantissa bits) represent the fraction after the radix point. FP32 uses 1 sign bit, 8 range bits and 23 precision bits. TensorFloat32 (TF32) uses up 19 bits and thus fits inside of 32 bits, but is used for faster computation. It uses 1 sign bit, 8 range bits and 10 precision bits. FP16 uses 1 sign bit, 5 range bits and 10 precision bits (same as TF32). BrainFloat16 (BFLOAT16) uses 1 sign bit, 8 range bits (same as FP32), and 7 precision bits. Deep learning models often use 32-bit floating point numbers to store the model parameters.

2.1.3 Quantization

Quantization is the process of mapping input values from a large set to output values in a smaller set. It forms the foundation of most lossy compression algorithms. Examples of quantization processes are rounding (defined as $Q(x) = \Delta \cdot \lfloor \frac{x}{\Delta} + \frac{1}{2} \rfloor$) and truncation (defined as $Q(x, n) = \frac{\lfloor 10^n \cdot x \rfloor}{10^n}$). Quantization is inherently non-linear and irreversible, because the process is a many-to-few mapping.

As seen in Section 2.1.2, the range and precision of the numerical representations can differ. In terms of quantization of numerical values, the large set of input values is the range of the datatype in which the source numerical value is stored, and the smaller set of output values is the range of the datatype in which the target numerical value is stored. A visualization of what this looks like is shown in Figure 2.2.

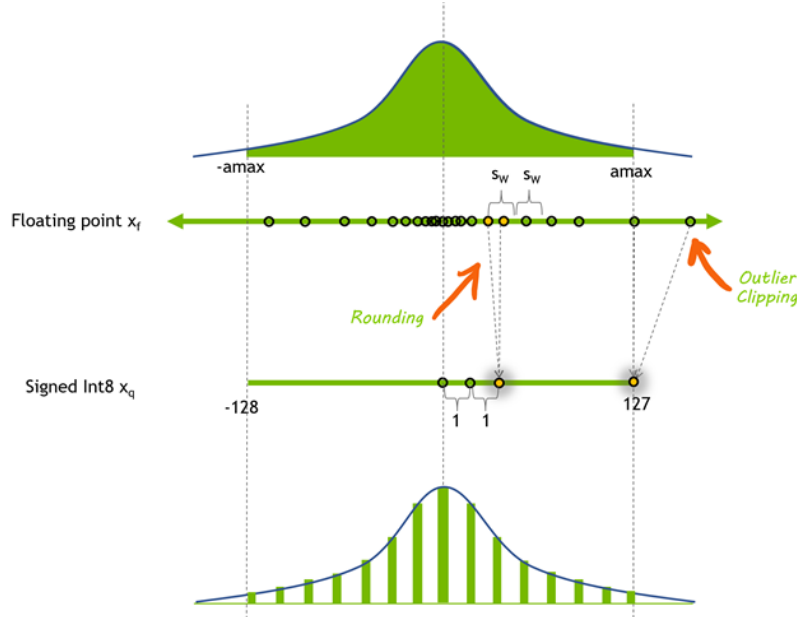


Figure 2.2: 8-bit signed integer quantization of a floating point tensor x_f . The symmetric dynamic range of $x_f[-amax, amax]$ is mapped through quantization to $[-128, 127]$ [2].

2.1.4 Post-training quantization

Transformers require a lot of computational power to train and perform inference with. Many people do not have the necessary resources to work with these transformer models, which imposes a limitation. Over time, many solutions have been proposed to overcome this limitation, such as post-training quantization, knowledge distillation, quantization-aware training, pruning, and low-rank factorization. This research focuses on post-training quantization.

Post-training quantization is a technique that compresses and speeds up models by quantizing the weights down from a higher bit precision range (e.g. 32-bit floating point numbers) to a lower bit precision range (e.g. 8-bit integers). The precision (or “expressivity”) of a model is reduced as a trade-off for reducing the storage-, memory, and computational footprint [13]. The goal is to find a lower-precision representation of the model, while maintaining as much of the model performance as possible.

Quantization is done after training since lower precision models are not continuous, which leads to the objective gradient of the model being discrete as well. Methods such as straight-through estimator have been proposed to overcome discrete gradients [14, 15].

2.1.5 Metrics

Several metrics can be used to measure the performance of a transformer model. General metrics, as described in Section 2.1.5.1, are universal metrics that can be used on any transformer model type. These metrics are an indication of the performance of the model. Specific architectures, such as decoder and sequence-to-sequence models, have specific metrics that measure the model performance. These metrics are described in Section 2.1.5.2 and Section 2.1.5.3. Alongside model quality, it is also interesting to measure the model performance in terms of computational complexity and efficiency. Section 2.1.5.4 presents metrics that can measure model efficiency. Section 2.1.5.5 describes additional metrics that do not fit in these categories, but are nonetheless interesting.

2.1.5.1 General metrics

In general, the following metrics can be used to measure performance:

- Accuracy - the percentage of correctly predicted tokens or sequences. $\text{accuracy} = \frac{\text{correct}}{\text{total}}$
- Precision - The ratio of correctly predicted observations of a label, compared to all the predictions of the label. $\text{precision} = \frac{\text{correctly predicted observations of label}}{\text{total predictions of label}}$

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- Recall - The ratio of correctly predicted observations of a label, compared to the total observations of the label. $\text{recall} = \frac{\text{correctly predicted observations of label}}{\text{total observations of label}}$
- F1 score - The harmonic mean of precision and recall. It is a balance between precision and recall. $\text{F1 score} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$
- Log-loss (or cross-entropy loss) [16, 17] - A measure of the performance of a classification model whose output is a probability value between 0 and 1. It increases as the predicted probability diverges from the actual label.
$$\text{Log-loss} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$
- Perplexity [18] - A measurement of how well a probabilistic model predicts a sample. It's often used to evaluate language models. $\text{PP} = 2^{-\frac{1}{N} \sum_{i=1}^N \log_2(p_i)}$
- Matthews Correlation Coefficient [19] - A balanced measure that takes into account true and false positives and negatives. It is used in binary classification and is especially useful with imbalanced datasets.
$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$
- Exact Match - The ratio of predictions that exactly match the ground truth. It is often used in text generation or question answering tasks.
$$\text{EM} = \frac{\text{number of exactly matching predictions}}{\text{total number of predictions}}$$

2.1.5.2 Decoder metrics

Decoder models have separate metrics to measure the performance of the model:

- BLEU score (Bilingual Evaluation Understudy) [20] - A metric for evaluating the quality of text generated by a model by comparing it to one or more reference texts. It measures the n-gram overlap between the generated and reference texts.
$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right)$$
where BP is the brevity penalty, p_n is the precision for n-grams, and w_n are weights.

2.1.5.3 Sequence-to-sequence metrics

Sequence-to-sequence models have separate metrics to measure the performance of the model. This is mostly related to translation quality, for which sequence-to-sequence models are often used.

- ROUGE score (Recall-Oriented Understudy for Gisting Evaluation) [21] - A set of metrics for evaluating text by comparing the overlap of n-grams, word sequences, and word pairs between the generated text and reference texts.

$$\text{ROUGE-N} = \frac{\sum_{\text{ref} \in \text{references}} \sum_{\text{n-gram} \in \text{ref}} \text{count}_{\text{match}}(\text{n-gram})}{\sum_{\text{ref} \in \text{references}} \sum_{\text{n-gram} \in \text{ref}} \text{count}(\text{n-gram})}$$

- METEOR score (Metric for Evaluation of Translation with Explicit Ordering) [22] - A metric that evaluates the alignment between generated text and reference text based on precision, recall, stemming, synonyms, and word order. $\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty})$ where F_{mean} is a combination of precision and recall.

- CIDEr (Consensus-based image description evaluation) [23] - A metric for evaluating the quality of captions by measuring consensus between candidate and reference captions using term frequency-inverse document frequency (TF-IDF) weighting.

$$\text{CIDEr} = \frac{1}{N} \sum_{n=1}^N \sum_{j=1}^M w_{ij} \cdot \text{IDF}(g_j) \text{ where } w_{ij} \text{ is the weight of the n-gram and } \text{IDF}(g_j) \text{ is its inverse document frequency.}$$

- SPICE (Semantic propositional image caption evaluation) [24] - A metric that evaluates image captions by comparing scene graphs of objects, attributes, and relationships between candidate and reference captions.

$$\text{SPICE} = \frac{|G_{\text{match}}|}{|G_{\text{gen}}| + |G_{\text{ref}}|} \text{ where } G_{\text{match}}, G_{\text{gen}}, \text{ and } G_{\text{ref}} \text{ are sets of matched, generated, and reference scene graph tuples, respectively.}$$

- Edit distance (Levenshtein) [25] - A measure of the similarity between two strings by counting the minimum number of single-character edits (insertions, deletions, substitutions) required to change one string into the other.

Edit Distance = min number of edits to convert one string into another

- BERTScore [26] - A metric that evaluates text similarity by computing cosine similarity between BERT embeddings of candidate and reference texts.

$$\text{BERTScore} = \frac{1}{N} \sum_{i=1}^N \text{cosine similarity}(e_{\text{candidate}}, e_{\text{reference}}) \text{ where } e \text{ represents BERT embeddings.}$$

- Word Error Rate - A metric for evaluating the performance of speech recognition systems by calculating the edit distance between the recognized and reference texts normalized by the number of words in the reference.

$$\text{WER} = \frac{\text{Substitutions} + \text{Insertions} + \text{Deletions}}{\text{Total words in reference}}$$

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2.1.5.4 Training and efficiency metrics

Apart from how well the model performs, there are also metrics to measure the efficiency of the model:

- Floating point operations (FLOPs) [27, 28, 29], measures the number of floating point operations needed for executing a specific task. FLOPs are an indicator of computational efficiency, with less operations being more efficient, and are hardware-agnostic. FLOPs are not an indicator of runtime, as parallelism might differ across algorithms.
- Training time - the total time it takes to train the model for a certain amount of epochs.
- Inference time [9, 30] is used to measure the runtime of an algorithm. The time taken by the model to process and generate an output for input data during inference or prediction is measured. This is primarily relevant for real-world applications, where the user interacting with a system would want a lower inference time, and is thus a good metric for evaluating the real-world performance when taking parallelism into account. Inference time is hardware-specific.
- Speed-up ratio [31, 32] is the ratio of the inference time between the baseline model and the accelerated model. This allows for a relative comparison which can be roughly compared across different hardware. Speed-up ratio can be approximated by the ratio of the number of transformer layers in the baseline model compared to the number of transformer layers used after acceleration.
- Memory usage - the amount of computer memory it takes to train a model or perform inference on a model, in MB or GB
- Convergence rate - how much closer we get to the global maximum with each timestep, batch, iteration or epoch.
- Number of parameters [33, 34] reports the total number of learnable weights or variables in a model. Parameters represent the weights and biases in the connections between neurons or attention layers. This directly reflects the storage cost of a model, which can be crucial for deployment on edge devices. It can also be a loose indicator for the computational cost and memory footprint during training and inference. In general, the number of parameters has a positive relationship with the expressivity of the model.

- Model size [35, 36, 37] is closely related to the number of parameters. It refers to the space that the model takes up on the disk or memory footprint required to store the entire model. The model size has a positive relationship with the number of parameters, as more parameters often lead to a larger model size. This is not always the case, as the data type and bit precision can affect the model size.
- Compression ratio [38, 39] is the ratio between the original model size of the uncompressed model compared to the model size of the compressed/reduced model. A high compression rate without significant loss of performance indicates that a compression method is efficient and effective.
- Parameter efficiency - the ratio between performance and amount of parameters that a model contains.

2.1.5.5 Additional metrics

- Robustness [40, 41, 42] measures the robustness of the model to adversarial attacks. Compressed models are significantly less robust on out-of-distribution data.
- The interpretability of a model could also be an important factor in some specific use cases. For example, in healthcare, it is desired to gain insight into how machine learning models come to their answer. Interpretability is hard to measure, and is thus a more subjective metric.

2.1.6 Benchmarking

The performance of a model on certain tasks can be measured using benchmarks. Comparing model performance between different models on a specific task is also possible to do using benchmarks.

For language models, most studies evaluate on common NLP benchmarks such as GLUE [43] and SuperGLUE [44], which are natural language understanding benchmarks. SQuAD [45] is used for machine reading comprehension. LAMBADA [46] is designed to evaluate the context-dependent understanding of language models. LAMA [47], StrategyQA [48], and HellaSwag [49] are all reasoning ability benchmarks.

2.1.7 Pareto front

The Pareto front, otherwise known as the Pareto frontier or the Pareto curve, is the set of all Pareto efficient solutions. A Pareto efficient solution is a solution for which there is no

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change possibly in at least one factor that results in a positive improvement. Pareto fronts are used to display the set of Pareto efficient configurations, enabling the user to make a decision based on trade-offs in the configuration itself.

2.1.8 Rate-distortion

Rate-distortion theory is a branch of information theory. It analyzes how much compression can be achieved using lossy compression methods, such as quantization. The rate usually constitutes the number of bits per data sample used to represent the compressed version of the original information. The definition of distortion varies, but in the most simple case it is defined as the error between the input and output signal. Higher distortion means lower quality. The distortion can be expressed using distortion functions, such as Hamming distortion, which is defined as follows:

$$d(x, \hat{x}) = \begin{cases} 0 & \text{if } x = \hat{x} \\ 1 & \text{if } x \neq \hat{x} \end{cases}$$

and Squared-error distortion, which is defined as follows:

$$d(x, \hat{x}) = (x - \hat{x})^2$$

The rate-distortion curve shows the reduction of quality (increase in distortion) when the rate is lowered, and vice versa.

2.2 State-of-the-art benchmarking methods

There are several benchmarking methods currently available. Chatbot arena [50] is a human evaluation benchmarking method. It evaluates the alignment with human preferences of chat-based large language models by allowing users to indicate a preferred answer between several model outputs. LiveBench [51] is a "challenging, contamination-free LLM benchmark". This benchmark aims to overcome the obstacle of test data contamination by creating a controlled environment for large language models to be evaluated, without the need for biased LLM judging and human crowd-sourcing. The benchmark contains questions across a multitude of topics, such as mathematics, news articles and datasets. LLM-QBench [52] benchmarks LLaMA-2 [53] and ChatGLM3 [54] using several benchmark tasks in order to formulate best practices for post-training quantization.

The difference between the state-of-the-art benchmarking methods and our solution however, is that the majority of these state-of-the-art benchmarking methods do not focus on

2.3 State-of-the-art quantization methods

measuring the effect of post-training quantization. Additionally, the majority of benchmarking methods focus on decoder models. LLM-QBench is very similar to this project, but the insights differ. This project aims to give a wide overview of the impact of post-training quantization on a wide range of models, while the aim of LLM-QBench is to formulate best practices based on a smaller experimental configuration, looking exclusively at decoder models and having a different research goal.

2.3 State-of-the-art quantization methods

LLM.int8() [8] is a weight quantization method. It uses vector-wise quantization with separate normalization constants for each inner product in the matrix multiplication to quantize most of the features in 8-bit integers, except for emergent outliers, which isolates the outlier feature dimensions into a 16-bit matrix multiplication. It is implemented in the BitsAndBytes [55] library.

SmoothQuant [56] is an 8-bit weight, 8-bit activation quantization method. It smooths the activation outliers by offline migrating the quantization difficult from activations to weights with a mathematically equivalent transformation.

OmniQuant [57] is a weight and activation quantization method. It comprises Learnable Weight Clipping (LWC) and Learnable Equivalent Transformation (LET). LWC modulates the extreme values of weights by optimizing the clipping threshold, while LET tackles activation outliers by shifting the challenge of quantization from activations to weights.

AWQ [58] (Activation-aware Weight Quantization) is a weight quantization method. It is based on the fact that not all weights are equally as important: protecting 1% of salient weights can greatly reduce quantization error. The method searches for the optimal per-channel scaling that protects the salient weights by observing the activation instead of the weights.

GPTQ [9] is a one-shot weight quantization method. The algorithm uses approximate second-order information to perform quantization in a "one-shot" manner by quantizing weights layer-by-layer, using a block-wise approach.

2.4 State-of-the-art models

2.4.1 Encoder models

RoBERTa [59] is an improvement on the BERT encoder model. The model is a replication study of BERT pre-training. It finds that BERT was significantly undertrained and pro-

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poses an improved recipe for training BERT models, which it calls RoBERTa. RoBERTa can match or exceed the performance of all of the post-BERT methods.

DeBERTa [60] is an encoder model architecture that improves the BERT and RoBERTa models using two novel techniques. The first is the disentangled attention mechanism, where each word is represented using two vectors that encode its content and position, respectively, and the attention weights among words are computed using disentangled matrices on their contents and relative positions, respectively. The second is an enhanced mask decoder which incorporates absolute positions in the decoding layer to predict the masked tokens in model pre-training. In addition, a new virtual adversarial training method is used for fine-tuning to improve model’s generalization on downstream tasks.

2.4.2 Decoder models

Claude¹ is a family of models developed by Anthropic. The models have been pre-trained to predict the next word in large amounts of text and then fine-tuned with Constitutional AI [61] to be helpful, honest, and harmless. The Claude 3 model family includes three state-of-the-art models in ascending order of capability: Claude 3 Haiku, Claude 3 Sonnet, and Claude 3 Opus.

GPT-4o [62], where o stands for "Omni" is a single, multi-modal neural network that has been trained on text, vision, and audio input. This means that it can describe what is shown on a computer screen, answer questions about the onscreen image, or describe what it sees if given access to a camera. It is developed by OpenAI. The GPT-4o models work similarly to the other GPT models, but their neural networks were trained on images and audio at the same time as text, which means that the model is adapted to process them as both inputs and outputs.

LLaMA 3 [63] (or Large Language model Meta AI) is a foundational large language model developed by Meta. It utilizes a transformer architecture, with the largest version containing over 10 trillion parameters. LLaMA is trained on a massive curated dataset, which enables it to predict the next word in a sequence and generate coherent and contextually relevant responses to prompts.

2.4.3 Sequence-to-sequence models

FLAN T5 [64] is a pre-trained language model that is trained on a collection of datasets phrased as instructions. This enables the model to respond better to instructions and

¹<https://www.anthropic.com/claude>

reduces the need for few-shot exemplars. The model is trained on a variety of instruction template types, including instructional templates for each task as given by the creators of the mixtures. The model is also trained on a mixture of both formats to enable both zero-shot and few-shot.

Aya 23 [65] is a family of multilingual instruction-tuned language models. It builds on the Cohere Command series, utilizing a standard decoder-only Transformer architecture. It uses parallel attention and feed-forward layers, SwiGLU [53] activation for better downstream performance, the use of rotary positional embeddings (RoPE) [66] for improved long-context handling, and a grouped query attention (GQA) mechanism [67] in the 8B model to reduce memory usage.

BART [11] is a denoising auto-encoder for pre-training sequence-to-sequence models. It is trained by corrupting text with an arbitrary noising function, and learning a model to reconstruct the original text. It uses a standard Transformer-based neural machine translation architecture which, despite its simplicity, can be seen as generalizing BERT (due to the bidirectional encoder), GPT (with the left-to-right decoder), and other pre-training schemes.

2. LITERATURE REVIEW

3

Comparative study design

3.1 Quantization methods

There are 4 quantization methods used in this comparative study: downcasting, linear quantization, half-quadratic quantization, 8-bit matrix multiplication. We choose to study solely on-the-fly quantization methods, as opposed to methods that require calibration like AWQ and GPTQ, to avoid the additional factor of calibration, as well as limited compatibility in some of these methods. An overview of quantization methods can be found in Appendix A.

3.1.1 Downcasting

Our baseline quantization method is downcasting. This method casts the model weights from a higher bit precision (floating point 32-bits) to a lower bit precision, without any additional methods or technology. This method reduces memory and compute footprint of a model, but does not take into account any quantization error. We use downcasting as our baseline, as it is the most simple way to quantize model weights to a lower bit precision.

Downcasting from FP32 to FP16 works as follows:

$$x_{\text{FP32}} = (-1)^s \times 2^{(e_{\text{FP32}}-127)} \times (1 + m_{\text{FP32}}) \quad (3.1)$$

where s is the sign bit, e_{FP32} is the 8-bit exponent, and m_{FP32} is the 23-bit mantissa.

$$x_{\text{FP16}} = (-1)^s \times 2^{(e_{\text{FP16}}-15)} \times (1 + m_{\text{FP16}}) \quad (3.2)$$

where s is the sign bit, e_{FP16} is the 5-bit exponent, and m_{FP16} is the 10-bit mantissa.

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To convert, the following definitions are used:

$$s_{\text{FP16}} = s_{\text{FP32}} \quad (3.3)$$

$$e_{\text{FP16}} = e_{\text{FP32}} - 127 + 15 \quad (3.4)$$

$$m_{\text{FP16}} = \left\lfloor \frac{m_{\text{FP32}}}{2^{13}} \right\rfloor \quad (3.5)$$

3.1.2 Linear quantization and per group quantization

Linear quantization maps the range of values in a higher bit precision, down to a range of values suitable for a lower bit precision. Every increment in the quantized range corresponds to a fixed-size increment in the original range. Per group quantization quantizes a group of parameters with the same quantization parameters. This allows the quantization error of a group of parameters to only affect the group of parameters itself.

In our study, we use the Quanto library [68] by Hugging Face to quantize the model weights. Quanto is designed to utilize simple quantization schemes that are adaptable across any modality. The library uses linear quantization and per group quantization. These quantization methods are explained in Section 3.1.2.1 and Section 3.1.2.2 respectively. The library supports weight and activation quantization, but we only utilize weight quantization due to calibration requirements.

3.1.2.1 Linear quantization

Given an input value x that falls within a range $[x_{\min}, x_{\max}]$, linear quantization works as follows:

$$Q = 2^b \quad (3.6)$$

where b is the number of bits used for quantization, and Q is the total number of quantization levels.

The step size Δ represents the distance between adjacent quantization levels.

$$\Delta = \frac{x_{\max} - x_{\min}}{Q - 1} \quad (3.7)$$

The input value x is mapped to a quantized value $q(x)$ using:

$$q(x) = \left\lfloor \frac{x - x_{\min}}{\Delta} + 0.5 \right\rfloor \quad (3.8)$$

The quantized value $q(x)$ can be mapped back to its corresponding continuous value:

$$x_q = x_{\min} + q(x) \times \Delta \quad (3.9)$$

3.1.2.2 Per group quantization

Given an input tensor X that is divided into G groups, per-group quantization works as follows:

$$X = \{X_1, X_2, \dots, X_G\} \quad (3.10)$$

where X_i represents data in the i -th group.

For each group X_i the minimum and maximum values are determined:

$$x_{\min,i} = \min(X_i), \quad x_{\max,i} = \max(X_i) \quad (3.11)$$

The quantization step size for the i -th group is calculated as follows:

$$\Delta_i = \frac{x_{\max,i} - x_{\min,i}}{Q - 1} \quad (3.12)$$

Each group is quantized separately. Each element $x \in X_i$ is quantized to a discrete level $q(x)$ as follows:

$$q(x) = \left\lfloor \frac{x - x_{\min,i}}{\Delta_i} + 0.5 \right\rfloor \quad (3.13)$$

The quantized value $q(x)$ for each group can be mapped back to its continuous form as follows:

$$x_{q,i} = x_{\min,i} + q(x) \times \Delta_i \quad (3.14)$$

3.1.3 Half-quadratic quantization

Half-quadratic quantization [69], or HQQ for short, is a method used to solve non-linear optimization problems by introducing an auxiliary variable that allows the original problem to be split into simpler, quadratic sub-problems. This process alternates between updating the auxiliary variable and the original variable iteratively, which gradually leads to a quantized solution.

We use the HQQ library¹ in this study, which is an implementation that does weight quantization.

3.1.4 LLM.int8() and QLoRA

In our study, we use the BitsAndBytes library [55] by Tim Dettmers and the BitsAndBytes foundation, to quantize the model weights. BitsAndBytes uses several quantization techniques. For 8-bit quantization, the library uses LLM.int8() [8]. There is already a small description of the functionality of LLM.int8() in Section 2.3.

¹<https://github.com/mobiusml/hqq/>

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For 4-int quantization, QLoRA [70] is utilized. QLoRA quantized the model parameters down to "NormalFloat4", which is introduced in the paper. It is designed to be optimal for normally distributed weights. NF4 captures more information by aligning the quantization bins with the distribution of the model's weights, improving precision compared to standard 4-bit quantization methods.

Additionally, QLoRA uses double quantization to further reduce memory consumption. In this process, the quantization constants are quantized, effectively compressing the data even more. This layered approach significantly reduces the average memory footprint, allowing large models to operate efficiently on limited hardware resources.

The name QLoRA stems from the method that freezes the quantized weights, and adds trainable low-rank adapters.

3.2 Baseline models

We use a range of models in this comparative study. We want to investigate the difference in performance across multiple axes, thus we have a wide configuration of models to analyse the quantization methods on. Model type is taken into account by analyzing encoder, decoder, and sequence-to-sequence models. Model size is taken into account by analyzing models with different numbers of parameters.

For our baselines, we choose to measure the performance of the base (pre-trained) model, and the model fine-tuned on the task. The difference in performance across a range of bit-precisions is then measured after quantization.

The models used in this study can be found in Table 3.1.

Model Name	Model Type	# Model Parameters
bert-base-uncased	Encoder	110M
bert-large-uncased	Encoder	336M
openai-community/gpt2	Decoder	137M
openai-community/gpt2-medium	Decoder	380M
facebook/bart-base	Sequence-to-sequence	139M
google/flan-t5-base	Sequence-to-sequence	248M

Table 3.1: The models used in this study, with model type and amount of parameters

Due to computational constraints, we have chosen to use smaller models (less than one billion parameters). The benchmarking environment is made in such a way that it would still be possible to benchmark state-of-the-art models such as Llama 3.1 [63], but since

3.3 Benchmarks and metrics

these constraints are put on the project, the older, smaller models are used. This possibly impacts the results and insights, as the model performance is naturally lower.

3.3 Benchmarks and metrics

The benchmarks used for this comparative study are GLUE [43] and SuperGLUE [44], which are natural language understanding benchmarks. They test the "understanding" of concepts that the model has. The tasks chosen for this study are all sequence classification tasks to be able to easily compare performance between several tasks, as opposed to having vastly different tasks. The benchmarks, with corresponding tasks¹, metrics and descriptions², can be found in Table 3.2.

Benchmark	Task	Metric	Description
GLUE	The Corpus of Linguistic Acceptability	Matthew's Correlation Coefficient	Determine if a sentence is grammatically correct or not
	The Stanford Sentiment Treebank	Accuracy	Determine if the sentence has a positive or negative sentiment
	Microsoft Research Paraphrase Corpus	F1 / Accuracy	Determine if two sentences are paraphrases from one another
	Quora Question Pairs	F1 / Accuracy	Determine if two questions are semantically equivalent or not
	MultiNLI	Accuracy	Determine if a sentence entails, contradicts or is unrelated to another sentence
	Question NLI	Accuracy	Determine if the answer to a question is contained in a second sentence or not
	Recognizing Textual Entailment	Accuracy	Determine if a sentence entails a given hypothesis or not
	Winograd NLI	Accuracy	Determine if a sentence with an anonymous pronoun and a sentence with this pronoun replaced are entailed or not
SuperGLUE	CommitmentBank	F1 / Accuracy	Given a text and a clause, predict how much the text commits to a the clause
	Choice of Plausible Alternatives	Accuracy	Given a premise sentence and two possible choices, the system must determine either the cause or effect of the premise from two possible choices
	Words in Context	Accuracy	Given two text snippets and a polysemous word that appears in both sentence, determine whether the word is used with the same sense in both sentences
	The Winograd Schema Challenge	Accuracy	Determine the correct referent of the pronoun from among the provided choices
	BoolQ	Accuracy	Answer a yes/no question about a text passage

Table 3.2: An overview of all the benchmarks, tasks, and corresponding metrics used in the experiments, with additional an task description

¹<https://gluebenchmark.com/tasks>

²<https://medium.com/nlplanet/two-minutes-nlp-glue-tasks-and-2022-leaderboard-517baedfa597>,
<https://medium.com/nlplanet/two-minutes-nlp-superglue-tasks-and-2022-leaderboard-492d8c849ed>

3. COMPARATIVE STUDY DESIGN

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Repository design

4.1 Methodology

We have developed a repository for benchmarking models and quantization methods on several benchmark tasks and datasets. This repository consists of several parts, which we will describe in the following section.

4.1.1 General setup

4.1.2 Experimental configuration

We use an experimental configuration, provided through a Python dictionary to provide the program with information on what benchmarks, benchmark tasks, quantization methods, number of bits, and models are used in an experiment. This dictionary also contains metadata information, such as what task type a benchmark task has, how many target labels a task has, what helper function needs to be called for benchmark pre-processing. The experimental configuration used in this study can be found in Appendix C.

4.1.3 Component functions

The component functions perform tasks such as tokenizing a dataset, loading in a model, evaluating a model, fine-tuning a model, quantizing a model, and running experiments.

- For loading in the models, the model data is obtained from the Hugging Face model hub.
- For tokenization, the corresponding tokenizer for the model from Hugging Face is loaded in.

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- For evaluation of the model, the `transformers.Trainer.evaluate` function¹ is used. A separate `TrainerEvaluation` class, which inherits from the `Trainer` class was created for this to work since quantized models are not loaded into the trainer class.
- For fine-tuning, the `Trainer` class is used.

4.1.4 Helper functions

Helper functions process the data from the benchmark tasks so that the dataset can be effectively used to benchmark.

Some benchmarks require online submission and thus have closed test set labeling. We create our own test split to ensure that we obtain evaluations from every model. This test split is equal to the validation split of the benchmark task. The training split is shuffled and divided into an 80% train split and a 20% test split.

Furthermore, we use oversampling on the datasets. Some datasets have a strong class imbalance, which caused issues during training: the model often quickly got stuck in a local minimum where the dominant learned strategy was predicting solely the class with the most data points. To prevent this stagnation during training, we chose to balance the amount of data points per class by sampling extra data points in the training data such that each class has an equal amount of data entries. We chose oversampling over undersampling since some of the datasets were already quite small. Oversampling is not applied on validation and test sets, so that the model is still evaluated on the original data distribution during training and evaluation.

Each benchmark also has a specific helper function. Decoder models need a prompt to perform benchmarks. To this end, each benchmark task is transformed into a prompt, to be used as input for the decoder models. The prompt format for each benchmark task can be found in Appendix B.

4.1.5 Metric functions

As described in Table 3.2, metrics are calculated for the benchmark tasks. These metrics are calculated using metric functions.

¹Documentation of the `Trainer` class: https://huggingface.co/docs/transformers/main_classes/trainer

4.2 Models

4.2.1 Model selection motivation

As mentioned in Section 3.2, the models are chosen with the computational restrictions in mind, though the repository is for the most part designed in a model-agnostic manner: the datasets and quantization methods require implementation, whilst models require no implementation beyond the AutoModel implementation inside of the Hugging Face ecosystem.

4.2.2 Sequence classification with decoder and sequence-to-sequence models

Decoder and sequence-to-sequence models do not natively support sequence classification due to the nature of their output format being a sequence. Encoder models have an additional classification head that classifies based on the embeddings that the model has created. The Hugging Face transformers library offers support for sequence classification for decoder and sequence-to-sequence models. This is done with a classification head which reads in the last generated token. For this to work, the prompt has to be formulated so that the classification is the last token. It is worth noting that even though this technique works, it is not exactly the same as a usual classification model, as the probability distribution of the decoder/sequence-to-sequence model is formed around the vocabulary of the model, instead of the classes in the classification task.

4.3 Quantization methods

As mentioned in Section 3.1, only on-the-fly post-training-quantization methods are used for this study, to eliminate the additional factor of calibration that some PTQ methods require. The models are quantized before training, to measure the effect on a pre-trained model, and after training to measure the effect on a fine-tuned model. The models are quantized to 2, 4, and 8 bits wherever possible. The models are also downcasted to 16 bits.

4.4 Reproducibility

To make the experiments reproducible, we added a set seed 25 to the code. Furthermore, all models and benchmark tasks are openly available to reproduce the results.

4. REPOSITORY DESIGN

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Experiments

5.1 Experimental setup

5.1.1 Models, datasets, quantization methods

The experimental setup in terms of models, datasets, and quantization methods used is described in Chapter 3.

5.1.2 Hyperparameters

There is minimal change in the hyperparameters across experiments to keep the experiments as controlled as possible. Most of the hyperparameter choices are the default Hugging Face hyperparameters.

A batch size of 4 is used for both training and test-set evaluation. The model is trained for 5 epochs, with evaluation being done at the end of each epoch. A base learning rate of $5e-5$ is used, using a linear learning rate scheduler. We use the torch implementation of AdamW as the optimizer.

The best model, according to the lowest evaluation loss, is loaded in and returned at the end of training.

5.1.3 Hardware

We ran the experiments on several NVIDIA GeForce RTX 4090 graphics cards from a data center, rented on the compute marketplace website `vast.ai`.

5.2 Results

The results will be shown in Tables 5.1 through 5.13 and Figures 5.1 through 5.32.

5. EXPERIMENTS

5.2.1 Tables

The following tables are an aggregate of each model result per task. The mean (μ), standard deviation (σ) and percent change compared to the baseline 32-bit precision model ($\% \Delta = \frac{\mu_x - \mu_{32bit}}{\mu_{32bit}}$) are displayed per metric. For the sake of completeness, detailed versions of each individual model performance on the benchmark tasks can be found in Appendix D.

Finetuned	Quantization Method	# Bits	Eval Loss			Eval Mcc			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	1.73	1.13	+0.00%	0.01	0.03	+0.00%	20.77	9.12	+0.00%	63.22	33.36	+0.00%	15.82	8.35	+0.00%
	BitsAndBytes	4	1.79	1.36	+3.65%	-0.00	0.02	-140.20%	25.34	10.48	+21.98%	49.66	21.66	-21.45%	12.43	5.42	-21.45%
	BitsAndBytes	8	1.62	1.09	-6.54%	0.00	0.03	-83.98%	44.33	13.68	+113.44%	26.10	8.76	-58.72%	6.53	2.19	-58.72%
	Downcasting	16	1.75	1.15	+0.98%	0.01	0.03	+8.31%	12.82	6.40	-38.28%	102.48	47.05	+62.09%	25.64	11.77	+62.09%
	HQQ	2	1.91	1.47	+10.40%	0.00	0.03	-65.83%	25.65	10.02	+23.50%	47.98	18.78	-24.11%	12.01	4.70	-24.11%
	HQQ	4	2.22	1.10	+28.36%	-0.00	0.03	-105.44%	24.18	9.75	+16.41%	51.23	20.24	-18.97%	12.82	5.06	-18.97%
	HQQ	8	2.22	1.09	+28.20%	-0.00	0.03	-122.72%	22.24	9.65	+7.08%	56.61	23.30	-10.47%	14.16	5.83	-10.47%
	Quanto	2	1.72	1.39	-0.54%	0.00	0.01	-95.22%	34.26	15.16	+64.93%	37.86	18.43	-40.12%	9.47	4.61	-40.12%
	Quanto	4	1.76	1.15	+2.00%	-0.00	0.03	-135.78%	32.60	14.09	+56.96%	39.40	18.57	-37.69%	9.86	4.65	-37.69%
	Quanto	8	1.72	1.12	-0.21%	0.01	0.02	-16.22%	22.08	9.15	+6.30%	57.14	25.79	-9.63%	14.30	6.45	-9.62%
Yes	None	32	0.73	0.10	+0.00%	0.16	0.20	+0.00%	20.03	9.16	+0.00%	65.58	33.54	+0.00%	16.41	8.39	+0.00%
	BitsAndBytes	4	0.73	0.09	-0.61%	0.17	0.21	+3.88%	25.37	10.36	+26.64%	49.31	21.12	-24.81%	12.34	5.29	-24.81%
	BitsAndBytes	8	0.84	0.33	+13.98%	0.11	0.16	-34.36%	42.03	13.31	+109.82%	27.70	9.36	-57.76%	6.93	2.34	-57.76%
	Downcasting	16	0.73	0.10	-0.03%	0.16	0.20	-1.73%	12.86	6.40	-35.78%	102.25	47.49	+55.92%	25.59	11.88	+55.91%
	HQQ	2	0.95	0.28	+29.63%	0.11	0.11	-34.08%	25.58	9.97	+27.72%	48.08	18.79	-26.68%	12.03	4.70	-26.68%
	HQQ	4	0.74	0.13	+0.67%	0.13	0.18	-20.45%	24.03	9.67	+19.98%	51.42	20.14	-21.58%	12.87	5.04	-21.58%
	HQQ	8	0.75	0.12	+1.67%	0.13	0.18	-20.48%	22.17	9.67	+10.66%	56.87	23.41	-13.28%	14.23	5.86	-13.28%
	Quanto	2	0.90	0.25	+22.49%	0.07	0.08	-59.25%	34.28	15.18	+71.15%	37.87	18.47	-42.25%	9.48	4.62	-42.25%
	Quanto	4	0.72	0.09	-1.31%	0.16	0.21	-1.34%	32.50	13.95	+62.23%	39.39	18.39	-39.94%	9.86	4.60	-39.94%
	Quanto	8	0.73	0.10	-0.12%	0.16	0.20	-0.87%	22.13	9.15	+10.50%	56.95	25.69	-13.16%	14.25	6.43	-13.16%

Table 5.1: Aggregated results on the glue/cola task

Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.34	0.02	+0.00%	1.30	0.19	+0.00%	121.33	47.64	+0.00%	95.64	37.55	+0.00%	23.91	9.39	+0.00%
	BitsAndBytes	4	0.34	0.02	+0.02%	1.30	0.19	-0.05%	178.50	63.32	+47.12%	62.90	22.31	-34.23%	15.73	5.58	-34.23%
	BitsAndBytes	8	0.34	0.01	+0.83%	1.19	0.08	-8.56%	465.59	143.10	+283.74%	23.28	7.15	-75.66%	5.82	1.79	-75.66%
	Downcasting	16	0.34	0.02	-0.03%	1.30	0.19	-0.00%	76.10	23.37	-37.28%	142.41	43.74	+48.90%	35.61	10.94	+48.90%
	HQQ	2	0.32	0.00	-5.17%	1.14	0.03	-12.57%	234.39	81.65	+93.18%	47.66	16.60	-50.17%	11.92	4.15	-50.17%
	HQQ	4	0.34	0.02	+0.15%	1.31	0.19	+0.26%	206.91	72.22	+70.53%	54.02	18.85	-43.52%	13.51	4.71	-43.52%
	HQQ	8	0.34	0.02	-0.03%	1.30	0.19	+0.02%	173.41	60.99	+42.92%	64.59	22.72	-32.46%	16.15	5.68	-32.47%
	Quanto	2	0.32	0.00	-4.39%	1.13	0.01	-13.59%	284.92	96.35	+134.83%	38.90	13.15	-59.33%	9.72	3.29	-59.33%
	Quanto	4	0.34	0.02	-0.23%	1.31	0.20	+0.59%	265.22	90.46	+118.60%	41.88	14.28	-56.21%	10.47	3.57	-56.21%
	Quanto	8	0.34	0.02	-0.05%	1.30	0.19	-0.02%	145.89	52.59	+20.25%	77.32	27.87	-19.16%	19.33	6.97	-19.16%
Yes	None	32	0.33	0.00	+0.00%	1.10	0.00	+0.00%	119.15	45.33	+0.00%	96.32	36.64	+0.00%	24.08	9.16	+0.00%
	BitsAndBytes	4	0.33	0.00	+0.00%	1.10	0.00	+0.00%	179.10	63.53	+50.32%	62.69	22.24	-34.92%	15.67	5.56	-34.91%
	BitsAndBytes	8	0.33	0.00	+0.00%	1.10	0.00	+0.00%	435.60	126.56	+265.59%	24.61	7.15	-74.45%	6.15	1.79	-74.45%
	Downcasting	16	0.33	0.00	+0.00%	1.10	0.00	-0.09%	76.32	23.66	-35.95%	142.29	44.12	+47.73%	35.58	11.03	+47.73%
	HQQ	2	0.32	0.00	-1.40%	1.10	0.00	+0.24%	235.81	83.32	+97.92%	47.56	16.80	-50.62%	11.89	4.20	-50.62%
	HQQ	4	0.33	0.00	+0.00%	1.10	0.00	+0.00%	205.85	71.69	+72.76%	54.26	18.90	-43.66%	13.57	4.73	-43.66%
	HQQ	8	0.33	0.00	+0.00%	1.10	0.00	-0.00%	172.55	60.52	+44.82%	64.86	22.75	-32.66%	16.22	5.69	-32.66%
	Quanto	2	0.32	0.00	-1.40%	1.10	0.00	+0.21%	284.31	96.53	+138.62%	39.02	13.25	-59.49%	9.76	3.31	-59.49%
	Quanto	4	0.33	0.00	+0.00%	1.10	0.00	+0.04%	265.33	90.04	+122.69%	41.80	14.19	-56.60%	10.45	3.55	-56.60%
	Quanto	8	0.33	0.00	+0.00%	1.10	0.00	+0.00%	144.74	51.97	+21.48%	77.85	27.95	-19.18%	19.46	6.99	-19.18%

Table 5.2: Aggregated results on the glue/mnli task

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Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.51	0.02	+0.00%	1.15	0.40	+0.00%	5.30	2.35	+0.00%	64.97	32.20	+0.00%	16.42	8.14	+0.00%
	BitsAndBytes	4	0.51	0.03	+0.59%	1.25	0.61	+9.08%	6.80	2.88	+28.36%	49.38	21.56	-24.00%	12.48	5.45	-24.00%
	BitsAndBytes	8	0.51	0.02	+0.12%	1.06	0.40	-7.53%	12.00	3.74	+126.28%	25.65	8.57	-60.51%	6.48	2.16	-60.51%
	Downcasting	16	0.51	0.02	+0.12%	1.15	0.41	+0.44%	3.39	1.69	-36.07%	102.86	47.17	+58.33%	25.99	11.92	+58.34%
	Quanto	2	0.49	0.03	-3.21%	1.08	0.52	-6.10%	9.06	4.05	+70.88%	38.14	18.66	-41.30%	9.64	4.72	-41.30%
	Quanto	4	0.51	0.02	+0.00%	1.16	0.41	+0.59%	8.61	3.77	+62.48%	39.69	18.74	-38.91%	10.03	4.74	-38.91%
	Quanto	8	0.51	0.02	+0.00%	1.15	0.40	-0.20%	5.81	2.40	+9.51%	57.55	25.47	-11.41%	14.54	6.44	-11.41%
Yes	None	32	0.59	0.11	+0.00%	0.69	0.01	+0.00%	5.26	2.40	+0.00%	66.00	32.96	+0.00%	16.68	8.33	+0.00%
	BitsAndBytes	4	0.58	0.11	-0.82%	0.70	0.01	+0.42%	6.79	2.89	+29.07%	49.55	21.60	-24.93%	12.52	5.46	-24.93%
	BitsAndBytes	8	0.57	0.12	-1.85%	0.70	0.01	+0.95%	11.21	3.19	+113.06%	26.98	8.26	-59.12%	6.82	2.09	-59.12%
	Downcasting	16	0.59	0.11	+0.10%	0.70	0.01	+0.11%	3.41	1.71	-35.18%	101.89	45.67	+54.38%	25.75	11.54	+54.38%
	Quanto	2	0.53	0.03	-9.97%	0.71	0.03	+2.05%	9.09	4.03	+72.80%	37.90	18.53	-42.58%	9.58	4.68	-42.58%
	Quanto	4	0.58	0.11	-1.23%	0.70	0.01	+0.47%	8.60	3.76	+63.43%	39.73	18.72	-39.80%	10.04	4.73	-39.80%
	Quanto	8	0.58	0.11	-0.10%	0.70	0.01	+0.20%	5.82	2.41	+10.62%	57.54	25.70	-12.82%	14.54	6.49	-12.82%

Table 5.6: Aggregated results on the glue/rte task

Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.50	0.01	+0.00%	1.11	0.51	+0.00%	17.08	6.72	+0.00%	60.81	26.33	+0.00%	15.20	6.58	+0.00%
	BitsAndBytes	4	0.50	0.01	-0.42%	1.18	0.50	+6.70%	19.53	7.98	+14.34%	53.37	21.49	-12.24%	13.34	5.37	-12.24%
	BitsAndBytes	8	0.50	0.01	+0.21%	1.19	0.54	+8.01%	37.45	16.73	+119.22%	28.47	12.78	-53.18%	7.12	3.19	-53.17%
	Downcasting	16	0.50	0.01	-0.02%	1.25	0.60	+12.92%	8.76	2.98	-48.73%	112.34	37.74	+84.74%	28.09	9.43	+84.74%
	Quanto	2	0.50	0.01	-0.82%	1.15	0.49	+3.69%	24.82	11.08	+45.28%	44.33	22.40	-27.10%	11.08	5.60	-27.10%
	Quanto	4	0.50	0.01	+0.21%	1.21	0.55	+9.54%	23.86	10.33	+39.66%	45.50	22.17	-25.18%	11.37	5.54	-25.18%
	Quanto	8	0.51	0.01	+0.44%	1.24	0.58	+12.06%	17.11	7.19	+0.17%	60.91	24.09	+0.17%	15.23	6.02	+0.17%
Yes	None	32	0.75	0.15	+0.00%	0.62	0.09	+0.00%	15.73	7.33	+0.00%	68.07	28.55	+0.00%	17.02	7.14	+0.00%
	BitsAndBytes	4	0.75	0.15	+0.00%	0.62	0.09	-0.29%	19.54	7.97	+24.27%	53.28	21.39	-21.73%	13.32	5.35	-21.73%
	BitsAndBytes	8	0.70	0.16	-6.03%	0.70	0.14	+12.56%	32.68	11.92	+107.76%	30.80	12.04	-54.75%	7.70	3.01	-54.75%
	Downcasting	16	0.75	0.15	-0.04%	0.62	0.09	+0.17%	8.72	3.02	-44.56%	113.25	38.45	+66.39%	28.31	9.61	+66.39%
	Quanto	2	0.72	0.15	-3.19%	0.67	0.10	+8.64%	24.79	11.04	+57.59%	44.38	22.42	-34.81%	11.09	5.61	-34.80%
	Quanto	4	0.74	0.15	-0.46%	0.62	0.09	-0.46%	23.75	10.22	+51.00%	45.61	22.11	-32.99%	11.40	5.53	-32.99%
	Quanto	8	0.75	0.15	-0.04%	0.62	0.09	+0.04%	17.06	7.25	+8.47%	61.35	24.55	-9.87%	15.34	6.14	-9.87%

Table 5.7: Aggregated results on the glue/sst2 task

Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.52	0.06	+0.00%	1.28	0.67	+0.00%	1.36	0.62	+0.00%	65.45	32.54	+0.00%	16.59	8.25	+0.00%
	BitsAndBytes	4	0.51	0.06	-3.14%	1.23	0.74	-3.70%	1.75	0.74	+28.52%	49.33	21.51	-24.62%	12.51	5.45	-24.62%
	BitsAndBytes	8	0.53	0.05	+1.79%	1.25	0.70	-2.42%	3.07	0.95	+126.34%	25.60	8.46	-60.89%	6.49	2.14	-60.89%
	Downcasting	16	0.52	0.06	+0.00%	1.28	0.67	+0.13%	0.86	0.43	-36.61%	103.43	46.43	+58.03%	26.22	11.77	+58.03%
	Quanto	2	0.53	0.07	+1.79%	1.35	0.94	+5.58%	2.34	1.03	+71.97%	37.84	18.50	-42.18%	9.59	4.69	-42.18%
	Quanto	4	0.50	0.07	-4.93%	1.27	0.65	-0.85%	2.22	0.98	+63.61%	39.49	18.67	-39.67%	10.01	4.73	-39.67%
	Quanto	8	0.53	0.07	+1.35%	1.28	0.67	-0.04%	1.49	0.62	+9.94%	57.36	25.45	-12.36%	14.54	6.45	-12.36%
Yes	None	32	0.50	0.06	+0.00%	0.70	0.02	+0.00%	1.35	0.61	+0.00%	65.28	31.42	+0.00%	16.55	7.97	+0.00%
	BitsAndBytes	4	0.45	0.04	-9.86%	0.71	0.02	+0.37%	1.75	0.74	+29.60%	49.27	21.52	-24.52%	12.49	5.46	-24.53%
	BitsAndBytes	8	0.50	0.06	+0.47%	0.70	0.02	-0.19%	2.97	0.85	+120.04%	26.14	8.10	-59.96%	6.63	2.05	-59.96%
	Downcasting	16	0.50	0.06	+0.47%	0.70	0.01	-0.03%	0.87	0.43	-35.69%	102.63	46.77	+57.22%	26.02	11.86	+57.22%
	Quanto	2	0.50	0.06	+0.94%	0.86	0.27	+21.49%	2.33	1.03	+72.89%	37.70	18.20	-42.24%	9.56	4.62	-42.24%
	Quanto	4	0.51	0.04	+2.35%	0.70	0.02	-0.07%	2.23	0.95	+65.21%	38.99	18.28	-40.28%	9.88	4.63	-40.28%
	Quanto	8	0.51	0.05	+1.88%	0.70	0.01	-0.02%	1.50	0.63	+11.16%	57.23	25.42	-12.33%	14.51	6.45	-12.33%

Table 5.8: Aggregated results on the glue/wnli task

5.2 Results

Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.43	0.09	+0.00%	1.10	0.38	+0.00%	79.99	37.99	+0.00%	54.98	31.59	+0.00%	13.75	7.90	+0.00%
	BitsAndBytes	4	0.45	0.10	+2.68%	1.44	0.77	+30.40%	81.95	31.17	+2.45%	47.40	20.38	-13.78%	11.86	5.10	-13.78%
	BitsAndBytes	8	0.43	0.08	-0.52%	0.99	0.27	-9.79%	146.51	54.50	+83.16%	26.02	11.07	-52.67%	6.51	2.77	-52.67%
	Downcasting	16	0.43	0.09	-0.08%	1.10	0.38	-0.05%	40.72	19.02	-49.09%	97.98	41.40	+78.20%	24.51	10.36	+78.20%
	Quanto	2	0.47	0.10	+8.52%	1.11	0.40	+0.34%	113.20	54.88	+41.52%	38.89	22.93	-29.26%	9.73	5.74	-29.26%
	Quanto	4	0.43	0.09	+0.15%	1.20	0.38	+9.08%	107.74	50.34	+34.69%	39.99	22.58	-27.27%	10.00	5.65	-27.27%
	Quanto	8	0.44	0.09	+0.49%	1.18	0.37	+6.94%	71.29	28.41	-10.88%	55.67	24.97	+1.25%	13.93	6.25	+1.25%
Yes	None	32	0.52	0.15	+0.00%	0.75	0.08	+0.00%	73.14	30.47	+0.00%	56.71	30.06	+0.00%	14.19	7.52	+0.00%
	BitsAndBytes	4	0.52	0.15	+0.31%	0.74	0.07	-1.09%	82.02	31.13	+12.13%	47.33	20.31	-16.54%	11.84	5.08	-16.54%
	BitsAndBytes	8	0.51	0.15	-1.63%	0.76	0.09	+1.21%	132.26	43.82	+80.82%	28.42	12.04	-49.88%	7.11	3.01	-49.88%
	Downcasting	16	0.52	0.15	+0.02%	0.75	0.08	+0.01%	40.78	18.94	-44.25%	97.62	40.99	+72.14%	24.42	10.25	+72.13%
	Quanto	2	0.53	0.11	+1.97%	0.77	0.15	+3.25%	113.18	54.81	+54.73%	38.86	22.86	-31.48%	9.72	5.72	-31.48%
	Quanto	4	0.50	0.15	-3.06%	0.76	0.07	+1.79%	107.59	50.24	+47.10%	40.01	22.55	-29.45%	10.01	5.64	-29.45%
	Quanto	8	0.50	0.16	-3.79%	0.76	0.08	+1.81%	71.35	28.47	-2.45%	55.64	24.99	-1.88%	13.92	6.25	-1.89%

Table 5.9: Aggregated results on the super_glue/boolq task

Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval F1 Score			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.45	0.06	+0.00%	0.28	0.05	+0.00%	1.77	0.84	+0.00%	1.06	0.43	+0.00%	63.21	26.91	+0.00%	15.80	6.73	+0.00%
	BitsAndBytes	4	0.43	0.08	-2.67%	0.28	0.06	-1.98%	1.76	0.71	-0.64%	1.37	0.54	+29.47%	48.50	19.88	-23.26%	12.13	4.97	-23.26%
	BitsAndBytes	8	0.42	0.06	-5.33%	0.29	0.05	+1.98%	1.69	0.80	-4.37%	2.46	0.94	+132.77%	26.55	11.10	-57.99%	6.64	2.77	-57.99%
	Downcasting	16	0.45	0.06	+0.00%	0.28	0.05	+0.00%	1.77	0.83	+0.10%	0.67	0.31	-36.20%	100.17	40.18	+58.47%	25.04	10.04	+58.48%
	Quanto	2	0.28	0.13	-36.67%	0.20	0.09	-27.65%	1.86	1.02	+5.46%	1.92	0.86	+81.69%	37.24	20.61	-41.08%	9.31	5.15	-41.08%
	Quanto	4	0.45	0.06	+0.00%	0.28	0.05	-0.00%	1.78	0.83	+0.98%	1.86	0.78	+75.97%	37.88	20.60	-40.07%	9.47	5.15	-40.07%
	Quanto	8	0.45	0.06	+0.00%	0.28	0.05	+0.00%	1.77	0.84	-0.02%	1.19	0.45	+12.15%	55.44	22.26	-12.30%	13.86	5.57	-12.30%
Yes	None	32	0.63	0.26	+0.00%	0.61	0.29	+0.00%	0.94	0.21	+0.00%	1.06	0.43	+0.00%	62.85	26.53	+0.00%	15.71	6.63	+0.00%
	BitsAndBytes	4	0.69	0.15	+8.96%	0.65	0.20	+6.46%	0.90	0.26	-4.18%	1.37	0.54	+29.27%	48.46	20.07	-22.90%	12.12	5.02	-22.90%
	BitsAndBytes	8	0.68	0.17	+7.08%	0.65	0.21	+5.80%	0.98	0.25	+4.24%	2.37	0.74	+122.40%	26.76	10.53	-57.43%	6.69	2.63	-57.43%
	Downcasting	16	0.70	0.14	+10.85%	0.66	0.19	+8.68%	0.94	0.21	-0.12%	0.68	0.32	-35.95%	99.66	40.21	+58.56%	24.91	10.05	+58.56%
	Quanto	2	0.46	0.29	-26.42%	0.41	0.33	-32.27%	1.19	0.19	+26.12%	1.99	0.85	+86.90%	36.25	20.96	-42.33%	9.06	5.24	-42.33%
	Quanto	4	0.67	0.18	+6.13%	0.63	0.23	+2.48%	0.96	0.20	+1.84%	1.83	0.79	+72.30%	38.41	20.37	-38.89%	9.60	5.09	-38.89%
	Quanto	8	0.70	0.14	+10.85%	0.66	0.19	+8.68%	0.94	0.21	-0.31%	1.18	0.44	+11.23%	55.45	22.20	-11.77%	13.86	5.55	-11.77%

Table 5.10: Aggregated results on the super_glue/cb task

Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.51	0.05	+0.00%	1.25	0.65	+0.00%	1.89	0.76	+0.00%	62.73	25.93	+0.00%	15.68	6.48	+0.00%
	BitsAndBytes	4	0.51	0.06	+0.33%	1.19	0.85	-4.94%	2.46	0.96	+29.76%	48.19	19.62	-23.18%	12.05	4.90	-23.18%
	BitsAndBytes	8	0.52	0.05	+1.64%	1.22	0.65	-2.65%	4.40	1.66	+132.43%	26.43	10.94	-57.87%	6.61	2.73	-57.87%
	Downcasting	16	0.51	0.05	+0.00%	1.26	0.66	+0.44%	1.21	0.56	-36.17%	99.96	39.90	+59.35%	24.99	9.98	+59.35%
	Quanto	2	0.51	0.05	+0.00%	1.34	0.90	+6.55%	3.40	1.52	+79.72%	37.54	20.65	-40.15%	9.39	5.16	-40.15%
	Quanto	4	0.51	0.05	-0.33%	1.24	0.63	-1.18%	3.23	1.40	+70.59%	38.92	20.46	-37.96%	9.73	5.11	-37.96%
	Quanto	8	0.51	0.05	+0.00%	1.26	0.65	+0.17%	2.11	0.79	+11.56%	55.35	21.91	-11.76%	13.84	5.48	-11.76%
Yes	None	32	0.46	0.03	+0.00%	0.78	0.07	+0.00%	1.90	0.77	+0.00%	62.61	25.88	+0.00%	15.65	6.47	+0.00%
	BitsAndBytes	4	0.45	0.04	-1.82%	0.78	0.06	-1.07%	2.45	0.96	+29.36%	48.37	19.93	-22.74%	12.09	4.98	-22.74%
	BitsAndBytes	8	0.48	0.05	+4.01%	0.77	0.08	-1.31%	4.22	1.36	+122.23%	26.90	10.57	-57.04%	6.72	2.64	-57.05%
	Downcasting	16	0.46	0.04	+0.73%	0.78	0.07	+0.02%	1.23	0.56	-35.23%	97.88	39.10	+56.32%	24.47	9.77	+56.32%
	Quanto	2	0.51	0.05	+12.41%	0.84	0.17	+6.58%	3.40	1.52	+79.10%	37.60	20.68	-39.95%	9.40	5.17	-39.95%
	Quanto	4	0.46	0.03	+0.73%	0.78	0.06	-0.28%	3.23	1.39	+70.04%	38.89	20.40	-37.89%	9.72	5.10	-37.89%
	Quanto	8	0.47	0.04	+2.19%	0.78	0.07	-0.02%	2.11	0.80	+11.49%	55.35	21.94	-11.61%	13.84	5.48	-11.61%

Table 5.11: Aggregated results on the super_glue/copa task

5. EXPERIMENTS

Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.50	0.00	+0.00%	1.29	0.60	+0.00%	12.12	4.85	+0.00%	62.48	26.06	+0.00%	15.67	6.54	+0.00%
	BitsAndBytes	4	0.50	0.00	-0.63%	1.27	0.71	-2.07%	15.68	6.13	+29.34%	48.25	19.82	-22.78%	12.10	4.97	-22.78%
	BitsAndBytes	8	0.50	0.01	-0.94%	1.22	0.58	-5.46%	28.10	10.56	+131.81%	26.37	10.81	-57.79%	6.61	2.71	-57.79%
	Downcasting	16	0.50	0.00	+0.00%	1.30	0.60	+0.24%	7.84	3.62	-35.34%	98.58	40.18	+57.79%	24.72	10.08	+57.79%
	Quanto	2	0.50	0.00	+0.21%	1.33	0.87	+2.46%	24.30	11.98	+100.49%	35.33	21.56	-43.45%	8.86	5.41	-43.44%
	Quanto	4	0.50	0.01	-0.73%	1.29	0.58	-0.61%	23.51	11.27	+93.95%	35.91	21.34	-42.52%	9.01	5.35	-42.52%
	Quanto	8	0.50	0.00	+0.16%	1.30	0.60	+0.25%	13.50	5.06	+11.42%	55.30	22.13	-11.48%	13.87	5.55	-11.48%
Yes	None	32	0.54	0.04	+0.00%	0.79	0.15	+0.00%	12.16	4.88	+0.00%	62.37	26.15	+0.00%	15.64	6.56	+0.00%
	BitsAndBytes	4	0.54	0.04	-0.24%	0.78	0.13	-2.10%	15.75	6.19	+29.59%	48.11	19.90	-22.87%	12.07	4.99	-22.86%
	BitsAndBytes	8	0.53	0.04	-0.97%	0.79	0.15	+0.03%	27.03	10.52	+122.33%	27.49	10.79	-55.93%	6.89	2.71	-55.93%
	Downcasting	16	0.54	0.04	+0.15%	0.79	0.15	-0.02%	7.85	3.60	-35.45%	98.11	39.53	+57.30%	24.60	9.91	+57.30%
	Quanto	2	0.53	0.03	-1.99%	0.72	0.03	-9.67%	25.09	12.17	+106.37%	34.37	21.73	-44.90%	8.62	5.45	-44.90%
	Quanto	4	0.53	0.04	-0.78%	0.80	0.18	+1.32%	23.64	10.80	+94.47%	35.39	21.30	-43.26%	8.88	5.34	-43.26%
	Quanto	8	0.54	0.04	+0.05%	0.79	0.15	-0.09%	13.46	5.02	+10.73%	55.39	22.04	-11.18%	13.89	5.53	-11.18%

Table 5.12: Aggregated results on the super_glue/wic task

Finetuned	Quantization Method	# Bits	Eval Accuracy			Eval Loss			Eval Runtime			Eval Samples Per Second			Eval Steps Per Second		
			μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ	μ	σ	% Δ
No	None	32	0.50	0.12	+0.00%	1.30	0.78	+0.00%	1.97	0.79	+0.00%	62.51	25.85	+0.00%	15.63	6.46	+0.00%
	BitsAndBytes	4	0.50	0.12	-0.32%	1.34	0.90	+3.51%	2.55	1.00	+29.10%	48.42	19.83	-22.54%	12.11	4.96	-22.54%
	BitsAndBytes	8	0.51	0.12	+1.92%	1.23	0.75	-5.12%	4.55	1.73	+130.58%	26.67	11.15	-57.33%	6.67	2.79	-57.33%
	Downcasting	16	0.50	0.12	+0.00%	1.29	0.78	-0.21%	1.26	0.58	-36.13%	99.65	40.09	+59.40%	24.91	10.02	+59.40%
	Quanto	2	0.47	0.12	-5.43%	1.43	1.07	+10.26%	3.53	1.59	+78.75%	37.69	20.67	-39.70%	9.42	5.17	-39.70%
	Quanto	4	0.51	0.12	+1.28%	1.28	0.74	-1.57%	3.35	1.45	+69.97%	38.90	20.37	-37.78%	9.72	5.09	-37.78%
	Quanto	8	0.50	0.12	+0.64%	1.30	0.78	+0.21%	2.20	0.83	+11.31%	55.44	21.97	-11.32%	13.86	5.49	-11.32%
Yes	None	32	0.50	0.10	+0.00%	0.70	0.02	+0.00%	1.97	0.79	+0.00%	62.51	25.64	+0.00%	15.63	6.41	+0.00%
	BitsAndBytes	4	0.53	0.11	+7.77%	0.69	0.01	-0.62%	2.56	1.01	+30.06%	48.18	19.93	-22.93%	12.04	4.98	-22.92%
	BitsAndBytes	8	0.51	0.10	+3.24%	0.70	0.02	-0.03%	4.50	1.58	+128.25%	26.44	10.40	-57.71%	6.61	2.60	-57.70%
	Downcasting	16	0.49	0.10	-0.65%	0.70	0.02	-0.00%	1.28	0.58	-35.26%	98.04	39.20	+56.85%	24.51	9.80	+56.85%
	Quanto	2	0.54	0.12	+9.06%	1.01	0.52	+45.00%	3.54	1.58	+79.50%	37.53	20.69	-39.95%	9.38	5.17	-39.95%
	Quanto	4	0.53	0.11	+6.15%	0.69	0.02	-1.48%	3.36	1.45	+70.32%	38.84	20.37	-37.87%	9.71	5.09	-37.87%
	Quanto	8	0.50	0.11	+0.32%	0.70	0.02	+0.09%	2.19	0.82	+11.27%	55.42	21.92	-11.34%	13.86	5.48	-11.33%

Table 5.13: Aggregated results on the super_glue/wsc task

5.2.2 Aggregate figures

The following figures are an aggregate of each model result per task. For each bit-precision and quantization method, a box-plot is presented, displaying the distribution of the results of each model. For the sake of completeness, detailed versions of each individual model performance on the benchmark tasks can be found in Appendix E.

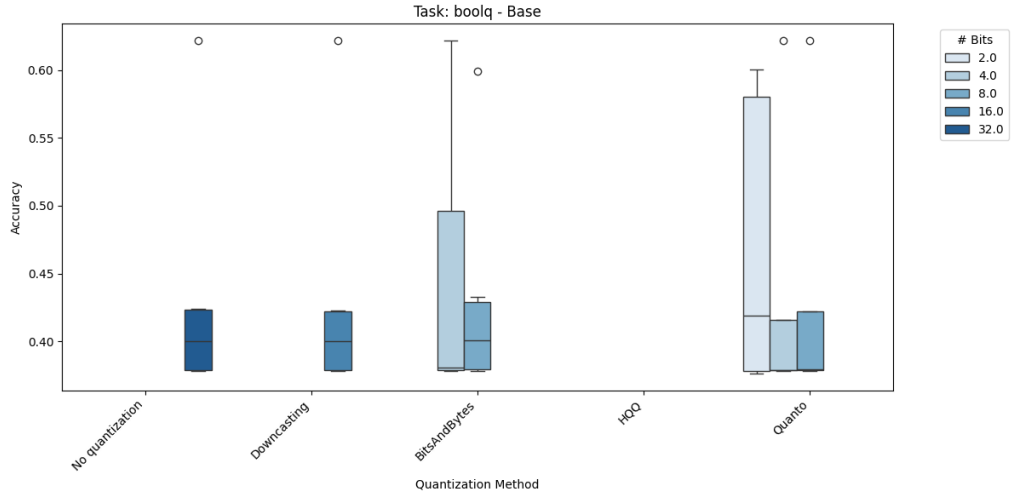


Figure 5.1: Distribution of Accuracy for boolq with base

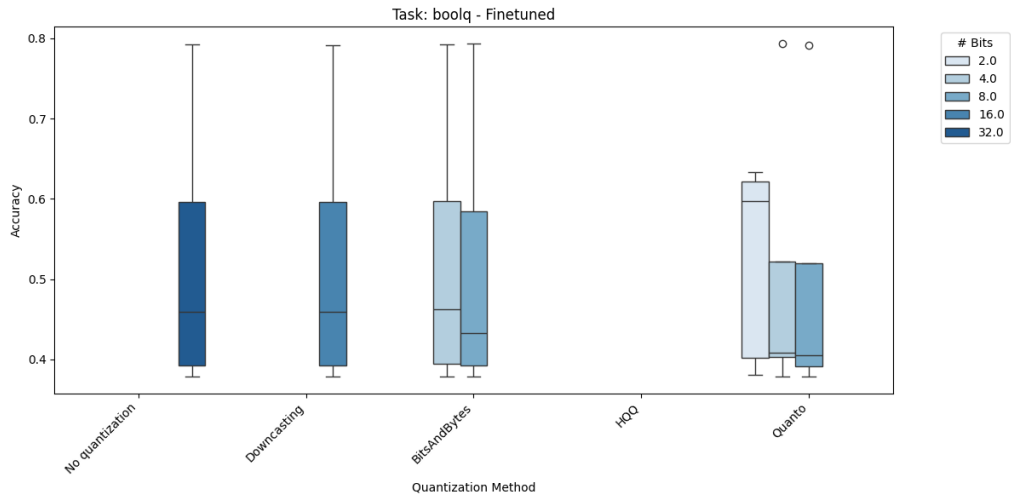


Figure 5.2: Distribution of Accuracy for boolq with finetuned

5. EXPERIMENTS

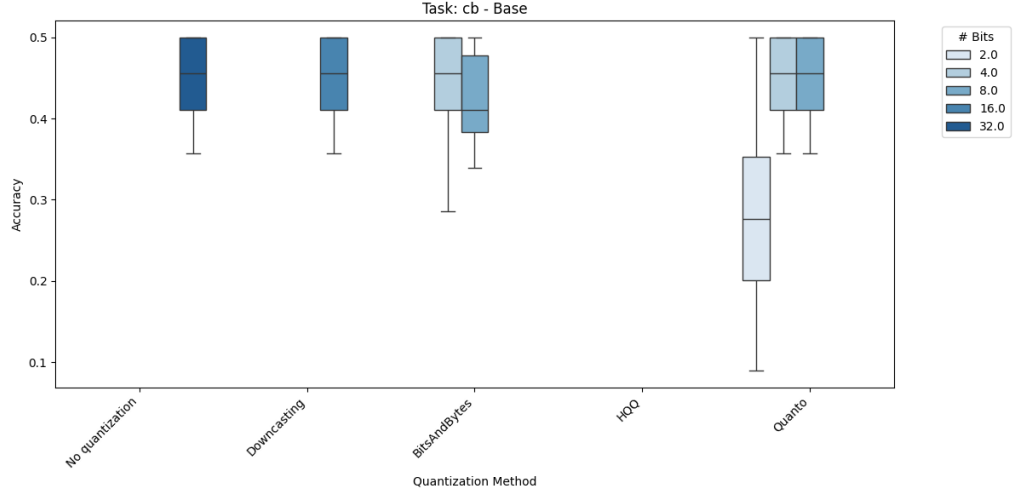


Figure 5.3: Distribution of Accuracy for cb with base

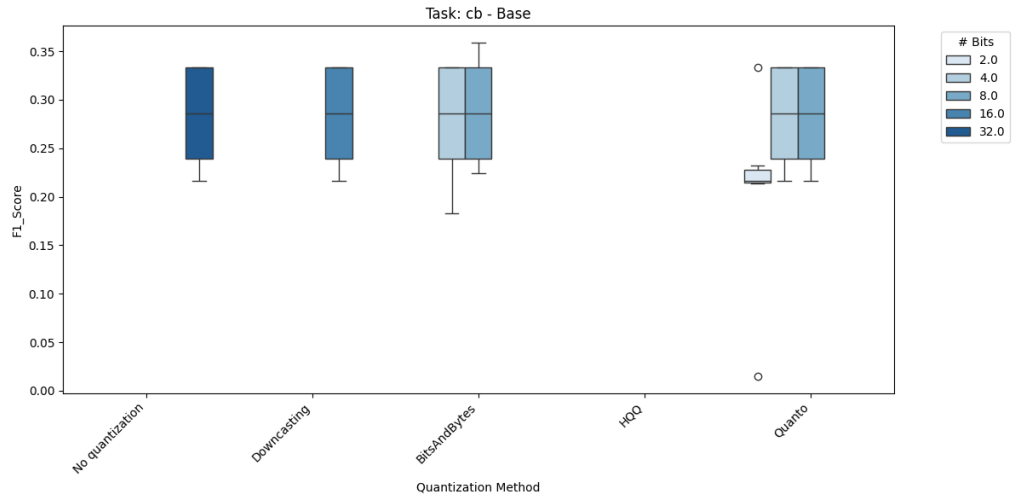


Figure 5.4: Distribution of F1_Score for cb with base

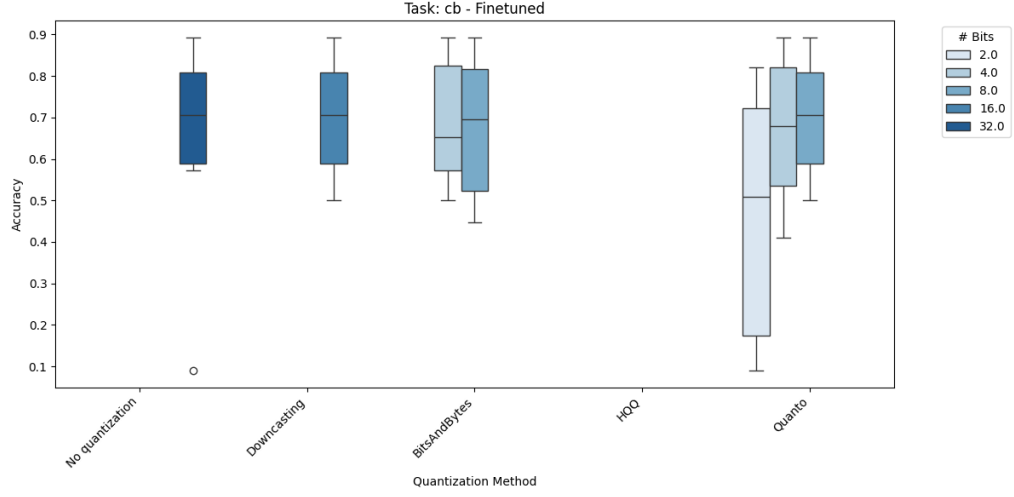


Figure 5.5: Distribution of Accuracy for cb with finetuned

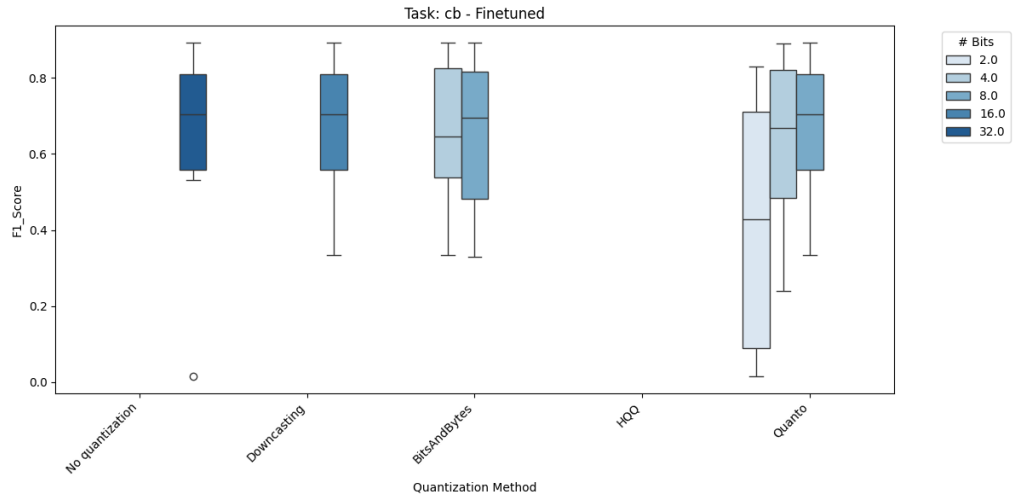


Figure 5.6: Distribution of F1_Score for cb with finetuned

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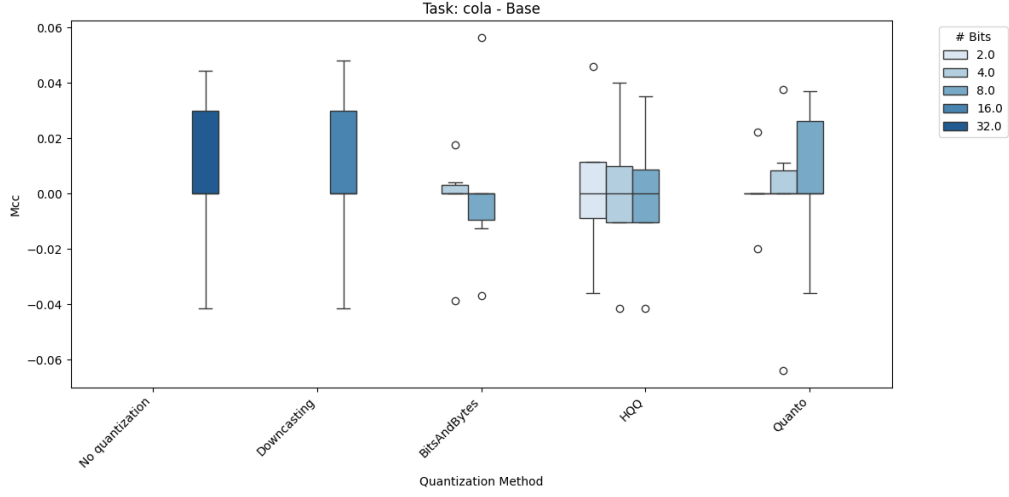


Figure 5.7: Distribution of Mcc for cola with base

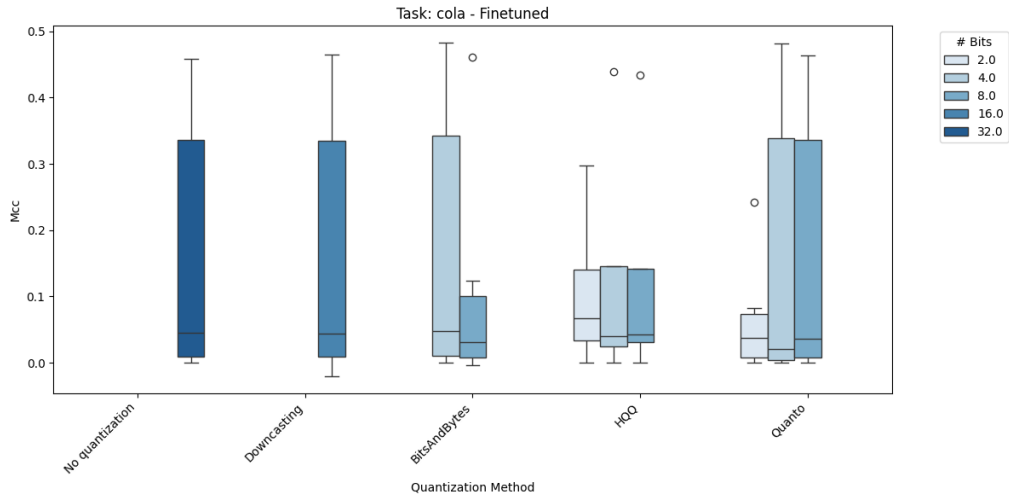


Figure 5.8: Distribution of Mcc for cola with finetuned

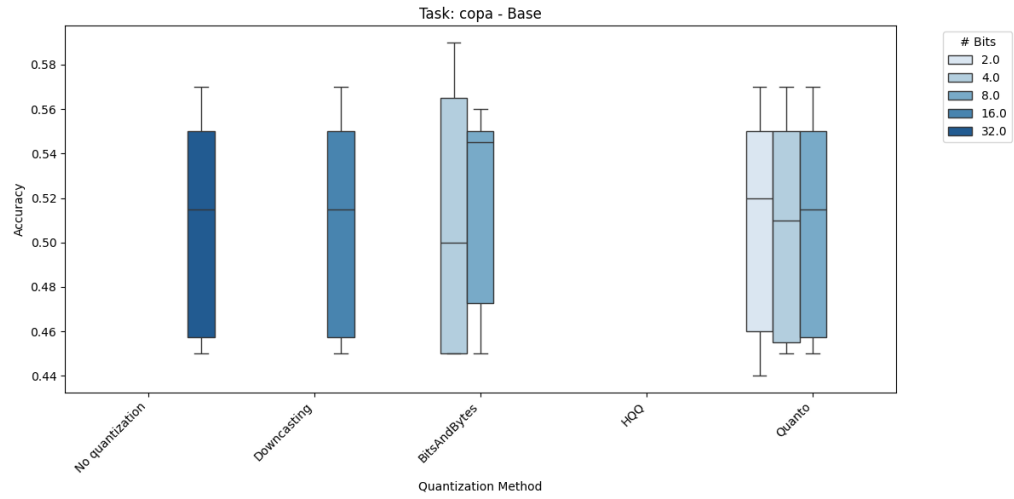


Figure 5.9: Distribution of Accuracy for copa with base

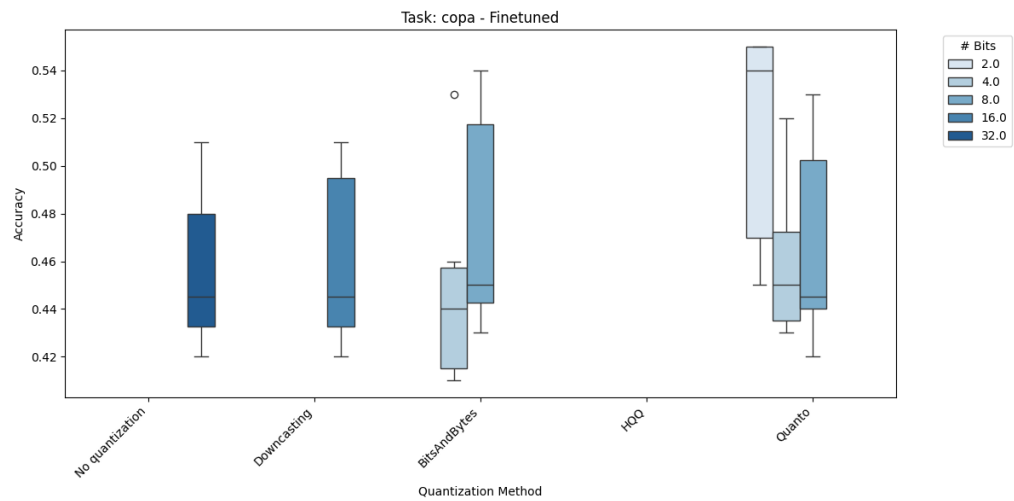


Figure 5.10: Distribution of Accuracy for copa with finetuned

5. EXPERIMENTS

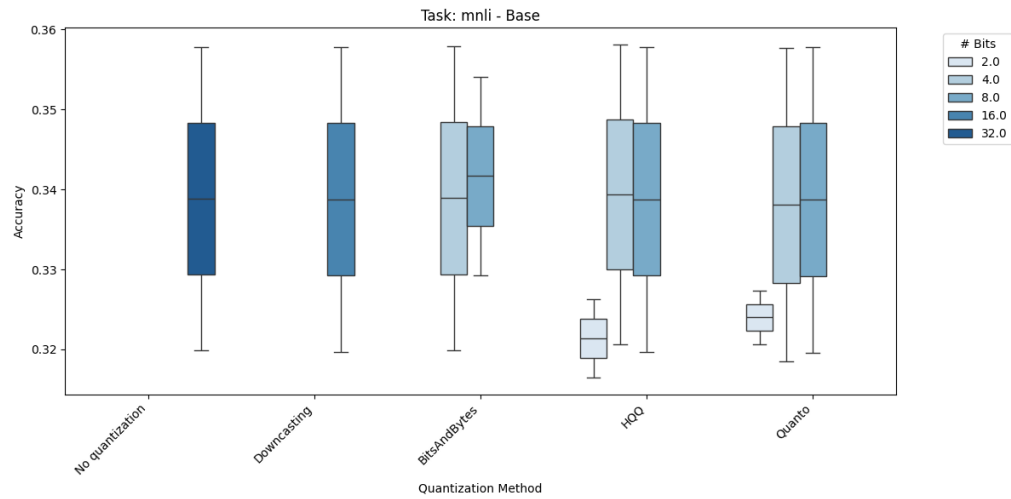


Figure 5.11: Distribution of Accuracy for mnli with base

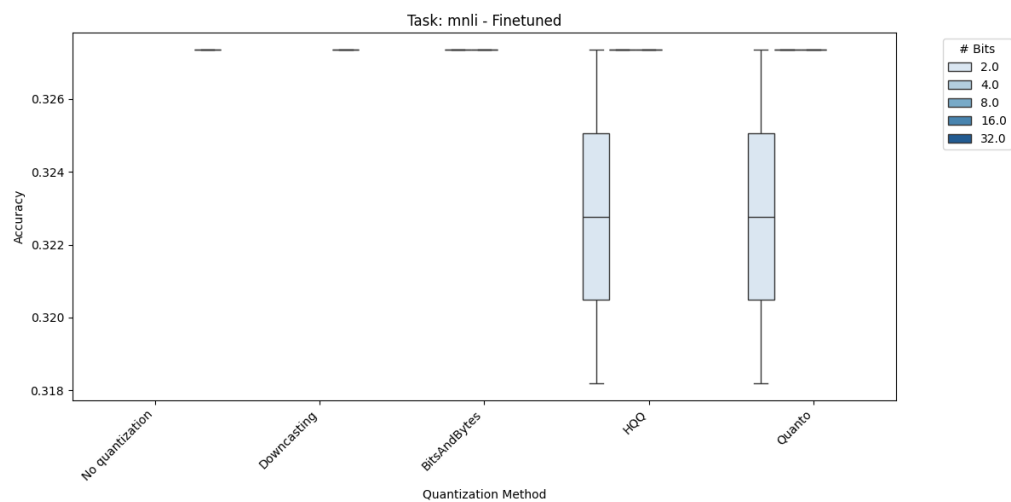


Figure 5.12: Distribution of Accuracy for mnli with finetuned

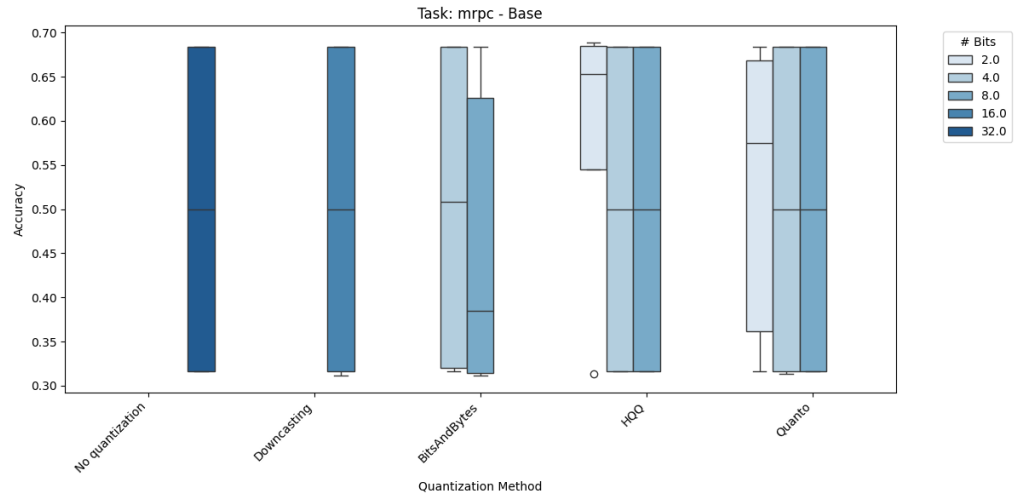


Figure 5.13: Distribution of Accuracy for mrpc with base

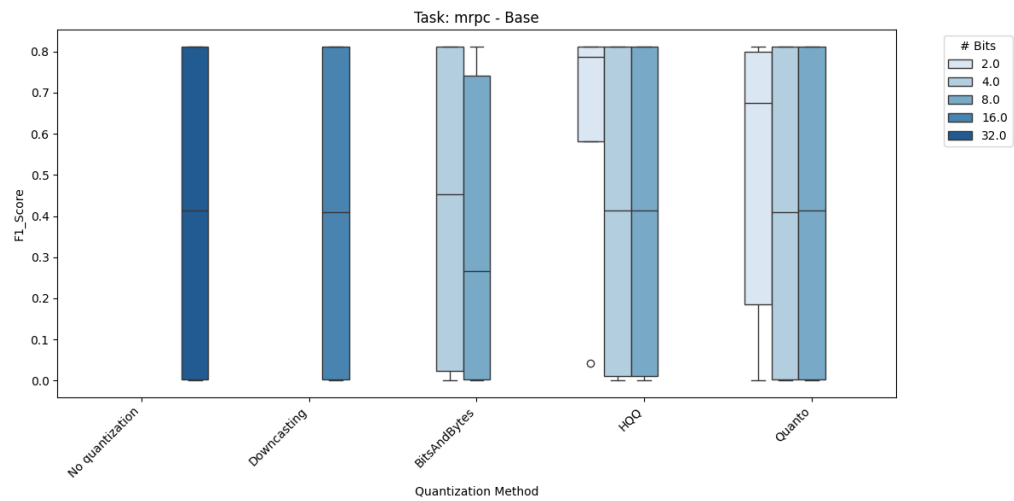


Figure 5.14: Distribution of F1_Score for mrpc with base

5. EXPERIMENTS

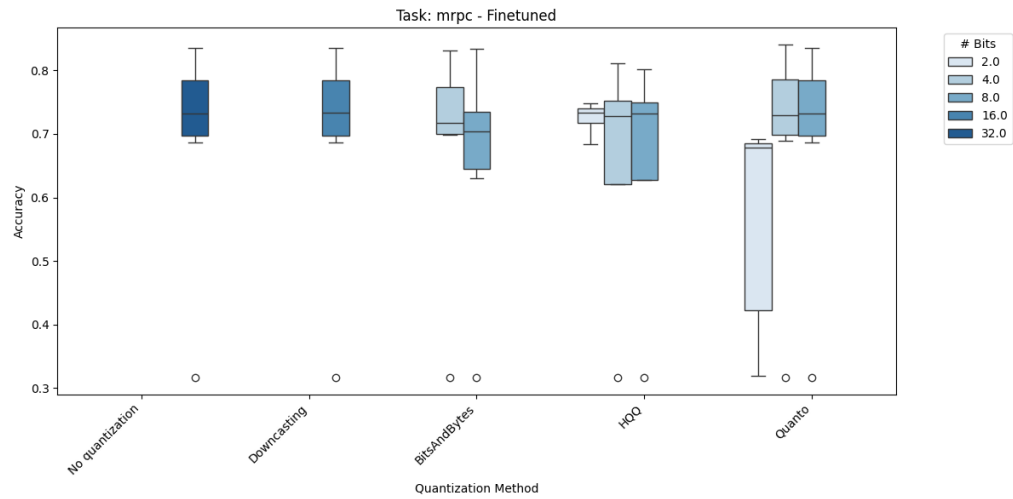


Figure 5.15: Distribution of Accuracy for mrpc with finetuned

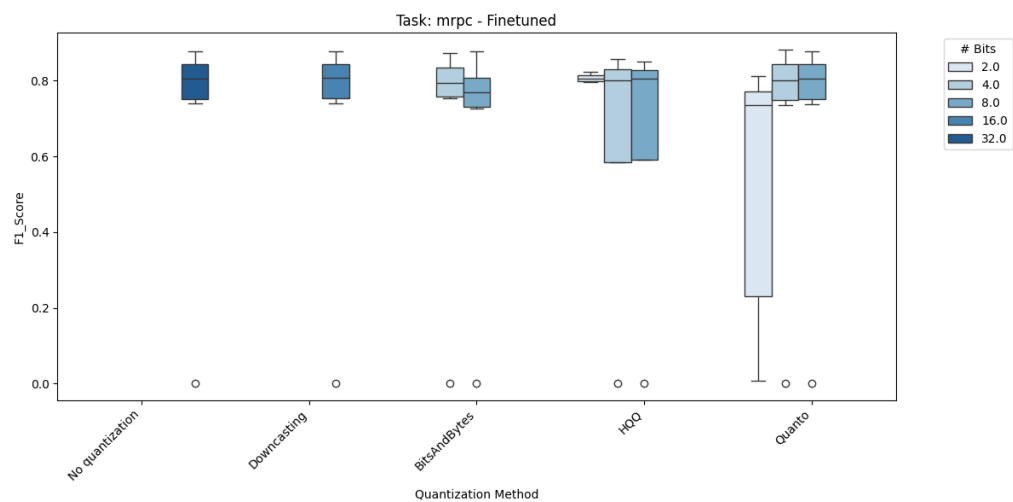


Figure 5.16: Distribution of F1_Score for mrpc with finetuned

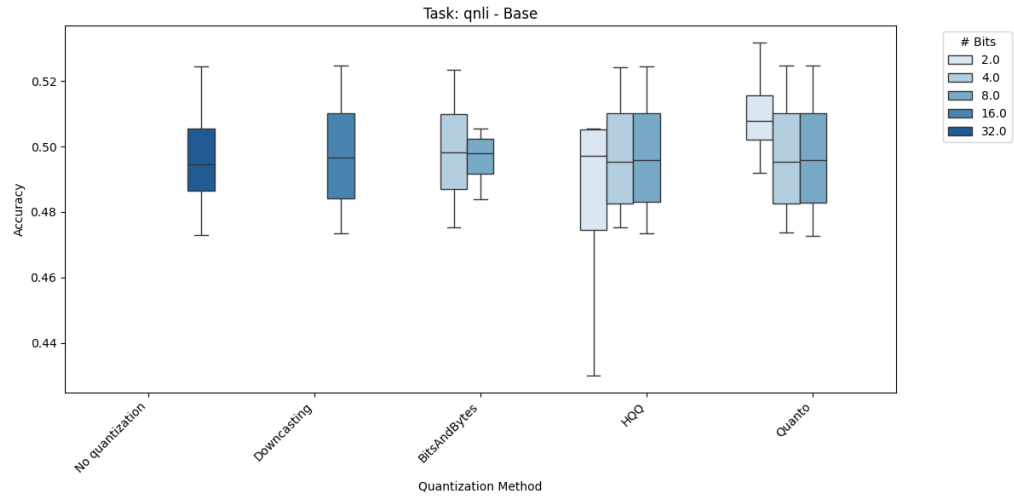


Figure 5.17: Distribution of Accuracy for qnli with base

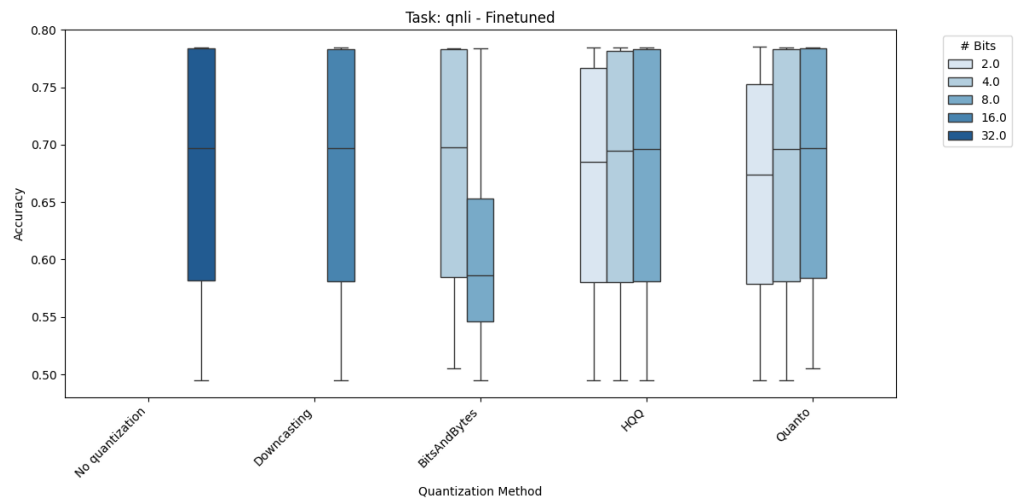


Figure 5.18: Distribution of Accuracy for qnli with finetuned

5. EXPERIMENTS

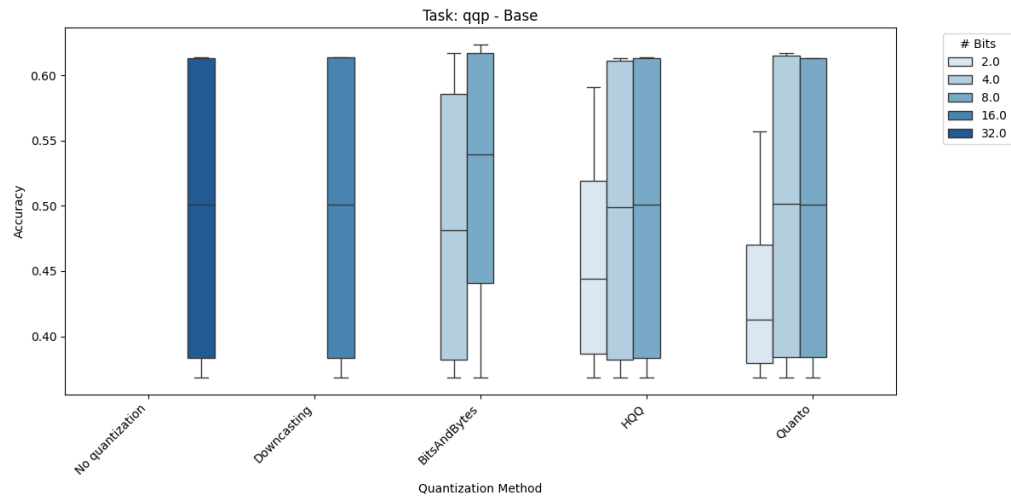


Figure 5.19: Distribution of Accuracy for qqp with base

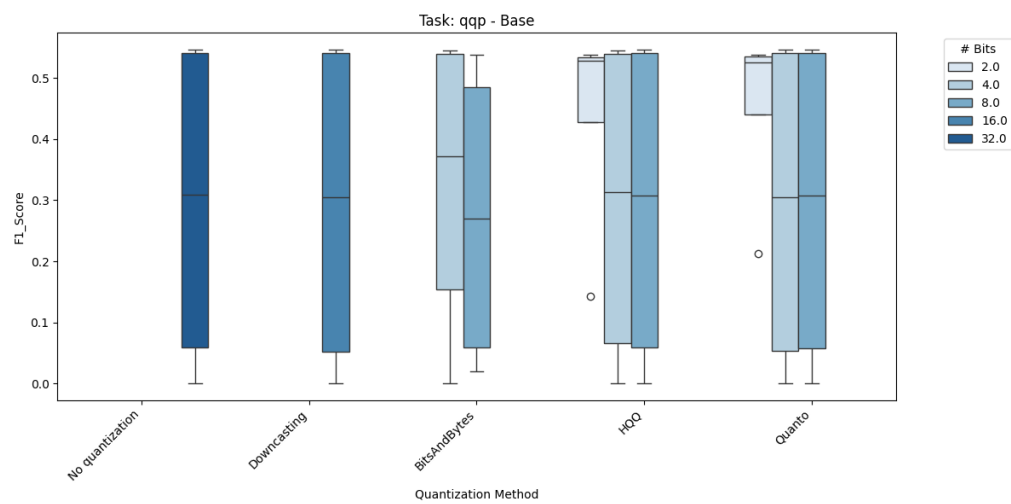


Figure 5.20: Distribution of F1_Score for qqp with base

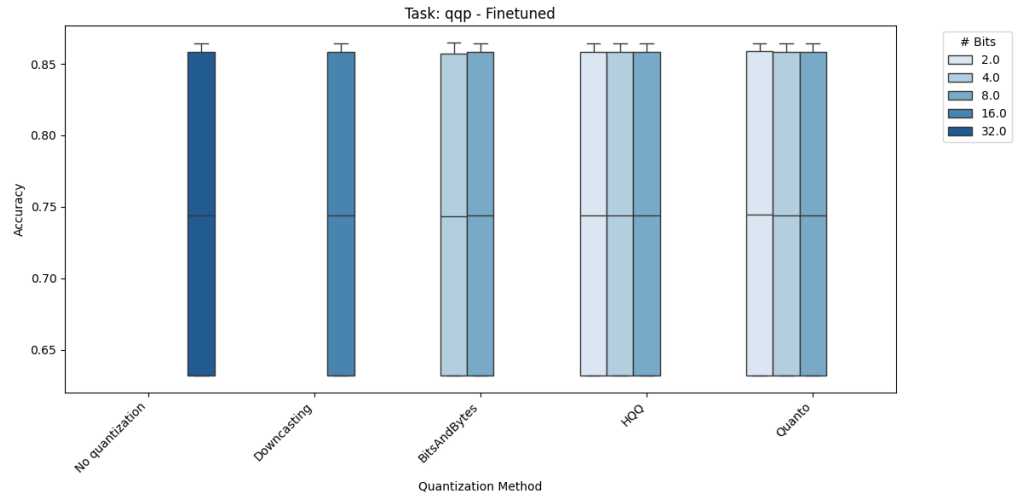


Figure 5.21: Distribution of Accuracy for qqp with finetuned

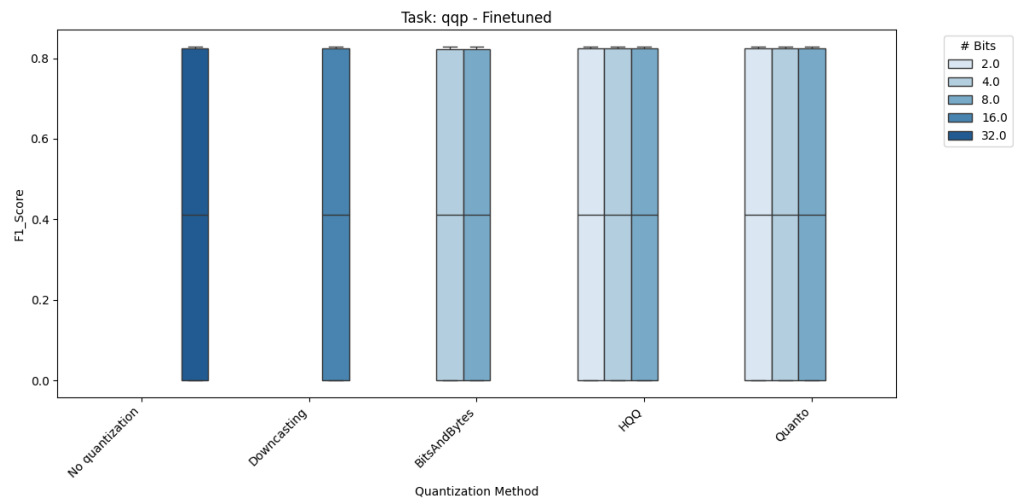


Figure 5.22: Distribution of F1_Score for qqp with finetuned

5. EXPERIMENTS

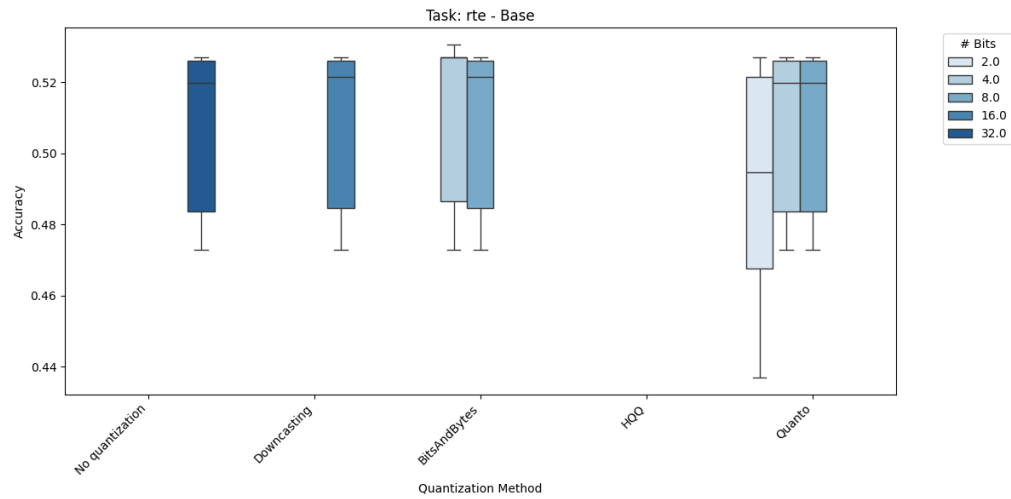


Figure 5.23: Distribution of Accuracy for rte with base

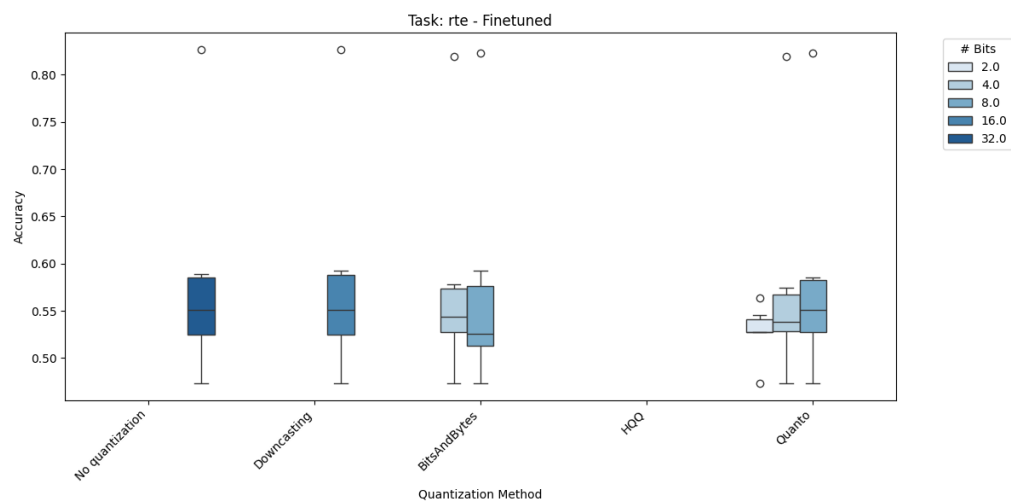


Figure 5.24: Distribution of Accuracy for rte with finetuned

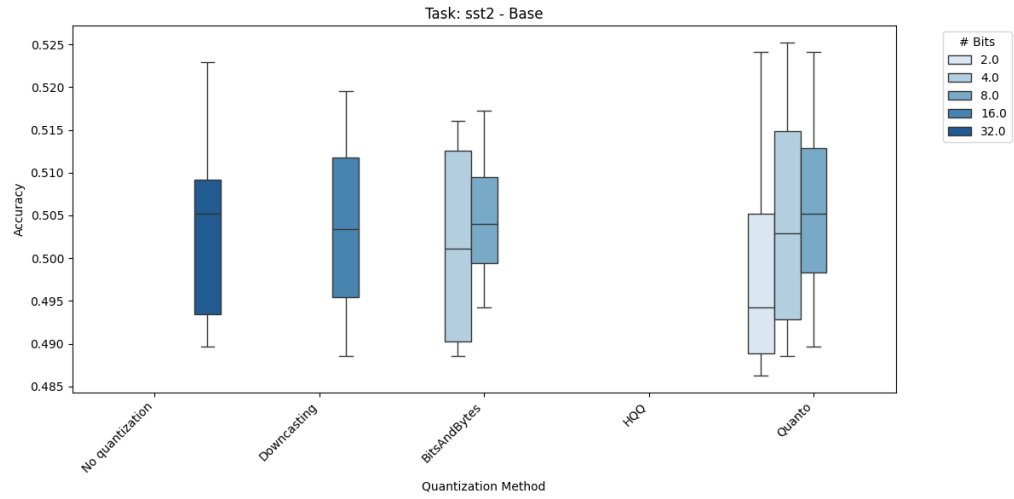


Figure 5.25: Distribution of Accuracy for sst2 with base

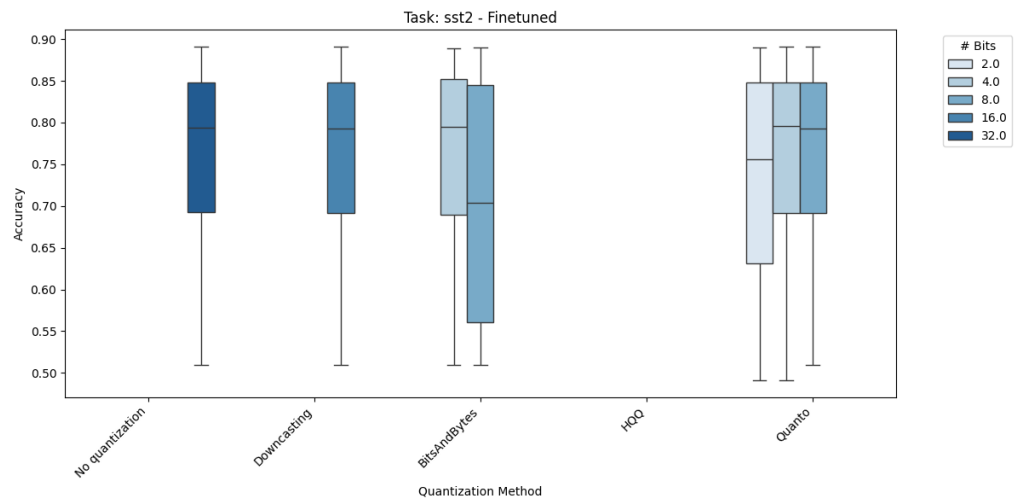


Figure 5.26: Distribution of Accuracy for sst2 with finetuned

5. EXPERIMENTS

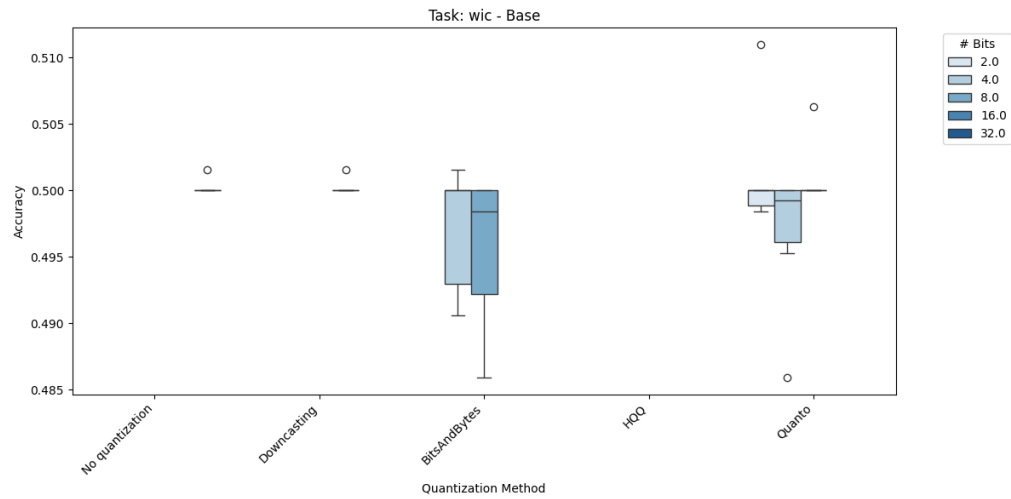


Figure 5.27: Distribution of Accuracy for wic with base

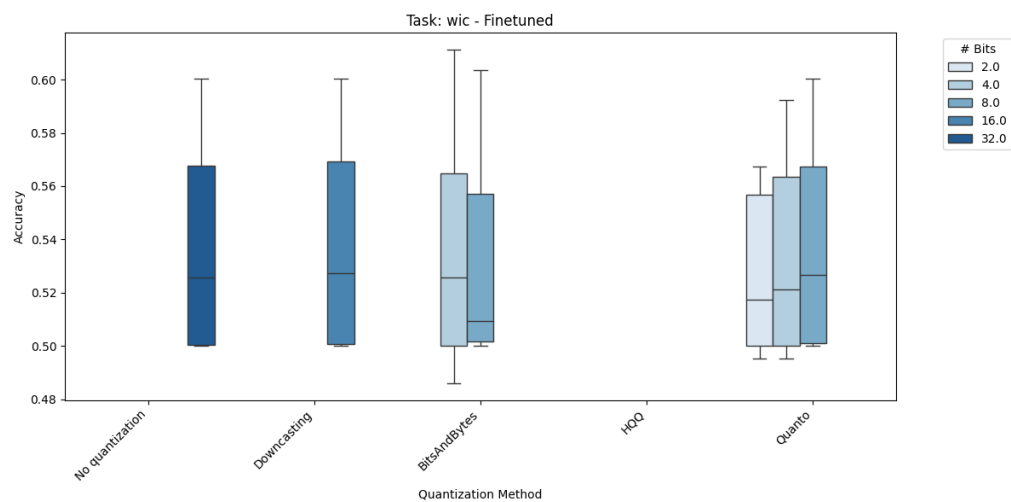


Figure 5.28: Distribution of Accuracy for wic with finetuned

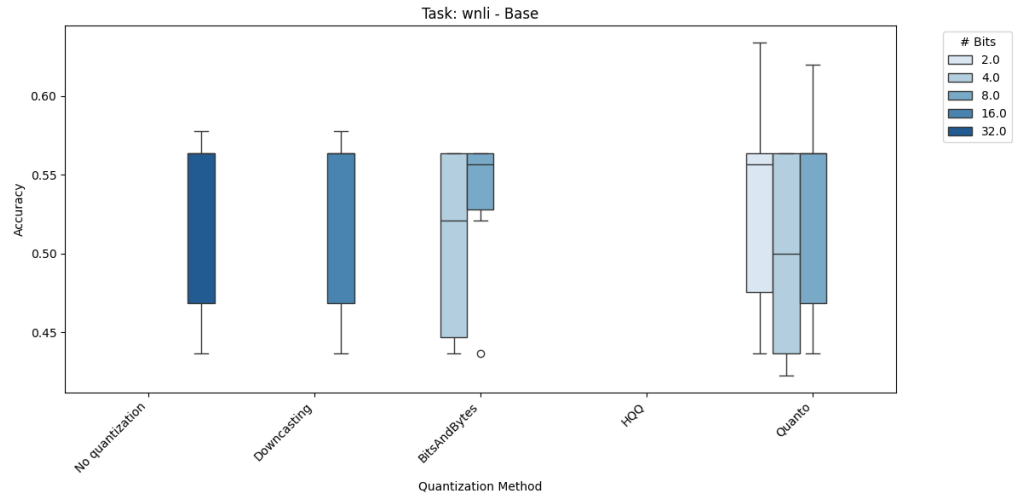


Figure 5.29: Distribution of Accuracy for wnli with base

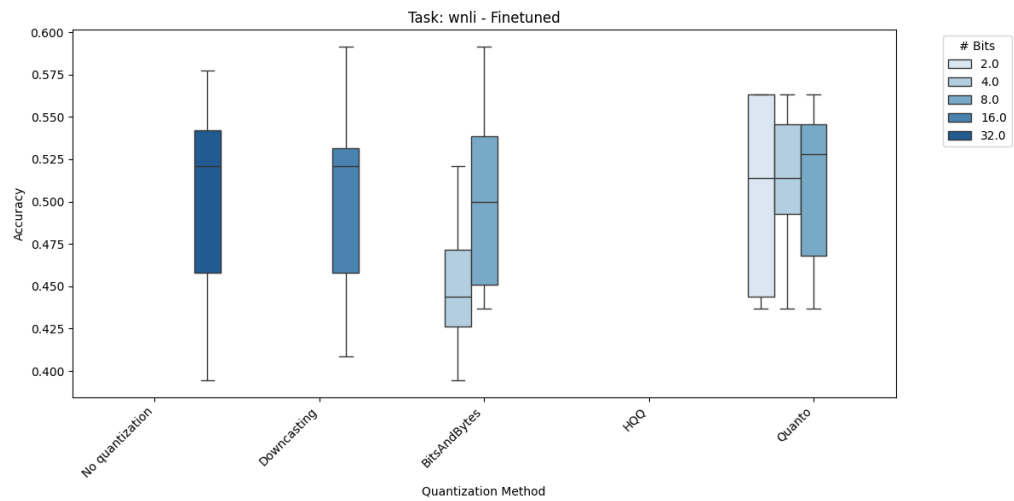


Figure 5.30: Distribution of Accuracy for wnli with finetuned

5. EXPERIMENTS

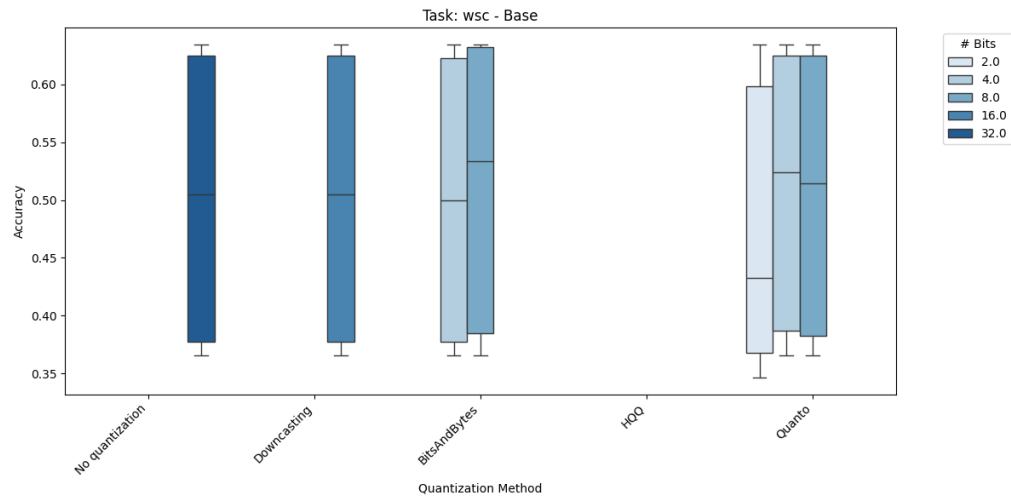


Figure 5.31: Distribution of Accuracy for wsc with base

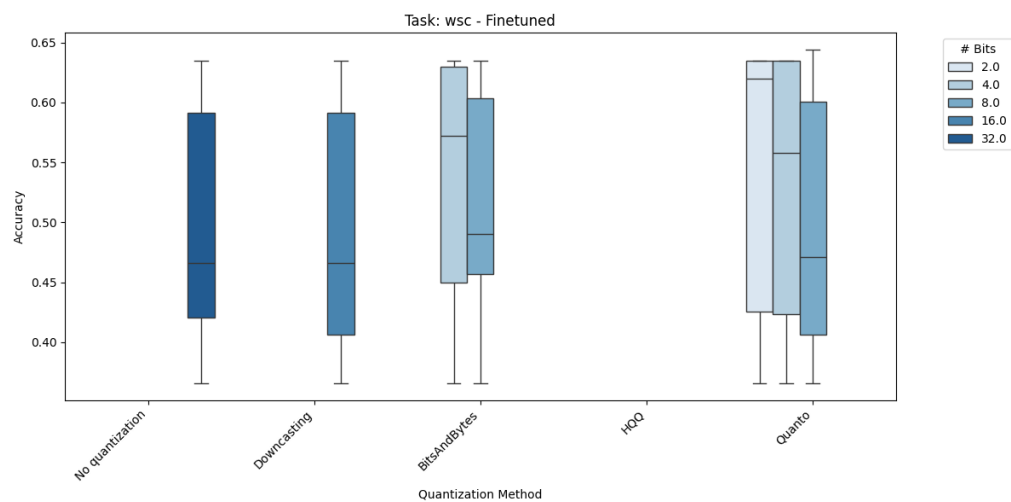


Figure 5.32: Distribution of Accuracy for wsc with finetuned

5.2.3 Pareto fronts

The following figures showcase a Pareto front (explained in Section 2.1.7) for each task. For each bit precision, the best model is highlighted. These figures show the optimal model at each bit precision per task.

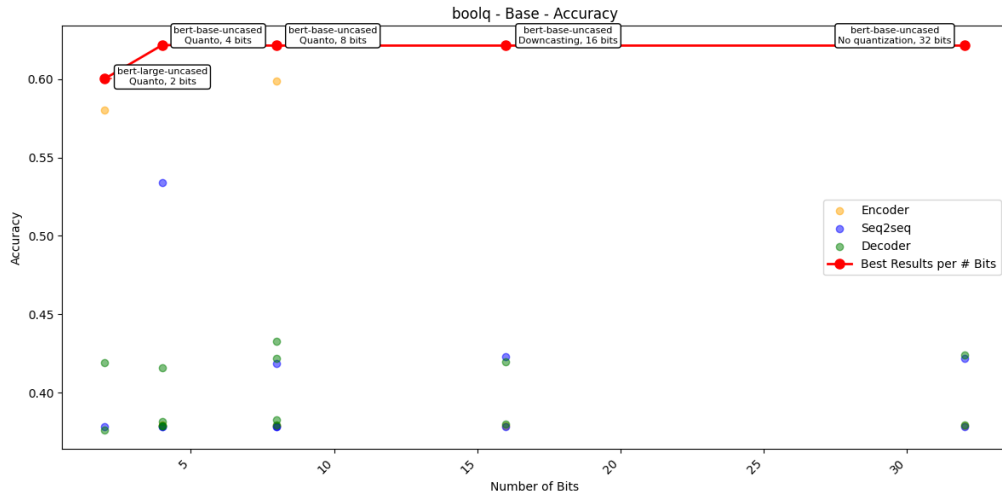


Figure 5.33: Pareto front for Accuracy in boolq with base

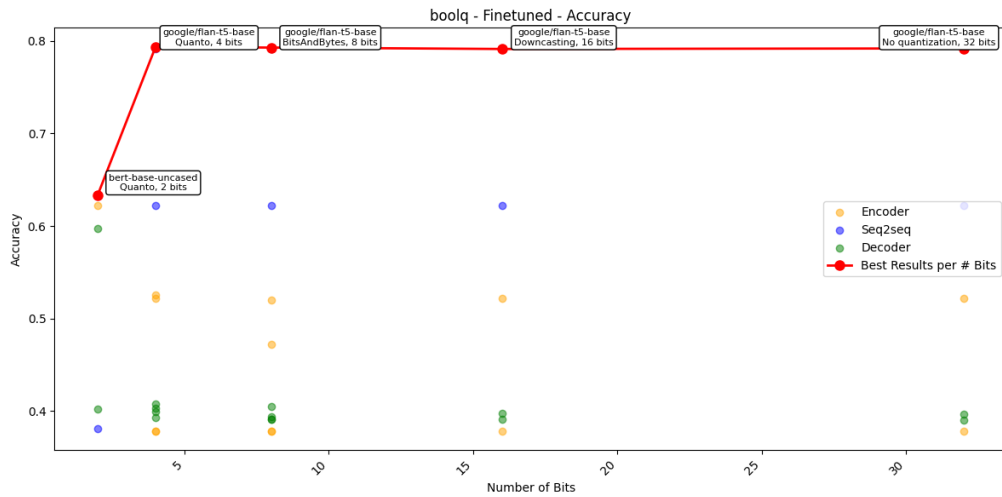


Figure 5.34: Pareto front for Accuracy in boolq with finetuned

5. EXPERIMENTS

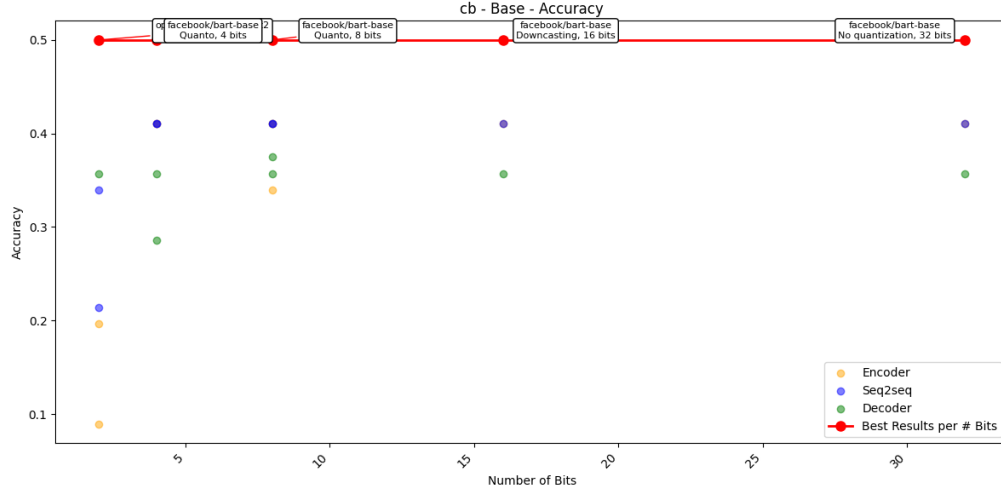


Figure 5.35: Pareto front for Accuracy in cb with base

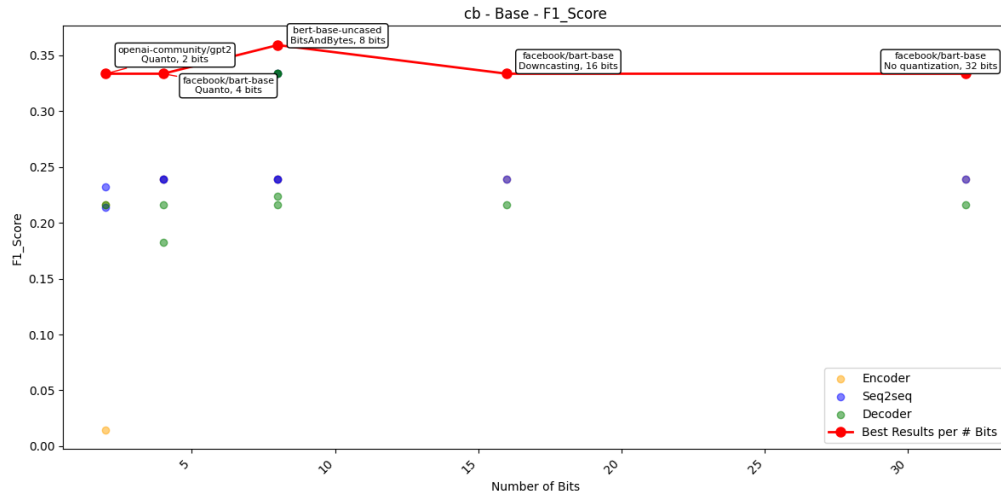


Figure 5.36: Pareto front for F1_Score in cb with base

5.2 Results

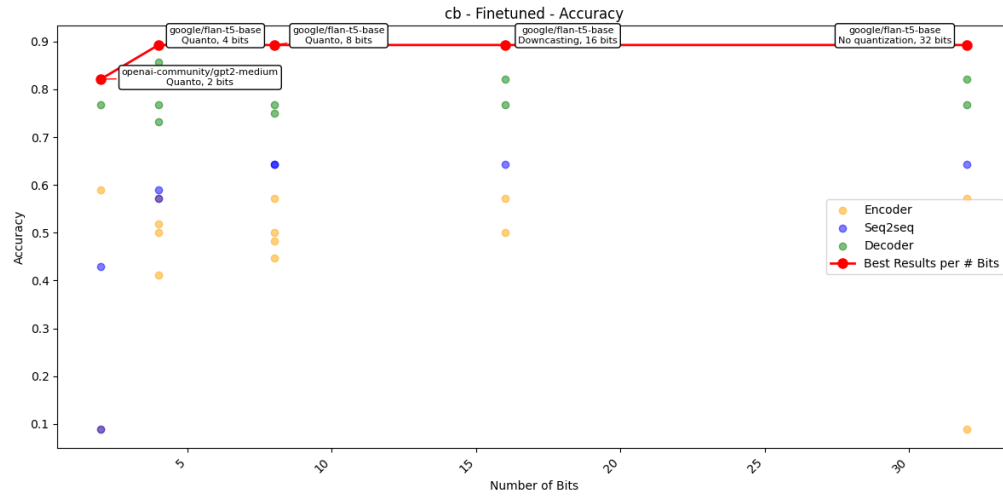


Figure 5.37: Pareto front for Accuracy in cb with finetuned

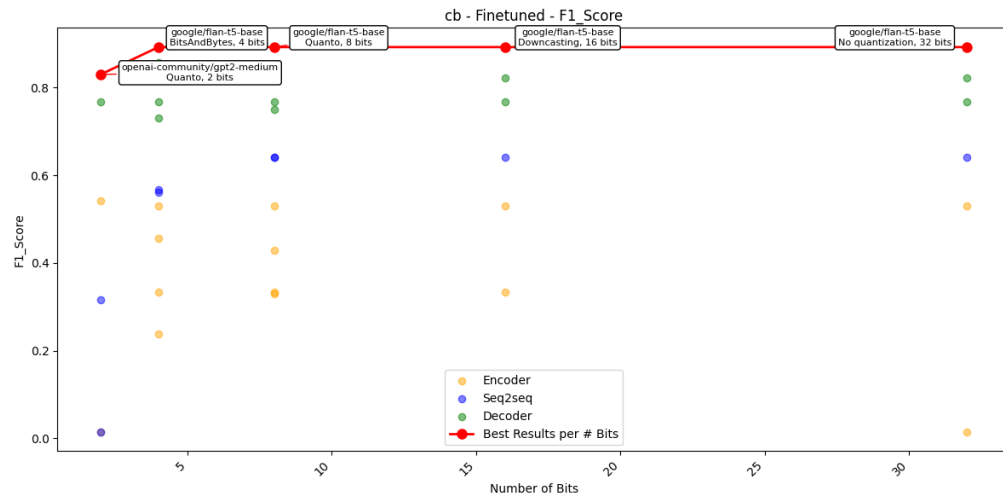


Figure 5.38: Pareto front for F1_Score in cb with finetuned

5. EXPERIMENTS

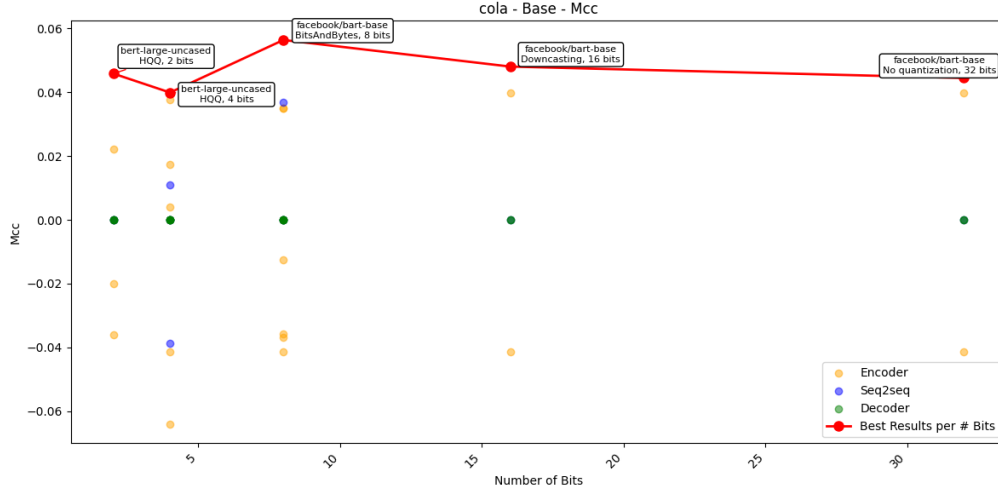


Figure 5.39: Pareto front for Mcc in cola with base

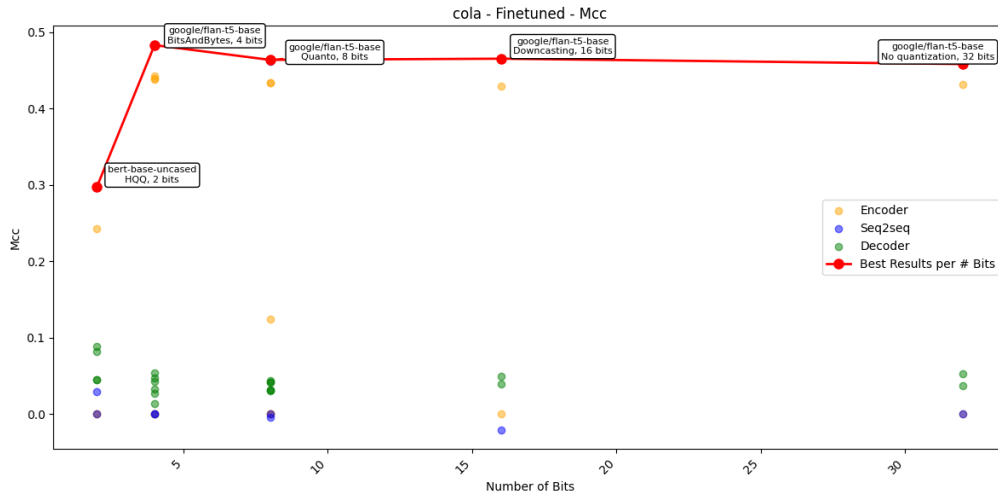


Figure 5.40: Pareto front for Mcc in cola with finetuned

5.2 Results

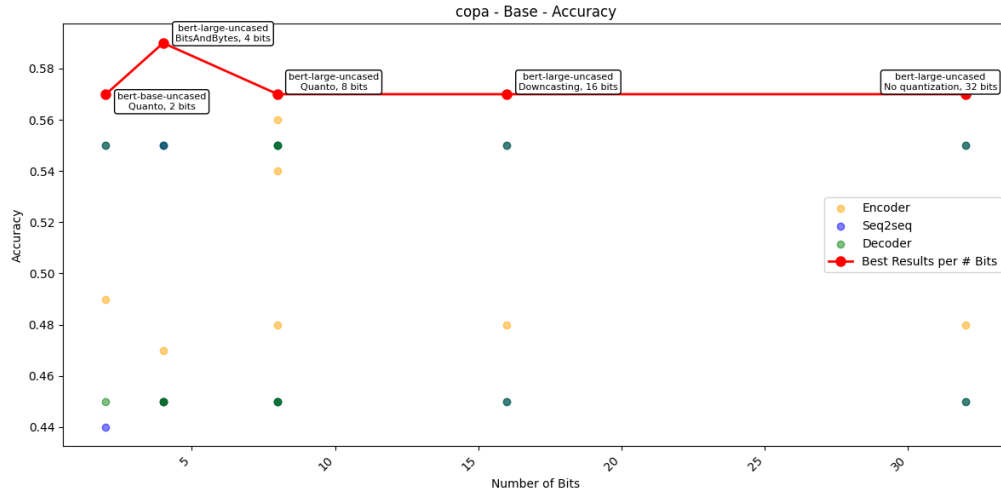


Figure 5.41: Pareto front for Accuracy in copa with base

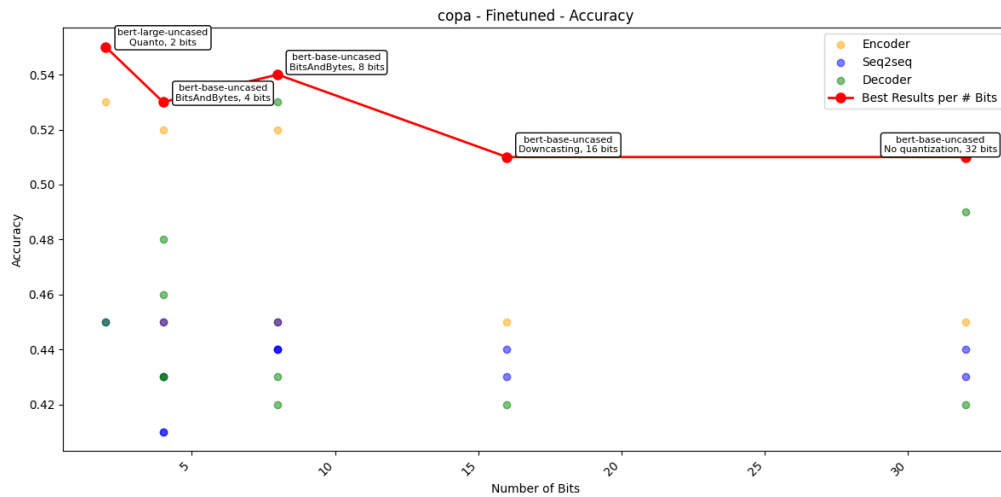


Figure 5.42: Pareto front for Accuracy in copa with finetuned

5. EXPERIMENTS

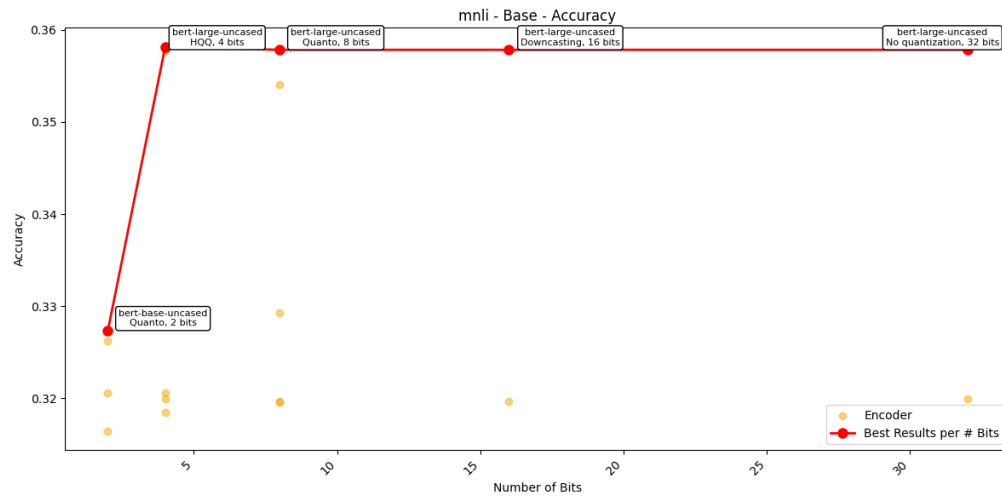


Figure 5.43: Pareto front for Accuracy in mnli with base

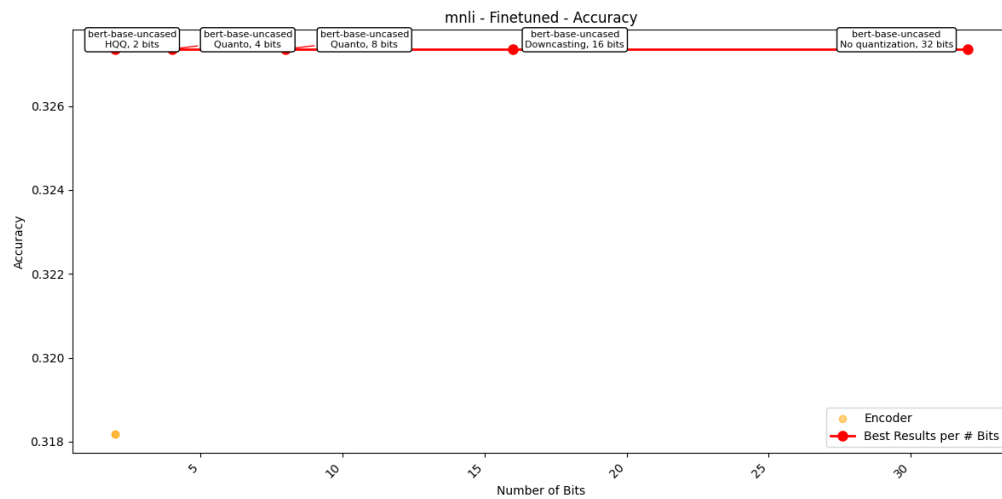


Figure 5.44: Pareto front for Accuracy in mnli with finetuned

5.2 Results

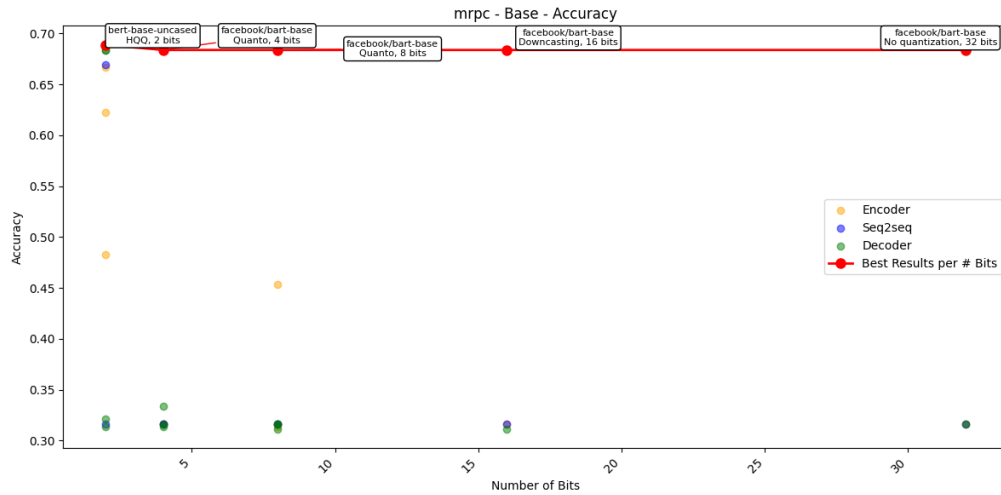


Figure 5.45: Pareto front for Accuracy in mrpc with base

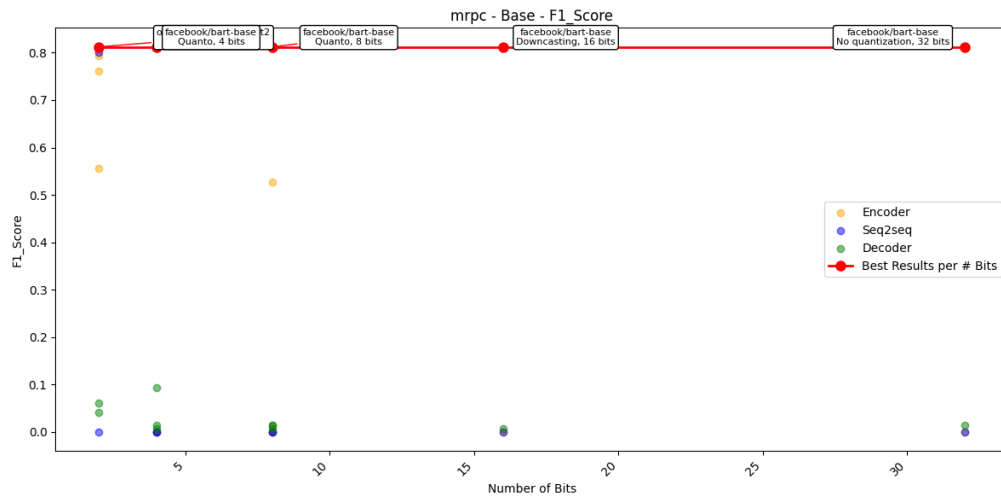


Figure 5.46: Pareto front for F1_Score in mrpc with base

5. EXPERIMENTS

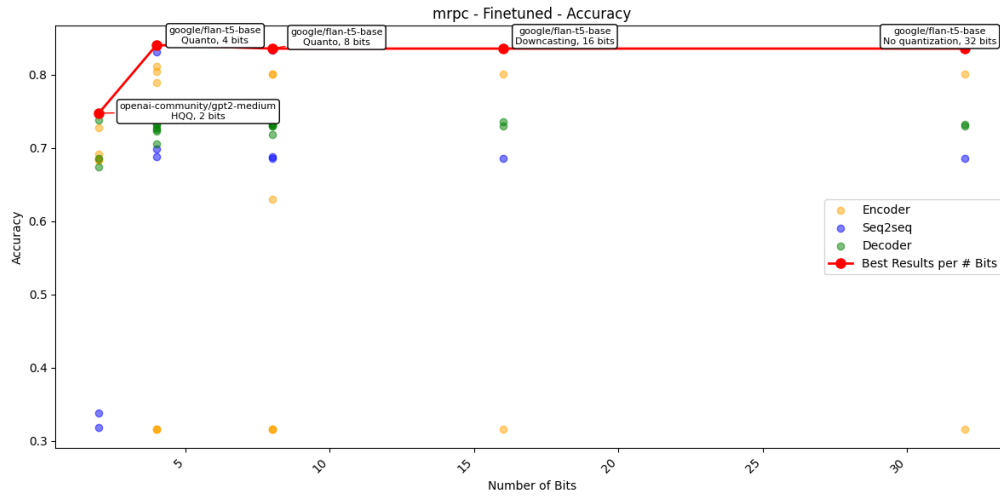


Figure 5.47: Pareto front for Accuracy in mrpc with finetuned

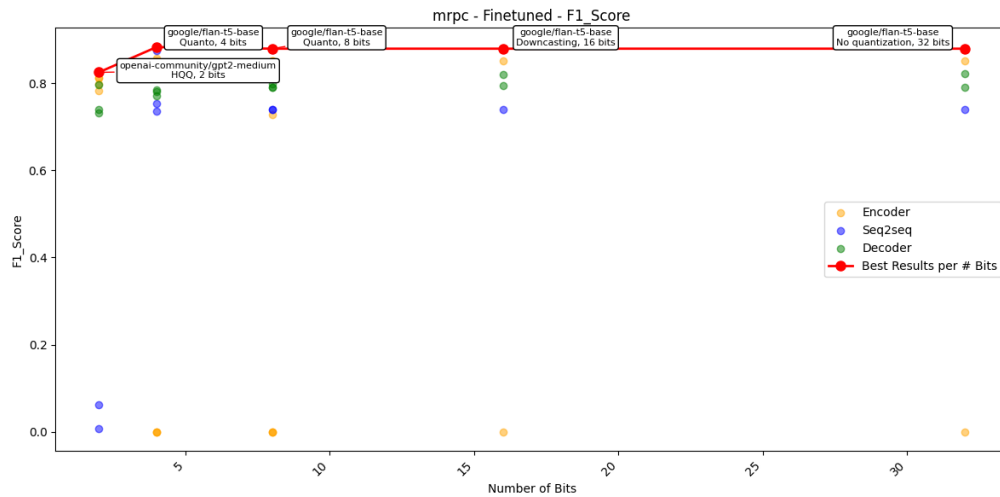


Figure 5.48: Pareto front for F1_Score in mrpc with finetuned

5.2 Results

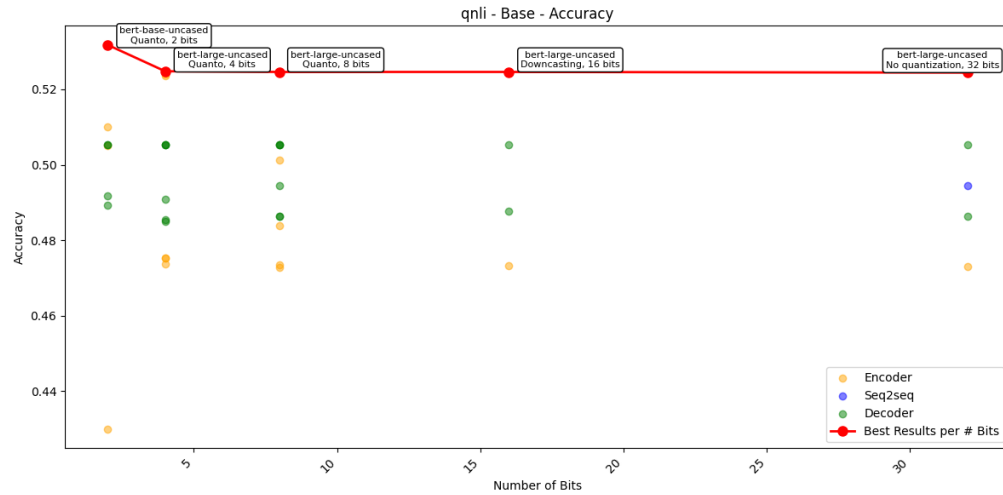


Figure 5.49: Pareto front for Accuracy in qnli with base

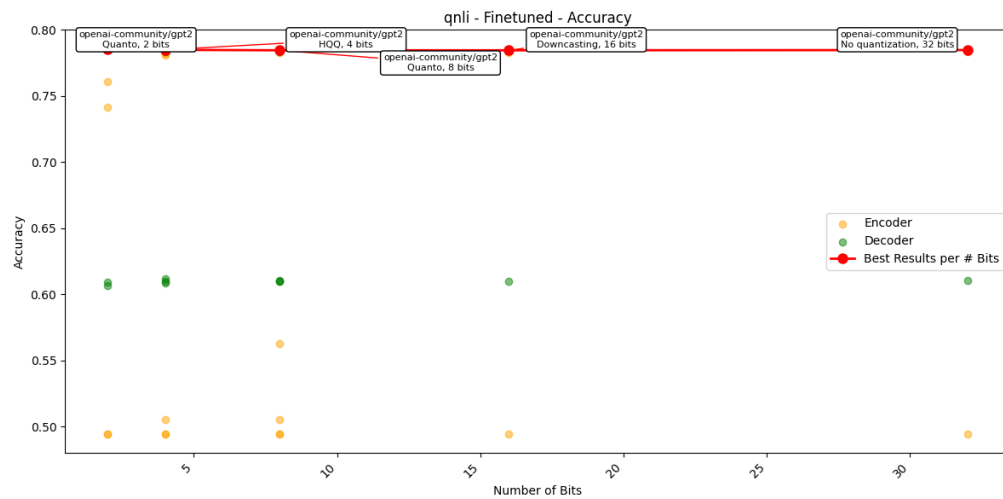


Figure 5.50: Pareto front for Accuracy in qnli with finetuned

5. EXPERIMENTS

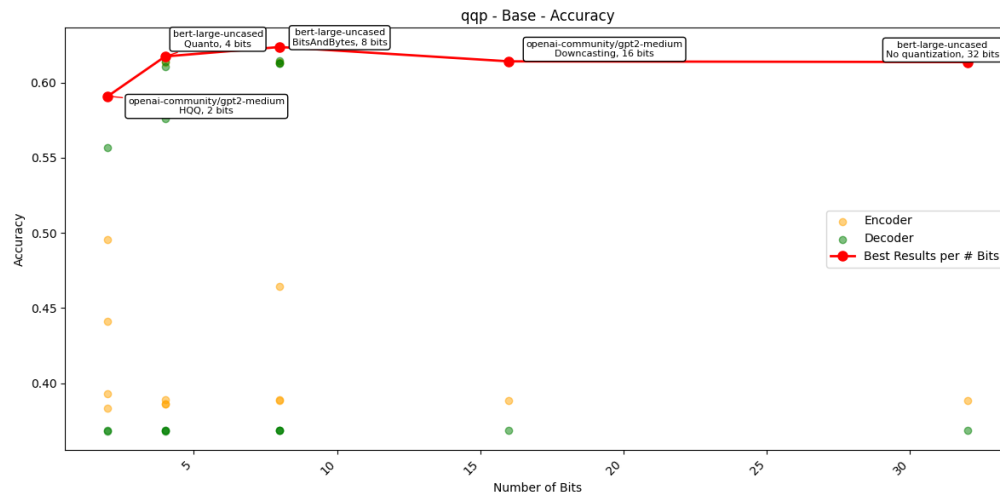


Figure 5.51: Pareto front for Accuracy in qqp with base

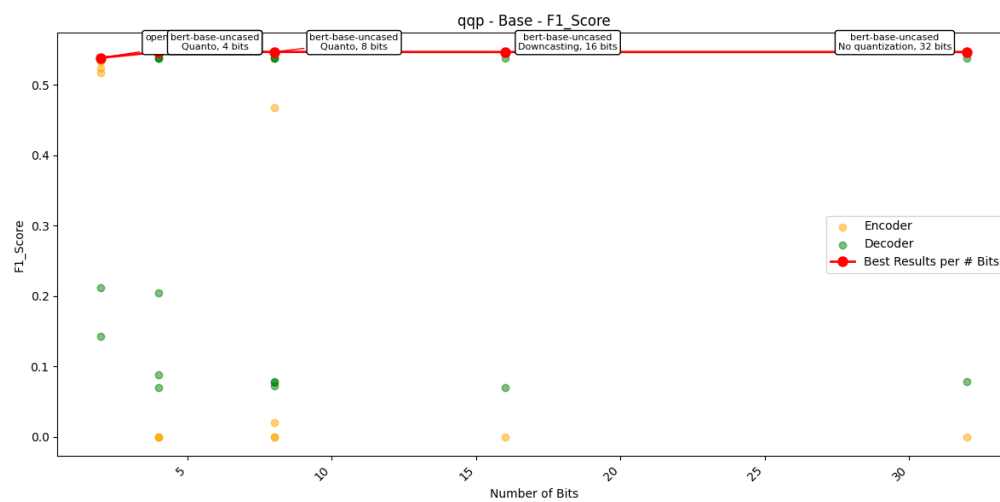


Figure 5.52: Pareto front for F1_Score in qqp with base

5.2 Results

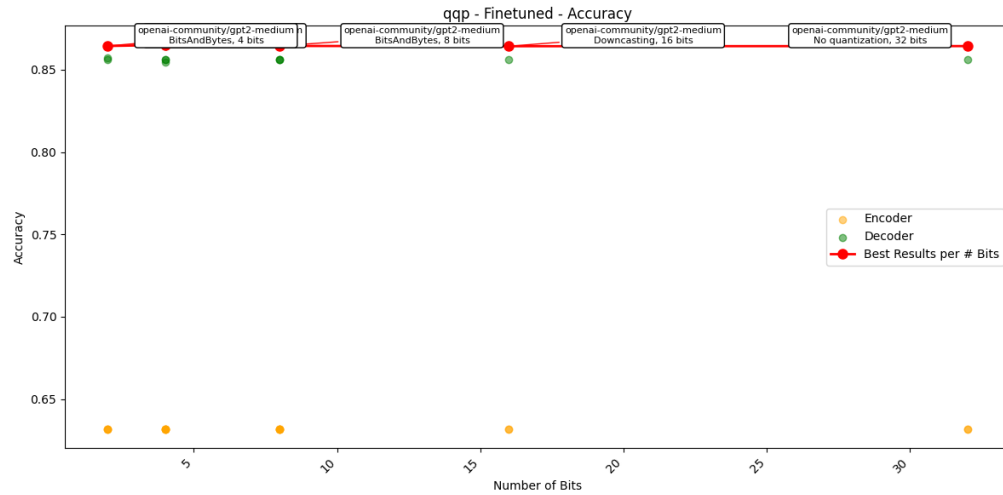


Figure 5.53: Pareto front for Accuracy in qqp with finetuned

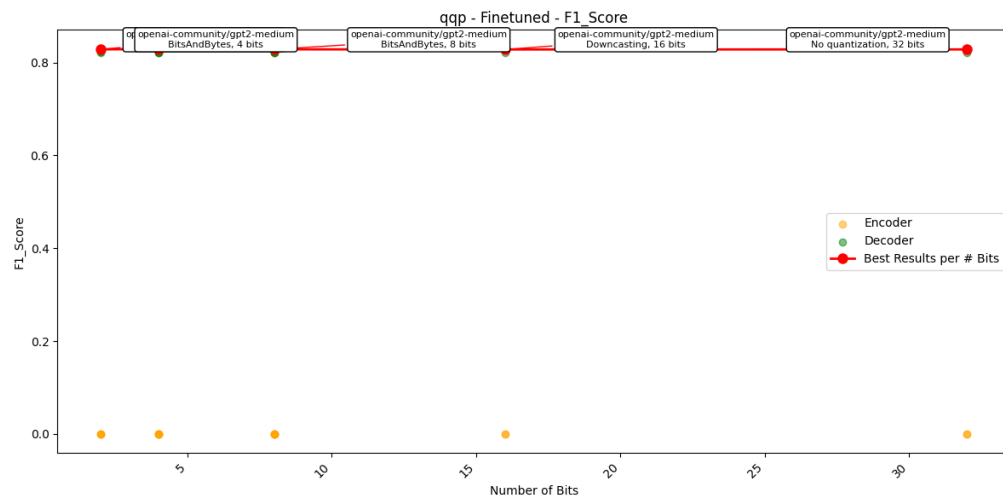


Figure 5.54: Pareto front for F1_Score in qqp with finetuned

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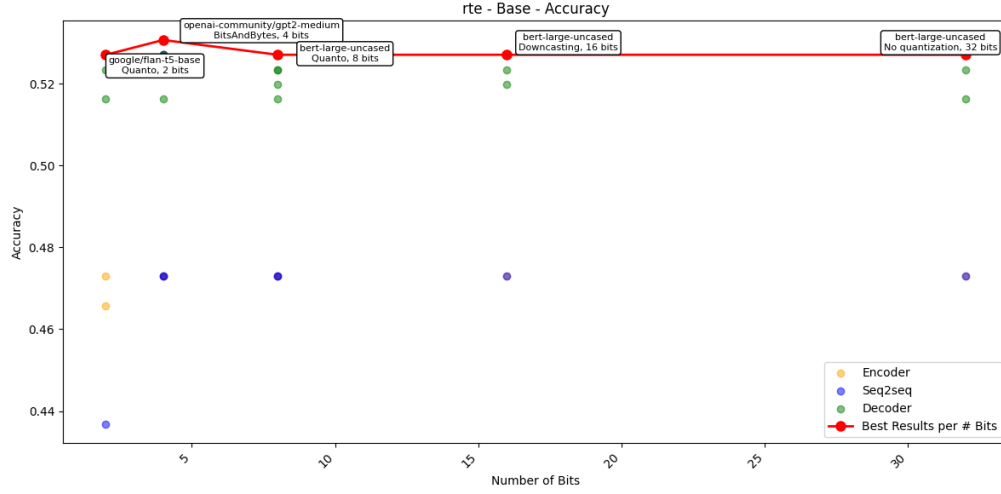


Figure 5.55: Pareto front for Accuracy in rte with base

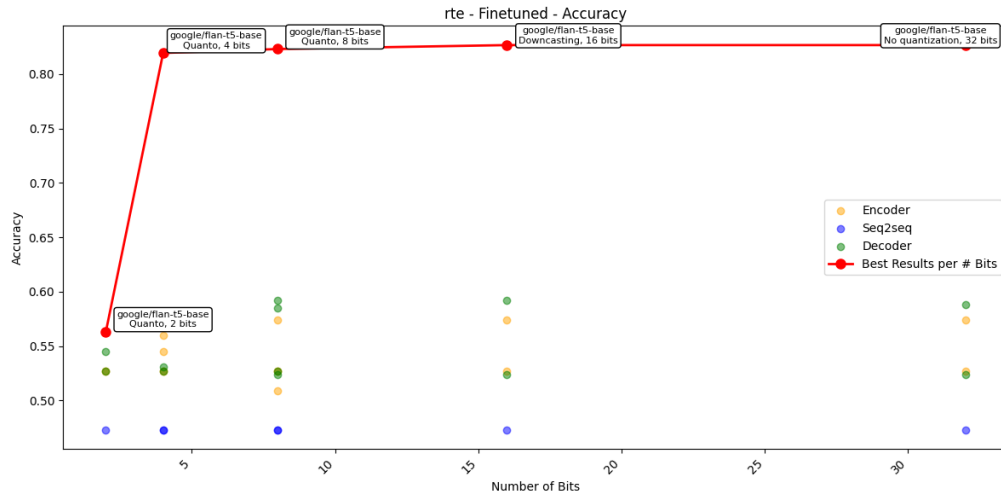


Figure 5.56: Pareto front for Accuracy in rte with finetuned

5.2 Results

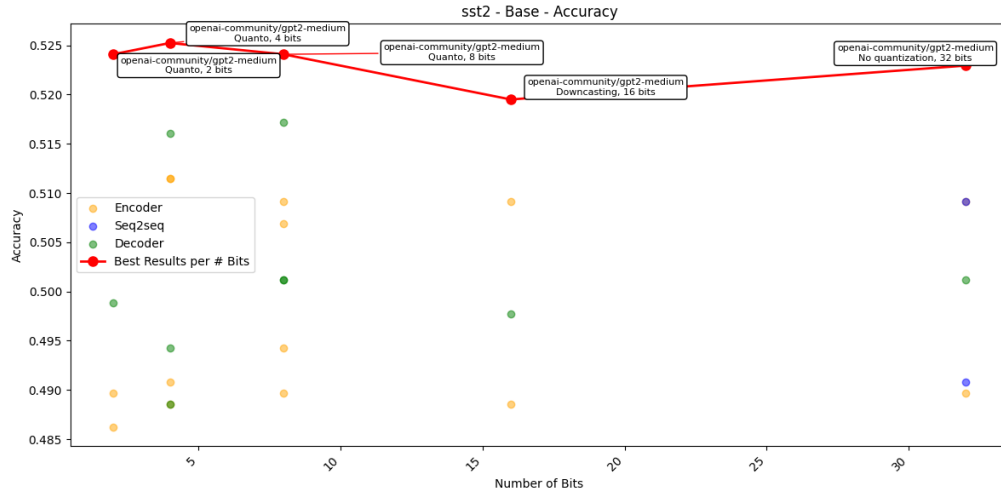


Figure 5.57: Pareto front for Accuracy in sst2 with base

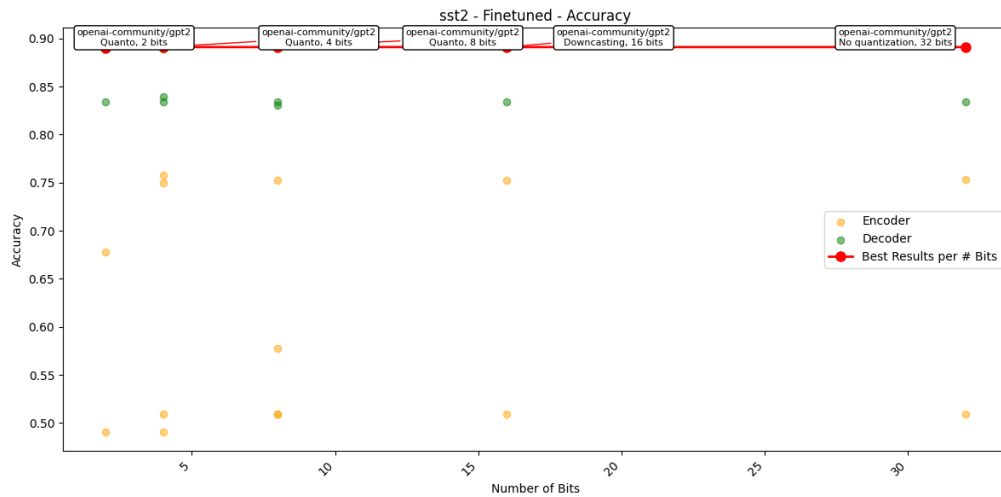


Figure 5.58: Pareto front for Accuracy in sst2 with finetuned

5. EXPERIMENTS

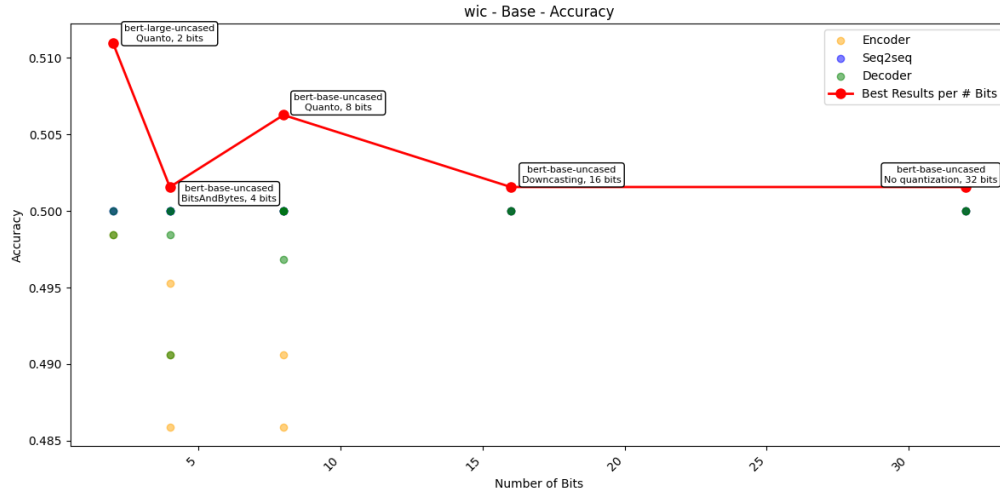


Figure 5.59: Pareto front for Accuracy in wic with base

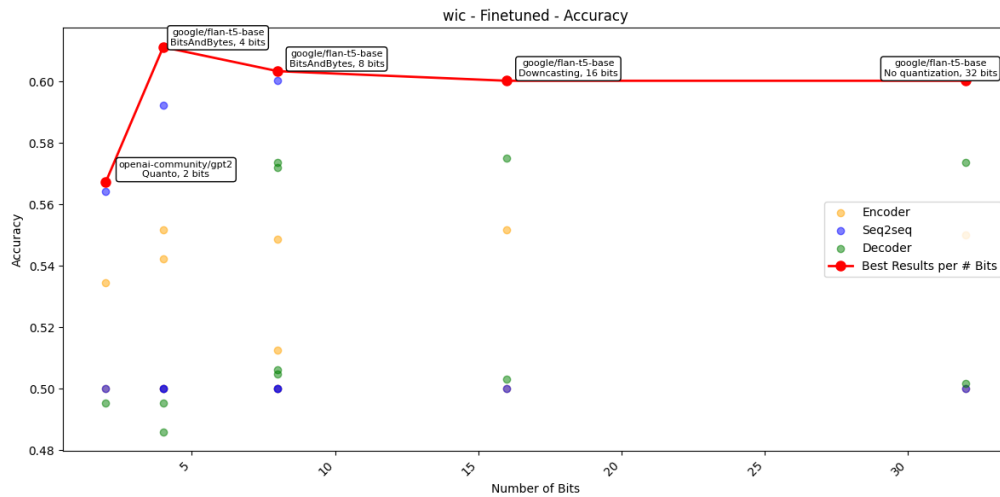


Figure 5.60: Pareto front for Accuracy in wic with finetuned

5.2 Results

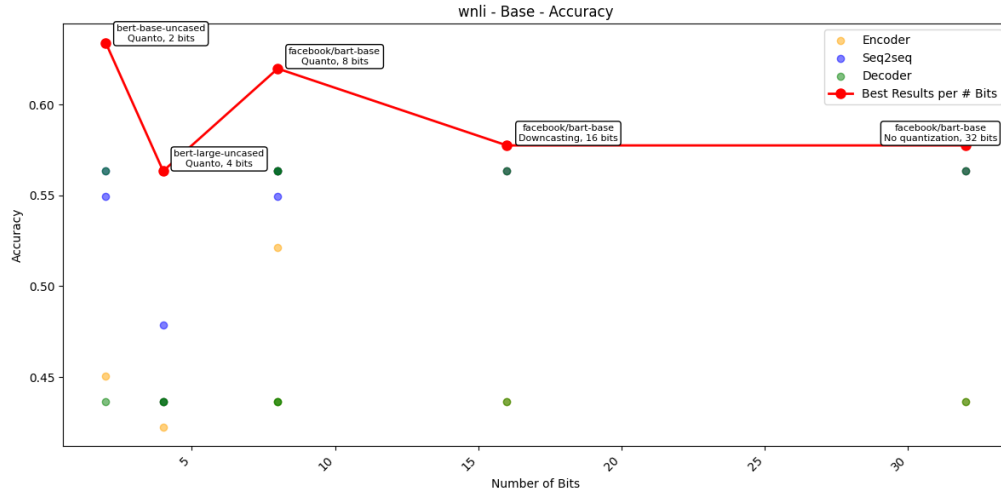


Figure 5.61: Pareto front for Accuracy in wnli with base

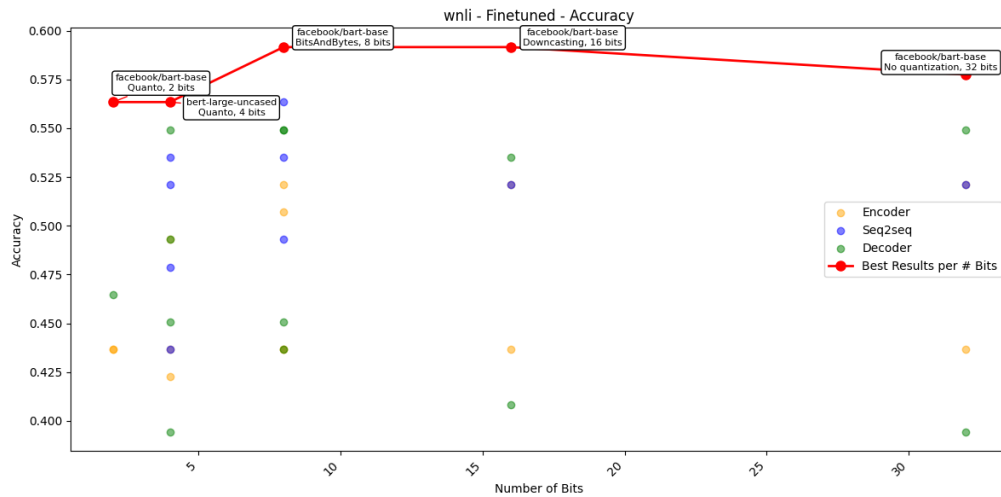


Figure 5.62: Pareto front for Accuracy in wnli with finetuned

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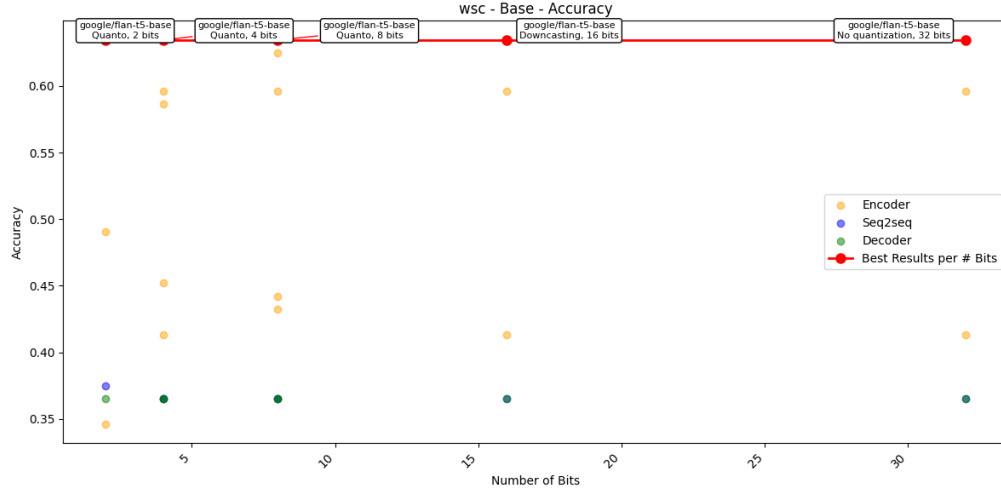


Figure 5.63: Pareto front for Accuracy in wsc with base

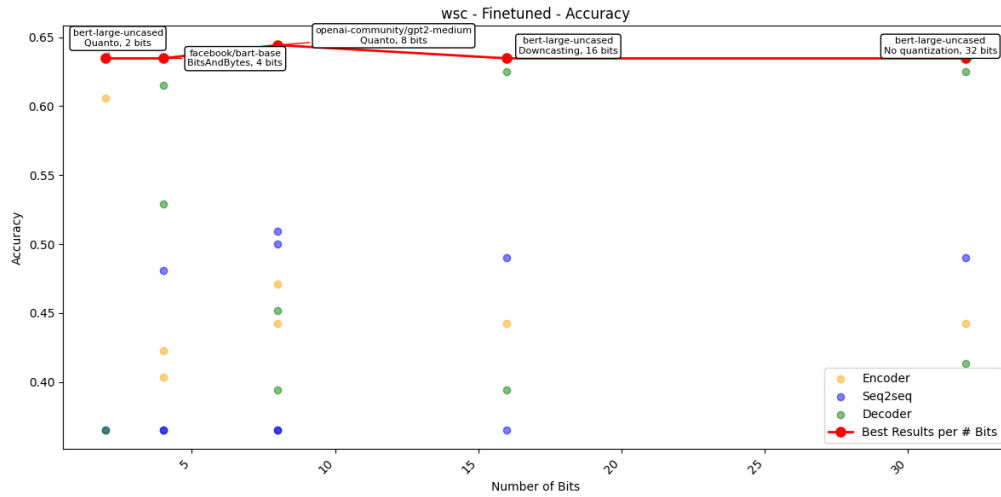


Figure 5.64: Pareto front for Accuracy in wsc with finetuned

5.2.4 Rate-distortion

The following figures showcase a rate-distortion "curve" (explained in Section 2.1.8) for each task. It is not a curve, as the data is discrete, compared to the usual rate-distortion functions, which are continuous. The figures show the average score and standard deviation per quantization method, per bit-precision, with the individual scores plotted as well with x-markers. The figures display the effect that the lower bit precision has on the performance of the model.

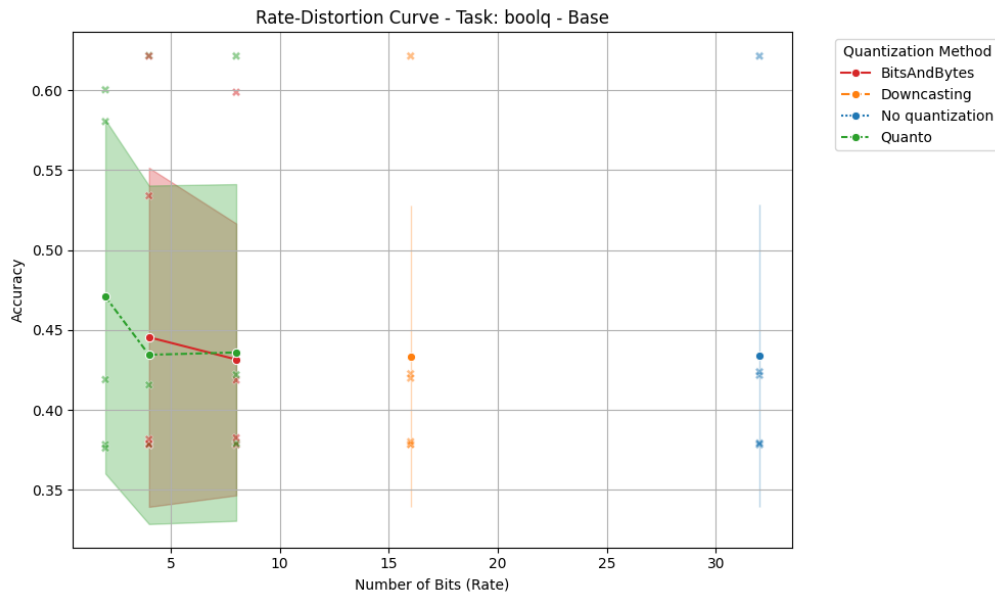


Figure 5.65: Rate distortion for Accuracy in boolq with base

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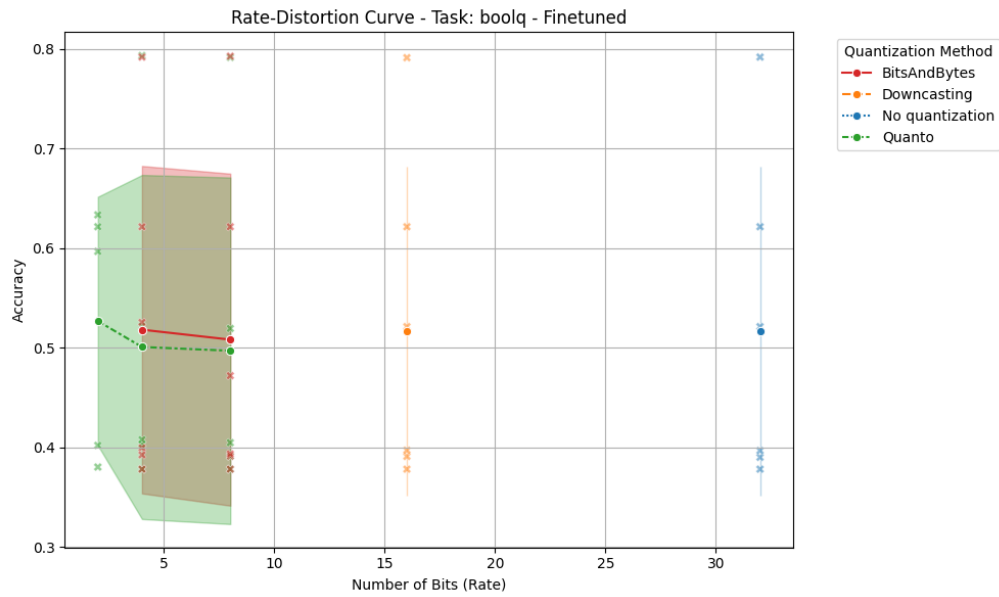


Figure 5.66: Rate distortion for Accuracy in boolq with finetuned

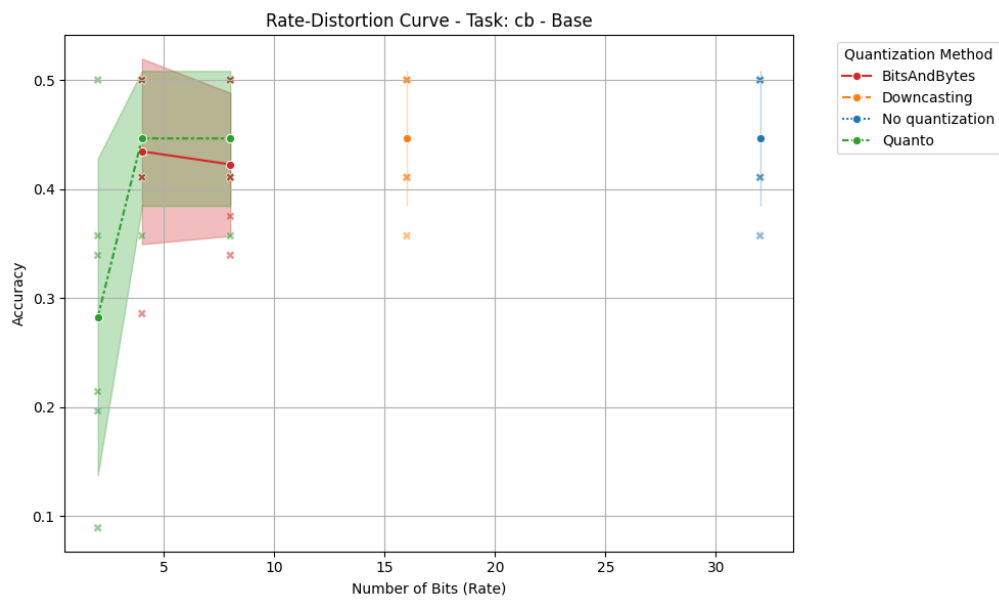


Figure 5.67: Rate distortion for Accuracy in cb with base

5.2 Results

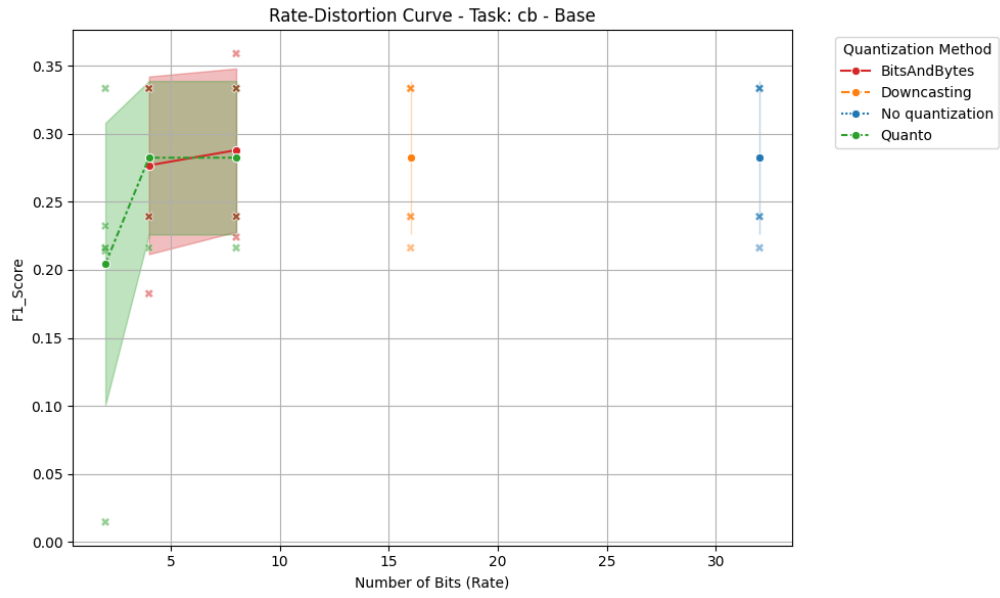


Figure 5.68: Rate distortion for F1_Score in cb with base

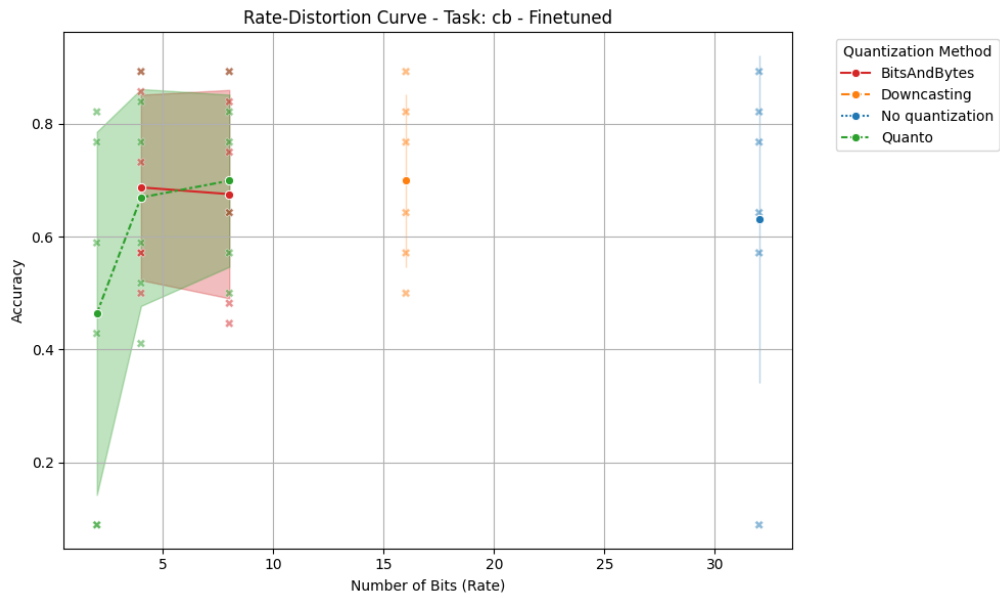


Figure 5.69: Rate distortion for Accuracy in cb with finetuned

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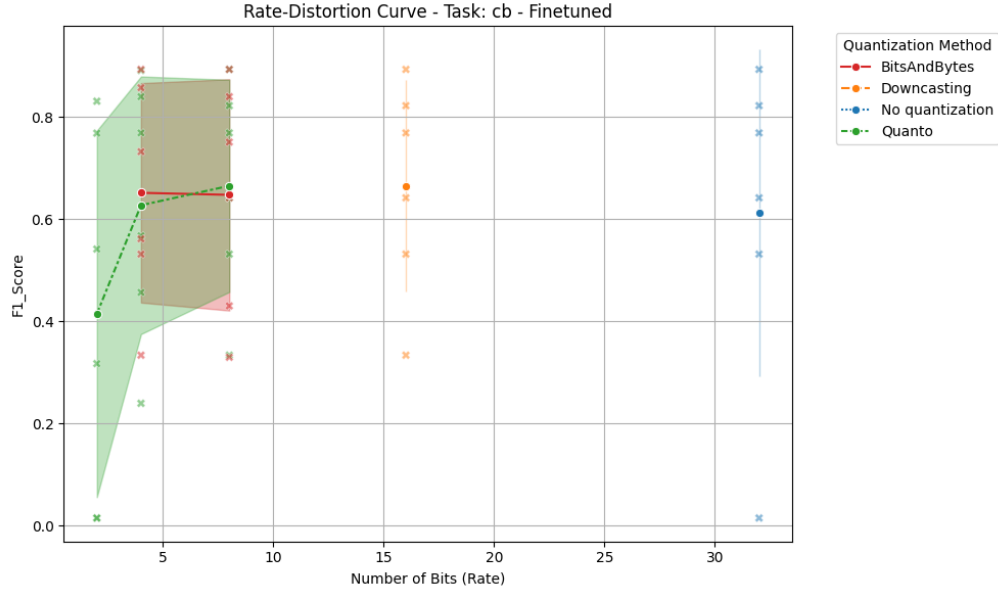


Figure 5.70: Rate distortion for F1_Score in cb with finetuned

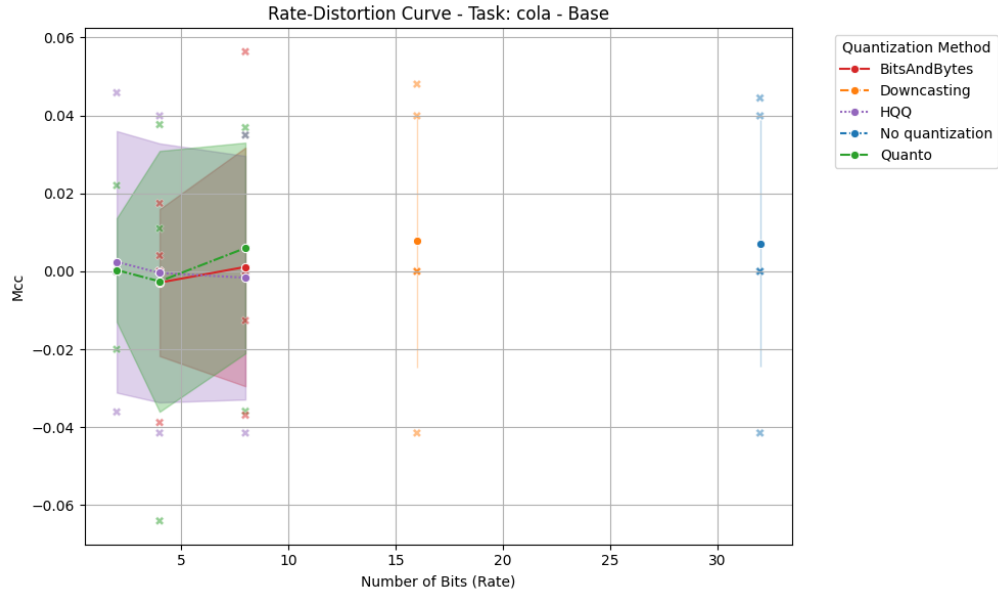


Figure 5.71: Rate distortion for Mcc in cola with base

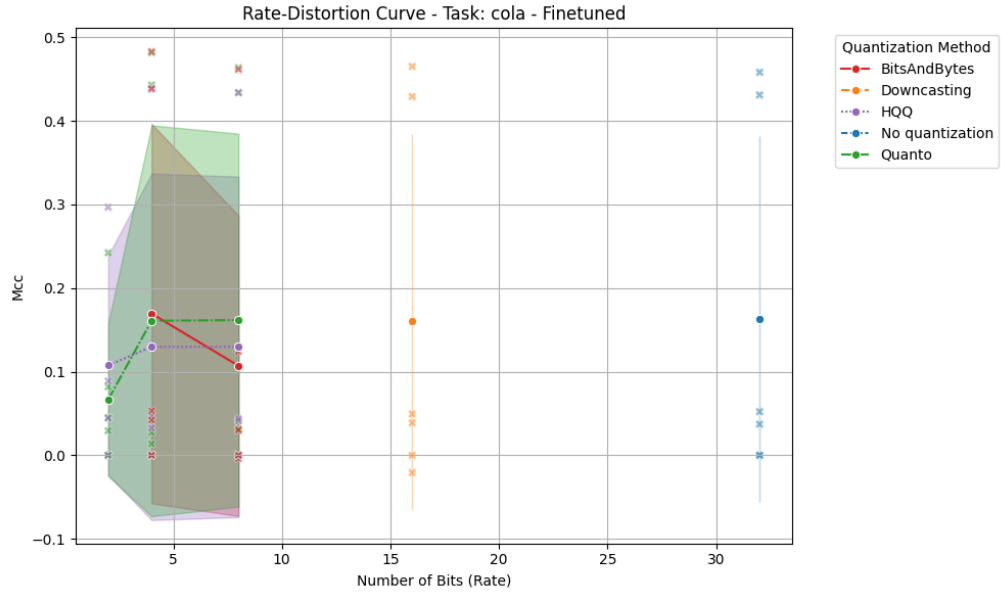


Figure 5.72: Rate distortion for Mcc in cola with finetuned

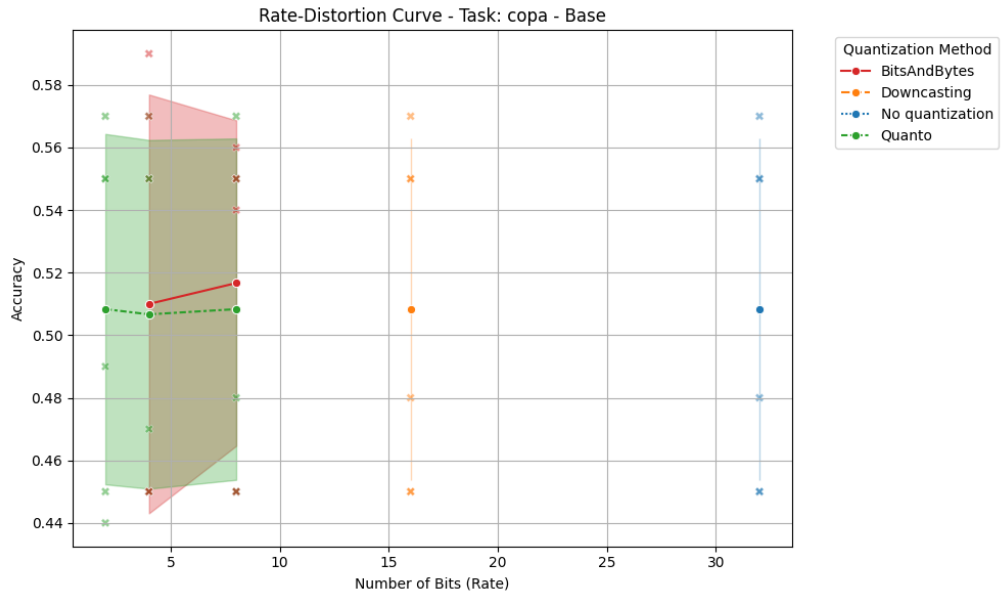


Figure 5.73: Rate distortion for Accuracy in copa with base

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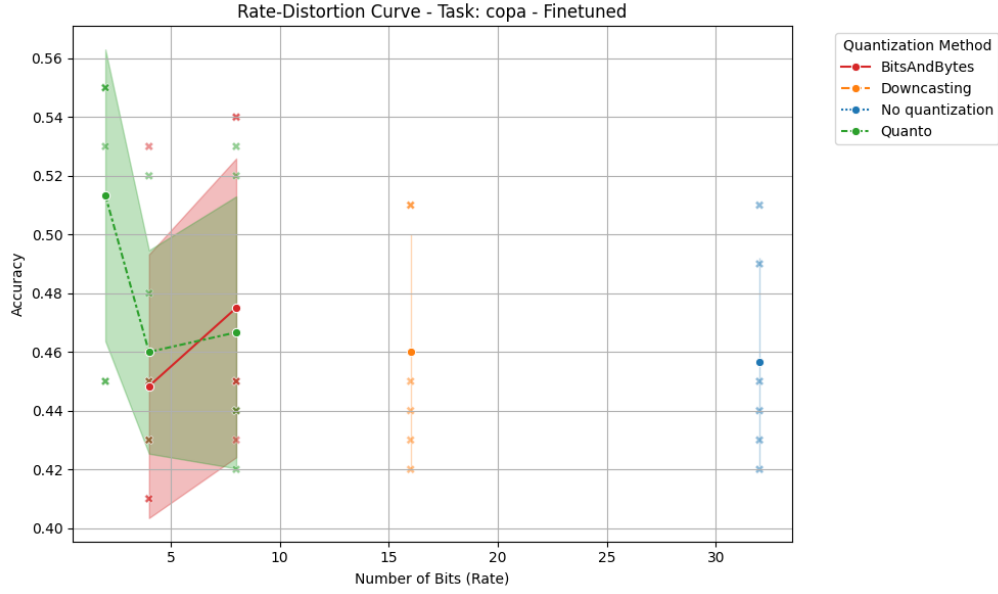


Figure 5.74: Rate distortion for Accuracy in copa with finetuned

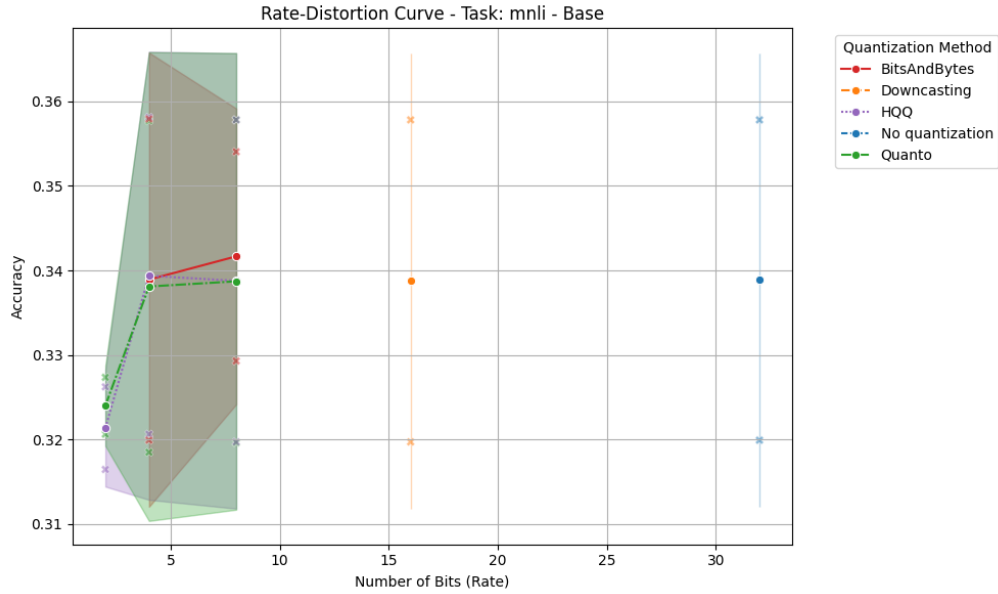


Figure 5.75: Rate distortion for Accuracy in mnli with base

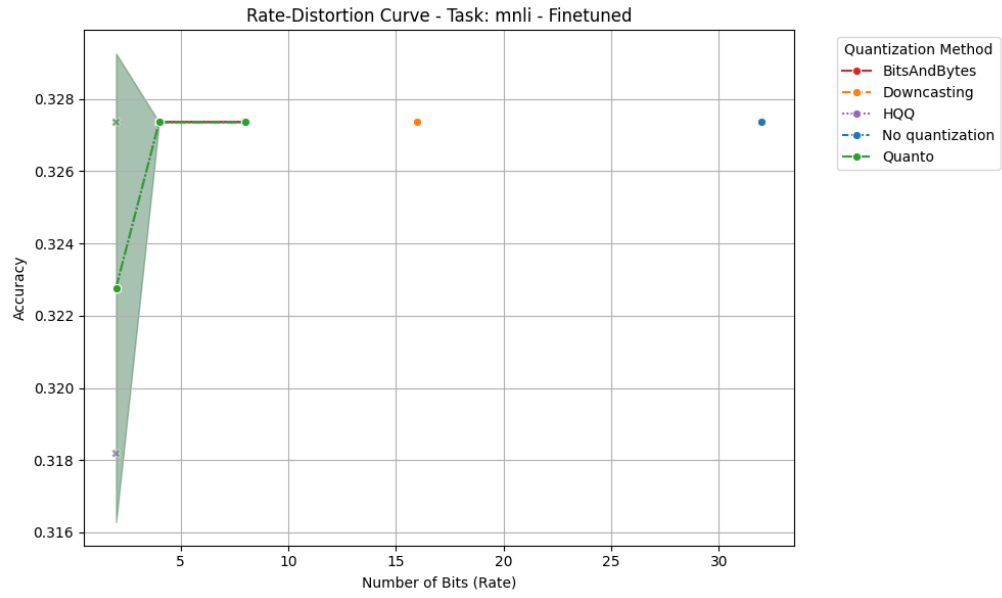


Figure 5.76: Rate distortion for Accuracy in mnli with finetuned

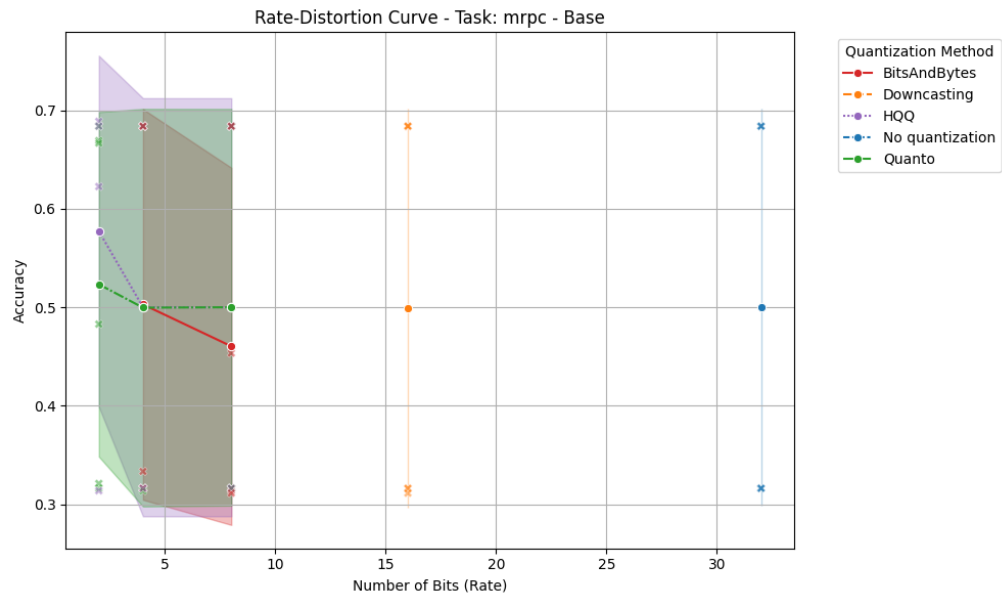


Figure 5.77: Rate distortion for Accuracy in mrpc with base

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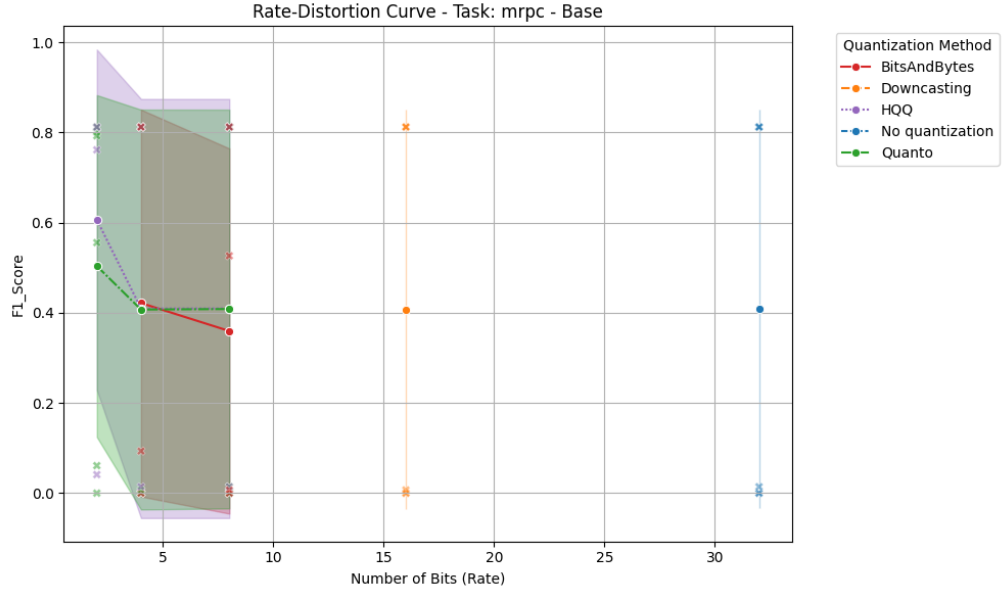


Figure 5.78: Rate distortion for F1_Score in mrpc with base

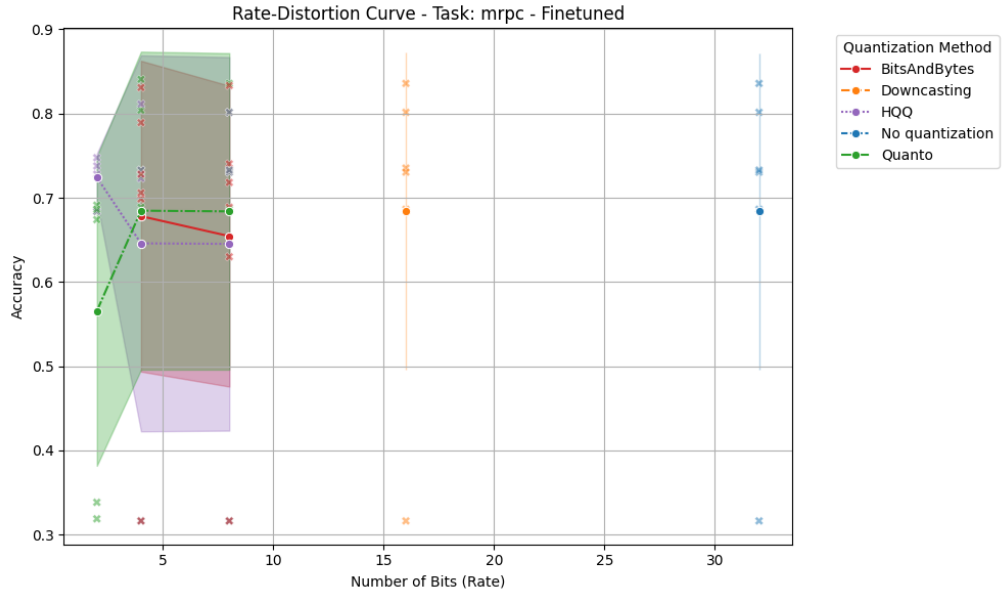


Figure 5.79: Rate distortion for Accuracy in mrpc with finetuned

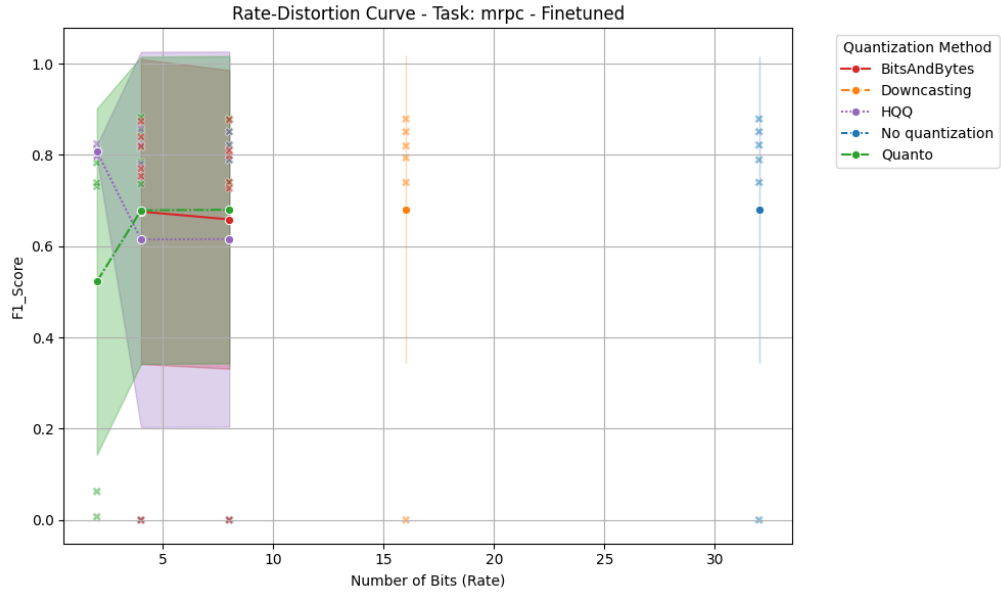


Figure 5.80: Rate distortion for F1_Score in mrpc with finetuned

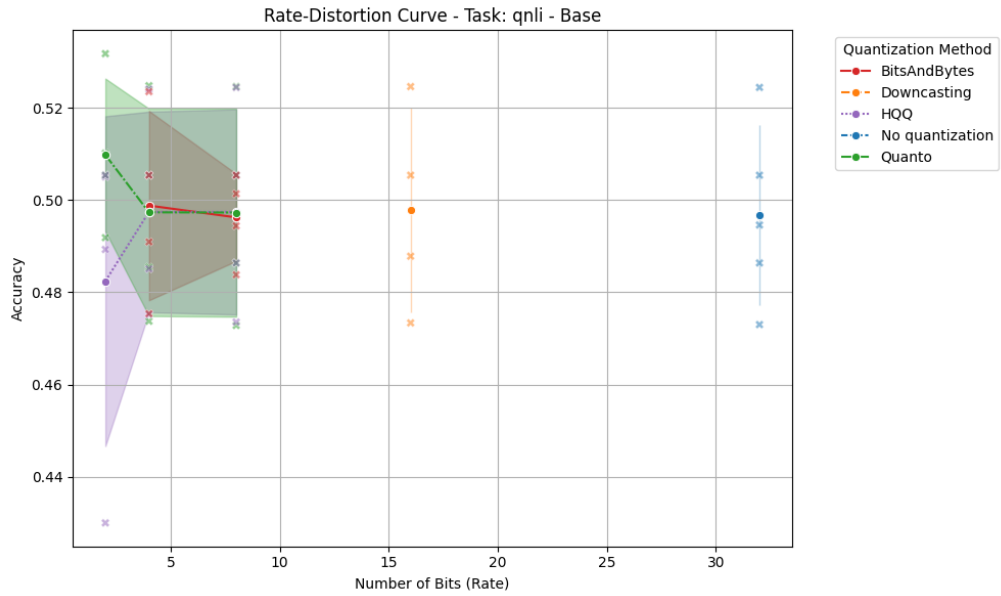


Figure 5.81: Rate distortion for Accuracy in qnli with base

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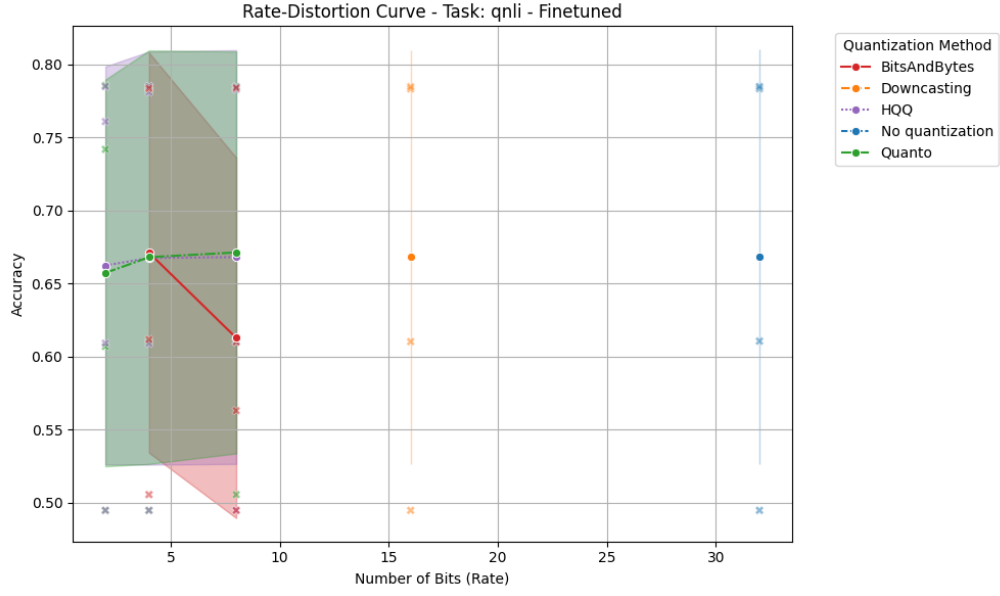


Figure 5.82: Rate distortion for Accuracy in qnli with finetuned

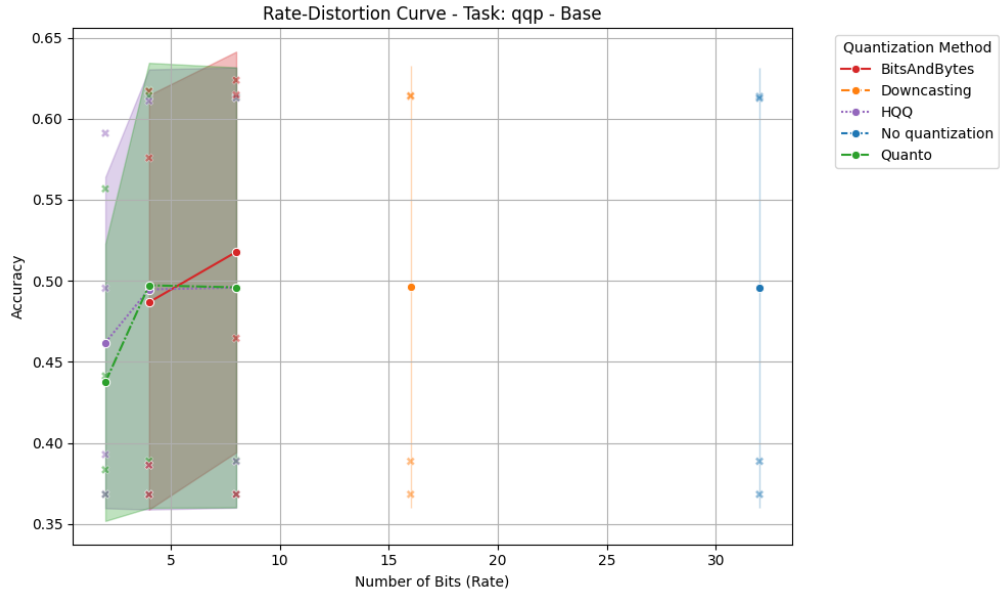


Figure 5.83: Rate distortion for Accuracy in qqp with base

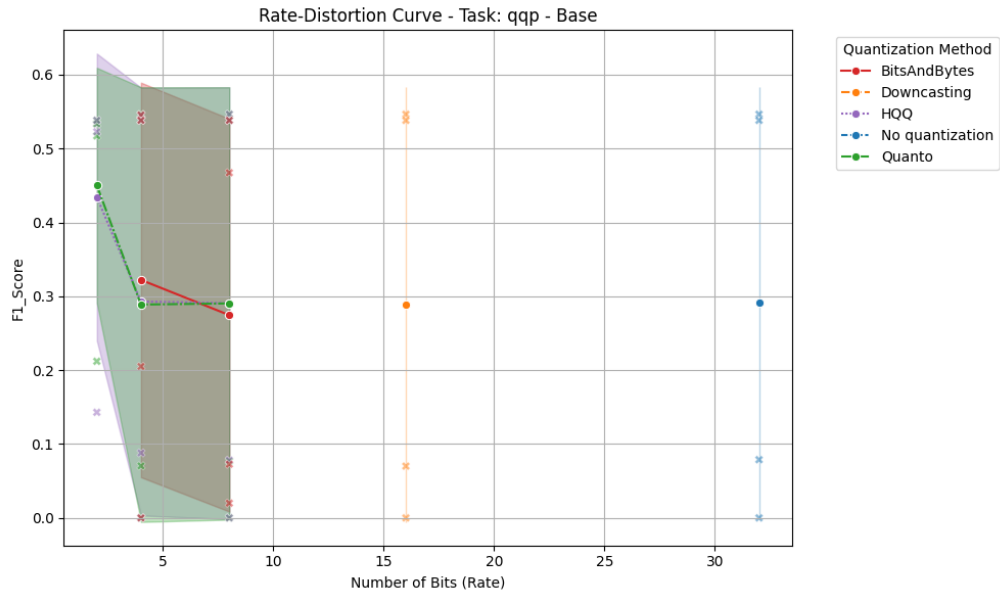


Figure 5.84: Rate distortion for F1_Score in qqp with base

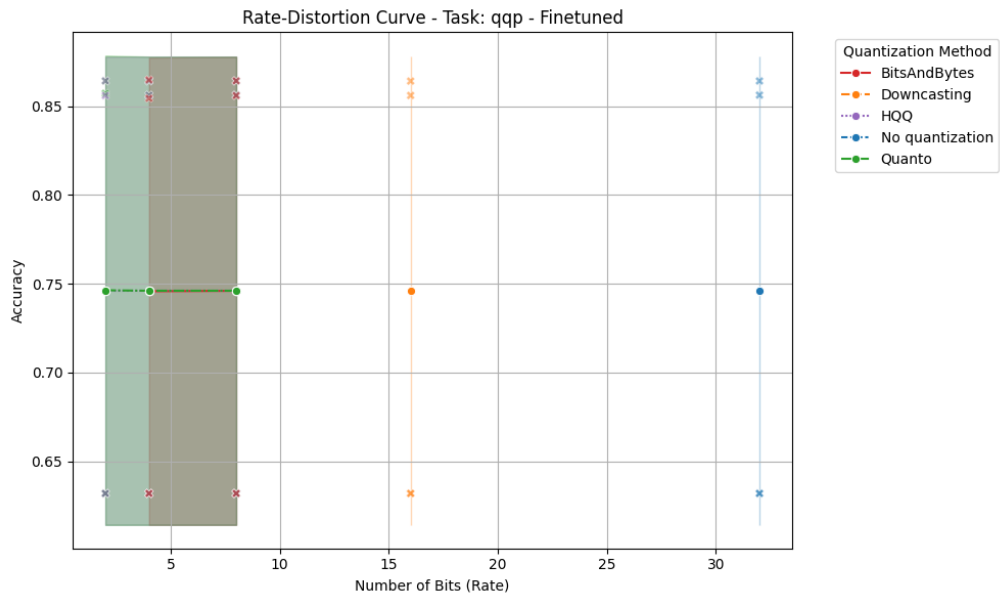


Figure 5.85: Rate distortion for Accuracy in qqp with finetuned

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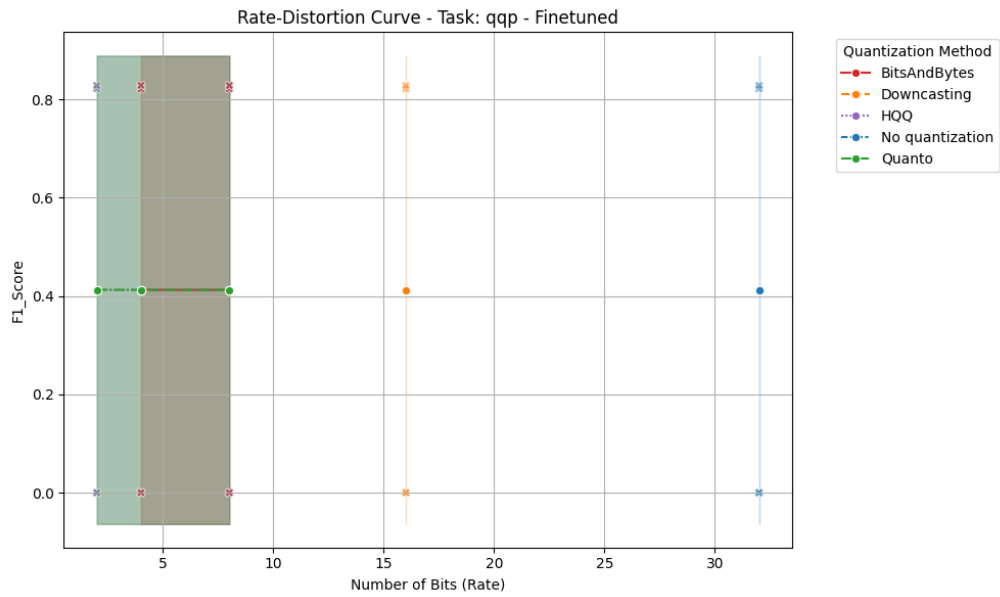


Figure 5.86: Rate distortion for F1_Score in qqp with finetuned

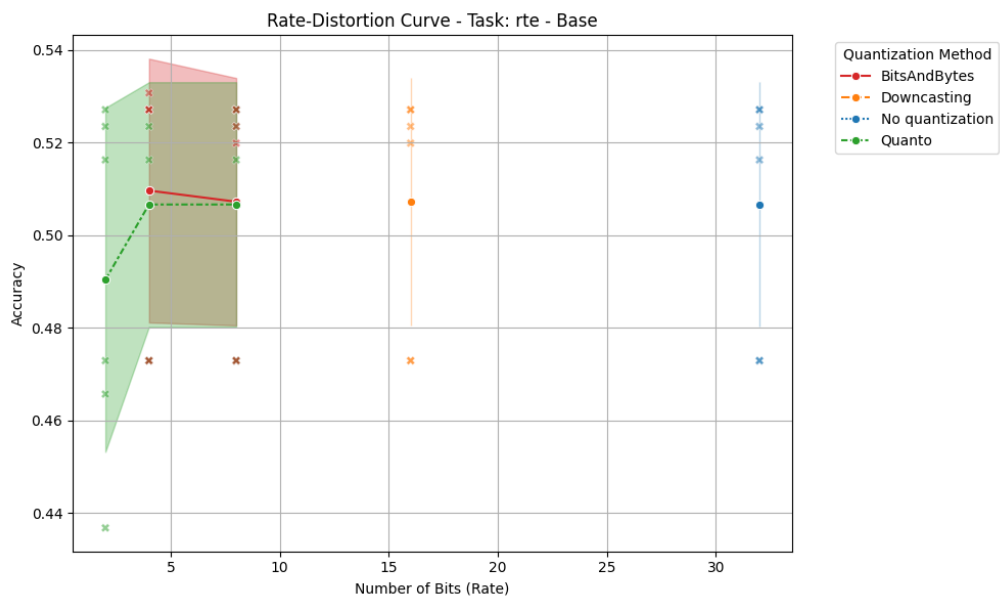


Figure 5.87: Rate distortion for Accuracy in rte with base

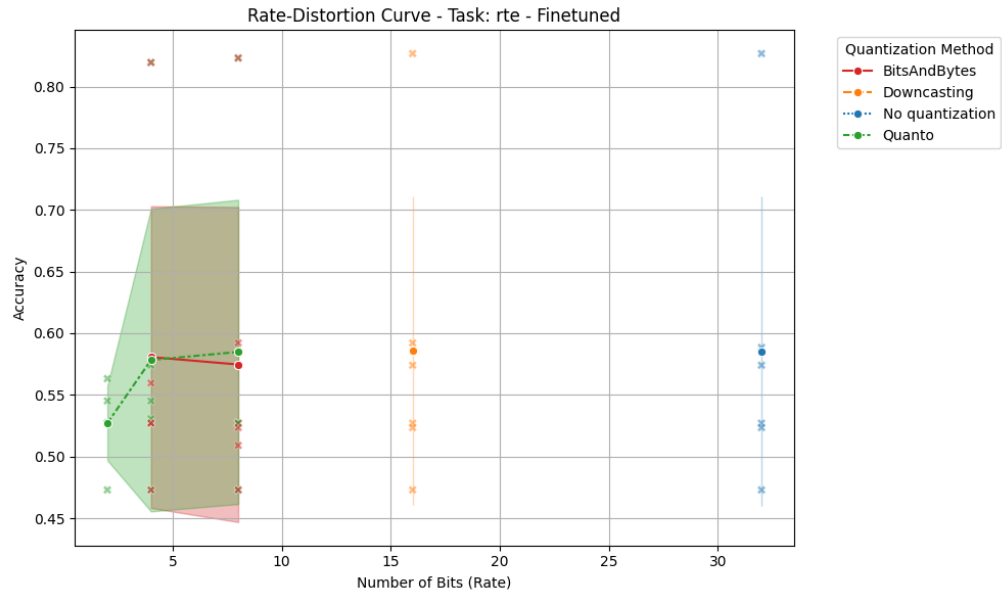


Figure 5.88: Rate distortion for Accuracy in rte with finetuned

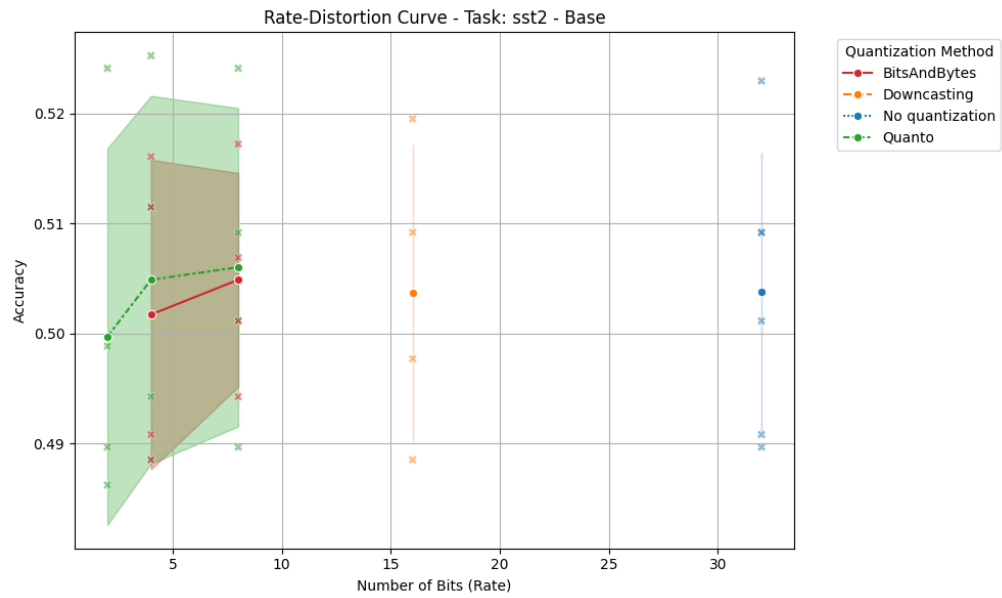


Figure 5.89: Rate distortion for Accuracy in sst2 with base

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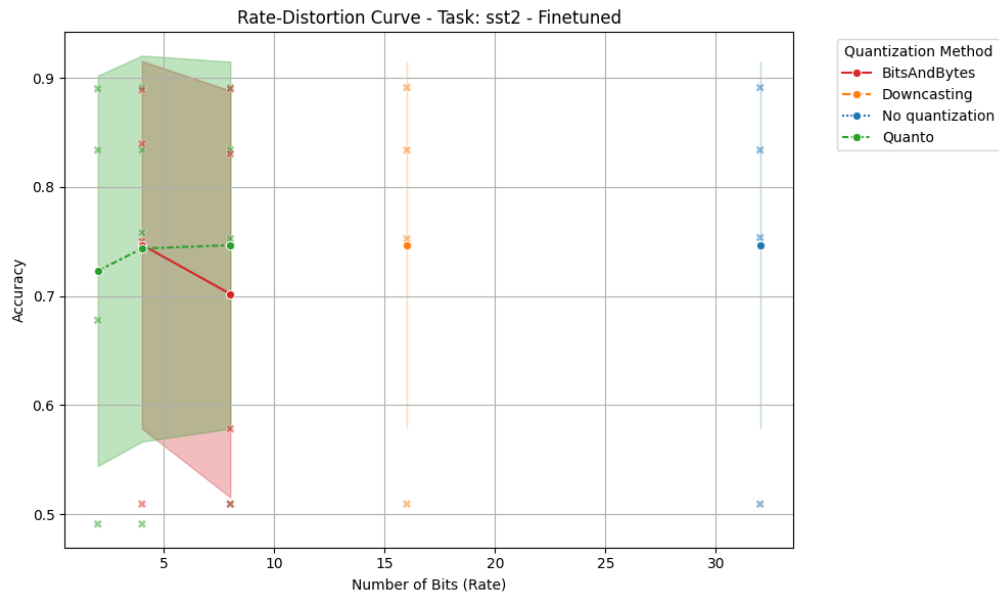


Figure 5.90: Rate distortion for Accuracy in sst2 with finetuned

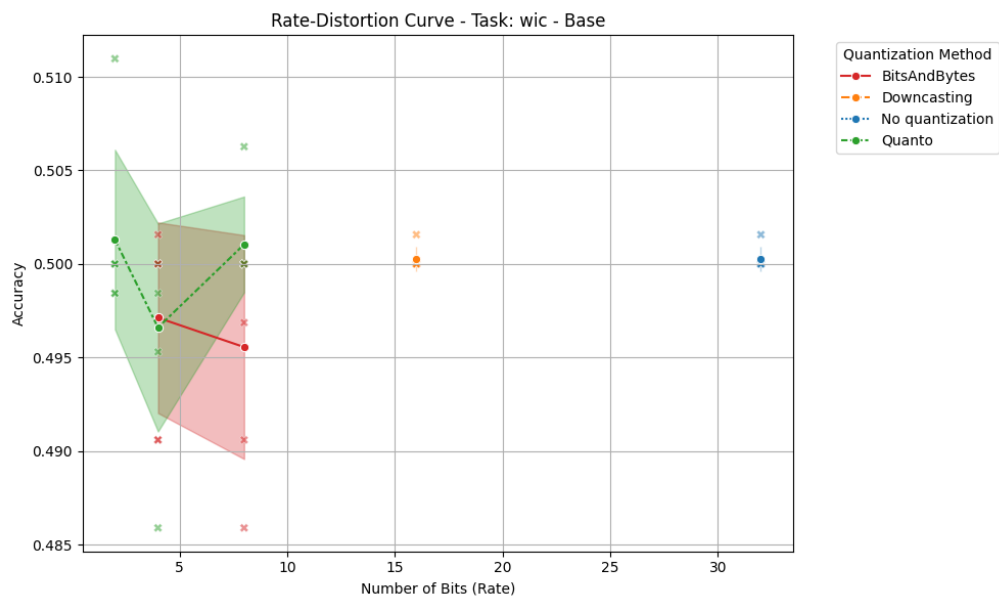


Figure 5.91: Rate distortion for Accuracy in wic with base

5.2 Results

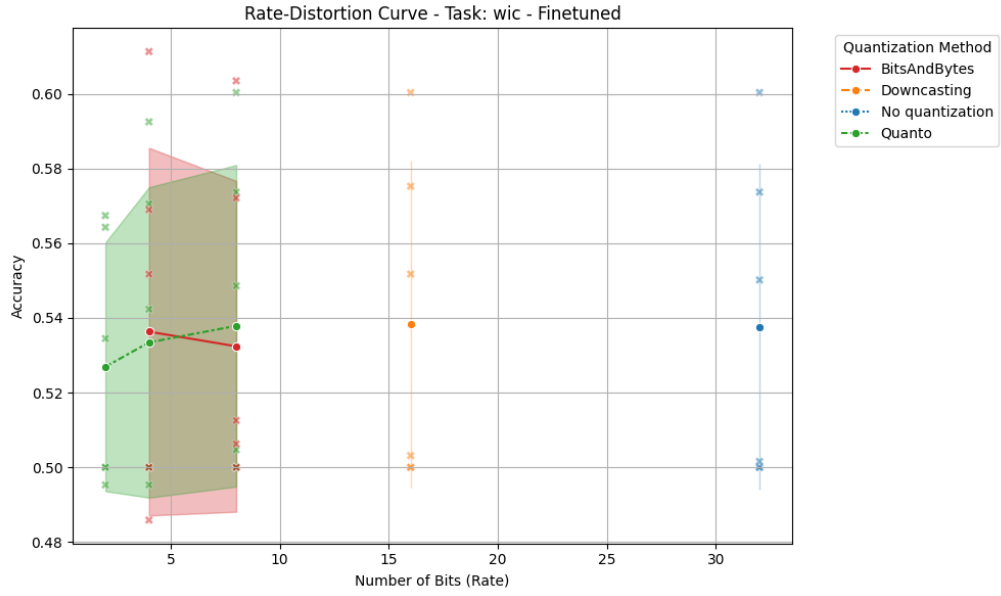


Figure 5.92: Rate distortion for Accuracy in wic with finetuned

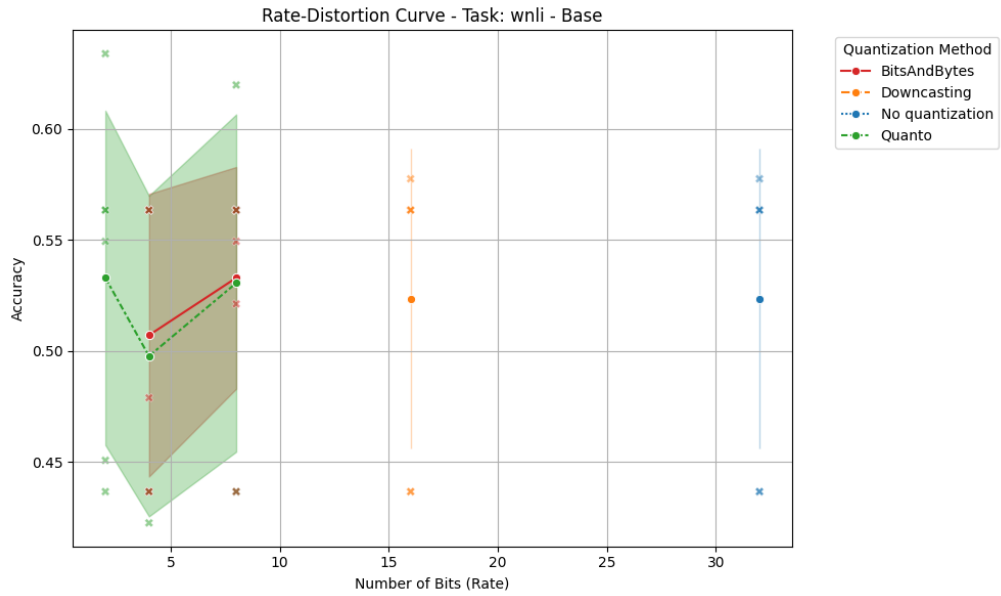


Figure 5.93: Rate distortion for Accuracy in wnli with base

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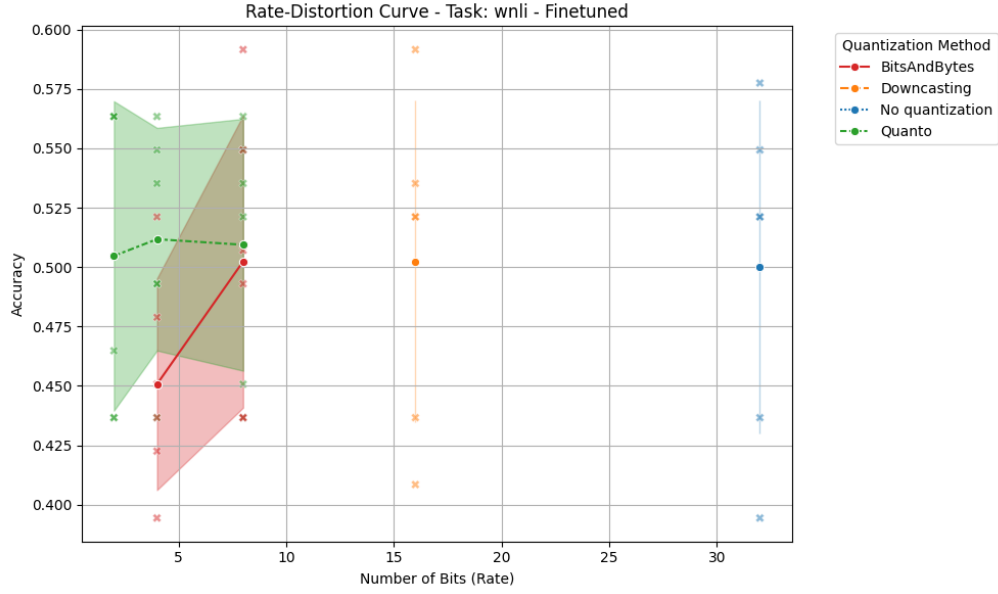


Figure 5.94: Rate distortion for Accuracy in wnli with finetuned

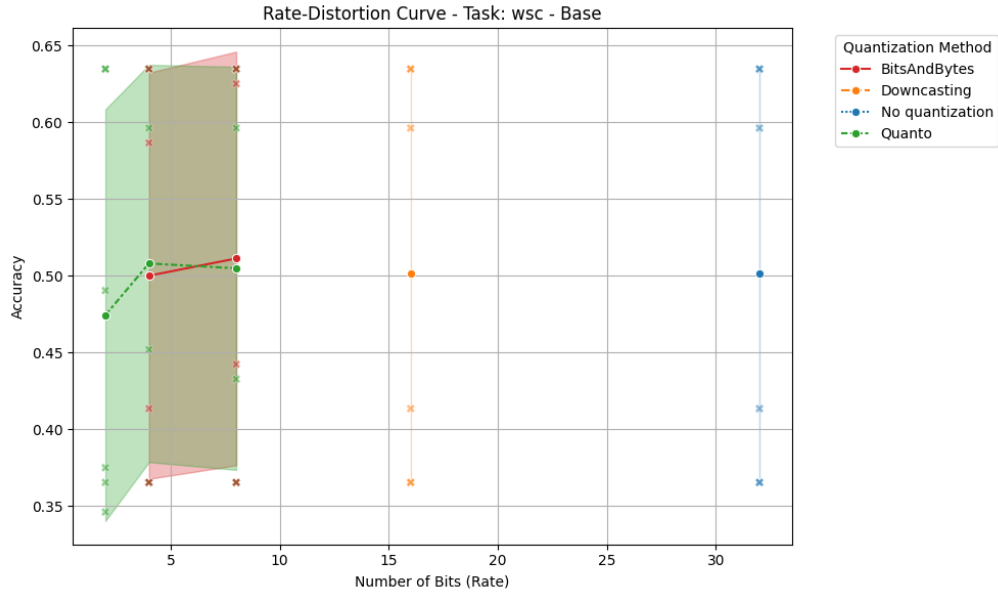


Figure 5.95: Rate distortion for Accuracy in wsc with base

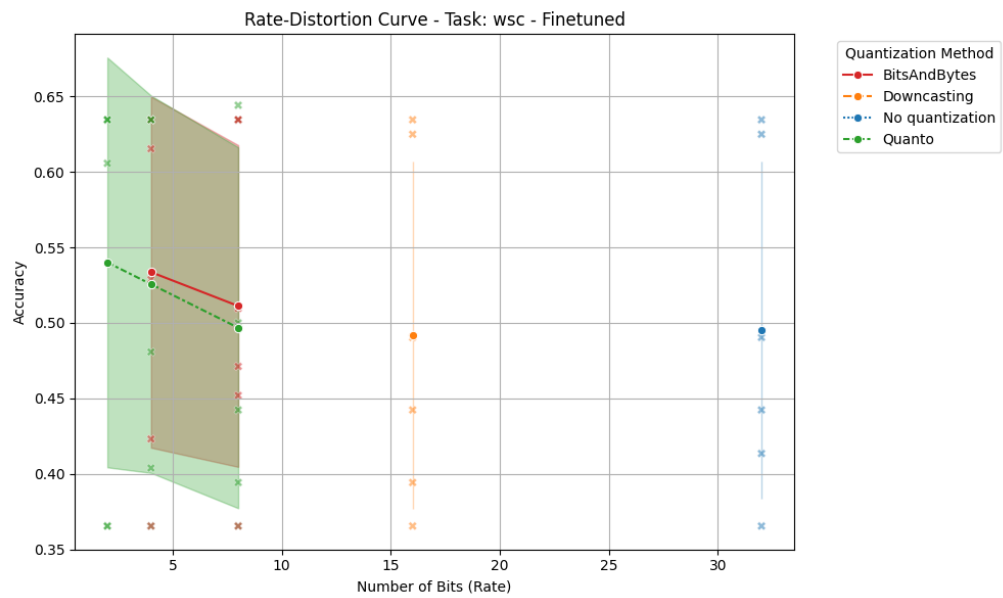


Figure 5.96: Rate distortion for Accuracy in wsc with finetuned

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5.3 Discussion

5.3.1 Remarks

5.3.1.1 Generalization

By investigating the confusion matrices of the model, we can see that in a lot of cases the base model predicts just a single class. In some cases, this is also present in the fine-tuned models. This is something to keep in mind when looking at the results. The confusion matrices are omitted as they provide very little insight other than the aforementioned effect.

5.3.1.2 Incomplete result set

Notably, some of the presented figures and tables in Section 5.2 are incomplete. Especially in the case of the glue/mnli task, for the HQQ quantization method, and the sequence-to-sequence models, there is data missing. There are unfortunately no decoder/sequence-to-sequence models that were compatible with the mnli task. The model was not able to load in.

At the time of writing, the HQQ method was still very new and not widely compatible with different deep learning models. In some cases, the method was not able to quantize a model. For the most part however, HQQ was able to fulfill its task.

Sequence-to-sequence models require about twice the computational power as any encoder or decoder model, as the model is a combination of both. This has unfortunately caused some of the sequence-to-sequence models to not be able to run the bigger benchmark tasks such as qqp.

For the sake of completeness, these figures and tables are still left in, as they could still present interesting results.

5.3.2 By bit precision

In the results, we see that there is a positive correlation between the amount of bits used in the model parameters, and the performance of the model. Post-training quantization has a negative effect on the performance of the model. Downcasting the model parameters from 32-bit floating point to 16-bit brain floating point does seem to preserve most of the performance of the full-precision model, in most cases having the least difference in performance with the 32-bit precision models, compared to lower bit precisions. 8-bit and even 4-bit precision models fare quite well, usually performing almost as well as the full

precision models. 2-bit quantization seems to be where we observe a generally significant drop in performance.

5.3.3 By task

The performance of each (quantized) model varies a lot between tasks. Especially the more difficult tasks of the SuperGLUE show that the models do not always align well with the task objective. When the foundational fine-tuned model does not perform well, the quantized models perform about similarly. The quantized models do not go below the task baseline.

5.3.4 By tuning

We can see that generally speaking, fine-tuning a model leads to significantly better results for the quantized models. Furthermore, it is interesting to note that the performance of a base model does not degrade as much after applying post-training quantization methods.

5.3.5 By quantization method

There does not seem to be a significant difference between the quantization methods. The bit precision seems to be the leading factor in the negative relationship between quantization and performance.

5.3.6 By performance

It is worth mentioning that lower bit-precision models have a worse inference speed, as seen in Section 5.2.1. In general, lower bit precision has a positive effect on the computational footprint of the compressed model, but these models generally run better on supported or specialized hardware such as Intel x86 CPUs [71]. Because the hardware on which the benchmarks are ran supports fast floating point computations, we can see that the performance metrics (runtime, samples per second, steps per second) are only better for the downcasted models.

5.3.7 By model type

As made visible in Section 5.2.3, the performance does seem to vary between model types. Notably, sequence-to-sequence models seem to generally perform really well when quantized to 4-bits or higher. When these models are quantized to 2-bits however, we see a drastic drop-off in performance (e.g. in Figure 5.47). To investigate the different model

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types further, it is possible to investigate Appendix D, as performance between models is presented.

6

Conclusion

6.1 Findings

As shown in Section 5.2, we observe a positive correlation between amount of bits used in the model weights and the model performance. Post-training quantization has a negative impact on performance, with a slight difference for 4 and 8 bits, and a significant difference for 2-bits. Downcasting, which is as simple as casting the weights to a lower bit precision such as brain-floating point 16-bits, is shown to have little impact on the performance of the model, while still compressing the model size and reducing the computational footprint. The effect that post-training quantization has on model performance varies on the task and whether the model is fine-tuned on the task. The difference between quantization methods is overshadowed by the effect that the reduction of bit precision has on the model performance.

Research question 1, "What kind of post-training quantization methods are available for large language models?", is answered by the literature review in Chapter 2 and further elaborated upon in Appendix A. There are a plethora of post-training quantization methods available for large language models.

Research question 2, "What is the effect of post-training quantization on encoder, decoder, and encoder/decoder models?", is answered by the figures in Section 5.2.3. There is a visible difference between the performance of the models, but this varies from task to task.

Research question 3 ("What is the effect of post-training quantization across different bit precisions on different large language models?"), 4 ("How do different large language models and post-training quantization methods compare in performance across different bit precisions?") and 5 ("How big is the effect that post-training quantization has on large

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language models?" are answered by the Pareto front (Section 5.2.3) and rate-distortion (Section 5.2.4) figures, and the aggregate tables (Section 5.2.1) with bit-precision in its axis. Furthermore, Appendix D and Appendix E illustrate how each model is affected by different post-training quantization methods.

Research question 6 is answered by looking at the rate distortion figures in Section 5.2.4. We can see that the reduction in performance is related to the bit-precision in which the model parameters are stored.

6.2 Future work

The work presented is constrained by the time and scale allocated for the project, but there are several interesting future steps to take in research.

6.2.1 Extended experimental setup

This research had a relatively small experimental setup, only containing 6 models and 2 benchmarks. It would be interesting to see how the effect of post-training quantization differs with higher-capacity models, varying architectures and on different tasks. Furthermore, it would be interesting to consider different types of language models for these experiments, such as Mamba [72], and observe the possible differences in output when quantized.

6.2.2 Computational limitations

The project does not include the state-of-the-art models due to computational limitations. As mentioned before, the repository was made from a model-agnostic design perspective. Given the opportunity to be able to run bigger models than the models presented, the result set of this project could be augmented with up-to-date metrics.

6.2.3 Calibration methods

As mentioned in Section 3.1, this study takes into account only on-the-fly quantization methods. There are a multitude of quantization methods that prefer or require calibration on a sample dataset to perform correctly, especially when the activation parameters are quantized. It would be interesting to widely compare these calibration methods with as well, and possibly study the effect that calibration, as opposed to on-the-fly, has on the performance.

6.2.4 Quantization aware training

Apart from post-training quantization, there is also quantization-aware training [73], which trains models so that the negative effect of quantization is reduced. Like calibration, this is another factor that can be studied.

6.2.5 Behavioral differences

Apart from the model performance, studying the behavior of quantized models compared to full-precision models can be of interest. Does the behavior change? And if so, in what way?

6.2.6 Edge devices

Quantization in machine learning is often used to be able to deploy models to edge devices. These devices might be constrained to battery power or computational power. It would be interesting to see what quantization has to offer for smaller (specialized) devices, and explore what the possibilities in this field could be.

6. CONCLUSION

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Appendix A

Overview of quantization methods

Quantization method	On the fly quantization	CPU	CUDA GPU	RoCm GPU (AMD)	Metal (Apple Silicon)	torch.compile() support	Number of bits	Supports fine-tuning through PEFT	Serializable with transformers	transformers support	Link to library
AQLM	N	Y	Y	N	N	Y	1 / 2	Y	Y	Y	https://github.com/Vals1994/AQLM
AWQ	N	N	Y	Y	N	?	4	Y	Y	Y	urlhttps://github.com/casper-hansen/AutoAWQ
bitsandbytes	Y	N	Y	N	N	N	4 / 8	Y	Y	Y	https://github.com/TimDettmers/bitsandbytes
EETQ	Y	N	Y	N	N	?	8	Y	Y	Y	https://github.com/WetEase-FuXi/EETQ
GGUF / GGML (llama.cpp)	Y	Y	Y	N	Y	N	1-8	N	?	?	https://github.com/ggerganov/llama.cpp
GPTQ	N	N	Y	Y	N	N	2 / 3 / 4 / 8	Y	Y	Y	https://github.com/AutoGPTQ/AutoGPTQ
HQQ	Y	Y	Y	N	N	Y	1 - 8	Y	N	Y	https://github.com/mobiusml/hqq/
Quanto	Y	Y	Y	N	Y	Y	2 / 4 / 8	N	N	Y	https://github.com/huggingface/quanto
FBGEMM_FPS	Y	N	Y	N	N	N	8	N	Y	Y	https://github.com/pytorch/FBGEMM
torchao	Y	?	Y	N	int4 only	?	4 / 8	Y	N	Y	https://github.com/pytorch/ao

Table A.1: Overview of quantization methods, comparing specific features and support [3].

A. OVERVIEW OF QUANTIZATION METHODS

Appendix B

Task prompts

Task prompts for the GLUE benchmark tasks can be found in Table B.1. Task prompts for the SuperGLUE benchmark tasks can be found in Table B.2.

Task	Prompt
cola	### Instruction: Is the following sentence grammatically correct? Output 1 if it is an acceptable grammatical sentence and 0 if it is not.
	### Sentence: {sentence}
	### Response:
mnli	### Instruction: Does the premise entail the hypothesis, contradict the hypothesis, or neither? Output 0 if the premise entails the hypothesis, output 2 if the premise contradicts the hypothesis, and output 1 if the premise neither entails or contradicts the hypothesis.
	### Premise: {premise}
	### Hypothesis: {hypothesis}
	### Response:

B. TASK PROMPTS

mrpc	### Instruction: Is sentence 1 semantically equivalent to sentence 2? Output 1 if they are equivalent and 0 if they are not. ### Sentence 1: {sentence1} ### Sentence 2: {sentence2} ### Response:
qnli	### Instruction: Does the context sentence contain the answer to the question? Output 1 if the answer is not in the context sentence and 0 if the answer is in the context sentence. ### Question: {question} ### Context sentence: {context} ### Response:
qqp	### Instruction: Is question 1 semantically equivalent to question 2? Output 1 if the questions are semantically equivalent and 0 if the questions are not semantically equivalent. ### Question 1: {question1} ### Question 2: {question2} ### Response:

rte	#### Instruction: Can the meaning of sentence 1 be entailed from sentence 2, or the other way around? Output 0 if the meaning of sentence 1 can be entailed from sentence 2, or the other way around, and 1 if the sentences cannot be entailed from each other.
	#### Sentence 1: {sentence1}
	#### Sentence 2: {sentence2}
	#### Response:
sst2	#### Instruction: Predict the sentiment of the sentence. Output 0 if the sentiment is negative, and 1 if the sentiment is positive.
	#### Sentence: {sentence}
	#### Response:
wnli	#### Instruction: Can sentence 2 be entailed from sentence 1. Output 1 if sentence 2 can be entailed from sentence 1, and output 0 if not.
	#### Sentence 1: {sentence1}
	#### Sentence 2: {sentence2}
	#### Response:

Table B.1: Prompts accompanying the GLUE benchmark tasks for inputting decoder and sequence-to-sequence models. This prompt is tokenized and used to obtain the output of the model.

B. TASK PROMPTS

Task	Prompt
boolq	### Instruction: Answer the question based on the context. Output 1 if the answer is yes, and 0 if the answer is no. ### Question: {question} ## Context: {context} ### Response:
cb	### Instruction: Determine whether the hypothesis is true, false, or neither based on the premise. Output 0 for true, 1 for false, and 2 for neither. ### Premise: {premise} ### Hypothesis: {hypothesis} ### Response:
copa	### Instruction: Choose the alternative that more plausibly answers the question based on the premise. Output 0 for the first alternative and 1 for the second alternative. ### Premise: {premise} ### Alternative 1: {alternative1} ### Alternative 2: {alternative2} ### Question: {question} ### Response:

rte	### Instruction: Does the premise entail the hypothesis? Output 1 if yes, and 0 if no. ### Premise: {premise} ### Hypothesis: {hypothesis} ### Response:
wic	### Instruction: Does the word have the same meaning in both sentences? Output 1 if yes, and 0 if no. ### Word: {word} ### Sentence 1: {sentence1} ### Sentence 2: {sentence2} ### Response:
wsc	### Instruction:Which text does the pronoun refer to? Output 0 if text 1, and output 1 if text 2. ### Sentence: {sentence} ### Text 1: {text1} ### Text 2: {text2} ### Response:

Table B.2: Prompts accompanying the SuperGLUE benchmark tasks for inputting decoder and sequence-to-sequence models. This prompt is tokenized and used to obtain the output of the model.

B. TASK PROMPTS

Appendix C

Experimental configuration

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                        "num_labels": 2
                    },
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                        "name": "mnli",
                        "task_type": "sequence_classification",
                        "metric": mf.compute_metrics_accuracy,
                        "num_labels": 3
                    },
                    {
                        "name": "mrpc",
                        "task_type":
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                    },
                    {
```

C. EXPERIMENTAL CONFIGURATION

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    },
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    },
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```

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    },
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    },
    {
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    }
  ]
}
```


C. EXPERIMENTAL CONFIGURATION

```
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},

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        "bert-large-uncased"
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    "decoder": [
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        "openai-community/gpt2-medium"
    ],
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        "facebook/bart-base",
```

```
        "google/flan-t5-base"  
    ]  
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}
```

C. EXPERIMENTAL CONFIGURATION

Appendix D

Detailed tables

We could not fit every individual result of the benchmarks in the main report, as the result set was simply too large to not overwhelm the reader. As a result, we present the entire result set in this appendix, as a courtesy to present the data underlying the aggregate results presented in the report. The results are displayed from Table D.1 to Table D.13. The memory footprint is in bytes. Every percentage is calculated from the full-precision (32 bit) model.

Table D.1: Results on the glue/cola task

Model Name	FT	Quantization Method	# Bits	Eval Loss	%	Eval Mcc	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.62	-	-0.04	-	7.97	-	130.81	-	32.73	-	437,943,304	-
		BitsAndBytes	4	0.64	1.92 %	0.00	109.77 %	12.17	52.67 %	85.68	-34.50 %	21.44	-34.50 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.65	3.48 %	-0.04	11.05 %	34.50	332.71 %	30.23	-76.89 %	7.57	-76.89 %	133,451,268	-69.53 %
		Downcasting	16	0.62	0.00 %	-0.04	0.00 %	5.59	-29.83 %	186.42	42.51 %	46.65	42.51 %	218,975,748	-50.00 %
		HQQ	2	0.69	10.96 %	-0.04	13.03 %	16.21	103.26 %	64.36	-50.80 %	16.11	-50.80 %	117,220,744	-73.23 %
		HQQ	4	0.63	0.41 %	-0.04	0.00 %	14.16	77.60 %	73.66	-43.69 %	18.43	-43.69 %	138,602,248	-68.35 %
		HQQ	8	0.62	-0.04 %	-0.04	0.00 %	11.83	48.35 %	88.18	-32.59 %	22.07	-32.59 %	181,365,256	-58.59 %
		Quanto	2	0.69	10.95 %	-0.02	51.78 %	20.02	151.08 %	52.10	-60.17 %	13.04	-60.17 %	437,943,896	0.00 %
		Quanto	4	0.64	2.69 %	-0.06	-54.39 %	18.51	132.12 %	56.36	-56.92 %	14.10	-56.92 %	437,943,896	0.00 %
		Quanto	8	0.62	-0.03 %	-0.04	13.44 %	9.90	24.18 %	105.34	-19.47 %	26.36	-19.47 %	437,943,896	0.00 %
	Yes		32	0.95	-	0.43	-	7.85	-	132.87	-	33.25	-	437,943,304	-
		BitsAndBytes	4	0.91	-3.39 %	0.44	1.67 %	12.35	57.36 %	84.44	-36.45 %	21.13	-36.45 %	90,689,028	-79.29 %
		BitsAndBytes	8	1.56	65.03 %	0.12	-71.21 %	33.41	325.57 %	31.22	-76.50 %	7.81	-76.50 %	133,451,268	-69.53 %
		Downcasting	16	0.95	0.07 %	0.43	-0.44 %	5.55	-29.35 %	188.08	41.55 %	47.06	41.55 %	218,975,748	-50.00 %
		HQQ	2	1.44	52.29 %	0.30	-31.13 %	16.23	106.81 %	64.25	-51.65 %	16.08	-51.65 %	117,220,744	-73.23 %
		HQQ	4	0.95	0.50 %	0.44	1.91 %	14.12	79.90 %	73.86	-44.41 %	18.48	-44.41 %	138,602,248	-68.35 %
		HQQ	8	0.95	0.09 %	0.43	0.64 %	11.82	50.57 %	88.24	-33.59 %	22.08	-33.59 %	181,365,256	-58.59 %
		Quanto	2	1.10	16.72 %	0.24	-43.80 %	19.89	153.42 %	52.43	-60.54 %	13.12	-60.54 %	437,943,896	0.00 %
		Quanto	4	0.93	-2.04 %	0.44	2.78 %	18.60	136.93 %	56.08	-57.79 %	14.03	-57.79 %	437,943,896	0.00 %
		Quanto	8	0.94	-0.52 %	0.43	0.64 %	9.92	26.35 %	105.16	-20.86 %	26.32	-20.86 %	437,943,896	0.00 %
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
bert-large-uncased	No		32	1.80	-	0.04	-	18.05	-	57.78	-	14.46	-	1,340,583,944	-
		BitsAndBytes	4	1.77	-1.26 %	0.02	-56.23 %	26.43	46.39 %	39.47	-31.69 %	9.88	-31.68 %	215,738,372	-83.91 %
		BitsAndBytes	8	1.27	-29.05 %	-0.01	-131.60 %	66.16	266.48 %	15.77	-72.71 %	3.94	-72.71 %	367,257,604	-72.60 %
		Downcasting	16	1.80	-0.02 %	0.04	0.00 %	10.92	-39.49 %	95.48	65.26 %	23.89	65.26 %	670,296,068	-50.00 %
		HQQ	2	0.72	-60.05 %	0.05	14.86 %	34.70	92.22 %	30.06	-47.98 %	7.52	-47.97 %	204,182,024	-84.77 %
		HQQ	4	1.79	-0.60 %	0.04	0.00 %	30.91	71.25 %	33.74	-41.61 %	8.44	-41.60 %	279,942,152	-79.12 %
		HQQ	8	1.80	0.23 %	0.03	-12.39 %	25.41	40.77 %	41.04	-28.96 %	10.27	-28.96 %	431,462,408	-67.82 %

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Model Name	FT	Quantization Method	# Bits	Eval Loss	%	Eval Mcc	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%	
		Quanto	4	0.63	0.83 %	0.01	-73.62 %	14.62	1.76 %	71.34	-1.73 %	17.85	-1.73 %	510,348,344	0.00 %	
		Quanto	8	0.62	0.06 %	0.04	-21.02 %	14.74	2.58 %	70.77	-2.51 %	17.71	-2.52 %	510,348,344	0.00 %	
		—	—	—	—	—	—	—	—	—	—	—	—	—	—	
openai-community gpt2-medium	No		32	3.16	-	0.00	-	35.60	-	29.29	-	7.33	-	1,444,466,784	-	
		BitsAndBytes	4	2.35	-25.44 %	0.00	-	36.55	2.66 %	28.54	-2.59 %	7.14	-2.58 %	281,831,472	-80.49 %	
		BitsAndBytes	8	3.14	-0.37 %	0.00	-	46.82	31.51 %	22.27	-23.96 %	5.57	-23.96 %	432,826,416	-70.04 %	
		Downcasting	16	3.24	2.56 %	0.00	-	17.31	-51.38 %	60.25	105.67 %	15.08	105.68 %	734,816,304	-49.13 %	
		HQQ	2	1.93	-38.93 %	0.00	-	36.59	2.77 %	28.50	-2.70 %	7.13	-2.69 %	1,444,459,104	-0.00 %	
		HQQ	4	2.99	-5.13 %	0.00	-	36.53	2.59 %	28.55	-2.53 %	7.14	-2.52 %	1,444,459,616	-0.00 %	
		HQQ	8	3.15	-0.25 %	0.00	-	36.54	2.62 %	28.55	-2.55 %	7.14	-2.55 %	1,444,460,640	-0.00 %	
		Quanto	2	2.43	-23.11 %	0.00	-	35.94	0.93 %	29.02	-0.93 %	7.26	-0.91 %	1,444,466,792	0.00 %	
		Quanto	4	3.22	2.02 %	0.00	-	35.86	0.72 %	29.08	-0.72 %	7.28	-0.71 %	1,444,466,792	0.00 %	
		Quanto	8	3.16	0.09 %	0.00	-	35.82	0.61 %	29.12	-0.61 %	7.29	-0.60 %	1,444,466,792	0.00 %	
	Yes		32	0.71	-	0.04	-	35.66	-	29.25	-	7.32	-	1,444,466,784	-	
		BitsAndBytes	4	0.71	-0.62 %	0.05	43.14 %	36.47	2.29 %	28.59	-2.24 %	7.16	-2.23 %	281,831,472	-80.49 %	
		BitsAndBytes	8	0.71	-0.14 %	0.03	-19.70 %	48.33	35.54 %	21.58	-26.22 %	5.40	-26.22 %	432,826,416	-70.04 %	
		Downcasting	16	0.71	-0.02 %	0.04	4.26 %	17.28	-51.53 %	60.34	106.31 %	15.10	106.31 %	734,816,304	-49.13 %	
		HQQ	2	0.78	9.03 %	0.04	19.26 %	36.49	2.33 %	28.58	-2.28 %	7.15	-2.28 %	1,444,459,104	-0.00 %	
		HQQ	4	0.69	-3.33 %	0.03	-12.32 %	36.59	2.60 %	28.51	-2.53 %	7.13	-2.53 %	1,444,459,616	-0.00 %	
		HQQ	8	0.72	0.28 %	0.04	17.85 %	36.49	2.33 %	28.58	-2.27 %	7.15	-2.27 %	1,444,460,640	-0.00 %	
		Quanto	2	0.85	19.14 %	0.05	22.00 %	36.03	1.04 %	28.95	-1.03 %	7.24	-1.02 %	1,444,466,792	0.00 %	
		Quanto	4	0.68	-4.12 %	0.03	-26.79 %	35.93	0.76 %	29.03	-0.75 %	7.26	-0.75 %	1,444,466,792	0.00 %	
		Quanto	8	0.71	-0.09 %	0.03	-15.44 %	35.85	0.53 %	29.09	-0.53 %	7.28	-0.52 %	1,444,466,792	0.00 %	

Table D.2: Results on the glue/mnli task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.32	-	1.12	-	73.69	-	133.20	-	33.30	-	437,946,380	-
		BitsAndBytes	4	0.32	0.00 %	1.11	-0.18 %	115.18	56.30 %	85.22	-36.02 %	21.31	-36.02 %	90,690,566	-79.29 %
		BitsAndBytes	8	0.33	2.93 %	1.11	-0.88 %	322.50	337.65 %	30.43	-77.15 %	7.61	-77.15 %	133,452,806	-69.53 %
		Downcasting	16	0.32	-0.06 %	1.12	0.01 %	52.73	-28.45 %	186.16	39.76 %	46.54	39.76 %	218,977,286	-50.00 %
		HQQ	2	0.32	-1.08 %	1.11	-0.94 %	152.74	107.27 %	64.26	-51.75 %	16.07	-51.76 %	117,220,940	-73.23 %
		HQQ	4	0.32	0.22 %	1.11	-0.28 %	134.69	82.78 %	72.87	-45.29 %	18.22	-45.29 %	138,602,636	-68.35 %
		HQQ	8	0.32	-0.06 %	1.12	-0.02 %	112.42	52.56 %	87.31	-34.45 %	21.83	-34.45 %	181,366,028	-58.59 %
		Quanto	2	0.33	2.32 %	1.12	0.34 %	188.57	155.91 %	52.05	-60.92 %	13.01	-60.93 %	437,946,972	0.00 %
		Quanto	4	0.32	-0.45 %	1.12	-0.11 %	174.76	137.16 %	56.16	-57.84 %	14.04	-57.84 %	437,946,972	0.00 %
		Quanto	8	0.32	-0.10 %	1.12	-0.05 %	93.31	26.62 %	105.19	-21.03 %	26.30	-21.03 %	437,946,972	0.00 %
	Yes		32	0.33	-	1.10	-	73.82	-	132.96	-	33.24	-	437,946,380	-
		BitsAndBytes	4	0.33	0.00 %	1.10	-0.00 %	115.57	56.56 %	84.92	-36.13 %	21.23	-36.13 %	90,690,566	-79.29 %
		BitsAndBytes	8	0.33	0.00 %	1.10	-0.00 %	309.04	318.65 %	31.76	-76.11 %	7.94	-76.11 %	133,452,806	-69.53 %
		Downcasting	16	0.33	0.00 %	1.10	-0.01 %	52.65	-28.67 %	186.40	40.19 %	46.60	40.19 %	218,977,286	-50.00 %
		HQQ	2	0.33	0.00 %	1.10	0.09 %	152.49	106.57 %	64.36	-51.59 %	16.09	-51.59 %	117,220,940	-73.23 %
		HQQ	4	0.33	0.00 %	1.10	0.00 %	134.15	81.73 %	73.16	-44.97 %	18.29	-44.97 %	138,602,636	-68.35 %
		HQQ	8	0.33	0.00 %	1.10	0.00 %	112.02	51.75 %	87.61	-34.10 %	21.91	-34.10 %	181,366,028	-58.59 %
		Quanto	2	0.32	-2.80 %	1.10	-0.00 %	187.78	154.37 %	52.27	-60.69 %	13.07	-60.69 %	437,946,972	0.00 %
		Quanto	4	0.33	0.00 %	1.10	0.00 %	175.29	137.46 %	55.99	-57.89 %	14.00	-57.89 %	437,946,972	0.00 %
		Quanto	8	0.33	0.00 %	1.10	0.00 %	92.77	25.67 %	105.80	-20.43 %	26.45	-20.43 %	437,946,972	0.00 %
bert-large-uncased	No		32	0.36	-	1.49	-	168.97	-	58.09	-	14.52	-	1,340,588,044	-
		BitsAndBytes	4	0.36	0.03 %	1.49	0.05 %	241.82	43.11 %	40.59	-30.12 %	10.15	-30.12 %	215,740,422	-83.91 %
		BitsAndBytes	8	0.35	-1.05 %	1.28	-14.30 %	608.69	260.23 %	16.12	-72.24 %	4.03	-72.24 %	367,259,654	-72.60 %
		Downcasting	16	0.36	0.00 %	1.49	-0.02 %	99.47	-41.13 %	98.67	69.87 %	24.67	69.87 %	670,298,118	-50.00 %
		HQQ	2	0.33	-8.83 %	1.17	-21.29 %	316.04	87.03 %	31.06	-46.53 %	7.76	-46.53 %	204,182,284	-84.77 %
		HQQ	4	0.36	0.09 %	1.50	0.66 %	279.13	65.19 %	35.16	-39.46 %	8.79	-39.46 %	279,942,668	-79.12 %
		HQQ	8	0.36	0.00 %	1.49	0.05 %	234.39	38.72 %	41.87	-27.91 %	10.47	-27.91 %	431,463,436	-67.82 %
		Quanto	2	0.32	-10.39 %	1.13	-24.04 %	381.27	125.64 %	25.74	-55.68 %	6.44	-55.68 %	1,340,589,212	0.00 %
		Quanto	4	0.36	-0.03 %	1.51	1.12 %	355.69	110.50 %	27.59	-52.49 %	6.90	-52.50 %	1,340,589,212	0.00 %
		Quanto	8	0.36	0.00 %	1.49	-0.00 %	198.48	17.46 %	49.45	-14.87 %	12.36	-14.87 %	1,340,589,212	0.00 %

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D. DETAILED TABLES

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
	Yes	BitsAndBytes	32	0.33	-	1.10	-	164.48	-	59.67	-	14.92	-	1,340,588,044	-
		BitsAndBytes	4	0.33	0.00 %	1.10	0.00 %	242.63	47.52 %	40.45	-32.21 %	10.11	-32.21 %	215,740,422	-83.91 %
		BitsAndBytes	8	0.33	0.00 %	1.10	0.00 %	562.16	241.78 %	17.46	-70.74 %	4.37	-70.74 %	367,259,654	-72.60 %
		Downcasting	16	0.33	0.00 %	1.10	-0.16 %	99.98	-39.21 %	98.17	64.51 %	24.55	64.51 %	670,298,118	-50.00 %
		HQQ	2	0.32	-2.80 %	1.10	0.40 %	319.14	94.03 %	30.75	-48.46 %	7.69	-48.47 %	204,182,284	-84.77 %
		HQQ	4	0.33	0.00 %	1.10	0.00 %	277.54	68.74 %	35.36	-40.74 %	8.84	-40.74 %	279,942,668	-79.12 %
		HQQ	8	0.33	0.00 %	1.10	-0.00 %	233.07	41.70 %	42.11	-29.43 %	10.53	-29.43 %	431,463,436	-67.82 %
		Quanto	2	0.33	0.00 %	1.10	0.42 %	380.85	131.55 %	25.77	-56.81 %	6.44	-56.81 %	1,340,589,212	0.00 %
		Quanto	4	0.33	0.00 %	1.10	0.07 %	355.37	116.06 %	27.62	-53.72 %	6.91	-53.71 %	1,340,589,212	0.00 %
		Quanto	8	0.33	0.00 %	1.10	0.00 %	196.71	19.60 %	49.90	-16.39 %	12.47	-16.39 %	1,340,589,212	0.00 %

Table D.3: Results on the glue/mrpc task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval F1 Score	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No	BitsAndBytes	32	0.68	-	0.81	-	0.66	-	3.02	-	134.98	-	33.75	-	437,943,304	-
		BitsAndBytes	4	0.68	0.00 %	0.81	0.00 %	0.66	0.12 %	4.65	53.69 %	87.83	-34.93 %	21.96	-34.93 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.45	-33.69 %	0.53	-35.17 %	0.71	8.29 %	12.92	327.42 %	31.58	-76.60 %	7.89	-76.60 %	133,451,268	-69.53 %
		Downcasting	16	0.68	0.00 %	0.81	0.00 %	0.66	0.01 %	2.11	-30.26 %	193.55	43.39 %	48.39	43.39 %	218,975,748	-50.00 %
		HQQ	2	0.69	0.72 %	0.81	-0.18 %	0.66	0.17 %	6.23	106.10 %	65.49	-51.48 %	16.37	-51.48 %	117,220,744	-73.23 %
		HQQ	4	0.68	0.00 %	0.81	0.00 %	0.65	-0.48 %	5.45	80.44 %	74.81	-44.58 %	18.70	-44.58 %	138,602,248	-68.35 %
		HQQ	8	0.68	0.00 %	0.81	0.00 %	0.66	0.02 %	4.54	50.34 %	89.79	-33.48 %	22.45	-33.48 %	181,365,256	-58.59 %
		Quanto	2	0.48	-29.39 %	0.56	-31.57 %	0.70	6.42 %	7.63	152.54 %	53.45	-60.40 %	13.36	-60.40 %	437,943,896	0.00 %
		Quanto	4	0.68	0.00 %	0.81	0.00 %	0.66	1.04 %	7.06	133.56 %	57.79	-57.19 %	14.45	-57.19 %	437,943,896	0.00 %
		Quanto	8	0.68	0.00 %	0.81	0.00 %	0.66	0.20 %	3.79	25.41 %	107.64	-20.26 %	26.91	-20.26 %	437,943,896	0.00 %
	Yes	BitsAndBytes	32	0.80	-	0.85	-	0.90	-	3.03	-	134.84	-	33.71	-	437,943,304	-
		BitsAndBytes	4	0.79	-1.53 %	0.84	-1.33 %	0.91	2.17 %	4.63	52.92 %	88.18	-34.61 %	22.05	-34.60 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.63	-21.41 %	0.73	-14.56 %	1.64	83.38 %	13.01	329.86 %	31.37	-76.74 %	7.84	-76.74 %	133,451,268	-69.53 %
		Downcasting	16	0.80	0.00 %	0.85	0.00 %	0.90	0.40 %	2.14	-29.42 %	191.05	41.68 %	47.76	41.68 %	218,975,748	-50.00 %
		HQQ	2	0.73	-9.17 %	0.80	-6.14 %	1.03	15.54 %	6.20	104.80 %	65.84	-51.17 %	16.46	-51.17 %	117,220,744	-73.23 %
		HQQ	4	0.81	1.22 %	0.86	0.74 %	0.84	-5.81 %	5.42	79.07 %	75.30	-44.16 %	18.82	-44.16 %	138,602,248	-68.35 %
		HQQ	8	0.80	0.00 %	0.85	0.00 %	0.89	-0.31 %	4.53	49.81 %	90.01	-33.25 %	22.50	-33.25 %	181,365,256	-58.59 %
		Quanto	2	0.69	-13.76 %	0.78	-8.00 %	1.07	19.45 %	7.59	150.83 %	53.76	-60.13 %	13.44	-60.13 %	437,943,896	0.00 %
		Quanto	4	0.80	0.31 %	0.85	0.12 %	0.91	1.13 %	7.07	133.60 %	57.72	-57.19 %	14.43	-57.19 %	437,943,896	0.00 %
		Quanto	8	0.80	0.00 %	0.85	0.00 %	0.89	-0.37 %	3.80	25.49 %	107.45	-20.32 %	26.86	-20.31 %	437,943,896	0.00 %
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
bert-large-uncased	No	BitsAndBytes	32	0.32	-	0.00	-	1.81	-	6.87	-	59.40	-	14.85	-	1,340,583,944	-
		BitsAndBytes	4	0.32	0.00 %	0.00	-	1.81	-0.01 %	10.15	47.81 %	40.19	-32.35 %	10.05	-32.34 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.31	-0.78 %	0.00	-	1.27	-30.05 %	25.45	270.59 %	16.03	-73.02 %	4.01	-73.02 %	367,257,604	-72.60 %
		Downcasting	16	0.32	0.00 %	0.00	-	1.81	0.07 %	4.07	-40.69 %	100.16	68.61 %	25.04	68.61 %	670,296,068	-50.00 %
		HQQ	2	0.62	96.90 %	0.76	-	0.72	-60.31 %	13.47	96.10 %	30.29	-49.01 %	7.57	-49.00 %	204,182,024	-84.77 %
		HQQ	4	0.32	0.00 %	0.00	-	1.82	0.49 %	11.76	71.28 %	34.68	-41.62 %	8.67	-41.62 %	279,942,152	-79.12 %
		HQQ	8	0.32	0.00 %	0.00	-	1.81	0.07 %	9.77	42.25 %	41.76	-29.70 %	10.44	-29.70 %	431,462,408	-67.82 %
		Quanto	2	0.67	110.85 %	0.79	-	0.64	-64.58 %	15.88	131.22 %	25.69	-56.75 %	6.42	-56.75 %	1,340,585,112	0.00 %
		Quanto	4	0.32	0.00 %	0.00	-	1.85	1.85 %	14.83	115.87 %	27.52	-53.68 %	6.88	-53.68 %	1,340,585,112	0.00 %
		Quanto	8	0.32	0.00 %	0.00	-	1.81	-0.09 %	8.33	21.23 %	49.00	-17.51 %	12.25	-17.51 %	1,340,585,112	0.00 %
	Yes	BitsAndBytes	32	0.32	-	0.00	-	0.70	-	6.85	-	59.57	-	14.89	-	1,340,583,944	-
		BitsAndBytes	4	0.32	0.00 %	0.00	-	0.70	0.02 %	10.10	47.45 %	40.40	-32.18 %	10.10	-32.18 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.32	0.00 %	0.00	-	0.70	0.02 %	24.09	251.74 %	16.93	-71.57 %	4.23	-71.57 %	367,257,604	-72.60 %
		Downcasting	16	0.32	0.00 %	0.00	-	0.70	0.10 %	4.17	-39.17 %	97.92	64.39 %	24.48	64.38 %	670,296,068	-50.00 %
		HQQ	2	0.68	116.28 %	0.81	-	0.63	-9.31 %	13.24	93.31 %	30.81	-48.27 %	7.70	-48.27 %	204,182,024	-84.77 %
		HQQ	4	0.32	0.00 %	0.00	-	0.70	0.30 %	11.60	69.38 %	35.17	-40.96 %	8.79	-40.96 %	279,942,152	-79.12 %
		HQQ	8	0.32	0.00 %	0.00	-	0.70	0.00 %	9.78	42.79 %	41.72	-29.97 %	10.43	-29.97 %	431,462,408	-67.82 %
		Quanto	2	0.68	116.28 %	0.81	-	0.66	-4.83 %	15.88	131.78 %	25.70	-56.86 %	6.42	-56.86 %	1,340,585,112	0.00 %
		Quanto	4	0.32	0.00 %	0.00	-	0.72	3.39 %	14.80	116.02 %	27.57	-53.71 %	6.89	-53.71 %	1,340,585,112	0.00 %
		Quanto	8	0.32	0.00 %	0.00	-	0.70	0.16 %	8.44	23.15 %	48.37	-18.80 %	12.09	-18.80 %	1,340,585,112	0.00 %
facebook bart-base	No	BitsAndBytes	32	0.68	-	0.81	-	0.64	-	6.46	-	63.12	-	15.78	-	560,050,184	-
		BitsAndBytes	4	0.68	0.00 %	0.81	0.00 %	0.64	-0.09 %	7.95	23.04 %	51.30	-18.73 %	12.82	-18.73 %	130,504,708	-76.70 %

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D. DETAILED TABLES

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval F1 Score	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
		Downcasting	16	0.73	-0.33 %	0.82	-0.24 %	0.55	-0.06 %	6.78	-51.39 %	60.19	105.70 %	15.05	105.70 %	734,816,304	-49.13 %
		HQQ	2	0.75	2.01 %	0.82	0.36 %	0.55	-0.22 %	14.29	2.51 %	28.55	-2.45 %	7.14	-2.46 %	1,444,459,104	-0.00 %
		HQQ	4	0.73	0.00 %	0.82	0.00 %	0.55	0.49 %	14.31	2.61 %	28.52	-2.54 %	7.13	-2.54 %	1,444,459,616	-0.00 %
		HQQ	8	0.73	0.00 %	0.82	0.00 %	0.55	-0.04 %	14.27	2.32 %	28.60	-2.27 %	7.15	-2.27 %	1,444,460,640	-0.00 %
		Quanto	2	0.67	-8.03 %	0.74	-10.09 %	0.61	11.18 %	14.05	0.76 %	29.04	-0.75 %	7.26	-0.75 %	1,444,466,792	0.00 %
		Quanto	4	0.73	0.00 %	0.82	-0.58 %	0.55	0.18 %	14.00	0.43 %	29.14	-0.43 %	7.28	-0.44 %	1,444,466,792	0.00 %
		Quanto	8	0.73	0.00 %	0.82	0.00 %	0.55	0.03 %	13.96	0.12 %	29.23	-0.12 %	7.31	-0.12 %	1,444,466,792	0.00 %

Table D.4: Results on the glue/qnli task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.47	-	0.71	-	40.45	-	135.04	-	33.77	-	437,943,304	-
		BitsAndBytes	4	0.48	0.50 %	0.71	-0.16 %	62.53	54.56 %	87.37	-35.30 %	21.85	-35.30 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.48	2.28 %	0.71	-0.98 %	174.06	330.27 %	31.39	-76.76 %	7.85	-76.76 %	133,451,268	-69.53 %
		Downcasting	16	0.47	0.08 %	0.71	0.03 %	28.73	-28.99 %	190.17	40.82 %	47.55	40.82 %	218,975,748	-50.00 %
		HQQ	2	0.43	-9.09 %	0.71	-0.76 %	83.47	106.34 %	65.45	-51.54 %	16.36	-51.54 %	117,220,744	-73.23 %
		HQQ	4	0.48	0.46 %	0.71	0.01 %	72.88	80.16 %	74.96	-44.49 %	18.74	-44.49 %	138,602,248	-68.35 %
		HQQ	8	0.47	0.12 %	0.71	-0.03 %	61.00	50.80 %	89.55	-33.69 %	22.39	-33.69 %	181,365,256	-58.59 %
		Quanto	2	0.53	12.42 %	0.69	-3.11 %	113.82	181.35 %	48.00	-64.46 %	12.00	-64.46 %	437,943,896	0.00 %
		Quanto	4	0.47	0.15 %	0.71	-0.49 %	105.43	160.61 %	51.82	-61.63 %	12.96	-61.63 %	437,943,896	0.00 %
		Quanto	8	0.47	-0.04 %	0.71	-0.13 %	50.63	25.16 %	107.90	-20.10 %	26.98	-20.10 %	437,943,896	0.00 %
	Yes		32	0.78	-	0.53	-	40.57	-	134.66	-	33.67	-	437,943,304	-
		BitsAndBytes	4	0.78	0.05 %	0.53	-0.31 %	62.44	53.91 %	87.49	-35.03 %	21.88	-35.03 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.56	-28.14 %	0.77	46.15 %	177.38	337.23 %	30.80	-77.13 %	7.70	-77.13 %	133,451,268	-69.53 %
		Downcasting	16	0.78	-0.02 %	0.53	-0.06 %	28.76	-29.10 %	189.93	41.05 %	47.49	41.05 %	218,975,748	-50.00 %
		HQQ	2	0.76	-2.87 %	0.55	4.95 %	83.51	105.84 %	65.42	-51.42 %	16.36	-51.42 %	117,220,744	-73.23 %
		HQQ	4	0.78	-0.30 %	0.53	-0.02 %	72.78	79.38 %	75.07	-44.25 %	18.77	-44.25 %	138,602,248	-68.35 %
		HQQ	8	0.78	-0.07 %	0.53	0.02 %	60.87	50.02 %	89.76	-33.34 %	22.44	-33.34 %	181,365,256	-58.59 %
		Quanto	2	0.74	-5.30 %	0.59	12.09 %	113.27	179.19 %	48.23	-64.18 %	12.06	-64.18 %	437,943,896	0.00 %
		Quanto	4	0.78	-0.12 %	0.53	-0.01 %	105.19	159.28 %	51.94	-61.43 %	12.99	-61.43 %	437,943,896	0.00 %
		Quanto	8	0.78	0.12 %	0.53	-0.32 %	50.81	25.23 %	107.53	-20.15 %	26.89	-20.15 %	437,943,896	0.00 %
bert-large-uncased	No		32	0.52	-	1.26	-	92.83	-	58.85	-	14.71	-	1,340,583,944	-
		BitsAndBytes	4	0.52	-0.17 %	1.27	0.40 %	134.75	45.15 %	40.54	-31.11 %	10.14	-31.11 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.50	-4.40 %	0.96	24.08 %	339.30	265.49 %	16.10	-72.64 %	4.03	-72.64 %	367,257,604	-72.60 %
		Downcasting	16	0.52	0.03 %	1.26	-0.02 %	55.06	-40.69 %	99.22	68.60 %	24.81	68.60 %	670,296,068	-50.00 %
		HQQ	2	0.51	-3.70 %	0.76	-39.81 %	175.58	89.13 %	31.11	-47.13 %	7.78	-47.13 %	204,182,024	-84.77 %
		HQQ	4	0.52	-0.07 %	1.27	0.33 %	154.94	66.90 %	35.26	-40.08 %	8.82	-40.09 %	279,942,152	-79.12 %
		HQQ	8	0.52	0.00 %	1.27	0.12 %	129.90	39.92 %	42.06	-28.53 %	10.52	-28.54 %	431,462,408	-67.82 %
		Quanto	2	0.51	-2.72 %	0.72	-43.35 %	211.55	127.88 %	25.82	-56.12 %	6.46	-56.12 %	1,340,585,112	0.00 %
		Quanto	4	0.52	0.07 %	1.27	0.46 %	198.08	113.38 %	27.58	-53.14 %	6.90	-53.14 %	1,340,585,112	0.00 %
		Quanto	8	0.52	0.03 %	1.26	-0.01 %	110.02	18.52 %	49.65	-15.62 %	12.41	-15.63 %	1,340,585,112	0.00 %
	Yes		32	0.49	-	0.69	-	92.41	-	59.11	-	14.78	-	1,340,583,944	-
		BitsAndBytes	4	0.51	2.18 %	0.69	0.03 %	134.65	45.70 %	40.57	-31.36 %	10.14	-31.36 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.49	0.00 %	0.69	0.03 %	312.06	237.68 %	17.51	-70.39 %	4.38	-70.39 %	367,257,604	-72.60 %
		Downcasting	16	0.49	0.00 %	0.69	0.14 %	55.60	-39.84 %	98.26	66.22 %	24.57	66.23 %	670,296,068	-50.00 %
		HQQ	2	0.49	0.00 %	0.70	1.52 %	175.86	90.30 %	31.06	-47.45 %	7.77	-47.45 %	204,182,024	-84.77 %
		HQQ	4	0.49	0.00 %	0.69	0.00 %	154.35	67.02 %	35.39	-40.13 %	8.85	-40.13 %	279,942,152	-79.12 %
		HQQ	8	0.49	0.00 %	0.69	0.00 %	129.47	40.10 %	42.20	-28.62 %	10.55	-28.62 %	431,462,408	-67.82 %
		Quanto	2	0.49	0.00 %	0.70	1.42 %	211.55	128.91 %	25.82	-56.31 %	6.46	-56.32 %	1,340,585,112	0.00 %
		Quanto	4	0.49	0.00 %	0.69	0.01 %	197.15	113.34 %	27.71	-53.13 %	6.93	-53.12 %	1,340,585,112	0.00 %
		Quanto	8	0.51	2.18 %	0.69	-0.00 %	109.73	18.74 %	49.78	-15.78 %	12.45	-15.78 %	1,340,585,112	0.00 %
google flan-t5-base	No		32	0.49	-	0.91	-	249.11	-	21.93	-	5.48	-	893,982,728	-
openai-community gpt2	No		32	0.51	-	1.66	-	75.52	-	72.34	-	18.09	-	510,348,336	-
		BitsAndBytes	4	0.51	0.00 %	2.33	39.86 %	79.05	4.68 %	69.10	-4.47 %	17.28	-4.47 %	134,063,640	-73.73 %

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
		BitsAndBytes	8	0.51	0.00 %	1.56	-6.40 %	128.58	70.26 %	42.49	-41.27 %	10.62	-41.26 %	176,530,968	-65.41 %
		Downcasting	16	0.51	0.00 %	1.66	-0.23 %	41.97	-44.42 %	130.16	79.93 %	32.55	79.93 %	261,465,624	-48.77 %
		HQQ	2	0.51	0.00 %	2.43	45.84 %	78.44	3.86 %	69.65	-3.72 %	17.41	-3.72 %	510,342,576	-0.00 %
		HQQ	4	0.51	0.00 %	1.61	-3.31 %	78.58	4.05 %	69.52	-3.89 %	17.38	-3.89 %	510,342,960	-0.00 %
		HQQ	8	0.51	0.00 %	1.67	0.43 %	78.22	3.57 %	69.84	-3.45 %	17.46	-3.45 %	510,343,728	-0.00 %
		Quanto	2	0.51	0.00 %	1.63	-2.27 %	76.65	1.50 %	71.27	-1.48 %	17.82	-1.48 %	510,348,344	0.00 %
		Quanto	4	0.51	0.00 %	1.66	-0.47 %	76.35	1.09 %	71.56	-1.08 %	17.89	-1.08 %	510,348,344	0.00 %
		Quanto	8	0.51	0.00 %	1.66	-0.14 %	75.75	0.30 %	72.12	-0.30 %	18.03	-0.30 %	510,348,344	0.00 %
	Yes	BitsAndBytes	32	0.78	-0.71 %	1.56	-1.55 %	74.94	6.28 %	72.90	-5.91 %	18.23	-5.91 %	510,348,336	-
		BitsAndBytes	4	0.78	-0.16 %	0.70	-1.55 %	79.64	6.28 %	68.59	-5.91 %	17.15	-5.91 %	134,063,640	-73.73 %
		BitsAndBytes	8	0.78	-0.14 %	0.71	0.09 %	113.08	50.89 %	48.31	-33.73 %	12.08	-33.73 %	176,530,968	-65.41 %
		Downcasting	16	0.78	-0.02 %	0.71	-0.18 %	42.04	-43.90 %	129.93	78.24 %	32.49	78.24 %	261,465,624	-48.77 %
		HQQ	2	0.78	0.00 %	0.71	-0.76 %	78.63	4.93 %	69.47	-4.70 %	17.37	-4.70 %	510,342,576	-0.00 %
		HQQ	4	0.78	0.02 %	0.72	0.58 %	78.44	4.67 %	69.64	-4.46 %	17.41	-4.47 %	510,342,960	-0.00 %
		HQQ	8	0.78	-0.02 %	0.71	-0.01 %	78.38	4.59 %	69.70	-4.39 %	17.43	-4.39 %	510,343,728	-0.00 %
		Quanto	2	0.79	0.09 %	0.73	1.98 %	76.78	2.46 %	71.15	-2.40 %	17.79	-2.40 %	510,348,344	0.00 %
		Quanto	4	0.78	-0.02 %	0.70	-1.36 %	77.18	2.98 %	70.78	-2.90 %	17.70	-2.90 %	510,348,344	0.00 %
		Quanto	8	0.78	-0.02 %	0.71	-0.05 %	76.03	1.45 %	71.85	-1.43 %	17.97	-1.43 %	510,348,344	0.00 %
openai-community gpt2-medium	No	BitsAndBytes	32	0.49	-1.03 %	1.15	11.06 %	186.53	2.72 %	29.29	-2.65 %	7.32	-2.65 %	1,444,466,784	-
		BitsAndBytes	4	0.49	0.94 %	1.15	11.06 %	191.61	2.72 %	28.51	-2.65 %	7.13	-2.65 %	281,831,472	-80.49 %
		BitsAndBytes	8	0.49	1.66 %	1.04	0.20 %	248.32	33.13 %	22.00	-24.88 %	5.50	-24.88 %	432,826,416	-70.04 %
		Downcasting	16	0.49	0.30 %	1.02	-0.91 %	90.58	-51.44 %	60.31	105.93 %	15.08	105.93 %	734,816,304	-49.13 %
		HQQ	2	0.49	0.60 %	0.90	-13.28 %	192.35	3.12 %	28.40	-3.03 %	7.10	-3.02 %	1,444,459,104	-0.00 %
		HQQ	4	0.49	-0.26 %	1.02	-1.36 %	192.30	3.09 %	28.41	-3.00 %	7.10	-2.99 %	1,444,459,616	-0.00 %
		HQQ	8	0.49	0.00 %	1.03	-0.22 %	191.81	2.83 %	28.48	-2.75 %	7.12	-2.74 %	1,444,460,640	-0.00 %
		Quanto	2	0.49	1.13 %	1.14	10.18 %	188.62	1.12 %	28.96	-1.11 %	7.24	-1.11 %	1,444,466,792	0.00 %
		Quanto	4	0.49	-0.19 %	1.03	-0.25 %	187.77	0.66 %	29.09	-0.66 %	7.28	-0.66 %	1,444,466,792	0.00 %
		Quanto	8	0.49	0.00 %	1.03	-0.24 %	187.46	0.50 %	29.14	-0.50 %	7.29	-0.49 %	1,444,466,792	0.00 %
	Yes	BitsAndBytes	32	0.61	-0.66 %	0.66	-0.22 %	186.58	2.99 %	29.28	-2.90 %	7.32	-2.90 %	1,444,466,784	-
		BitsAndBytes	4	0.61	0.18 %	0.66	-0.22 %	192.15	2.99 %	28.43	-2.90 %	7.11	-2.90 %	281,831,472	-80.49 %
		BitsAndBytes	8	0.61	-0.12 %	0.66	0.00 %	211.34	13.27 %	25.85	-11.72 %	6.46	-11.72 %	432,826,416	-70.04 %
		Downcasting	16	0.61	-0.06 %	0.66	-0.00 %	90.83	-51.32 %	60.15	105.42 %	15.04	105.44 %	734,816,304	-49.13 %
		HQQ	2	0.61	-0.24 %	0.66	0.21 %	191.79	2.79 %	28.48	-2.72 %	7.12	-2.72 %	1,444,459,104	-0.00 %
		HQQ	4	0.61	-0.27 %	0.66	0.08 %	192.61	3.23 %	28.36	-3.13 %	7.09	-3.13 %	1,444,459,616	-0.00 %
		HQQ	8	0.61	-0.06 %	0.66	-0.01 %	192.08	2.95 %	28.44	-2.87 %	7.11	-2.87 %	1,444,460,640	-0.00 %
		Quanto	2	0.61	-0.60 %	0.66	0.49 %	188.00	0.76 %	29.06	-0.76 %	7.27	-0.75 %	1,444,466,792	0.00 %
		Quanto	4	0.61	-0.06 %	0.66	0.05 %	187.57	0.53 %	29.12	-0.53 %	7.28	-0.52 %	1,444,466,792	0.00 %
		Quanto	8	0.61	0.00 %	0.66	-0.01 %	187.96	0.74 %	29.06	-0.74 %	7.27	-0.74 %	1,444,466,792	0.00 %

Table D.5: Results on the glue/qqp task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval F1 Score	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No	BitsAndBytes	32	0.39	-0.55 %	0.55	-0.18 %	0.72	-0.26 %	297.28	55.78 %	136.00	-35.81 %	34.00	-35.81 %	437,943,304	-
		BitsAndBytes	4	0.39	-0.62 %	0.55	-0.18 %	0.72	-0.26 %	463.10	55.78 %	87.30	-35.81 %	21.83	-35.81 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.46	19.51 %	0.47	-14.49 %	0.71	-2.29 %	1295.66	335.85 %	31.20	-77.06 %	7.80	-77.06 %	133,451,268	-69.53 %
		Downcasting	16	0.39	-0.01 %	0.55	-0.00 %	0.72	0.04 %	211.01	-29.02 %	191.60	40.88 %	47.90	40.88 %	218,975,748	-50.00 %
		HQQ	2	0.50	27.45 %	0.53	-2.41 %	0.70	-3.64 %	617.54	107.73 %	65.47	-51.86 %	16.37	-51.86 %	117,220,744	-73.23 %
		HQQ	4	0.39	-0.60 %	0.55	-0.17 %	0.73	0.77 %	540.05	81.67 %	74.86	-44.95 %	18.72	-44.95 %	138,602,248	-68.35 %
		HQQ	8	0.39	-0.01 %	0.55	-0.00 %	0.72	-0.02 %	452.49	52.21 %	89.35	-34.30 %	22.34	-34.30 %	181,365,256	-58.59 %
		Quanto	2	0.44	13.53 %	0.52	-5.35 %	0.70	-2.37 %	750.30	152.39 %	53.88	-60.38 %	13.47	-60.38 %	437,943,896	0.00 %
		Quanto	4	0.39	0.06 %	0.55	0.02 %	0.71	-1.23 %	695.95	134.11 %	58.09	-57.29 %	14.52	-57.28 %	437,943,896	0.00 %
		Quanto	8	0.39	0.10 %	0.55	0.03 %	0.72	-0.22 %	375.66	26.37 %	107.62	-20.87 %	26.91	-20.87 %	437,943,896	0.00 %
	Yes	BitsAndBytes	32	0.63	-0.00 %	0.00	-0.69 %	0.69	0.13 %	296.38	56.48 %	136.41	-36.09 %	34.10	-36.09 %	437,943,304	-
		BitsAndBytes	4	0.63	0.00 %	0.00	-0.69 %	0.69	0.06 %	463.77	56.48 %	87.18	-36.09 %	21.80	-36.09 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.63	0.00 %	0.00	-0.69 %	0.69	0.09 %	1111.32	274.97 %	36.38	-73.33 %	9.10	-73.33 %	133,451,268	-69.53 %
		Downcasting	16	0.63	0.00 %	0.00	-0.69 %	0.69	0.09 %	210.86	-28.85 %	191.73	40.55 %	47.94	40.55 %	218,975,748	-50.00 %
		HQQ	2	0.63	0.00 %	0.00	-0.69 %	0.69	0.07 %	619.30	108.96 %	65.28	-52.14 %	16.32	-52.14 %	117,220,744	-73.23 %
		HQQ	4	0.63	0.00 %	0.00	-0.69 %	0.69	-0.01 %	540.23	82.28 %	74.84	-45.14 %	18.71	-45.14 %	138,602,248	-68.35 %

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval F1 Score	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
		HQQ	8	0.86	0.00 %	0.83	0.00 %	0.48	-0.05 %	1420.01	2.47 %	28.47	-2.41 %	7.12	-2.41 %	1,444,460,640	-0.00 %
		Quanto	2	0.86	0.00 %	0.83	0.01 %	0.49	0.65 %	1397.37	0.84 %	28.93	-0.83 %	7.23	-0.82 %	1,444,466,792	0.00 %
		Quanto	4	0.86	0.00 %	0.83	0.00 %	0.48	0.21 %	1396.11	0.75 %	28.96	-0.74 %	7.24	-0.74 %	1,444,466,792	0.00 %
		Quanto	8	0.86	0.00 %	0.83	0.00 %	0.48	-0.03 %	1391.63	0.43 %	29.05	-0.43 %	7.26	-0.43 %	1,444,466,792	0.00 %

Table D.6: Results on the glue/rte task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.47	-	0.95	-	2.14	-	129.17	-	32.64	-	437,943,304	-
		BitsAndBytes	4	0.47	0.00 %	0.94	-1.28 %	3.23	50.69 %	85.72	-33.64 %	21.66	-33.64 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.47	0.00 %	0.79	-17.54 %	9.14	326.24 %	30.30	-76.54 %	7.66	-76.54 %	133,451,268	-69.53 %
		Downcasting	16	0.47	0.00 %	0.95	0.04 %	1.48	-31.13 %	187.56	45.20 %	47.40	45.20 %	218,975,748	-50.00 %
		Quanto	2	0.47	-1.53 %	0.70	-26.25 %	5.32	148.06 %	52.07	-59.69 %	13.16	-59.69 %	437,943,896	0.00 %
		Quanto	4	0.47	0.00 %	0.93	-2.53 %	4.91	129.05 %	56.39	-56.34 %	14.25	-56.34 %	437,943,896	0.00 %
		Quanto	8	0.47	0.00 %	0.95	-0.03 %	2.66	24.21 %	103.99	-19.49 %	26.28	-19.49 %	437,943,896	0.00 %
	Yes		32	0.57	-	0.68	-	2.11	-	131.44	-	33.22	-	437,943,304	-
		BitsAndBytes	4	0.56	-2.52 %	0.69	0.88 %	3.23	53.45 %	85.66	-34.83 %	21.65	-34.83 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.51	-11.32 %	0.70	2.82 %	9.04	329.01 %	30.64	-76.69 %	7.74	-76.69 %	133,451,268	-69.53 %
		Downcasting	16	0.57	0.00 %	0.68	0.03 %	1.52	-28.01 %	182.58	38.91 %	46.14	38.91 %	218,975,748	-50.00 %
		Quanto	2	0.53	-8.18 %	0.71	4.63 %	5.43	157.86 %	50.97	-61.22 %	12.88	-61.22 %	437,943,896	0.00 %
		Quanto	4	0.55	-5.03 %	0.69	0.92 %	4.92	133.31 %	56.33	-57.14 %	14.24	-57.14 %	437,943,896	0.00 %
		Quanto	8	0.57	0.00 %	0.68	0.04 %	2.65	25.61 %	104.64	-20.39 %	26.44	-20.39 %	437,943,896	0.00 %
bert-large-uncased	No		32	0.53	-	1.27	-	4.84	-	57.27	-	14.47	-	1,340,583,944	-
		BitsAndBytes	4	0.53	0.00 %	1.27	-0.01 %	7.05	45.70 %	39.31	-31.37 %	9.93	-31.36 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.53	0.00 %	0.95	-25.83 %	17.75	267.01 %	15.60	-72.75 %	3.94	-72.75 %	367,257,604	-72.60 %
		Downcasting	16	0.53	0.00 %	1.28	0.04 %	2.94	-39.27 %	94.30	64.67 %	23.83	64.67 %	670,296,068	-50.00 %
		Quanto	2	0.47	-10.27 %	0.71	-43.97 %	11.17	130.91 %	24.80	-56.69 %	6.27	-56.69 %	1,340,585,112	0.00 %
		Quanto	4	0.53	0.00 %	1.29	1.34 %	10.26	112.04 %	27.01	-52.84 %	6.83	-52.84 %	1,340,585,112	0.00 %
		Quanto	8	0.53	0.00 %	1.27	-0.07 %	5.74	18.71 %	48.24	-15.76 %	12.19	-15.76 %	1,340,585,112	0.00 %
	Yes		32	0.53	-	0.69	-	4.75	-	58.33	-	14.74	-	1,340,583,944	-
		BitsAndBytes	4	0.53	0.00 %	0.69	-0.01 %	7.06	48.76 %	39.21	-32.78 %	9.91	-32.78 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.53	0.00 %	0.69	-0.01 %	15.19	219.86 %	18.24	-68.74 %	4.61	-68.73 %	367,257,604	-72.60 %
		Downcasting	16	0.53	0.00 %	0.69	0.22 %	2.90	-38.93 %	95.52	63.76 %	24.14	63.75 %	670,296,068	-50.00 %
		Quanto	2	0.53	0.00 %	0.69	0.08 %	11.09	133.60 %	24.97	-57.19 %	6.31	-57.19 %	1,340,585,112	0.00 %
		Quanto	4	0.53	0.00 %	0.69	0.01 %	10.22	115.28 %	27.10	-53.55 %	6.85	-53.55 %	1,340,585,112	0.00 %
		Quanto	8	0.53	0.00 %	0.69	-0.01 %	5.75	21.18 %	48.14	-17.48 %	12.16	-17.48 %	1,340,585,112	0.00 %
facebook bart-base	No		32	0.47	-	0.72	-	4.39	-	63.04	-	15.93	-	560,050,184	-
		BitsAndBytes	4	0.47	0.00 %	0.73	1.31 %	5.32	20.95 %	52.12	-17.32 %	13.17	-17.33 %	130,504,708	-76.70 %
		BitsAndBytes	8	0.47	0.00 %	0.72	-0.32 %	10.14	130.75 %	27.32	-56.66 %	6.90	-56.66 %	180,344,836	-67.80 %
		Downcasting	16	0.47	0.00 %	0.72	-0.04 %	2.67	-39.16 %	103.60	64.35 %	26.18	64.36 %	280,025,092	-50.00 %
		Quanto	2	0.44	-7.63 %	0.73	1.30 %	8.41	91.44 %	32.93	-47.76 %	8.32	-47.77 %	560,050,968	0.00 %
		Quanto	4	0.47	0.00 %	0.73	1.56 %	7.99	81.93 %	34.65	-45.03 %	8.76	-45.03 %	560,050,968	0.00 %
		Quanto	8	0.47	0.00 %	0.72	0.00 %	4.82	9.65 %	57.49	-8.80 %	14.53	-8.80 %	560,050,968	0.00 %
	Yes		32	0.47	-	0.71	-	4.23	-	65.56	-	16.57	-	560,050,184	-
		BitsAndBytes	4	0.47	0.00 %	0.71	0.20 %	5.22	23.43 %	53.11	-18.98 %	13.42	-18.98 %	130,504,708	-76.70 %
		BitsAndBytes	8	0.47	0.00 %	0.71	-0.08 %	9.63	127.90 %	28.77	-56.12 %	7.27	-56.12 %	180,344,836	-67.80 %
		Downcasting	16	0.47	0.00 %	0.71	-0.02 %	2.68	-36.61 %	103.42	57.76 %	26.14	57.76 %	280,025,092	-50.00 %
		Quanto	2	0.47	0.00 %	0.70	-1.91 %	8.47	100.53 %	32.69	-50.13 %	8.26	-50.13 %	560,050,968	0.00 %
		Quanto	4	0.47	0.00 %	0.71	-0.55 %	7.94	87.96 %	34.88	-46.80 %	8.81	-46.79 %	560,050,968	0.00 %
		Quanto	8	0.47	0.00 %	0.71	-0.03 %	4.86	14.95 %	57.03	-13.01 %	14.41	-13.00 %	560,050,968	0.00 %
google flan-t5-base	No		32	0.53	-	0.71	-	7.33	-	37.77	-	9.54	-	893,982,728	-
		BitsAndBytes	4	0.53	0.00 %	0.71	1.24 %	10.72	46.17 %	25.84	-31.59 %	6.53	-31.59 %	280,955,908	-68.57 %
		BitsAndBytes	8	0.53	0.00 %	0.71	-0.01 %	15.26	108.00 %	18.16	-51.92 %	4.59	-51.93 %	361,466,884	-59.57 %
		Downcasting	16	0.53	0.00 %	0.71	-0.01 %	6.52	-11.13 %	42.50	12.52 %	10.74	12.52 %	446,991,364	-50.00 %

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D. DETAILED TABLES

(Continued from previous page)																
Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%	
		Quanto	2	0.53	0.00 %	0.71	0.63 %	16.28	121.95 %	17.02	-54.94 %	4.30	-54.95 %	893,984,472	0.00 %	
		Quanto	4	0.53	0.00 %	0.70	-0.76 %	15.33	109.05 %	18.07	-52.17 %	4.57	-52.17 %	893,984,472	0.00 %	
		Quanto	8	0.53	0.00 %	0.71	-0.05 %	8.53	16.25 %	32.49	-13.98 %	8.21	-13.98 %	893,984,472	0.00 %	
	Yes	BitsAndBytes	32	0.83	-0.70 %	0.70	1.22 %	7.33	10.68 %	37.78	-25.93 %	9.55	-31.36 %	893,982,728	-	
		BitsAndBytes	4	0.82	-0.87 %	0.70	1.22 %	10.68	45.68 %	25.93	-31.35 %	6.55	-31.36 %	280,955,908	-68.57 %	
		BitsAndBytes	8	0.82	-0.44 %	0.72	3.04 %	15.25	108.00 %	18.16	-51.92 %	4.59	-51.93 %	361,466,884	-59.57 %	
		Downcasting	16	0.83	0.00 %	0.70	0.37 %	6.61	-9.83 %	41.90	10.90 %	10.59	10.90 %	446,991,364	-50.00 %	
		Quanto	2	0.56	-31.88 %	0.68	-2.08 %	16.30	122.32 %	16.99	-55.02 %	4.29	-55.03 %	893,984,472	0.00 %	
		Quanto	4	0.82	-0.87 %	0.72	2.75 %	15.28	108.38 %	18.13	-52.01 %	4.58	-52.01 %	893,984,472	0.00 %	
		Quanto	8	0.82	-0.44 %	0.70	1.24 %	8.53	16.38 %	32.46	-14.08 %	8.20	-14.08 %	893,984,472	0.00 %	
openai-community gpt2	No	BitsAndBytes	32	0.52	-1.85 %	1.85	-	3.81	-	72.74	-	18.38	-	510,348,336	-	
		BitsAndBytes	4	0.53	0.69 %	2.52	36.36 %	4.16	9.37 %	66.51	-8.57 %	16.81	-8.57 %	134,063,640	-73.73 %	
		BitsAndBytes	8	0.52	0.00 %	1.76	-4.74 %	6.73	76.74 %	41.16	-43.42 %	10.40	-43.42 %	176,530,968	-65.41 %	
		Downcasting	16	0.52	0.00 %	1.86	0.98 %	2.15	-43.47 %	128.68	76.90 %	32.52	76.91 %	261,465,624	-48.77 %	
		Quanto	2	0.52	0.00 %	1.93	4.30 %	3.83	0.51 %	72.37	-0.51 %	18.29	-0.51 %	510,348,344	0.00 %	
		Quanto	4	0.52	0.00 %	1.87	1.05 %	3.82	0.43 %	72.43	-0.43 %	18.30	-0.42 %	510,348,344	0.00 %	
		Quanto	8	0.52	0.00 %	1.83	-0.81 %	3.78	-0.82 %	73.34	0.83 %	18.53	0.83 %	510,348,344	0.00 %	
	Yes	BitsAndBytes	32	0.59	-0.69 %	0.69	-	3.78	-	73.34	-	18.53	-	510,348,336	-	
		BitsAndBytes	4	0.58	-1.84 %	0.69	0.14 %	4.15	9.92 %	66.72	-9.03 %	16.86	-9.02 %	134,063,640	-73.73 %	
		BitsAndBytes	8	0.59	0.61 %	0.69	-0.00 %	6.58	74.14 %	42.12	-42.57 %	10.64	-42.57 %	176,530,968	-65.41 %	
		Downcasting	16	0.59	0.61 %	0.69	0.02 %	2.17	-42.44 %	127.42	73.73 %	32.20	73.74 %	261,465,624	-48.77 %	
		Quanto	2	0.55	-7.36 %	0.78	11.93 %	3.83	1.39 %	72.33	-1.38 %	18.28	-1.37 %	510,348,344	0.00 %	
		Quanto	4	0.57	-2.45 %	0.69	-0.24 %	3.82	1.19 %	72.48	-1.18 %	18.32	-1.18 %	510,348,344	0.00 %	
		Quanto	8	0.58	-0.61 %	0.69	-0.04 %	3.78	0.03 %	73.32	-0.03 %	18.53	-0.02 %	510,348,344	0.00 %	
openai-community gpt2-medium	No	BitsAndBytes	32	0.52	-1.40 %	1.40	-	9.29	-	29.82	-	7.54	-	1,444,466,784	-	
		BitsAndBytes	4	0.53	2.80 %	1.34	-3.63 %	10.35	11.41 %	26.76	-10.25 %	6.76	-10.25 %	281,831,472	-80.49 %	
		BitsAndBytes	8	0.52	0.70 %	1.46	4.77 %	12.96	39.47 %	21.38	-28.30 %	5.40	-28.29 %	432,826,416	-70.04 %	
		Downcasting	16	0.52	0.70 %	1.41	0.82 %	4.58	-50.75 %	60.55	103.05 %	15.30	103.07 %	734,816,304	-49.13 %	
		Quanto	2	0.52	0.00 %	1.69	21.27 %	9.35	0.60 %	29.64	-0.60 %	7.49	-0.60 %	1,444,466,792	0.00 %	
		Quanto	4	0.52	0.00 %	1.42	1.61 %	9.36	0.77 %	29.59	-0.77 %	7.48	-0.76 %	1,444,466,792	0.00 %	
		Quanto	8	0.52	0.00 %	1.40	0.21 %	9.30	0.16 %	29.77	-0.16 %	7.52	-0.15 %	1,444,466,792	0.00 %	
	Yes	BitsAndBytes	32	0.52	-0.69 %	0.69	-	9.37	-	29.56	-	7.47	-	1,444,466,784	-	
		BitsAndBytes	4	0.53	0.69 %	0.69	0.07 %	10.39	10.89 %	26.66	-9.82 %	6.74	-9.83 %	281,831,472	-80.49 %	
		BitsAndBytes	8	0.52	0.00 %	0.69	0.00 %	11.56	23.33 %	23.97	-18.91 %	6.06	-18.92 %	432,826,416	-70.04 %	
		Downcasting	16	0.52	0.00 %	0.69	0.04 %	4.58	-51.15 %	60.51	104.72 %	15.29	104.71 %	734,816,304	-49.13 %	
		Quanto	2	0.53	0.69 %	0.69	-0.17 %	9.41	0.40 %	29.44	-0.40 %	7.44	-0.40 %	1,444,466,792	0.00 %	
		Quanto	4	0.53	1.38 %	0.69	-0.04 %	9.40	0.29 %	29.47	-0.29 %	7.45	-0.29 %	1,444,466,792	0.00 %	
		Quanto	8	0.53	0.69 %	0.69	0.02 %	9.34	-0.29 %	29.65	0.29 %	7.49	0.29 %	1,444,466,792	0.00 %	

Table D.7: Results on the glue/sst2 task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%	
bert-base-uncased	No	BitsAndBytes	32	0.51	-0.73 %	0.73	-	8.11	-	107.46	-	26.87	-	437,943,304	-	
		BitsAndBytes	4	0.51	0.45 %	0.71	-3.12 %	12.11	49.22 %	72.01	-32.99 %	18.00	-32.99 %	90,689,028	-79.29 %	
		BitsAndBytes	8	0.51	-0.45 %	0.72	-1.25 %	33.54	313.30 %	26.00	-75.80 %	6.50	-75.81 %	133,451,268	-69.53 %	
		Downcasting	16	0.51	0.00 %	0.73	0.09 %	6.06	-25.36 %	143.97	33.97 %	35.99	33.97 %	218,975,748	-50.00 %	
		Quanto	2	0.49	-3.83 %	0.70	-4.23 %	20.00	146.47 %	43.60	-59.43 %	10.90	-59.43 %	437,943,896	0.00 %	
		Quanto	4	0.51	0.45 %	0.71	-3.79 %	18.71	130.57 %	46.61	-56.63 %	11.65	-56.63 %	437,943,896	0.00 %	
		Quanto	8	0.51	0.00 %	0.73	-0.08 %	9.96	22.74 %	87.55	-18.53 %	21.89	-18.53 %	437,943,896	0.00 %	
	Yes	BitsAndBytes	32	0.75	-0.58 %	0.58	1.33 %	8.16	12.18 %	106.84	-	26.71	-	437,943,304	-	
		BitsAndBytes	4	0.75	-0.46 %	0.58	1.33 %	12.18	49.21 %	71.61	-32.98 %	17.90	-32.98 %	90,689,028	-79.29 %	
		BitsAndBytes	8	0.58	-23.29 %	0.88	53.44 %	29.89	266.30 %	29.17	-72.70 %	7.29	-72.70 %	133,451,268	-69.53 %	
		Downcasting	16	0.75	-0.15 %	0.58	0.42 %	5.84	-28.47 %	149.38	39.81 %	37.34	39.81 %	218,975,748	-50.00 %	
		Quanto	2	0.68	-10.05 %	0.75	30.78 %	19.95	144.51 %	43.70	-59.10 %	10.93	-59.10 %	437,943,896	0.00 %	
		Quanto	4	0.76	0.61 %	0.57	-1.75 %	18.60	127.85 %	46.89	-56.11 %	11.72	-56.11 %	437,943,896	0.00 %	

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
		Quanto	8	0.75	-0.15 %	0.58	0.18 %	9.78	19.81 %	89.18	-16.53 %	22.30	-16.53 %	437,943,896	0.00 %
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
bert-large-uncased	No		32	0.49	-	0.87	-	17.03	-	51.21	-	12.80	-	1,340,583,944	-
		BitsAndBytes	4	0.49	0.23 %	0.87	-0.00 %	25.36	48.92 %	34.39	-32.85 %	8.60	-32.85 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.49	0.94 %	0.77	-10.44 %	64.11	276.46 %	13.60	-73.44 %	3.40	-73.43 %	367,257,604	-72.60 %
		Downcasting	16	0.49	-0.23 %	0.87	0.01 %	10.87	-36.14 %	80.19	56.60 %	20.05	56.60 %	670,296,068	-50.00 %
		Quanto	2	0.49	-0.70 %	0.70	-19.43 %	41.10	141.36 %	21.22	-58.57 %	5.30	-58.57 %	1,340,585,112	0.00 %
		Quanto	4	0.49	-0.23 %	0.87	0.69 %	38.57	126.53 %	22.61	-55.85 %	5.65	-55.86 %	1,340,585,112	0.00 %
		Quanto	8	0.49	0.00 %	0.87	0.22 %	20.59	20.92 %	42.35	-17.30 %	10.59	-17.30 %	1,340,585,112	0.00 %
	Yes		32	0.51	-	0.69	-	16.94	-	51.47	-	12.87	-	1,340,583,944	-
		BitsAndBytes	4	0.51	0.00 %	0.69	0.05 %	25.26	49.07 %	34.53	-32.92 %	8.63	-32.92 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.51	0.00 %	0.69	-0.02 %	50.75	199.52 %	17.18	-66.61 %	4.30	-66.61 %	367,257,604	-72.60 %
		Downcasting	16	0.51	0.00 %	0.69	0.14 %	10.88	-35.76 %	80.11	55.66 %	20.03	55.66 %	670,296,068	-50.00 %
		Quanto	2	0.49	-3.60 %	0.72	3.28 %	40.98	141.86 %	21.28	-58.65 %	5.32	-58.65 %	1,340,585,112	0.00 %
		Quanto	4	0.49	-3.60 %	0.69	0.04 %	38.25	125.74 %	22.80	-55.70 %	5.70	-55.70 %	1,340,585,112	0.00 %
		Quanto	8	0.51	0.00 %	0.69	0.00 %	20.54	21.26 %	42.44	-17.53 %	10.61	-17.53 %	1,340,585,112	0.00 %
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
facebook bart-base	No		32	0.51	-	0.77	-	15.69	-	55.57	-	13.89	-	560,050,184	-
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
google flan-t5-base	No		32	0.49	-	0.91	-	23.90	-	36.48	-	9.12	-	893,982,728	-
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
openai-community gpt2	No		32	0.50	-	2.21	-	10.64	-	81.99	-	20.50	-	510,348,336	-
		BitsAndBytes	4	0.49	-2.52 %	2.00	-9.70 %	11.27	5.93 %	77.40	-5.60 %	19.35	-5.60 %	134,063,640	-73.73 %
		BitsAndBytes	8	0.50	0.00 %	2.08	-5.95 %	17.84	67.74 %	48.88	-40.39 %	12.22	-40.38 %	176,530,968	-65.41 %
		Downcasting	16	0.50	-0.69 %	2.25	1.77 %	5.61	-47.24 %	155.40	89.53 %	38.85	89.54 %	261,465,624	-48.77 %
		Quanto	2	0.50	-0.46 %	1.89	-14.32 %	10.82	1.70 %	80.62	-1.67 %	20.16	-1.67 %	510,348,344	0.00 %
		Quanto	4	0.49	-1.37 %	2.11	-4.28 %	10.78	1.32 %	80.92	-1.30 %	20.23	-1.30 %	510,348,344	0.00 %
		Quanto	8	0.50	0.00 %	2.21	-0.15 %	10.67	0.33 %	81.72	-0.33 %	20.43	-0.33 %	510,348,344	0.00 %
	Yes		32	0.89	-	0.49	-	10.66	-	81.83	-	20.46	-	510,348,336	-
		BitsAndBytes	4	0.89	-0.26 %	0.49	-0.83 %	11.27	5.74 %	77.39	-5.43 %	19.35	-5.43 %	134,063,640	-73.73 %
		BitsAndBytes	8	0.89	-0.13 %	0.49	0.46 %	17.39	63.18 %	50.15	-38.72 %	12.54	-38.72 %	176,530,968	-65.41 %
		Downcasting	16	0.89	0.00 %	0.49	0.17 %	5.68	-46.74 %	153.63	87.74 %	38.41	87.74 %	261,465,624	-48.77 %
		Quanto	2	0.89	-0.13 %	0.50	1.51 %	10.80	1.40 %	80.70	-1.38 %	20.18	-1.38 %	510,348,344	0.00 %
		Quanto	4	0.89	0.00 %	0.49	-0.74 %	10.78	1.16 %	80.89	-1.15 %	20.22	-1.15 %	510,348,344	0.00 %
		Quanto	8	0.89	0.00 %	0.49	0.06 %	10.66	0.06 %	81.78	-0.06 %	20.45	-0.06 %	510,348,344	0.00 %
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
openai-community gpt2-medium	No		32	0.52	-	1.15	-	27.12	-	32.15	-	8.04	-	1,444,466,784	-
		BitsAndBytes	4	0.52	-1.32 %	1.15	-0.27 %	29.39	8.37 %	29.66	-7.73 %	7.42	-7.73 %	281,831,472	-80.49 %
		BitsAndBytes	8	0.52	-1.10 %	1.20	4.22 %	34.31	26.50 %	25.41	-20.95 %	6.35	-20.94 %	432,826,416	-70.04 %
		Downcasting	16	0.52	-0.66 %	1.15	-0.43 %	12.49	-53.95 %	69.81	117.16 %	17.45	117.17 %	734,816,304	-49.13 %
		Quanto	2	0.52	0.22 %	1.29	12.33 %	27.36	0.86 %	31.88	-0.85 %	7.97	-0.85 %	1,444,466,792	0.00 %
		Quanto	4	0.53	0.44 %	1.15	0.15 %	27.37	0.90 %	31.86	-0.89 %	7.97	-0.88 %	1,444,466,792	0.00 %
		Quanto	8	0.52	0.22 %	1.15	-0.10 %	27.22	0.37 %	32.03	-0.37 %	8.01	-0.36 %	1,444,466,792	0.00 %
	Yes		32	0.83	-	0.72	-	27.15	-	32.12	-	8.03	-	1,444,466,784	-
		BitsAndBytes	4	0.84	0.69 %	0.71	-1.54 %	29.48	8.57 %	29.58	-7.89 %	7.40	-7.88 %	281,831,472	-80.49 %
		BitsAndBytes	8	0.83	-0.41 %	0.72	0.16 %	32.67	20.34 %	26.69	-16.90 %	6.67	-16.90 %	432,826,416	-70.04 %
		Downcasting	16	0.83	0.00 %	0.72	-0.01 %	12.48	-54.04 %	69.88	117.58 %	17.47	117.59 %	734,816,304	-49.13 %
		Quanto	2	0.83	0.00 %	0.72	0.90 %	27.41	0.94 %	31.82	-0.93 %	7.95	-0.93 %	1,444,466,792	0.00 %
		Quanto	4	0.83	0.00 %	0.72	0.28 %	27.37	0.81 %	31.86	-0.81 %	7.96	-0.80 %	1,444,466,792	0.00 %
		Quanto	8	0.83	0.00 %	0.72	-0.06 %	27.26	0.39 %	31.99	-0.39 %	8.00	-0.39 %	1,444,466,792	0.00 %

D. DETAILED TABLES

Table D.8: Results on the glue/wnli task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.44	-	0.72	-	0.55	-	129.81	-	32.91	-	437,943,304	-
		BitsAndBytes	4	0.44	0.00 %	0.71	-0.32 %	0.83	51.35 %	85.76	-33.94 %	21.74	-33.94 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.52	19.35 %	0.69	-3.33 %	2.34	328.35 %	30.30	-76.66 %	7.68	-76.66 %	133,451,268	-69.53 %
		Downcasting	16	0.44	0.00 %	0.72	-0.01 %	0.38	-30.07 %	185.64	43.01 %	47.06	43.01 %	218,975,748	-50.00 %
		Quanto	2	0.63	45.16 %	0.68	-4.88 %	1.38	152.80 %	51.34	-60.45 %	13.02	-60.45 %	437,943,896	0.00 %
		Quanto	4	0.42	-3.23 %	0.71	-0.88 %	1.27	131.86 %	55.98	-56.88 %	14.19	-56.88 %	437,943,896	0.00 %
		Quanto	8	0.44	0.00 %	0.71	-0.16 %	0.68	24.94 %	103.89	-19.97 %	26.34	-19.97 %	437,943,896	0.00 %
	Yes		32	0.52	-	0.70	-	0.56	-	126.78	-	32.14	-	437,943,304	-
		BitsAndBytes	4	0.42	-18.92 %	0.70	0.32 %	0.83	48.50 %	85.38	-32.66 %	21.65	-32.66 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.51	-2.70 %	0.69	-0.94 %	2.35	318.95 %	30.26	-76.13 %	7.67	-76.13 %	133,451,268	-69.53 %
		Downcasting	16	0.52	0.00 %	0.70	-0.05 %	0.38	-32.18 %	186.95	47.46 %	47.40	47.46 %	218,975,748	-50.00 %
		Quanto	2	0.44	-16.22 %	0.71	1.51 %	1.40	149.93 %	50.73	-59.99 %	12.86	-59.99 %	437,943,896	0.00 %
		Quanto	4	0.49	-5.41 %	0.70	0.13 %	1.33	136.80 %	53.54	-57.77 %	13.57	-57.77 %	437,943,896	0.00 %
		Quanto	8	0.52	0.00 %	0.70	0.01 %	0.69	22.54 %	103.46	-18.39 %	26.23	-18.39 %	437,943,896	0.00 %
bert-large-uncased	No		32	0.56	-	1.17	-	1.24	-	57.24	-	14.51	-	1,340,583,944	-
		BitsAndBytes	4	0.56	0.00 %	1.17	-0.59 %	1.81	45.96 %	39.22	-31.49 %	9.94	-31.48 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.56	0.00 %	0.88	-24.95 %	4.56	267.80 %	15.56	-72.81 %	3.95	-72.81 %	367,257,604	-72.60 %
		Downcasting	16	0.56	0.00 %	1.18	0.01 %	0.74	-40.60 %	96.38	68.36 %	24.43	68.36 %	670,296,068	-50.00 %
		Quanto	2	0.45	-20.00 %	0.73	-38.27 %	2.90	133.99 %	24.46	-57.26 %	6.20	-57.26 %	1,340,585,112	0.00 %
		Quanto	4	0.56	0.00 %	1.19	0.97 %	2.68	115.99 %	26.50	-53.70 %	6.72	-53.70 %	1,340,585,112	0.00 %
		Quanto	8	0.56	0.00 %	1.17	-0.01 %	1.50	21.31 %	47.19	-17.56 %	11.96	-17.56 %	1,340,585,112	0.00 %
	Yes		32	0.44	-	0.69	-	1.23	-	57.73	-	14.63	-	1,340,583,944	-
		BitsAndBytes	4	0.44	0.00 %	0.69	0.03 %	1.81	46.87 %	39.30	-31.91 %	9.96	-31.92 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.44	0.00 %	0.69	0.03 %	4.19	241.00 %	16.93	-70.67 %	4.29	-70.67 %	367,257,604	-72.60 %
		Downcasting	16	0.44	0.00 %	0.70	0.10 %	0.76	-38.58 %	94.00	62.83 %	23.83	62.83 %	670,296,068	-50.00 %
		Quanto	2	0.44	0.00 %	0.74	6.25 %	2.87	132.97 %	24.78	-57.08 %	6.28	-57.08 %	1,340,585,112	0.00 %
		Quanto	4	0.56	29.03 %	0.69	-0.36 %	2.68	118.27 %	26.45	-54.18 %	6.71	-54.19 %	1,340,585,112	0.00 %
		Quanto	8	0.44	0.00 %	0.70	0.08 %	1.49	20.86 %	47.77	-17.26 %	12.11	-17.25 %	1,340,585,112	0.00 %
facebook bart-base	No		32	0.58	-	0.69	-	1.08	-	65.92	-	16.71	-	560,050,184	-
		BitsAndBytes	4	0.48	-17.07 %	0.70	0.63 %	1.35	25.14 %	52.68	-20.09 %	13.35	-20.09 %	130,504,708	-76.70 %
		BitsAndBytes	8	0.55	-4.88 %	0.70	0.49 %	2.63	143.72 %	27.05	-58.97 %	6.86	-58.97 %	180,344,836	-67.80 %
		Downcasting	16	0.58	0.00 %	0.69	0.01 %	0.66	-38.65 %	107.44	62.99 %	27.24	63.00 %	280,025,092	-50.00 %
		Quanto	2	0.55	-4.88 %	0.69	-0.45 %	2.16	100.53 %	32.87	-50.14 %	8.33	-50.13 %	560,050,968	0.00 %
		Quanto	4	0.44	-24.39 %	0.72	3.54 %	2.03	88.53 %	34.96	-46.96 %	8.86	-46.96 %	560,050,968	0.00 %
		Quanto	8	0.62	7.32 %	0.69	-0.11 %	1.23	13.77 %	57.94	-12.10 %	14.69	-12.11 %	560,050,968	0.00 %
	Yes		32	0.58	-	0.69	-	1.07	-	66.35	-	16.82	-	560,050,184	-
		BitsAndBytes	4	0.48	-17.07 %	0.69	0.24 %	1.37	28.02 %	51.83	-21.88 %	13.14	-21.88 %	130,504,708	-76.70 %
		BitsAndBytes	8	0.59	2.44 %	0.69	-0.04 %	2.54	137.83 %	27.90	-57.95 %	7.07	-57.95 %	180,344,836	-67.80 %
		Downcasting	16	0.59	2.44 %	0.69	0.03 %	0.67	-37.26 %	105.77	59.40 %	26.81	59.40 %	280,025,092	-50.00 %
		Quanto	2	0.56	-2.44 %	0.69	0.24 %	2.15	101.38 %	32.95	-50.34 %	8.35	-50.34 %	560,050,968	0.00 %
		Quanto	4	0.44	-24.39 %	0.69	0.39 %	2.07	93.78 %	34.24	-48.39 %	8.68	-48.39 %	560,050,968	0.00 %
		Quanto	8	0.56	-2.44 %	0.69	-0.00 %	1.23	14.66 %	57.87	-12.79 %	14.67	-12.79 %	560,050,968	0.00 %
google flan-t5-base	No		32	0.56	-	0.80	-	1.91	-	37.09	-	9.40	-	893,982,728	-
		BitsAndBytes	4	0.56	0.00 %	0.80	0.12 %	2.74	43.03 %	25.93	-30.09 %	6.57	-30.09 %	280,955,908	-68.57 %
		BitsAndBytes	8	0.56	0.00 %	0.80	0.24 %	3.81	98.89 %	18.65	-49.72 %	4.73	-49.72 %	361,466,884	-59.57 %
		Downcasting	16	0.56	0.00 %	0.80	0.14 %	1.65	-13.70 %	42.97	15.88 %	10.89	15.88 %	446,991,364	-50.00 %
		Quanto	2	0.56	0.00 %	0.74	-6.89 %	4.16	117.38 %	17.06	-54.00 %	4.33	-54.00 %	893,984,472	0.00 %
		Quanto	4	0.56	0.00 %	0.77	-3.03 %	3.95	106.44 %	17.96	-51.56 %	4.55	-51.56 %	893,984,472	0.00 %
		Quanto	8	0.56	0.00 %	0.80	0.08 %	2.18	13.74 %	32.60	-12.08 %	8.27	-12.08 %	893,984,472	0.00 %
	Yes		32	0.52	-	0.70	-	1.87	-	37.93	-	9.62	-	893,982,728	-
		BitsAndBytes	4	0.52	0.00 %	0.70	-0.36 %	2.77	47.74 %	25.67	-32.31 %	6.51	-32.31 %	280,955,908	-68.57 %
		BitsAndBytes	8	0.49	-5.41 %	0.70	0.13 %	3.78	101.74 %	18.80	-50.43 %	4.77	-50.44 %	361,466,884	-59.57 %
		Downcasting	16	0.52	0.00 %	0.70	0.04 %	1.66	-11.37 %	42.79	12.82 %	10.85	12.82 %	446,991,364	-50.00 %
		Quanto	2	0.56	8.11 %	0.69	-1.58 %	4.17	122.88 %	17.02	-55.13 %	4.31	-55.14 %	893,984,472	0.00 %
		Quanto	4	0.54	2.70 %	0.70	-0.24 %	3.90	108.34 %	18.20	-52.00 %	4.62	-52.01 %	893,984,472	0.00 %
		Quanto	8	0.54	2.70 %	0.70	0.03 %	2.24	19.50 %	31.74	-16.32 %	8.05	-16.33 %	893,984,472	0.00 %

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
openai-community-gpt2	No		32	0.44	-	2.53	-	0.97	-	73.00	-	18.51	-	510,348,336	-
		BitsAndBytes	4	0.44	0.00 %	2.83	12.10 %	1.08	10.95 %	65.80	-9.87 %	16.68	-9.87 %	134,063,640	-73.73 %
		BitsAndBytes	8	0.44	0.00 %	2.46	-2.66 %	1.73	78.32 %	40.94	-43.92 %	10.38	-43.92 %	176,530,968	-65.41 %
		Downcasting	16	0.44	0.00 %	2.53	-0.09 %	0.56	-42.89 %	127.84	75.11 %	32.41	75.11 %	261,465,624	-48.77 %
		Quanto	2	0.44	0.00 %	3.05	20.66 %	0.99	1.46 %	71.96	-1.43 %	18.24	-1.43 %	510,348,344	0.00 %
		Quanto	4	0.44	0.00 %	2.44	-3.34 %	0.98	1.24 %	72.11	-1.22 %	18.28	-1.22 %	510,348,344	0.00 %
		Quanto	8	0.44	0.00 %	2.52	-0.14 %	0.97	0.25 %	72.83	-0.24 %	18.46	-0.24 %	510,348,344	0.00 %
	Yes		32	0.55	-	0.74	-	0.97	-	73.27	-	18.57	-	510,348,336	-
		BitsAndBytes	4	0.45	17.95 %	0.74	0.88 %	1.06	9.83 %	66.71	-8.95 %	16.91	-8.95 %	134,063,640	-73.73 %
		BitsAndBytes	8	0.55	0.00 %	0.73	-0.11 %	1.74	79.09 %	40.91	-44.16 %	10.37	-44.16 %	176,530,968	-65.41 %
		Downcasting	16	0.54	-2.56 %	0.74	-0.07 %	0.56	-41.89 %	126.08	72.08 %	31.96	72.08 %	261,465,624	-48.77 %
		Quanto	2	0.46	15.38 %	0.87	17.68 %	1.00	2.69 %	71.35	-2.62 %	18.09	-2.62 %	510,348,344	0.00 %
		Quanto	4	0.49	10.26 %	0.74	0.76 %	0.99	1.75 %	72.01	-1.72 %	18.25	-1.72 %	510,348,344	0.00 %
		Quanto	8	0.55	0.00 %	0.74	0.03 %	0.97	0.61 %	72.83	-0.60 %	18.46	-0.60 %	510,348,344	0.00 %
openai-community-gpt2-medium	No		32	0.56	-	1.78	-	2.40	-	29.63	-	7.51	-	1,444,466,784	-
		BitsAndBytes	4	0.56	0.00 %	1.19	-32.94 %	2.67	11.34 %	26.61	-10.18 %	6.75	-10.18 %	281,831,472	-80.49 %
		BitsAndBytes	8	0.56	0.00 %	1.97	10.81 %	3.37	40.60 %	21.07	-28.88 %	5.34	-28.87 %	432,826,416	-70.04 %
		Downcasting	16	0.56	0.00 %	1.79	0.63 %	1.18	-50.86 %	60.30	103.49 %	15.29	103.50 %	734,816,304	-49.13 %
		Quanto	2	0.56	0.00 %	2.23	25.26 %	2.42	0.91 %	29.36	-0.91 %	7.44	-0.91 %	1,444,466,792	0.00 %
		Quanto	4	0.56	0.00 %	1.79	0.76 %	2.42	0.81 %	29.39	-0.81 %	7.45	-0.80 %	1,444,466,792	0.00 %
		Quanto	8	0.56	0.00 %	1.78	0.12 %	2.39	-0.18 %	29.68	0.18 %	7.53	0.19 %	1,444,466,792	0.00 %
	Yes		32	0.39	-	0.71	-	2.40	-	29.61	-	7.51	-	1,444,466,784	-
		BitsAndBytes	4	0.39	0.00 %	0.72	1.10 %	2.66	10.85 %	26.71	-9.79 %	6.77	-9.79 %	281,831,472	-80.49 %
		BitsAndBytes	8	0.44	10.71 %	0.71	-0.21 %	3.22	34.42 %	22.03	-25.61 %	5.58	-25.61 %	432,826,416	-70.04 %
		Downcasting	16	0.41	3.57 %	0.71	-0.19 %	1.18	-50.81 %	60.20	103.29 %	15.26	103.29 %	734,816,304	-49.13 %
		Quanto	2	0.56	42.86 %	1.44	103.37 %	2.41	0.72 %	29.40	-0.71 %	7.45	-0.72 %	1,444,466,792	0.00 %
		Quanto	4	0.55	39.29 %	0.70	-1.11 %	2.41	0.48 %	29.47	-0.48 %	7.47	-0.48 %	1,444,466,792	0.00 %
		Quanto	8	0.45	14.29 %	0.71	-0.27 %	2.39	-0.29 %	29.70	0.29 %	7.53	0.29 %	1,444,466,792	0.00 %

Table D.9: Results on the super_glue/boolq task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.62	-	0.74	-	29.47	-	110.97	-	27.76	-	437,943,304	-
		BitsAndBytes	4	0.62	0.00 %	0.73	-1.42 %	44.87	52.28 %	72.87	-34.33 %	18.23	-34.33 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.60	-3.64 %	0.70	-5.59 %	123.83	320.22 %	26.41	-76.20 %	6.61	-76.20 %	133,451,268	-69.53 %
		Downcasting	16	0.62	0.00 %	0.74	0.02 %	21.52	-26.97 %	151.96	36.94 %	38.01	36.94 %	218,975,748	-50.00 %
		Quanto	2	0.58	-6.59 %	0.68	-7.37 %	74.78	153.79 %	43.73	-60.60 %	10.94	-60.60 %	437,943,896	0.00 %
		Quanto	4	0.62	0.05 %	0.72	-1.65 %	69.94	137.33 %	46.76	-57.86 %	11.70	-57.87 %	437,943,896	0.00 %
		Quanto	8	0.62	0.00 %	0.74	-0.03 %	36.34	23.34 %	89.97	-18.92 %	22.51	-18.92 %	437,943,896	0.00 %
	Yes		32	0.52	-	0.78	-	29.45	-	111.03	-	27.77	-	437,943,304	-
		BitsAndBytes	4	0.53	0.82 %	0.78	-0.89 %	44.86	52.33 %	72.89	-34.35 %	18.23	-34.35 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.47	-9.44 %	0.84	6.84 %	124.42	322.44 %	26.28	-76.33 %	6.58	-76.33 %	133,451,268	-69.53 %
		Downcasting	16	0.52	0.00 %	0.78	-0.03 %	21.56	-26.79 %	151.66	36.60 %	37.94	36.60 %	218,975,748	-50.00 %
		Quanto	2	0.63	21.52 %	0.69	-12.31 %	74.80	153.97 %	43.72	-60.62 %	10.94	-60.63 %	437,943,896	0.00 %
		Quanto	4	0.52	0.12 %	0.81	3.06 %	69.91	137.37 %	46.77	-57.87 %	11.70	-57.87 %	437,943,896	0.00 %
		Quanto	8	0.52	-0.35 %	0.78	0.06 %	36.37	23.49 %	89.91	-19.02 %	22.49	-19.02 %	437,943,896	0.00 %
bert-large-uncased	No		32	0.38	-	1.65	-	64.01	-	51.08	-	12.78	-	1,340,583,944	-
		BitsAndBytes	4	0.38	0.16 %	1.65	0.04 %	95.70	49.49 %	34.17	-33.11 %	8.55	-33.11 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.38	0.00 %	1.13	-31.18 %	243.68	280.67 %	13.42	-73.73 %	3.36	-73.73 %	367,257,604	-72.60 %
		Downcasting	16	0.38	0.00 %	1.65	0.01 %	41.09	-35.81 %	79.58	55.79 %	19.91	55.79 %	670,296,068	-50.00 %
		Quanto	2	0.60	58.56 %	0.67	-59.08 %	154.17	140.84 %	21.21	-58.48 %	5.31	-58.48 %	1,340,585,112	0.00 %
		Quanto	4	0.38	0.00 %	1.67	1.42 %	144.90	126.36 %	22.57	-55.82 %	5.64	-55.83 %	1,340,585,112	0.00 %
		Quanto	8	0.38	0.08 %	1.65	-0.07 %	77.29	20.74 %	42.31	-17.18 %	10.58	-17.18 %	1,340,585,112	0.00 %
	Yes		32	0.38	-	0.70	-	64.06	-	51.04	-	12.77	-	1,340,583,944	-
		BitsAndBytes	4	0.38	0.00 %	0.70	0.08 %	95.54	49.13 %	34.23	-32.94 %	8.56	-32.95 %	215,738,372	-83.91 %

(Continued on next page)

D. DETAILED TABLES

(Continued from previous page)																
Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%	
		BitsAndBytes	8	0.38	0.00 %	0.70	0.01 %	193.46	201.98 %	16.90	-66.89 %	4.23	-66.89 %	367,257,604	-72.60 %	
		Downcasting	16	0.38	0.00 %	0.70	0.12 %	41.06	-35.91 %	79.64	56.02 %	19.92	56.01 %	670,296,068	-50.00 %	
		Quanto	2	0.62	64.35 %	0.69	-0.66 %	154.03	140.42 %	21.23	-58.41 %	5.31	-58.41 %	1,340,585,112	0.00 %	
		Quanto	4	0.38	0.00 %	0.70	1.13 %	144.22	125.11 %	22.67	-55.58 %	5.67	-55.58 %	1,340,585,112	0.00 %	
		Quanto	8	0.38	0.00 %	0.69	-0.11 %	77.33	20.71 %	42.29	-17.16 %	10.58	-17.16 %	1,340,585,112	0.00 %	
facebook bart-base	No		32	0.42	-	0.71	-	131.65	-	24.84	-	6.21	-	560,050,184	-	
		BitsAndBytes	4	0.53	26.61 %	0.69	-2.67 %	74.70	-43.26 %	43.78	76.24 %	10.95	76.23 %	130,504,708	-76.70 %	
		BitsAndBytes	8	0.42	-0.73 %	0.71	0.29 %	130.15	-1.14 %	25.12	1.15 %	6.29	1.14 %	180,344,836	-67.80 %	
		Downcasting	16	0.42	0.22 %	0.71	-0.01 %	35.92	-72.72 %	91.04	266.52 %	22.77	266.50 %	280,025,092	-50.00 %	
	Yes		32	0.62	-	0.68	-	98.50	-	33.20	-	8.30	-	560,050,184	-	
		BitsAndBytes	4	0.62	0.00 %	0.68	-0.20 %	74.86	-24.00 %	43.68	31.57 %	10.93	31.57 %	130,504,708	-76.70 %	
		BitsAndBytes	8	0.62	0.00 %	0.68	-0.12 %	116.09	17.86 %	28.17	-15.15 %	7.05	-15.15 %	180,344,836	-67.80 %	
		Downcasting	16	0.62	0.00 %	0.68	0.08 %	36.09	-63.35 %	90.59	172.88 %	22.66	172.87 %	280,025,092	-50.00 %	
google flan-t5-base	No		32	0.38	-	0.81	-	113.26	-	28.87	-	7.22	-	893,982,728	-	
		BitsAndBytes	4	0.38	-0.08 %	0.83	2.41 %	126.15	11.38 %	25.92	-10.22 %	6.48	-10.22 %	280,955,908	-68.57 %	
		BitsAndBytes	8	0.38	0.00 %	0.81	0.06 %	180.48	59.35 %	18.12	-37.24 %	4.53	-37.25 %	361,466,884	-59.57 %	
		Downcasting	16	0.38	0.00 %	0.81	-0.00 %	77.65	-31.44 %	42.11	45.86 %	10.53	45.86 %	446,991,364	-50.00 %	
		Quanto	2	0.38	0.00 %	1.11	36.49 %	193.84	71.14 %	16.87	-41.57 %	4.22	-41.57 %	893,984,472	0.00 %	
		Quanto	4	0.38	-0.08 %	0.84	3.17 %	180.67	59.52 %	18.10	-37.31 %	4.53	-37.30 %	893,984,472	0.00 %	
		Quanto	8	0.38	0.00 %	0.81	-0.07 %	100.58	-11.20 %	32.51	12.61 %	8.13	12.61 %	893,984,472	0.00 %	
	Yes		32	0.79	-	0.91	-	104.84	-	31.19	-	7.80	-	893,982,728	-	
		BitsAndBytes	4	0.79	0.00 %	0.87	-4.20 %	125.97	20.15 %	25.96	-16.77 %	6.49	-16.76 %	280,955,908	-68.57 %	
		BitsAndBytes	8	0.79	0.12 %	0.91	0.16 %	179.41	71.12 %	18.23	-41.56 %	4.56	-41.57 %	361,466,884	-59.57 %	
		Downcasting	16	0.79	-0.08 %	0.91	-0.08 %	77.58	-26.01 %	42.15	35.15 %	10.54	35.16 %	446,991,364	-50.00 %	
		Quanto	2	0.38	-51.97 %	1.06	17.38 %	193.78	84.83 %	16.88	-45.90 %	4.22	-45.90 %	893,984,472	0.00 %	
		Quanto	4	0.79	0.19 %	0.88	-3.14 %	180.66	72.32 %	18.10	-41.97 %	4.53	-41.96 %	893,984,472	0.00 %	
		Quanto	8	0.79	-0.04 %	0.91	0.15 %	100.82	-3.84 %	32.44	3.99 %	8.11	4.00 %	893,984,472	0.00 %	
openai-community gpt2	No		32	0.38	-	1.53	-	39.90	-	81.95	-	20.50	-	510,348,336	-	
		BitsAndBytes	4	0.38	-0.16 %	2.79	81.86 %	42.24	5.85 %	77.42	-5.53 %	19.37	-5.53 %	134,063,640	-73.73 %	
		BitsAndBytes	8	0.38	0.89 %	1.40	-8.50 %	67.28	68.59 %	48.61	-40.69 %	12.16	-40.68 %	176,530,968	-65.41 %	
		Downcasting	16	0.38	0.24 %	1.52	-0.51 %	21.32	-46.56 %	153.36	87.14 %	38.36	87.14 %	261,465,624	-48.77 %	
		Quanto	2	0.38	-0.81 %	1.70	10.85 %	40.45	1.38 %	80.83	-1.36 %	20.22	-1.36 %	510,348,344	0.00 %	
		Quanto	4	0.38	-0.16 %	1.58	3.21 %	40.54	1.60 %	80.66	-1.57 %	20.18	-1.57 %	510,348,344	0.00 %	
		Quanto	8	0.38	0.00 %	1.52	-0.84 %	40.12	0.53 %	81.52	-0.53 %	20.39	-0.53 %	510,348,344	0.00 %	
	Yes		32	0.39	-	0.71	-	40.00	-	81.75	-	20.45	-	510,348,336	-	
		BitsAndBytes	4	0.40	2.43 %	0.71	-0.59 %	42.42	6.04 %	77.09	-5.69 %	19.29	-5.69 %	134,063,640	-73.73 %	
		BitsAndBytes	8	0.39	0.31 %	0.71	-0.04 %	61.15	52.87 %	53.47	-34.59 %	13.38	-34.59 %	176,530,968	-65.41 %	
		Downcasting	16	0.39	0.24 %	0.71	-0.02 %	21.54	-46.16 %	151.82	85.72 %	37.98	85.73 %	261,465,624	-48.77 %	
		Quanto	2	0.40	3.06 %	0.74	3.35 %	40.55	1.37 %	80.64	-1.36 %	20.17	-1.35 %	510,348,344	0.00 %	
		Quanto	4	0.40	3.29 %	0.71	-0.39 %	40.56	1.39 %	80.62	-1.37 %	20.17	-1.37 %	510,348,344	0.00 %	
		Quanto	8	0.39	0.31 %	0.71	-0.02 %	40.09	0.22 %	81.57	-0.22 %	20.40	-0.22 %	510,348,344	0.00 %	
openai-community gpt2-medium	No		32	0.42	-	1.18	-	101.63	-	32.17	-	8.05	-	1,444,466,784	-	
		BitsAndBytes	4	0.38	-9.96 %	1.94	65.13 %	108.04	6.30 %	30.27	-5.93 %	7.57	-5.93 %	281,831,472	-80.49 %	
		BitsAndBytes	8	0.43	2.09 %	1.21	3.03 %	133.62	31.48 %	24.47	-23.94 %	6.12	-23.93 %	432,826,416	-70.04 %	
		Downcasting	16	0.42	-0.94 %	1.18	0.34 %	46.84	-53.91 %	69.81	116.99 %	17.46	117.00 %	734,816,304	-49.13 %	
		Quanto	2	0.42	-1.15 %	1.37	16.25 %	102.73	1.08 %	31.83	-1.07 %	7.96	-1.06 %	1,444,466,792	0.00 %	
		Quanto	4	0.42	-1.95 %	1.20	1.73 %	102.64	0.99 %	31.86	-0.98 %	7.97	-0.98 %	1,444,466,792	0.00 %	
		Quanto	8	0.42	-0.43 %	1.18	0.29 %	102.11	0.46 %	32.03	-0.46 %	8.01	-0.46 %	1,444,466,792	0.00 %	
	Yes		32	0.40	-	0.71	-	102.01	-	32.05	-	8.02	-	1,444,466,784	-	
		BitsAndBytes	4	0.39	-1.08 %	0.71	0.14 %	108.46	6.32 %	30.15	-5.94 %	7.54	-5.95 %	281,831,472	-80.49 %	
		BitsAndBytes	8	0.39	-0.85 %	0.71	0.07 %	119.03	16.68 %	27.47	-14.30 %	6.87	-14.30 %	432,826,416	-70.04 %	
		Downcasting	16	0.40	0.08 %	0.71	0.04 %	46.82	-54.11 %	69.84	117.89 %	17.47	117.88 %	734,816,304	-49.13 %	
		Quanto	2	0.60	50.39 %	0.68	-3.88 %	102.72	0.70 %	31.83	-0.69 %	7.96	-0.70 %	1,444,466,792	0.00 %	
		Quanto	4	0.41	2.77 %	0.71	-0.29 %	102.62	0.60 %	31.86	-0.59 %	7.97	-0.60 %	1,444,466,792	0.00 %	
		Quanto	8	0.40	2.00 %	0.71	-0.23 %	102.15	0.13 %	32.01	-0.13 %	8.01	-0.14 %	1,444,466,792	0.00 %	

Table D.10: Results on the super_glue/cb task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval F1 Score	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.50	-	0.33	-	1.03	-	0.50	-	111.08	-	27.77	-	437,946,380	-
		BitsAndBytes	4	0.50	0.00 %	0.33	0.00 %	1.03	0.17 %	0.79	55.82 %	71.29	-35.82 %	17.82	-35.83 %	90,690,566	-79.29 %
		BitsAndBytes	8	0.34	-32.14 %	0.36	7.68 %	1.11	8.55 %	2.13	322.81 %	26.27	-76.35 %	6.57	-76.35 %	133,452,806	-69.53 %
		Downcasting	16	0.50	0.00 %	0.33	0.00 %	1.03	0.01 %	0.38	-25.31 %	148.74	33.90 %	37.19	33.90 %	218,977,286	50.00 %
		Quanto	2	0.09	-82.14 %	0.01	-95.61 %	1.22	18.49 %	1.31	160.03 %	42.72	-61.54 %	10.68	-61.54 %	437,946,972	0.00 %
		Quanto	4	0.50	0.00 %	0.33	0.00 %	1.02	-0.15 %	1.22	142.83 %	45.75	-58.82 %	11.44	-58.82 %	437,946,972	0.00 %
		Quanto	8	0.50	0.00 %	0.33	0.00 %	1.03	-0.01 %	0.64	26.01 %	88.16	-20.64 %	22.04	-20.64 %	437,946,972	0.00 %
	Yes		32	0.57	-	0.53	-	1.11	-	0.51	-	109.47	-	27.37	-	437,946,380	-
		BitsAndBytes	4	0.57	0.00 %	0.53	0.00 %	1.05	-5.13 %	0.78	52.56 %	71.75	-34.46 %	17.94	-34.46 %	90,690,566	-79.29 %
		BitsAndBytes	8	0.45	-21.87 %	0.43	-19.07 %	1.34	21.17 %	2.13	317.10 %	26.24	-76.03 %	6.56	-76.03 %	133,452,806	-69.53 %
		Downcasting	16	0.57	0.00 %	0.53	0.00 %	1.11	-0.06 %	0.38	-26.08 %	148.08	35.27 %	37.02	35.27 %	218,977,286	50.00 %
		Quanto	2	0.59	3.13 %	0.54	1.86 %	0.96	-12.91 %	1.32	157.23 %	42.55	-61.13 %	10.64	-61.13 %	437,946,972	0.00 %
		Quanto	4	0.52	-9.37 %	0.46	-14.10 %	1.14	2.69 %	1.22	139.11 %	45.78	-58.18 %	11.45	-58.18 %	437,946,972	0.00 %
		Quanto	8	0.57	0.00 %	0.53	0.00 %	1.11	-0.01 %	0.63	23.87 %	88.36	-19.28 %	22.09	-19.28 %	437,946,972	0.00 %
bert-large-uncased	No		32	0.41	-	0.24	-	1.86	-	1.10	-	50.70	-	12.67	-	1,340,588,044	-
		BitsAndBytes	4	0.41	0.00 %	0.24	0.00 %	1.86	-0.02 %	1.65	49.29 %	33.96	-33.02 %	8.49	-33.01 %	215,740,422	-83.91 %
		BitsAndBytes	8	0.41	0.00 %	0.24	0.00 %	1.46	-21.63 %	4.17	277.58 %	13.43	-73.51 %	3.36	-73.51 %	367,259,654	-72.60 %
		Downcasting	16	0.41	0.00 %	0.24	0.00 %	1.86	0.02 %	0.70	-36.28 %	79.57	56.95 %	19.89	56.95 %	670,298,118	50.00 %
		Quanto	2	0.20	-52.17 %	0.22	-9.75 %	1.17	-36.77 %	2.70	144.49 %	20.74	-59.10 %	5.18	-59.10 %	1,340,589,212	0.00 %
		Quanto	4	0.41	0.00 %	0.24	0.00 %	1.90	2.52 %	2.52	128.03 %	22.23	-56.14 %	5.56	-56.15 %	1,340,589,212	0.00 %
		Quanto	8	0.41	0.00 %	0.24	0.00 %	1.86	0.04 %	1.35	22.67 %	41.33	-18.48 %	10.33	-18.48 %	1,340,589,212	0.00 %
	Yes		32	0.09	-	0.01	-	1.10	-	1.11	-	50.35	-	12.59	-	1,340,588,044	-
		BitsAndBytes	4	0.50	460.00 %	0.33	2177.33 %	1.10	-0.61 %	1.65	48.53 %	33.90	-32.68 %	8.47	-32.68 %	215,740,422	-83.91 %
		BitsAndBytes	8	0.48	440.00 %	0.33	2149.56 %	1.10	-0.08 %	3.44	209.27 %	16.28	-67.67 %	4.07	-67.67 %	367,259,654	-72.60 %
		Downcasting	16	0.50	460.00 %	0.33	2177.33 %	1.10	-0.06 %	0.71	-35.84 %	78.47	55.87 %	19.62	55.86 %	670,298,118	50.00 %
		Quanto	2	0.09	0.00 %	0.01	0.00 %	1.18	6.89 %	2.69	141.61 %	20.84	-58.61 %	5.21	-58.62 %	1,340,589,212	0.00 %
		Quanto	4	0.41	360.00 %	0.24	1533.87 %	1.09	-1.50 %	2.52	126.90 %	22.19	-55.93 %	5.55	-55.93 %	1,340,589,212	0.00 %
		Quanto	8	0.50	460.00 %	0.33	2177.33 %	1.10	-0.16 %	1.35	21.55 %	41.42	-17.73 %	10.36	-17.73 %	1,340,589,212	0.00 %
facebook bart-base	No		32	0.50	-	0.33	-	1.03	-	0.85	-	65.58	-	16.39	-	560,053,260	-
		BitsAndBytes	4	0.50	0.00 %	0.33	0.00 %	1.03	-0.49 %	1.06	23.65 %	53.03	-19.13 %	13.26	-19.13 %	130,506,246	-76.70 %
		BitsAndBytes	8	0.50	0.00 %	0.33	0.00 %	1.04	0.07 %	2.02	137.06 %	27.66	-57.82 %	6.92	-57.82 %	180,346,374	-67.80 %
		Downcasting	16	0.50	0.00 %	0.33	0.00 %	1.03	0.02 %	0.53	-38.41 %	106.47	62.35 %	26.62	62.36 %	280,026,630	50.00 %
		Quanto	2	0.21	-57.14 %	0.23	-30.33 %	1.19	15.04 %	1.76	105.83 %	31.86	-51.42 %	7.96	-51.42 %	560,054,044	0.00 %
		Quanto	4	0.50	0.00 %	0.33	0.00 %	1.09	5.21 %	1.89	121.58 %	29.59	-54.87 %	7.40	-54.87 %	560,054,044	0.00 %
		Quanto	8	0.50	0.00 %	0.33	0.00 %	1.03	-0.23 %	0.96	12.99 %	58.04	-11.50 %	14.51	-11.50 %	560,054,044	0.00 %
	Yes		32	0.64	-	0.64	-	1.21	-	0.84	-	66.73	-	16.68	-	560,053,260	-
		BitsAndBytes	4	0.57	-11.11 %	0.56	-12.50 %	1.23	2.02 %	1.07	27.30 %	52.42	-21.45 %	13.11	-21.45 %	130,506,246	-76.70 %
		BitsAndBytes	8	0.64	0.00 %	0.64	0.00 %	1.20	-0.72 %	2.02	141.06 %	27.68	-58.52 %	6.92	-58.52 %	180,346,374	-67.80 %
		Downcasting	16	0.64	0.00 %	0.64	0.00 %	1.21	-0.29 %	0.52	-37.56 %	106.86	60.13 %	26.71	60.13 %	280,026,630	50.00 %
		Quanto	2	0.43	-33.33 %	0.32	-50.59 %	1.10	-8.90 %	2.15	155.76 %	26.09	-60.90 %	6.52	-60.90 %	560,054,044	0.00 %
		Quanto	4	0.59	-8.33 %	0.57	-11.58 %	1.24	2.26 %	1.69	101.84 %	33.06	-50.46 %	8.27	-50.46 %	560,054,044	0.00 %
		Quanto	8	0.64	0.00 %	0.64	0.00 %	1.20	-0.41 %	0.98	16.47 %	57.29	-14.15 %	14.32	-14.15 %	560,054,044	0.00 %
google flan-t5-base	No		32	0.41	-	0.24	-	1.28	-	1.44	-	38.85	-	9.71	-	893,985,804	-
		BitsAndBytes	4	0.41	0.00 %	0.24	0.00 %	1.28	0.10 %	2.11	46.33 %	26.55	-31.66 %	6.64	-31.66 %	280,957,446	-68.57 %
		BitsAndBytes	8	0.41	0.00 %	0.24	0.00 %	1.28	0.18 %	3.02	109.85 %	18.52	-52.35 %	4.63	-52.34 %	361,468,422	-59.57 %
		Downcasting	16	0.41	0.00 %	0.24	0.00 %	1.28	0.01 %	1.27	-11.97 %	44.14	13.60 %	11.03	13.60 %	446,992,902	50.00 %
		Quanto	2	0.34	-17.39 %	0.21	-10.60 %	1.33	3.49 %	3.29	128.50 %	17.00	-56.24 %	4.25	-56.23 %	893,987,548	0.00 %
		Quanto	4	0.41	0.00 %	0.24	0.00 %	1.29	0.64 %	3.07	113.00 %	18.24	-53.05 %	4.56	-53.05 %	893,987,548	0.00 %
		Quanto	8	0.41	0.00 %	0.24	0.00 %	1.29	0.35 %	1.71	18.90 %	32.68	-15.89 %	8.17	-15.89 %	893,987,548	0.00 %
	Yes		32	0.89	-	0.89	-	0.70	-	1.47	-	38.10	-	9.53	-	893,985,804	-
		BitsAndBytes	4	0.89	0.00 %	0.89	0.00 %	0.63	-9.37 %	2.12	44.53 %	26.36	-30.81 %	6.59	-30.81 %	280,957,446	-68.57 %
		BitsAndBytes	8	0.89	0.00 %	0.89	0.00 %	0.71	1.70 %	3.06	108.40 %	18.28	-52.02 %	4.57	-52.02 %	361,468,422	-59.57 %
		Downcasting	16	0.89	0.00 %	0.89	0.00 %	0.70	-0.16 %	1.29	-12.02 %	43.31	13.66 %	10.83	13.65 %	446,992,902	50.00 %
		Quanto	2	0.09	-90.00 %	0.01	-98.36 %	1.48	111.88 %	3.30	124.53 %	16.97	-55.46 %	4.24	-55.46 %	893,987,548	0.00 %
		Quanto	4	0.89	0.00 %	0.89	-0.23 %	0.76	8.47 %	3.09	110.10 %	18.14	-52.41 %	4.53	-52.40 %	893,987,548	0.00 %
		Quanto	8	0.89	0.00 %	0.89	0.00 %	0.70	0.98 %	1.69	14.75 %	33.20	-12.85 %	8.30	-12.86 %	893,987,548	0.00 %

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D. DETAILED TABLES

(Continued from previous page)																	
Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval F1 Score	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
openai-community gpt2	No		32	0.50	-	0.33	-	1.96	-	0.69	-	81.09	-	20.27	-	510,351,408	-
		BitsAndBytes	4	0.50	0.00 %	0.33	0.00 %	2.54	29.77 %	0.73	6.01 %	76.49	-5.67 %	19.12	-5.67 %	134,065,176	73.73 %
		BitsAndBytes	8	0.50	0.00 %	0.33	0.00 %	1.89	-3.45 %	1.15	66.03 %	48.84	-39.77 %	12.21	-39.77 %	176,532,504	-65.41 %
		Downcasting	16	0.50	0.00 %	0.33	0.00 %	1.97	0.74 %	0.37	-46.93 %	152.81	88.44 %	38.20	88.44 %	261,467,160	-48.77 %
		Quanto	2	0.50	0.00 %	0.33	0.00 %	2.33	19.19 %	0.71	2.13 %	79.40	-2.09 %	19.85	-2.09 %	510,351,416	0.00 %
		Quanto	4	0.50	0.00 %	0.33	0.00 %	1.95	-0.39 %	0.70	1.75 %	79.69	-1.73 %	19.92	-1.73 %	510,351,416	0.00 %
		Quanto	8	0.50	0.00 %	0.33	0.00 %	1.95	-0.15 %	0.70	0.75 %	80.48	-0.75 %	20.12	-0.75 %	510,351,416	0.00 %
	Yes		32	0.77	-	0.77	-	0.87	-	0.70	-	80.52	-	20.13	-	510,351,408	-
		BitsAndBytes	4	0.73	-4.65 %	0.73	-4.78 %	0.90	3.23 %	0.73	4.86 %	76.78	-4.64 %	19.20	-4.64 %	134,065,176	73.73 %
		BitsAndBytes	8	0.75	-2.33 %	0.75	-2.34 %	0.89	1.66 %	1.16	66.08 %	48.48	-39.79 %	12.12	-39.78 %	176,532,504	-65.41 %
		Downcasting	16	0.77	0.00 %	0.77	0.00 %	0.87	-0.24 %	0.37	-47.06 %	152.08	88.88 %	38.02	88.88 %	261,467,160	-48.77 %
		Quanto	2	0.77	0.00 %	0.77	-0.06 %	1.00	14.13 %	0.71	1.38 %	79.42	-1.37 %	19.85	-1.37 %	510,351,416	0.00 %
		Quanto	4	0.77	0.00 %	0.77	0.00 %	0.83	-5.01 %	0.70	1.27 %	79.52	-1.24 %	19.88	-1.24 %	510,351,416	0.00 %
		Quanto	8	0.77	0.00 %	0.77	0.00 %	0.87	-0.17 %	0.70	0.06 %	80.47	-0.06 %	20.12	-0.05 %	510,351,416	0.00 %
openai-community gpt2-medium	No		32	0.36	-	0.22	-	3.45	-	1.75	-	31.96	-	7.99	-	1,444,470,880	-
		BitsAndBytes	4	0.29	-20.00 %	0.18	-15.56 %	2.80	-18.79 %	1.89	7.60 %	29.70	-7.06 %	7.42	-7.06 %	281,833,520	-80.49 %
		BitsAndBytes	8	0.38	5.00 %	0.22	3.64 %	3.36	-2.47 %	2.28	29.87 %	24.61	-23.00 %	6.15	-22.99 %	432,828,464	-70.04 %
		Downcasting	16	0.36	0.00 %	0.22	0.00 %	3.44	-0.13 %	0.81	-53.89 %	69.31	116.88 %	17.33	116.89 %	734,818,352	-49.13 %
		Quanto	2	0.36	0.00 %	0.22	0.00 %	3.94	14.38 %	1.76	0.68 %	31.74	-0.67 %	7.93	-0.68 %	1,444,470,888	0.00 %
		Quanto	4	0.36	0.00 %	0.22	0.00 %	3.45	0.11 %	1.76	0.48 %	31.80	-0.48 %	7.95	-0.48 %	1,444,470,888	0.00 %
		Quanto	8	0.36	0.00 %	0.22	0.00 %	3.45	-0.06 %	1.75	0.06 %	31.94	-0.06 %	7.98	-0.06 %	1,444,470,888	0.00 %
	Yes		32	0.82	-	0.82	-	0.66	-	1.75	-	31.95	-	7.99	-	1,444,470,880	-
		BitsAndBytes	4	0.86	4.35 %	0.86	4.27 %	0.50	-24.22 %	1.89	8.10 %	29.56	-7.49 %	7.39	-7.50 %	281,833,520	-80.49 %
		BitsAndBytes	8	0.84	2.17 %	0.84	2.14 %	0.65	-1.67 %	2.38	35.60 %	23.56	-26.25 %	5.89	-26.25 %	432,828,464	-70.04 %
		Downcasting	16	0.82	0.00 %	0.82	0.00 %	0.66	0.15 %	0.81	-53.80 %	69.15	116.44 %	17.29	116.42 %	734,818,352	-49.13 %
		Quanto	2	0.82	0.00 %	0.83	1.05 %	1.41	113.28 %	1.77	1.06 %	31.62	-1.05 %	7.90	-1.05 %	1,444,470,888	0.00 %
		Quanto	4	0.84	2.17 %	0.84	2.14 %	0.71	7.31 %	1.76	0.51 %	31.79	-0.50 %	7.95	-0.51 %	1,444,470,888	0.00 %
		Quanto	8	0.82	0.00 %	0.82	0.00 %	0.64	-2.43 %	1.75	-0.05 %	31.97	0.06 %	7.99	0.05 %	1,444,470,888	0.00 %

Table D.11: Results on the super_glue/copa task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.48	-	0.70	-	0.93	-	107.27	-	26.82	-	437,943,304	-
		BitsAndBytes	4	0.45	-6.25 %	0.70	0.54 %	1.42	52.53 %	70.33	-34.44 %	17.58	-34.44 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.54	12.50 %	0.68	-2.25 %	3.87	314.75 %	25.86	-75.89 %	6.47	-75.89 %	133,451,268	-69.53 %
		Downcasting	16	0.48	0.00 %	0.70	0.01 %	0.69	-25.55 %	144.10	34.33 %	36.02	34.33 %	218,975,748	-50.00 %
		Quanto	2	0.57	18.75 %	0.68	-2.04 %	2.36	153.14 %	42.38	-60.50 %	10.59	-60.50 %	437,943,896	0.00 %
		Quanto	4	0.47	-2.08 %	0.70	-0.01 %	2.20	135.60 %	45.53	-57.56 %	11.38	-57.55 %	437,943,896	0.00 %
		Quanto	8	0.48	0.00 %	0.70	-0.05 %	1.15	23.88 %	86.59	-19.28 %	21.65	-19.27 %	437,943,896	0.00 %
	Yes		32	0.51	-	0.74	-	0.93	-	107.08	-	26.77	-	437,943,304	-
		BitsAndBytes	4	0.53	3.92 %	0.73	-1.78 %	1.41	51.32 %	70.76	-33.92 %	17.69	-33.91 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.54	5.88 %	0.69	-7.88 %	3.86	312.97 %	25.93	-75.79 %	6.48	-75.79 %	133,451,268	-69.53 %
		Downcasting	16	0.51	0.00 %	0.74	0.03 %	0.69	-25.75 %	144.22	34.68 %	36.05	34.68 %	218,975,748	-50.00 %
		Quanto	2	0.53	3.92 %	0.70	-6.44 %	2.35	151.90 %	42.51	-60.30 %	10.63	-60.30 %	437,943,896	0.00 %
		Quanto	4	0.52	1.96 %	0.73	-1.85 %	2.19	134.90 %	45.59	-57.43 %	11.40	-57.43 %	437,943,896	0.00 %
		Quanto	8	0.52	1.96 %	0.74	0.08 %	1.16	23.71 %	86.56	-19.17 %	21.64	-19.17 %	437,943,896	0.00 %
bert-large-uncased	No		32	0.57	-	1.14	-	2.00	-	50.02	-	12.51	-	1,340,583,944	-
		BitsAndBytes	4	0.59	3.51 %	1.11	-2.50 %	2.97	48.41 %	33.71	-32.62 %	8.43	-32.61 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.56	-1.75 %	0.91	-19.87 %	7.43	271.68 %	13.46	-73.10 %	3.37	-73.09 %	367,257,604	-72.60 %
		Downcasting	16	0.57	0.00 %	1.14	-0.00 %	1.27	-36.51 %	78.79	57.50 %	19.70	57.51 %	670,296,068	-50.00 %
		Quanto	2	0.49	-14.04 %	0.72	-36.91 %	4.80	140.30 %	20.82	-58.38 %	5.20	-58.38 %	1,340,585,112	0.00 %
		Quanto	4	0.57	0.00 %	1.12	-1.22 %	4.53	126.68 %	22.07	-55.88 %	5.52	-55.88 %	1,340,585,112	0.00 %
		Quanto	8	0.57	0.00 %	1.14	-0.01 %	2.43	21.51 %	41.17	-17.70 %	10.29	-17.70 %	1,340,585,112	0.00 %
	Yes		32	0.45	-	0.69	-	2.00	-	50.12	-	12.53	-	1,340,583,944	-
		BitsAndBytes	4	0.45	0.00 %	0.70	0.42 %	2.96	48.26 %	33.81	-32.55 %	8.45	-32.55 %	215,738,372	-83.91 %

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
		BitsAndBytes	8	0.45	0.00 %	0.69	-0.00 %	6.31	216.24 %	15.85	-68.38 %	3.96	-68.38 %	367,257,604	-72.60 %
		Downcasting	16	0.45	0.00 %	0.69	0.08 %	1.29	-35.40 %	77.58	54.79 %	19.40	54.78 %	670,296,068	-50.00 %
		Quanto	2	0.55	22.22 %	0.69	-0.65 %	4.79	139.96 %	20.89	-58.33 %	5.22	-58.33 %	1,340,585,112	0.00 %
		Quanto	4	0.45	0.00 %	0.69	0.07 %	4.53	126.88 %	22.09	-55.92 %	5.52	-55.93 %	1,340,585,112	0.00 %
		Quanto	8	0.45	0.00 %	0.69	-0.06 %	2.42	21.55 %	41.24	-17.73 %	10.31	-17.73 %	1,340,585,112	0.00 %
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
facebook bart-base	No	BitsAndBytes	32	0.45	-	0.74	-	1.50	-	66.82	-	16.71	-	560,050,184	-
		BitsAndBytes	4	0.45	0.00 %	0.72	-2.63 %	1.89	26.29 %	52.91	-20.82 %	13.23	-20.82 %	130,504,708	-76.70 %
		BitsAndBytes	8	0.45	0.00 %	0.75	1.69 %	3.64	142.94 %	27.51	-58.84 %	6.88	-58.84 %	180,344,836	-67.80 %
		Downcasting	16	0.45	0.00 %	0.73	-0.16 %	0.92	-38.60 %	108.83	62.86 %	27.21	62.86 %	280,025,092	-50.00 %
		Quanto	2	0.44	-2.22 %	0.72	-1.80 %	2.99	100.11 %	33.39	-50.03 %	8.35	-50.03 %	560,050,968	0.00 %
		Quanto	4	0.45	0.00 %	0.75	1.70 %	2.81	87.83 %	35.58	-46.76 %	8.89	-46.76 %	560,050,968	0.00 %
		Quanto	8	0.45	0.00 %	0.74	1.26 %	1.72	15.25 %	57.98	-13.23 %	14.50	-13.23 %	560,050,968	0.00 %
	Yes	BitsAndBytes	32	0.44	-	0.86	-	1.50	-	66.79	-	16.70	-	560,050,184	-
		BitsAndBytes	4	0.41	-6.82 %	0.84	-2.80 %	1.91	27.29 %	52.47	-21.44 %	13.12	-21.44 %	130,504,708	-76.70 %
		BitsAndBytes	8	0.44	0.00 %	0.87	0.76 %	3.55	137.38 %	28.14	-57.88 %	7.03	-57.88 %	180,344,836	-67.80 %
		Downcasting	16	0.44	0.00 %	0.86	-0.07 %	0.99	-34.09 %	101.33	51.72 %	25.33	51.72 %	280,025,092	-50.00 %
		Quanto	2	0.45	2.27 %	0.80	-7.80 %	3.00	100.06 %	33.39	-50.01 %	8.35	-50.02 %	560,050,968	0.00 %
		Quanto	4	0.45	2.27 %	0.84	-3.25 %	2.83	88.88 %	35.36	-47.06 %	8.84	-47.06 %	560,050,968	0.00 %
		Quanto	8	0.44	0.00 %	0.87	0.33 %	1.72	14.70 %	58.23	-12.82 %	14.56	-12.82 %	560,050,968	0.00 %
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
google flan-t5-base	No	BitsAndBytes	32	0.55	-	0.74	-	2.59	-	38.60	-	9.65	-	893,982,728	-
		BitsAndBytes	4	0.55	0.00 %	0.75	1.46 %	3.78	45.84 %	26.47	-31.43 %	6.62	-31.43 %	280,955,908	-68.57 %
		BitsAndBytes	8	0.55	0.00 %	0.74	0.34 %	5.31	104.95 %	18.83	-51.21 %	4.71	-51.21 %	361,466,884	-59.57 %
		Downcasting	16	0.55	0.00 %	0.74	0.01 %	2.28	-11.89 %	43.81	13.50 %	10.95	13.49 %	446,991,364	-50.00 %
		Quanto	2	0.55	0.00 %	0.77	3.47 %	5.85	125.93 %	17.08	-55.74 %	4.27	-55.74 %	893,984,472	0.00 %
		Quanto	4	0.55	0.00 %	0.73	-1.39 %	5.45	110.28 %	18.36	-52.44 %	4.59	-52.45 %	893,984,472	0.00 %
		Quanto	8	0.55	0.00 %	0.74	0.30 %	3.00	15.91 %	33.30	-13.72 %	8.32	-13.73 %	893,984,472	0.00 %
	Yes	BitsAndBytes	32	0.43	-	0.86	-	2.60	-	38.39	-	9.60	-	893,982,728	-
		BitsAndBytes	4	0.41	-4.65 %	0.86	-0.80 %	3.78	45.18 %	26.44	-31.12 %	6.61	-31.12 %	280,955,908	-68.57 %
		BitsAndBytes	8	0.45	4.65 %	0.86	-0.35 %	5.44	108.66 %	18.40	-52.08 %	4.60	-52.08 %	361,466,884	-59.57 %
		Downcasting	16	0.43	0.00 %	0.87	0.13 %	2.30	-11.53 %	43.39	13.03 %	10.85	13.04 %	446,991,364	-50.00 %
		Quanto	2	0.55	27.91 %	0.70	-18.77 %	5.85	124.71 %	17.08	-55.50 %	4.27	-55.50 %	893,984,472	0.00 %
		Quanto	4	0.43	0.00 %	0.85	-1.90 %	5.41	107.71 %	18.48	-51.86 %	4.62	-51.86 %	893,984,472	0.00 %
		Quanto	8	0.44	2.33 %	0.86	-0.04 %	3.03	16.13 %	33.06	-13.89 %	8.26	-13.89 %	893,984,472	0.00 %
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
openai-community gpt2	No	BitsAndBytes	32	0.45	-	2.45	-	1.23	-	81.53	-	20.38	-	510,348,336	-
		BitsAndBytes	4	0.45	0.00 %	3.07	25.30 %	1.32	7.28 %	76.00	-6.79 %	19.00	-6.79 %	134,063,640	-73.73 %
		BitsAndBytes	8	0.45	0.00 %	2.36	-3.72 %	2.06	68.19 %	48.48	-40.54 %	12.12	-40.54 %	176,530,968	-65.41 %
		Downcasting	16	0.45	0.00 %	2.47	0.62 %	0.65	-47.30 %	154.71	89.76 %	38.68	89.76 %	261,465,624	-48.77 %
		Quanto	2	0.45	0.00 %	2.99	22.02 %	1.25	2.16 %	79.81	-2.11 %	19.95	-2.12 %	510,348,344	0.00 %
		Quanto	4	0.45	0.00 %	2.36	-3.57 %	1.25	1.74 %	80.14	-1.71 %	20.03	-1.71 %	510,348,344	0.00 %
		Quanto	8	0.45	0.00 %	2.45	-0.04 %	1.23	0.54 %	81.09	-0.53 %	20.27	-0.53 %	510,348,344	0.00 %
	Yes	BitsAndBytes	32	0.42	-	0.84	-	1.23	-	81.23	-	20.31	-	510,348,336	-
		BitsAndBytes	4	0.43	2.38 %	0.82	-2.01 %	1.30	5.43 %	77.06	-5.14 %	19.26	-5.15 %	134,063,640	-73.73 %
		BitsAndBytes	8	0.43	2.38 %	0.83	-0.09 %	2.06	67.38 %	48.53	-40.26 %	12.13	-40.26 %	176,530,968	-65.41 %
		Downcasting	16	0.42	0.00 %	0.84	0.05 %	0.66	-46.25 %	151.12	86.02 %	37.78	86.02 %	261,465,624	-48.77 %
		Quanto	2	0.45	7.14 %	1.02	22.24 %	1.25	1.65 %	79.91	-1.63 %	19.98	-1.62 %	510,348,344	0.00 %
		Quanto	4	0.43	2.38 %	0.80	-3.69 %	1.25	1.54 %	80.01	-1.51 %	20.00	-1.52 %	510,348,344	0.00 %
		Quanto	8	0.42	0.00 %	0.83	-0.09 %	1.23	0.21 %	81.06	-0.21 %	20.27	-0.21 %	510,348,344	0.00 %
		—	—	—	—	—	—	—	—	—	—	—	—	—	—
openai-community gpt2-medium	No	BitsAndBytes	32	0.55	-	1.75	-	3.11	-	32.13	-	8.03	-	1,444,466,784	-
		BitsAndBytes	4	0.57	3.64 %	0.80	-54.60 %	3.36	8.12 %	29.72	-7.51 %	7.43	-7.51 %	281,831,472	-80.49 %
		BitsAndBytes	8	0.55	0.00 %	1.87	6.77 %	4.09	31.52 %	24.43	-23.96 %	6.11	-23.96 %	432,826,416	-70.04 %
		Downcasting	16	0.55	0.00 %	1.77	1.08 %	1.44	-53.79 %	69.53	116.38 %	17.38	116.37 %	734,816,304	-49.13 %
		Quanto	2	0.55	0.00 %	2.13	21.39 %	3.15	1.12 %	31.78	-1.10 %	7.94	-1.11 %	1,444,466,792	0.00 %
		Quanto	4	0.55	0.00 %	1.76	0.60 %	3.14	0.91 %	31.84	-0.90 %	7.96	-0.91 %	1,444,466,792	0.00 %
		Quanto	8	0.55	0.00 %	1.76	0.16 %	3.13	0.44 %	31.99	-0.44 %	8.00	-0.44 %	1,444,466,792	0.00 %
	Yes	BitsAndBytes	32	0.49	-	0.70	-	3.12	-	32.06	-	8.02	-	1,444,466,784	-
		BitsAndBytes	4	0.46	-6.12 %	0.71	1.13 %	3.37	7.91 %	29.71	-7.33 %	7.43	-7.32 %	281,831,472	-80.49 %

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D. DETAILED TABLES

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
		BitsAndBytes	8	0.54	10.20 %	0.70	-0.81 %	4.08	30.70 %	24.53	-23.49 %	6.13	-23.48 %	432,826,416	-70.04 %
		Downcasting	16	0.51	4.08 %	0.70	-0.10 %	1.44	-53.94 %	69.61	117.13 %	17.40	117.14 %	734,816,304	-49.13 %
		Quanto	2	0.55	12.24 %	1.11	57.68 %	3.14	0.78 %	31.81	-0.78 %	7.95	-0.77 %	1,444,466,792	0.00 %
		Quanto	4	0.48	-2.04 %	0.78	10.74 %	3.14	0.80 %	31.80	-0.80 %	7.95	-0.80 %	1,444,466,792	0.00 %
		Quanto	8	0.53	8.16 %	0.70	-0.43 %	3.13	0.42 %	31.93	-0.42 %	7.98	-0.41 %	1,444,466,792	0.00 %

Table D.12: Results on the super_glue/wic task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%
bert-base-uncased	No		32	0.50	-	0.70	-	5.91	-	107.96	-	27.07	-	437,943,304	-
		BitsAndBytes	4	0.50	0.00 %	0.70	-0.11 %	9.03	52.75 %	70.68	-34.53 %	17.73	-34.53 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.49	-3.12 %	0.71	2.21 %	24.62	316.53 %	25.92	-75.99 %	6.50	-75.99 %	133,451,268	-69.53 %
		Downcasting	16	0.50	0.00 %	0.70	-0.02 %	4.38	-25.89 %	145.68	34.94 %	36.53	34.94 %	218,975,748	-50.00 %
		Quanto	2	0.50	-0.62 %	0.70	-0.17 %	18.16	207.22 %	35.14	-67.45 %	8.81	-67.45 %	437,943,896	0.00 %
		Quanto	4	0.49	-3.12 %	0.70	-0.09 %	18.42	211.69 %	34.64	-67.92 %	8.69	-67.92 %	437,943,896	0.00 %
		Quanto	8	0.51	0.94 %	0.70	-0.04 %	7.29	23.33 %	87.54	-18.91 %	21.95	-18.91 %	437,943,896	0.00 %
	Yes		32	0.55	-	0.83	-	5.91	-	108.03	-	27.09	-	437,943,304	-
		BitsAndBytes	4	0.55	0.28 %	0.82	-1.36 %	9.03	52.84 %	70.68	-34.57 %	17.73	-34.57 %	90,689,028	-79.29 %
		BitsAndBytes	8	0.51	-6.84 %	0.84	1.46 %	24.45	313.93 %	26.10	-75.84 %	6.54	-75.84 %	133,451,268	-69.53 %
		Downcasting	16	0.55	0.28 %	0.83	0.03 %	4.40	-25.43 %	144.86	34.09 %	36.33	34.09 %	218,975,748	-50.00 %
		Quanto	2	0.53	-2.85 %	0.71	-14.77 %	21.37	261.84 %	29.86	-72.36 %	7.49	-72.36 %	437,943,896	0.00 %
		Quanto	4	0.54	-1.42 %	0.82	-1.28 %	20.53	247.60 %	31.08	-71.23 %	7.79	-71.23 %	437,943,896	0.00 %
		Quanto	8	0.55	-0.28 %	0.83	-0.03 %	7.30	23.62 %	87.39	-19.11 %	21.91	-19.11 %	437,943,896	0.00 %
bert-large-uncased	No		32	0.50	-	1.30	-	12.72	-	50.16	-	12.58	-	1,340,583,944	-
		BitsAndBytes	4	0.49	-1.88 %	1.31	0.84 %	18.90	48.59 %	33.76	-32.70 %	8.47	-32.70 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.49	-1.88 %	0.95	-26.76 %	47.31	271.94 %	13.49	-73.11 %	3.38	-73.12 %	367,257,604	-72.60 %
		Downcasting	16	0.50	0.00 %	1.30	0.00 %	8.22	-35.43 %	77.62	54.75 %	19.47	54.75 %	670,296,068	-50.00 %
		Quanto	2	0.51	2.19 %	0.73	-43.60 %	42.89	237.20 %	14.88	-70.34 %	3.73	-70.34 %	1,340,585,112	0.00 %
		Quanto	4	0.50	-0.94 %	1.32	1.64 %	41.69	227.74 %	15.30	-69.49 %	3.84	-69.49 %	1,340,585,112	0.00 %
		Quanto	8	0.50	0.00 %	1.30	-0.01 %	15.37	20.85 %	41.51	-17.25 %	10.41	-17.26 %	1,340,585,112	0.00 %
	Yes		32	0.50	-	0.69	-	12.73	-	50.11	-	12.57	-	1,340,583,944	-
		BitsAndBytes	4	0.50	0.00 %	0.69	0.02 %	19.02	49.38 %	33.55	-33.05 %	8.41	-33.05 %	215,738,372	-83.91 %
		BitsAndBytes	8	0.50	0.00 %	0.69	0.02 %	45.74	259.21 %	13.95	-72.16 %	3.50	-72.16 %	367,257,604	-72.60 %
		Downcasting	16	0.50	0.00 %	0.69	-0.29 %	8.22	-35.41 %	77.58	54.82 %	19.45	54.82 %	670,296,068	-50.00 %
		Quanto	2	0.50	0.00 %	0.69	0.03 %	44.47	249.23 %	14.35	-71.37 %	3.60	-71.37 %	1,340,585,112	0.00 %
		Quanto	4	0.50	0.00 %	0.69	0.05 %	40.39	217.21 %	15.80	-68.47 %	3.96	-68.47 %	1,340,585,112	0.00 %
		Quanto	8	0.50	0.00 %	0.69	0.00 %	15.38	20.83 %	41.47	-17.24 %	10.40	-17.24 %	1,340,585,112	0.00 %
facebook bart-base	No		32	0.50	-	0.75	-	9.89	-	64.49	-	16.17	-	560,050,184	-
		BitsAndBytes	4	0.50	0.00 %	0.79	5.95 %	12.27	24.01 %	52.01	-19.36 %	13.04	-19.36 %	130,504,708	-76.70 %
		BitsAndBytes	8	0.50	0.00 %	0.76	1.84 %	23.36	136.17 %	27.31	-57.66 %	6.85	-57.65 %	180,344,836	-67.80 %
		Downcasting	16	0.50	0.00 %	0.75	-0.06 %	6.29	-36.46 %	101.49	57.37 %	25.45	57.37 %	280,025,092	-50.00 %
		Quanto	2	0.50	0.00 %	0.72	-3.28 %	19.49	97.03 %	32.73	-49.25 %	8.21	-49.25 %	560,050,968	0.00 %
		Quanto	4	0.50	0.00 %	0.77	3.20 %	18.31	85.07 %	34.85	-45.96 %	8.74	-45.96 %	560,050,968	0.00 %
		Quanto	8	0.50	0.00 %	0.76	1.95 %	11.31	14.37 %	56.39	-12.56 %	14.14	-12.56 %	560,050,968	0.00 %
	Yes		32	0.50	-	0.69	-	9.95	-	64.13	-	16.08	-	560,050,184	-
		BitsAndBytes	4	0.50	0.00 %	0.69	0.07 %	12.29	23.58 %	51.89	-19.08 %	13.01	-19.08 %	130,504,708	-76.70 %
		BitsAndBytes	8	0.50	0.00 %	0.69	0.00 %	18.96	90.59 %	33.65	-47.53 %	8.44	-47.53 %	180,344,836	-67.80 %
		Downcasting	16	0.50	0.00 %	0.69	0.01 %	6.30	-36.70 %	101.31	57.97 %	25.41	57.98 %	280,025,092	-50.00 %
		Quanto	2	0.50	0.00 %	0.75	8.86 %	19.50	96.02 %	32.72	-48.98 %	8.20	-48.99 %	560,050,968	0.00 %
		Quanto	4	0.50	0.00 %	0.70	0.48 %	18.31	84.01 %	34.85	-45.65 %	8.74	-45.65 %	560,050,968	0.00 %
		Quanto	8	0.50	0.00 %	0.69	0.01 %	11.21	12.73 %	56.89	-11.29 %	14.27	-11.29 %	560,050,968	0.00 %
google flan-t5-base	No		32	0.50	-	0.90	-	16.54	-	38.59	-	9.68	-	893,982,728	-
		BitsAndBytes	4	0.50	0.00 %	0.91	0.59 %	24.22	46.48 %	26.34	-31.73 %	6.61	-31.73 %	280,955,908	-68.57 %
		BitsAndBytes	8	0.50	0.00 %	0.91	0.70 %	34.37	107.83 %	18.57	-51.89 %	4.66	-51.88 %	361,466,884	-59.57 %

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%	
		Downcasting	16	0.50	0.00 %	0.91	0.09 %	14.81	-10.42 %	43.07	11.63 %	10.80	11.64 %	446,991,364	50.00 %	-
		Quanto	2	0.50	0.00 %	0.79	-12.81 %	37.30	125.61 %	17.10	-55.68 %	4.29	-55.67 %	893,984,472	0.00 %	-
		Quanto	4	0.50	0.00 %	0.88	-2.51 %	34.66	109.60 %	18.41	-52.29 %	4.62	-52.28 %	893,984,472	0.00 %	-
		Quanto	8	0.50	0.00 %	0.91	0.36 %	19.26	16.48 %	33.13	-14.15 %	8.31	-14.14 %	893,984,472	0.00 %	-
	Yes	Downcasting	32	0.60	-1.12 %	-	-	16.66	-	38.30	-	9.60	-	893,982,728	-	-
		BitsAndBytes	4	0.61	1.83 %	1.04	-7.26 %	24.26	45.61 %	26.30	-31.32 %	6.60	-31.32 %	280,955,908	-68.57 %	-
		BitsAndBytes	8	0.60	0.52 %	1.11	-1.06 %	34.50	107.07 %	18.50	-51.71 %	4.64	-51.71 %	361,466,884	-59.57 %	-
		Downcasting	16	0.60	0.00 %	1.12	0.10 %	14.81	-11.12 %	43.09	12.51 %	10.81	12.52 %	446,991,364	50.00 %	-
		Quanto	2	0.56	-6.01 %	0.68	-39.31 %	37.25	123.62 %	17.13	-55.28 %	4.29	-55.28 %	893,984,472	0.00 %	-
		Quanto	4	0.59	-1.31 %	1.19	6.08 %	34.69	108.23 %	18.39	-51.98 %	4.61	-51.98 %	893,984,472	0.00 %	-
		Quanto	8	0.60	0.00 %	1.11	-0.37 %	19.14	14.86 %	33.34	-12.94 %	8.36	-12.93 %	893,984,472	0.00 %	-
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
openai-community gpt2	No	Downcasting	32	0.50	-2.38 %	-	-	7.83	-	81.51	-	20.44	-	510,348,336	-	-
		BitsAndBytes	4	0.50	0.00 %	2.78	17.01 %	8.31	6.15 %	76.78	-5.79 %	19.26	-5.80 %	134,063,640	-73.73 %	-
		BitsAndBytes	8	0.50	0.00 %	2.29	-3.46 %	13.27	69.56 %	48.07	-41.02 %	12.05	-41.03 %	176,530,968	-65.41 %	-
		Downcasting	16	0.50	0.00 %	2.37	-0.05 %	4.14	-47.13 %	154.18	89.16 %	38.66	89.15 %	261,465,624	-48.77 %	-
		Quanto	2	0.50	0.00 %	2.90	21.98 %	7.95	1.54 %	80.27	-1.52 %	20.13	-1.52 %	510,348,344	0.00 %	-
		Quanto	4	0.50	0.00 %	2.28	-4.06 %	7.93	1.34 %	80.43	-1.32 %	20.17	-1.33 %	510,348,344	0.00 %	-
		Quanto	8	0.50	0.00 %	2.37	-0.06 %	7.85	0.29 %	81.27	-0.29 %	20.38	-0.29 %	510,348,344	0.00 %	-
	Yes	Downcasting	32	0.57	-0.73 %	-	-	7.82	-	81.56	-	20.45	-	510,348,336	-	-
		BitsAndBytes	4	0.57	-0.82 %	0.72	-1.14 %	8.32	6.31 %	76.72	-5.94 %	19.24	-5.94 %	134,063,640	-73.73 %	-
		BitsAndBytes	8	0.57	-0.27 %	0.73	0.13 %	13.50	72.61 %	47.25	-42.07 %	11.85	-42.07 %	176,530,968	-65.41 %	-
		Downcasting	16	0.58	0.27 %	0.73	-0.08 %	4.19	-46.40 %	152.15	86.55 %	38.16	86.56 %	261,465,624	-48.77 %	-
		Quanto	2	0.57	-1.09 %	0.77	5.54 %	7.95	1.64 %	80.25	-1.61 %	20.12	-1.61 %	510,348,344	0.00 %	-
		Quanto	4	0.57	-0.55 %	0.73	0.25 %	7.94	1.54 %	80.32	-1.52 %	20.14	-1.52 %	510,348,344	0.00 %	-
		Quanto	8	0.57	0.00 %	0.73	0.02 %	7.86	0.46 %	81.19	-0.46 %	20.36	-0.46 %	510,348,344	0.00 %	-
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
openai-community gpt2-medium	No	Downcasting	32	0.50	-1.74 %	-	-	19.84	-	32.16	-	8.07	-	1,444,466,784	-	-
		BitsAndBytes	4	0.49	-1.88 %	1.11	-35.92 %	21.33	7.54 %	29.91	-7.01 %	7.50	-7.02 %	281,831,472	-80.49 %	-
		BitsAndBytes	8	0.50	-0.63 %	1.71	-1.72 %	25.65	29.32 %	24.87	-22.67 %	6.24	-22.68 %	432,826,416	-70.04 %	-
		Downcasting	16	0.50	0.00 %	1.76	1.13 %	9.19	-53.68 %	69.45	115.91 %	17.42	115.92 %	734,816,304	-49.13 %	-
		Quanto	2	0.50	-0.31 %	2.11	21.61 %	20.01	0.88 %	31.88	-0.87 %	8.00	-0.87 %	1,444,466,792	0.00 %	-
		Quanto	4	0.50	-0.31 %	1.77	1.54 %	20.04	1.03 %	31.83	-1.03 %	7.98	-1.02 %	1,444,466,792	0.00 %	-
		Quanto	8	0.50	0.00 %	1.74	0.20 %	19.94	0.52 %	32.00	-0.52 %	8.03	-0.51 %	1,444,466,792	0.00 %	-
	Yes	Downcasting	32	0.50	-0.69 %	-	-	19.88	-	32.10	-	8.05	-	1,444,466,784	-	-
		BitsAndBytes	4	0.49	-3.12 %	0.69	0.02 %	21.61	8.75 %	29.52	-8.04 %	7.40	-8.04 %	281,831,472	-80.49 %	-
		BitsAndBytes	8	0.51	0.94 %	0.69	0.02 %	25.04	25.97 %	25.48	-20.62 %	6.39	-20.62 %	432,826,416	-70.04 %	-
		Downcasting	16	0.50	0.31 %	0.69	-0.00 %	9.16	-53.92 %	69.67	117.03 %	17.47	117.03 %	734,816,304	-49.13 %	-
		Quanto	2	0.50	-1.25 %	0.69	0.09 %	19.99	0.59 %	31.91	-0.59 %	8.00	-0.58 %	1,444,466,792	0.00 %	-
		Quanto	4	0.50	-1.25 %	0.69	0.01 %	20.00	0.63 %	31.90	-0.62 %	8.00	-0.62 %	1,444,466,792	0.00 %	-
		Quanto	8	0.50	0.63 %	0.69	-0.00 %	19.88	0.02 %	32.09	-0.02 %	8.05	-0.01 %	1,444,466,792	0.00 %	-

Table D.13: Results on the super_glue/wsc task

Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%	
bert-base-uncased	No	Downcasting	32	0.41	-0.71 %	-	-	0.97	-	107.06	-	26.77	-	437,943,304	-	-
		BitsAndBytes	4	0.41	0.00 %	0.71	-0.43 %	1.47	50.99 %	70.91	-33.77 %	17.73	-33.77 %	90,689,028	-79.29 %	-
		BitsAndBytes	8	0.44	6.98 %	0.71	-0.36 %	4.00	312.21 %	25.97	-75.74 %	6.49	-75.74 %	133,451,268	-69.53 %	-
		Downcasting	16	0.41	0.00 %	0.71	-0.03 %	0.72	-25.75 %	144.19	34.68 %	36.05	34.68 %	218,975,748	-50.00 %	-
		Quanto	2	0.49	18.60 %	0.70	-1.75 %	2.42	149.45 %	42.92	-59.91 %	10.73	-59.91 %	437,943,896	0.00 %	-
		Quanto	4	0.45	9.30 %	0.70	-1.42 %	2.28	134.97 %	45.56	-57.44 %	11.39	-57.44 %	437,943,896	0.00 %	-
		Quanto	8	0.43	4.65 %	0.71	-0.16 %	1.20	23.48 %	86.70	-19.01 %	21.68	-19.01 %	437,943,896	0.00 %	-
	Yes	Downcasting	32	0.44	-0.70 %	-	-	0.98	-	106.30	-	26.57	-	437,943,304	-	-
		BitsAndBytes	4	0.42	-4.35 %	0.70	0.23 %	1.47	50.22 %	70.76	-33.44 %	17.69	-33.43 %	90,689,028	-79.29 %	-
		BitsAndBytes	8	0.47	6.52 %	0.70	0.02 %	4.02	311.25 %	25.85	-75.69 %	6.46	-75.68 %	133,451,268	-69.53 %	-
		Downcasting	16	0.44	0.00 %	0.70	0.02 %	0.73	-25.34 %	142.38	33.94 %	35.59	33.94 %	218,975,748	-50.00 %	-
		Quanto	2	0.61	36.96 %	0.68	-3.31 %	2.45	150.82 %	42.38	-60.13 %	10.60	-60.13 %	437,943,896	0.00 %	-

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D. DETAILED TABLES

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%	
		Quanto	4	0.40	-8.70	0.70	0.56	2.28	133.26	45.57	-57.13	11.39	-57.13	437,943,896	0.00	
		Quanto	8	0.44	0.00	0.70	-0.02	1.20	22.64	86.68	-18.46	21.67	-18.46	437,943,896	0.00	
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
bert-large-uncased	No		32	0.60	-	1.03	-	2.09	-	49.67	-	12.42	-	1,340,583,944	-	
		BitsAndBytes	4	0.59	-1.61	1.03	0.46	3.08	46.96	33.80	-31.95	8.45	-31.95	215,738,372	-83.91	
		BitsAndBytes	8	0.62	4.84	0.75	-26.87	7.75	270.14	13.42	-72.98	3.35	-72.98	367,257,604	-72.60	
		Downcasting	16	0.60	0.00	1.02	-0.09	1.32	-36.72	78.49	58.02	19.62	58.02	670,296,068	-50.00	
		Quanto	2	0.35	-41.94	0.76	-26.20	4.99	138.37	20.84	-58.05	5.21	-58.04	1,340,585,112	0.00	
		Quanto	4	0.60	0.00	1.01	-1.19	4.68	123.30	22.24	-55.22	5.56	-55.22	1,340,585,112	0.00	
		Quanto	8	0.60	0.00	1.02	-0.22	2.52	20.56	41.20	-17.05	10.30	-17.06	1,340,585,112	0.00	
	Yes		32	0.63	-	0.68	-	2.09	-	49.70	-	12.43	-	1,340,583,944	-	
		BitsAndBytes	4	0.63	0.00	0.67	-0.83	3.07	46.63	33.90	-31.80	8.47	-31.80	215,738,372	-83.91	
		BitsAndBytes	8	0.63	0.00	0.68	-0.03	7.33	250.12	14.20	-71.44	3.55	-71.44	367,257,604	-72.60	
		Downcasting	16	0.63	0.00	0.68	-0.00	1.35	-35.33	76.86	54.63	19.21	54.64	670,296,068	-50.00	
		Quanto	2	0.63	0.00	0.67	-0.87	4.97	137.63	20.92	-57.92	5.23	-57.92	1,340,585,112	0.00	
		Quanto	4	0.63	0.00	0.68	0.02	4.69	124.14	22.18	-55.38	5.54	-55.38	1,340,585,112	0.00	
		Quanto	8	0.63	0.00	0.68	0.03	2.52	20.54	41.24	-17.03	10.31	-17.04	1,340,585,112	0.00	
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
facebook bart-base	No		32	0.37	-	0.76	-	1.58	-	65.91	-	16.48	-	560,050,184	-	
		BitsAndBytes	4	0.37	0.00	0.78	2.47	1.96	24.50	52.94	-19.68	13.23	-19.68	130,504,708	-76.70	
		BitsAndBytes	8	0.37	0.00	0.78	3.17	3.74	137.30	27.77	-57.86	6.94	-57.86	180,344,836	-67.80	
		Downcasting	16	0.37	0.00	0.76	-0.16	0.98	-37.75	105.87	60.65	26.47	60.65	280,025,092	-50.00	
		Quanto	2	0.38	2.63	0.76	0.42	3.09	95.91	33.64	-48.96	8.41	-48.96	560,050,968	0.00	
		Quanto	4	0.37	0.00	0.78	2.81	2.94	86.18	35.40	-46.29	8.85	-46.29	560,050,968	0.00	
		Quanto	8	0.37	0.00	0.78	2.68	1.78	12.67	58.49	-11.24	14.62	-11.25	560,050,968	0.00	
	Yes		32	0.49	-	0.69	-	1.55	-	66.94	-	16.74	-	560,050,184	-	
		BitsAndBytes	4	0.63	29.41	0.68	-1.27	2.01	29.53	51.68	-22.80	12.92	-22.80	130,504,708	-76.70	
		BitsAndBytes	8	0.51	3.92	0.69	-0.14	3.72	139.30	27.98	-58.21	6.99	-58.21	180,344,836	-67.80	
		Downcasting	16	0.49	0.00	0.69	-0.02	1.01	-35.19	103.28	54.28	25.82	54.28	280,025,092	-50.00	
		Quanto	2	0.37	-25.49	0.78	13.20	3.15	102.46	33.07	-50.60	8.27	-50.60	560,050,968	0.00	
		Quanto	4	0.48	-1.96	0.69	-0.02	2.96	90.65	35.11	-47.55	8.78	-47.55	560,050,968	0.00	
		Quanto	8	0.50	1.96	0.69	-0.09	1.79	15.00	58.22	-13.04	14.55	-13.04	560,050,968	0.00	
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
google flan-t5-base	No		32	0.63	-	0.70	-	2.68	-	38.77	-	9.69	-	893,982,728	-	
		BitsAndBytes	4	0.63	0.00	0.71	0.13	3.94	46.74	26.42	-31.85	6.61	-31.85	280,955,908	-68.57	
		BitsAndBytes	8	0.63	0.00	0.71	0.27	5.46	103.66	19.04	-50.90	4.76	-50.90	361,466,884	-59.57	
		Downcasting	16	0.63	0.00	0.71	0.07	2.37	-11.67	43.89	13.22	10.97	13.21	446,991,364	-50.00	
		Quanto	2	0.63	0.00	0.67	-5.03	6.08	126.67	17.10	-55.88	4.28	-55.88	893,984,472	0.00	
		Quanto	4	0.63	0.00	0.70	-1.18	5.65	110.71	18.40	-52.54	4.60	-52.54	893,984,472	0.00	
		Quanto	8	0.63	0.00	0.71	0.13	3.14	16.98	33.14	-14.51	8.29	-14.52	893,984,472	0.00	
	Yes		32	0.37	-	0.71	-	2.68	-	38.74	-	9.68	-	893,982,728	-	
		BitsAndBytes	4	0.37	0.00	0.71	-1.01	3.96	47.32	26.29	-32.12	6.57	-32.13	280,955,908	-68.57	
		BitsAndBytes	8	0.37	0.00	0.71	-0.05	5.40	101.21	19.25	-50.30	4.81	-50.30	361,466,884	-59.57	
		Downcasting	16	0.37	0.00	0.71	-0.02	2.39	-10.93	43.49	12.27	10.87	12.27	446,991,364	-50.00	
		Quanto	2	0.63	73.68	0.66	-6.72	6.10	127.17	17.05	-55.98	4.26	-55.98	893,984,472	0.00	
		Quanto	4	0.37	0.00	0.72	0.96	5.65	110.42	18.41	-52.48	4.60	-52.48	893,984,472	0.00	
		Quanto	8	0.37	0.00	0.71	-0.07	3.12	16.22	33.33	-13.96	8.33	-13.95	893,984,472	0.00	
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
openai-community gpt2	No		32	0.37	-	2.85	-	1.28	-	81.54	-	20.39	-	510,348,336	-	
		BitsAndBytes	4	0.37	0.00	3.22	13.23	1.36	6.48	76.58	-6.09	19.14	-6.09	134,063,640	-73.73	
		BitsAndBytes	8	0.37	0.00	2.71	-4.68	2.12	65.88	49.16	-39.71	12.29	-39.71	176,530,968	-65.41	
		Downcasting	16	0.37	0.00	2.85	0.08	0.67	-47.69	155.88	91.17	38.97	91.17	261,465,624	-48.77	
		Quanto	2	0.37	0.00	3.49	22.56	1.30	2.12	79.84	-2.08	19.96	-2.08	510,348,344	0.00	
		Quanto	4	0.37	0.00	2.72	-4.32	1.30	1.99	79.95	-1.95	19.99	-1.95	510,348,344	0.00	
		Quanto	8	0.37	0.00	2.84	-0.07	1.28	0.54	81.11	-0.53	20.28	-0.53	510,348,344	0.00	
	Yes		32	0.41	-	0.72	-	1.28	-	81.27	-	20.32	-	510,348,336	-	
		BitsAndBytes	4	0.62	48.84	0.69	-3.99	1.35	5.59	76.97	-5.30	19.24	-5.30	134,063,640	-73.73	
		BitsAndBytes	8	0.45	9.30	0.72	0.45	2.19	71.51	47.38	-41.70	11.85	-41.70	176,530,968	-65.41	
		Downcasting	16	0.39	-4.65	0.72	0.00	0.68	-46.78	152.72	87.92	38.18	87.92	261,465,624	-48.77	
		Quanto	2	0.37	-11.63	2.10	192.98	1.30	1.69	79.92	-1.66	19.98	-1.66	510,348,344	0.00	
		Quanto	4	0.63	53.49	0.65	-8.75	1.30	1.70	79.91	-1.67	19.98	-1.67	510,348,344	0.00	

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Model Name	FT	Quantization Method	# Bits	Eval Accuracy	%	Eval Loss	%	Eval Runtime	%	Eval Samples Per Second	%	Eval Steps Per Second	%	Memory Footprint	%	
		Quanto	8	0.39	-4.65 %	0.72	0.85 %	1.28	0.22 %	81.09	-0.22 %	20.27	-0.22 %	510,348,344	0.00 %	
		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
openai-community gpt2-medium	No		32	0.63	-	1.73	-	3.24	-	32.15	-	8.04	-	1,444,466,784	-	-
		BitsAndBytes	4	0.63	0.00 %	1.61	-7.18 %	3.48	7.51 %	29.90	-6.99 %	7.47	-6.98 %	281,831,472	-80.49 %	
		BitsAndBytes	8	0.63	0.00 %	1.72	-0.76 %	4.21	30.28 %	24.68	-23.24 %	6.17	-23.23 %	432,826,416	-70.04 %	
		Downcasting	16	0.63	0.00 %	1.72	-0.97 %	1.49	-53.81 %	69.59	116.48 %	17.40	116.49 %	734,816,304	-49.13 %	
		Quanto	2	0.63	0.00 %	2.20	27.09 %	3.27	1.02 %	31.82	-1.00 %	7.96	-1.00 %	1,444,466,792	0.00 %	
		Quanto	4	0.63	0.00 %	1.74	0.55 %	3.27	1.05 %	31.81	-1.04 %	7.95	-1.03 %	1,444,466,792	0.00 %	
		Quanto	8	0.63	0.00 %	1.73	0.03 %	3.25	0.55 %	31.97	-0.54 %	7.99	-0.54 %	1,444,466,792	0.00 %	
	Yes		32	0.62	-	0.67	-	3.24	-	32.08	-	8.02	-	1,444,466,784	-	-
		BitsAndBytes	4	0.53	-15.38 %	0.69	3.38 %	3.53	8.91 %	29.46	-8.18 %	7.36	-8.18 %	281,831,472	-80.49 %	
		BitsAndBytes	8	0.63	1.54 %	0.67	-0.48 %	4.34	33.86 %	23.97	-25.30 %	5.99	-25.29 %	432,826,416	-70.04 %	
		Downcasting	16	0.62	0.00 %	0.67	0.00 %	1.50	-53.85 %	69.51	116.68 %	17.38	116.68 %	734,816,304	-49.13 %	
		Quanto	2	0.63	1.54 %	1.15	71.83 %	3.26	0.70 %	31.86	-0.70 %	7.96	-0.70 %	1,444,466,792	0.00 %	
		Quanto	4	0.63	1.54 %	0.66	-1.48 %	3.27	0.73 %	31.85	-0.73 %	7.96	-0.72 %	1,444,466,792	0.00 %	
		Quanto	8	0.64	3.08 %	0.67	-0.19 %	3.25	0.35 %	31.97	-0.35 %	7.99	-0.34 %	1,444,466,792	0.00 %	

D. DETAILED TABLES

Appendix E

Detailed figures

As mentioned in Appendix D, the individual results of the benchmarks do not fit in the main report. This section presents a graphical representation of the entire result set. The results are displayed from Figure E.1 to Figure E.168.

E. DETAILED FIGURES

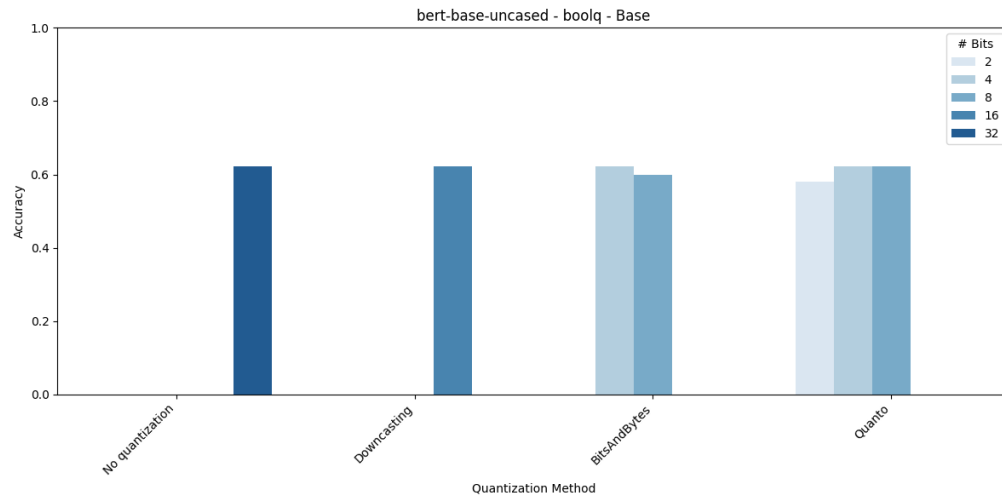


Figure E.1: Results of base bert-base-uncased on boolq using Accuracy

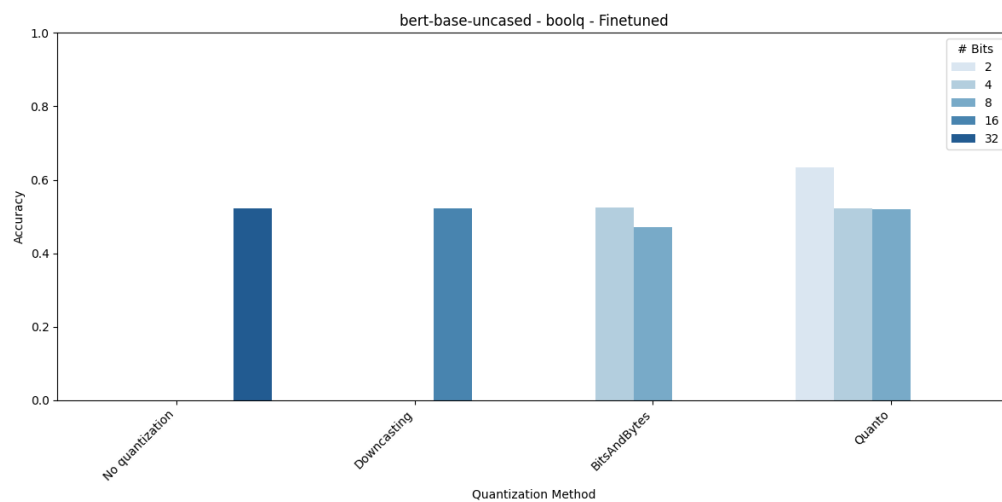


Figure E.2: Results of finetuned bert-base-uncased on boolq using Accuracy

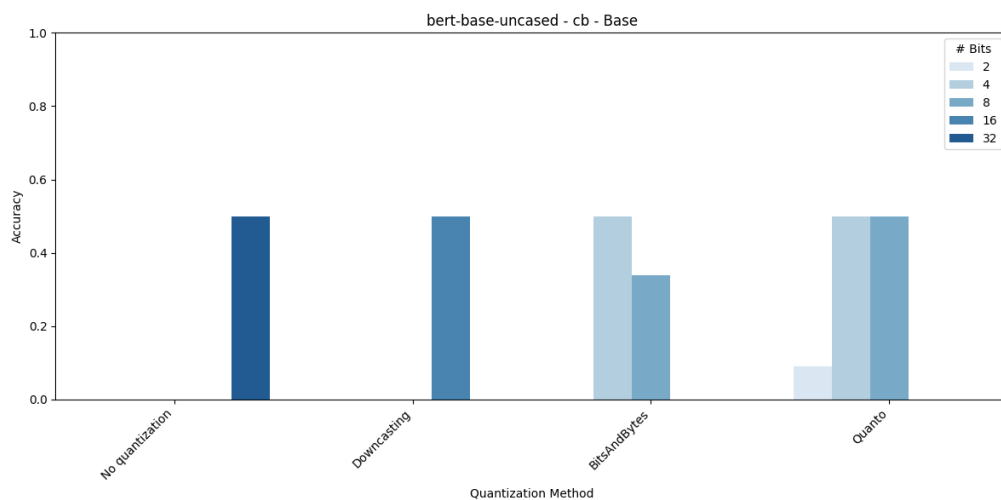


Figure E.3: Results of base bert-base-uncased on cb using Accuracy

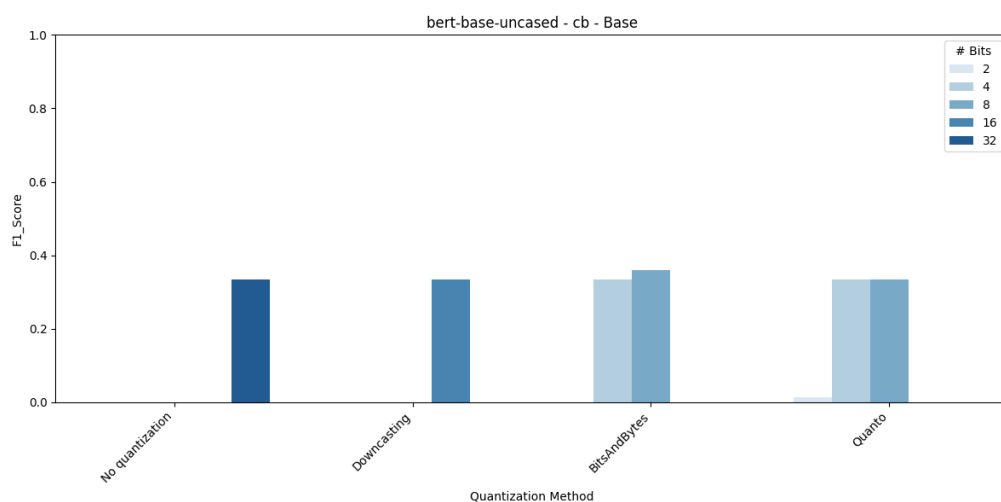


Figure E.4: Results of base bert-base-uncased on cb using F1_Score

E. DETAILED FIGURES

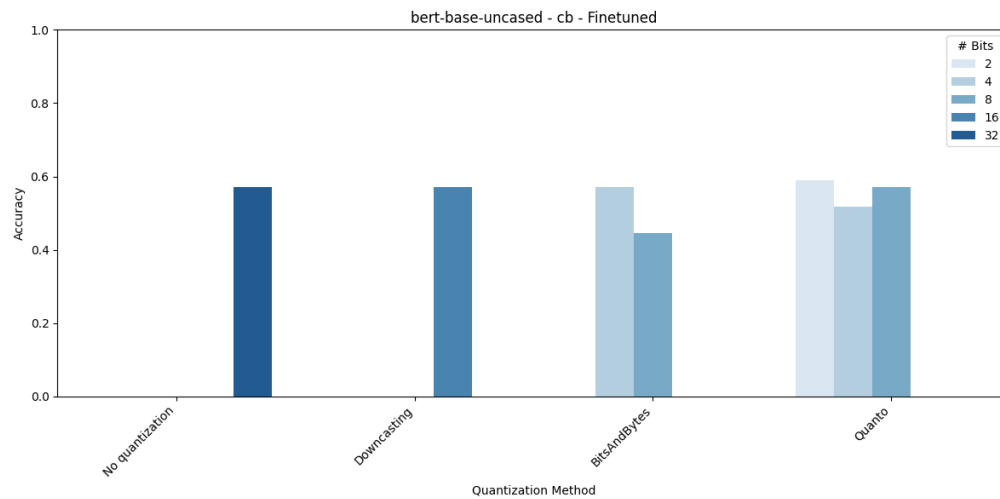


Figure E.5: Results of finetuned bert-base-uncased on cb using Accuracy

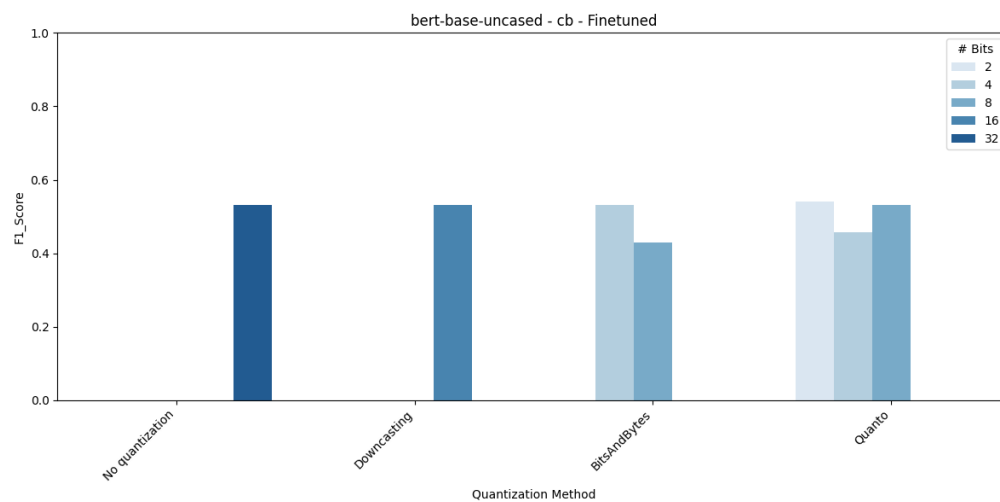


Figure E.6: Results of finetuned bert-base-uncased on cb using F1_Score

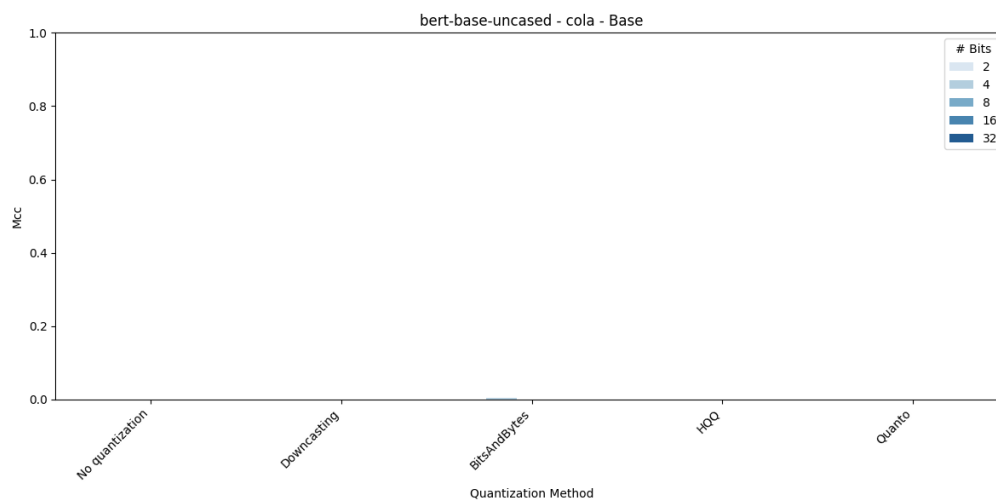


Figure E.7: Results of base bert-base-uncased on cola using Mcc

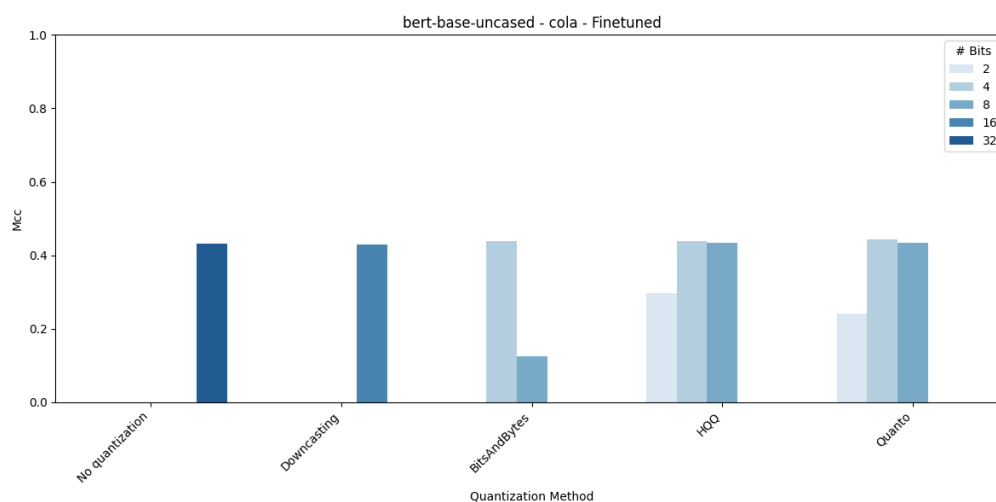


Figure E.8: Results of finetuned bert-base-uncased on cola using Mcc

E. DETAILED FIGURES

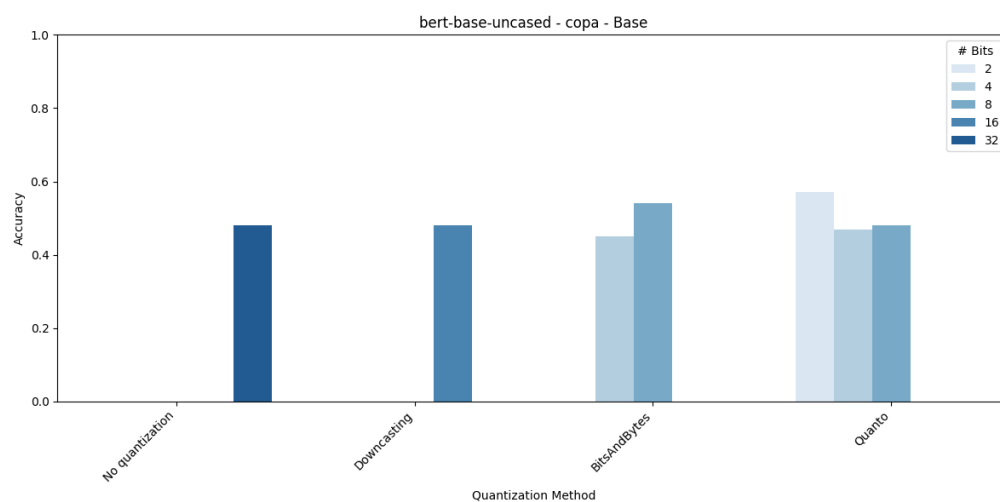


Figure E.9: Results of base bert-base-uncased on copa using Accuracy

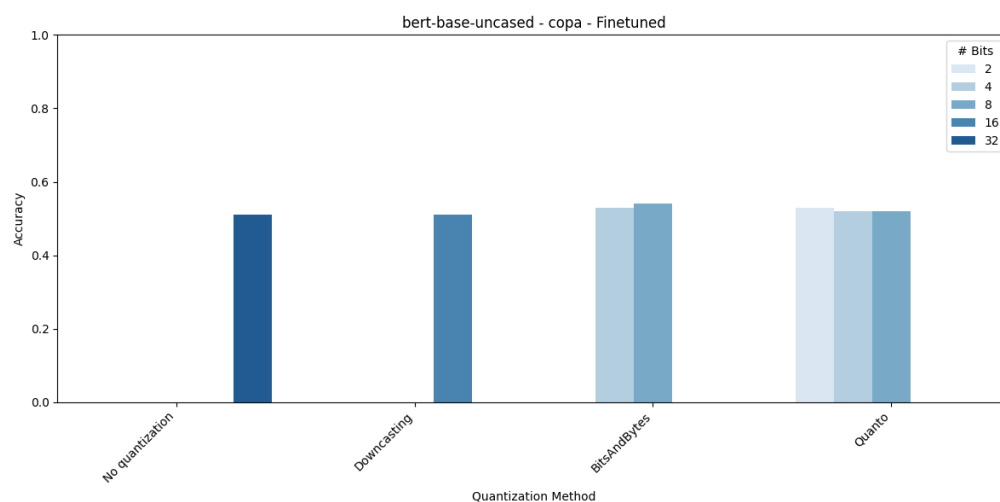


Figure E.10: Results of finetuned bert-base-uncased on copa using Accuracy

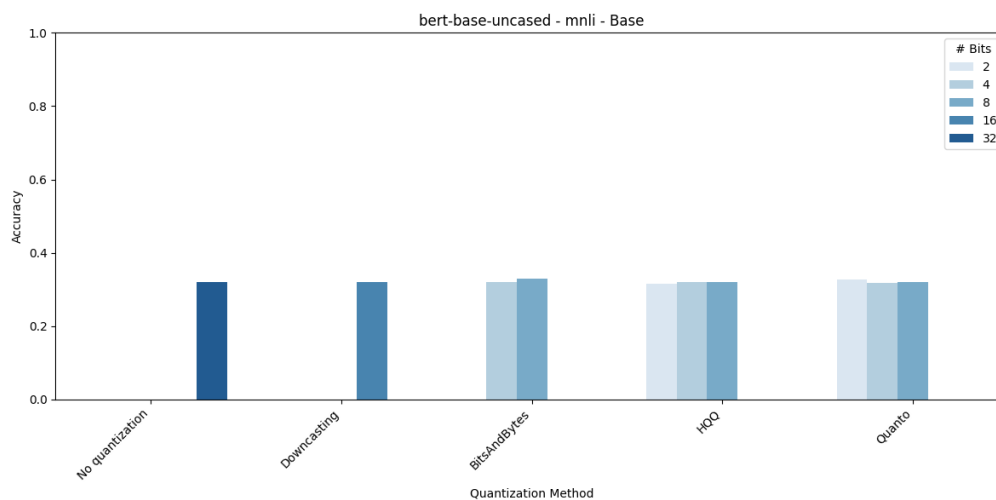


Figure E.11: Results of base bert-base-uncased on mnli using Accuracy

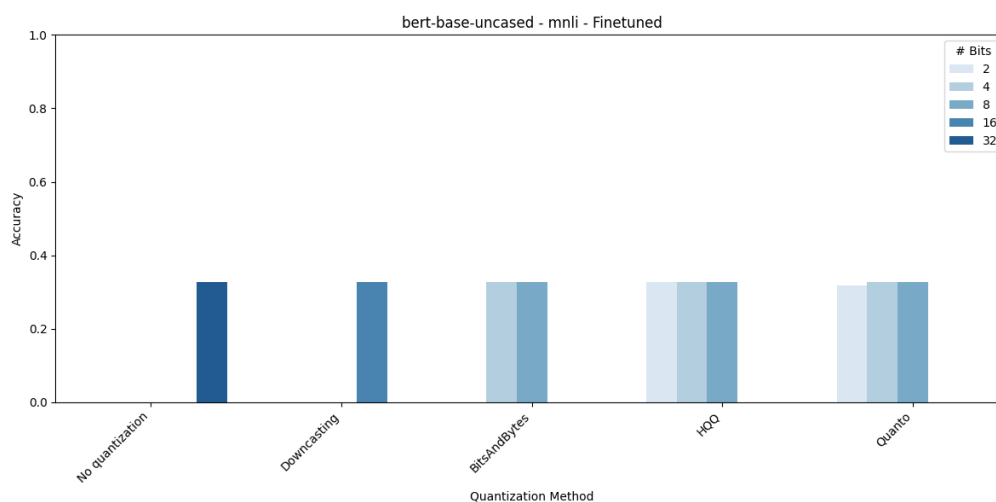


Figure E.12: Results of finetuned bert-base-uncased on mnli using Accuracy

E. DETAILED FIGURES

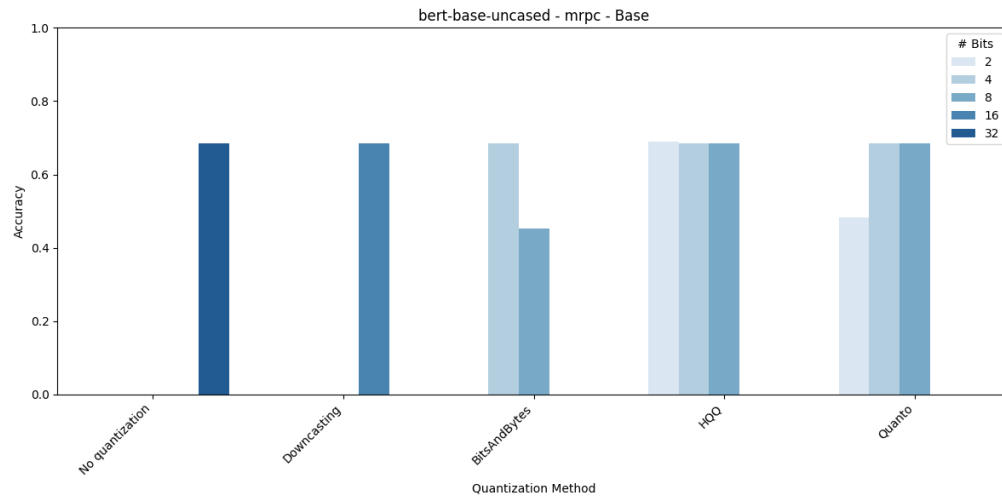


Figure E.13: Results of base bert-base-uncased on mrpc using Accuracy

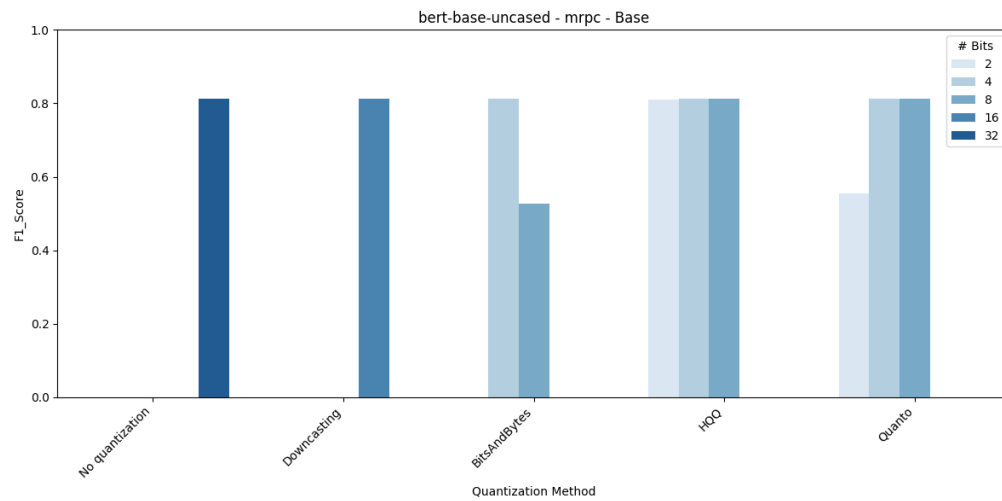


Figure E.14: Results of base bert-base-uncased on mrpc using F1_Score

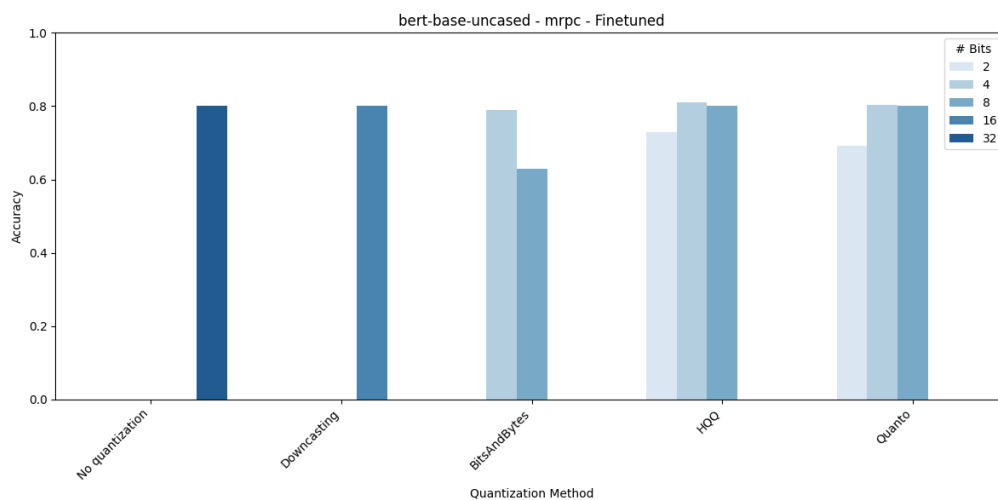


Figure E.15: Results of finetuned bert-base-uncased on mrpc using Accuracy

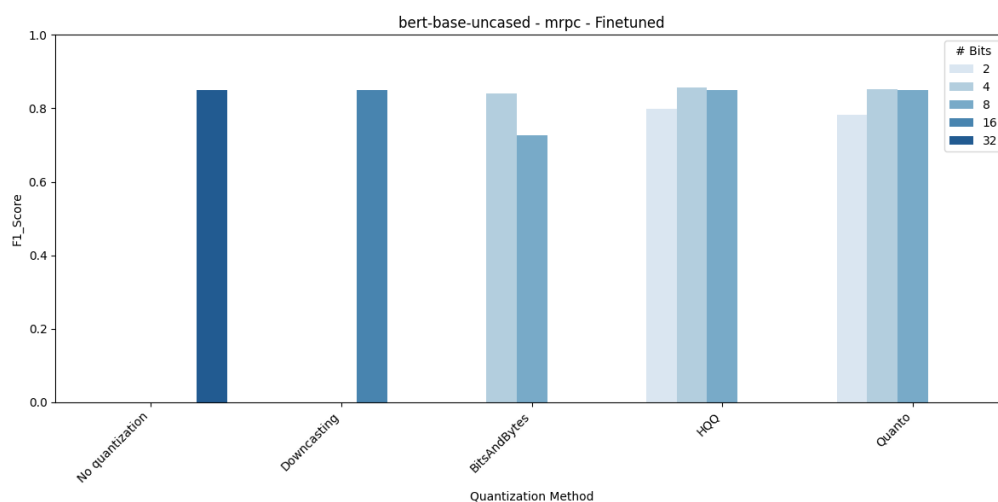


Figure E.16: Results of finetuned bert-base-uncased on mrpc using F1_Score

E. DETAILED FIGURES

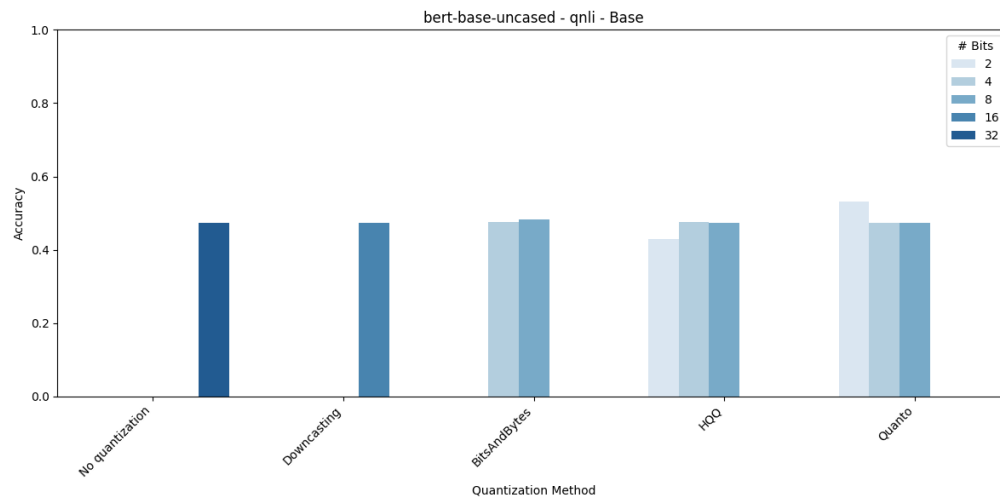


Figure E.17: Results of base bert-base-uncased on qnli using Accuracy

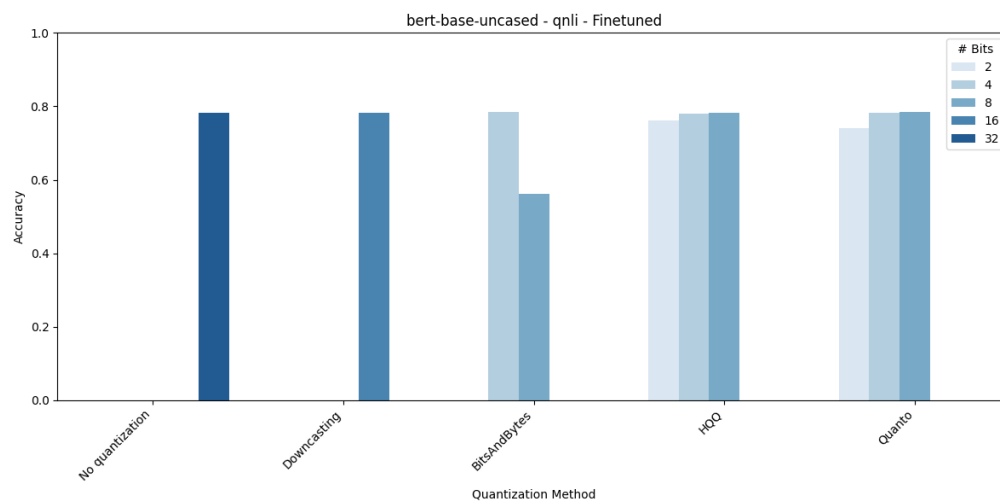


Figure E.18: Results of finetuned bert-base-uncased on qnli using Accuracy

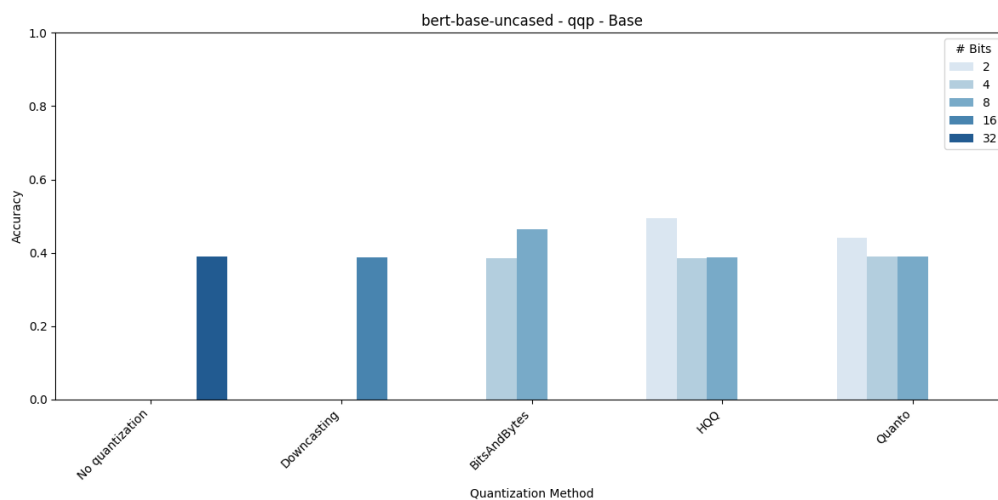


Figure E.19: Results of base bert-base-uncased on qqp using Accuracy

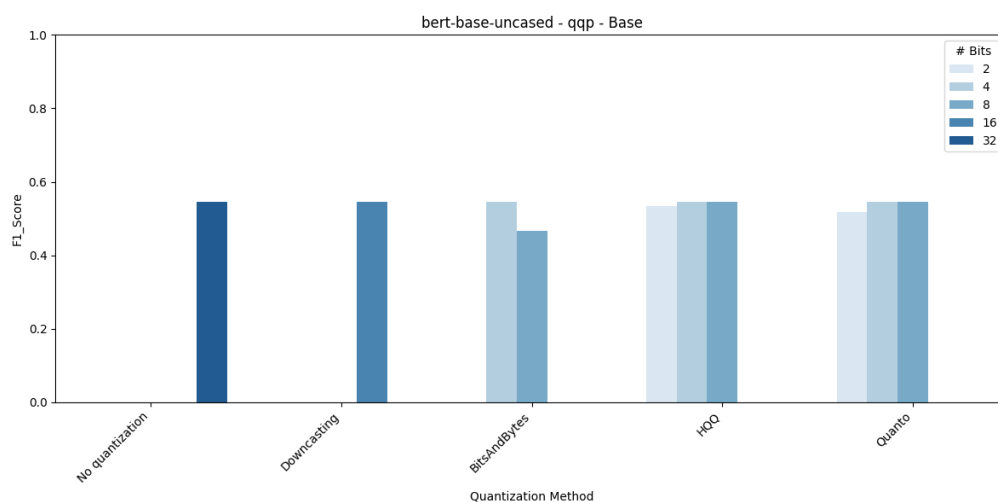


Figure E.20: Results of base bert-base-uncased on qqp using F1_Score

E. DETAILED FIGURES

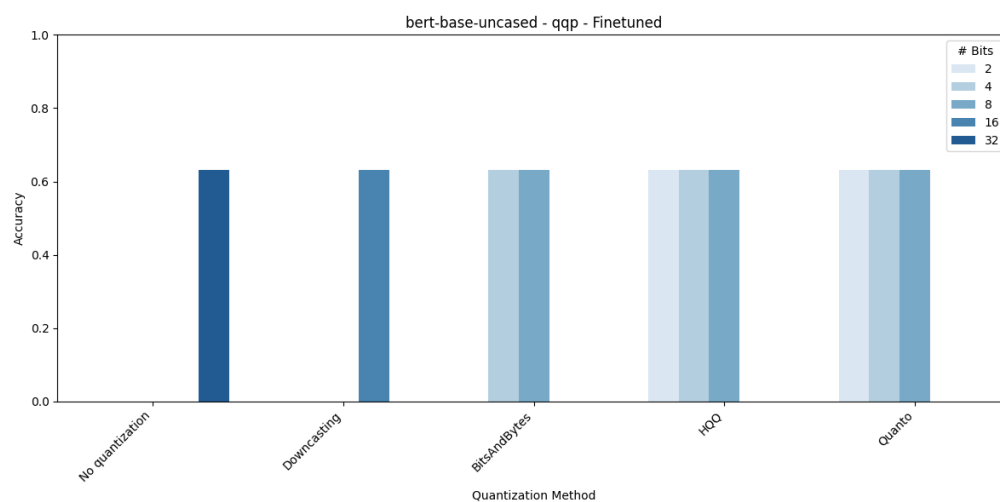


Figure E.21: Results of finetuned bert-base-uncased on qqp using Accuracy

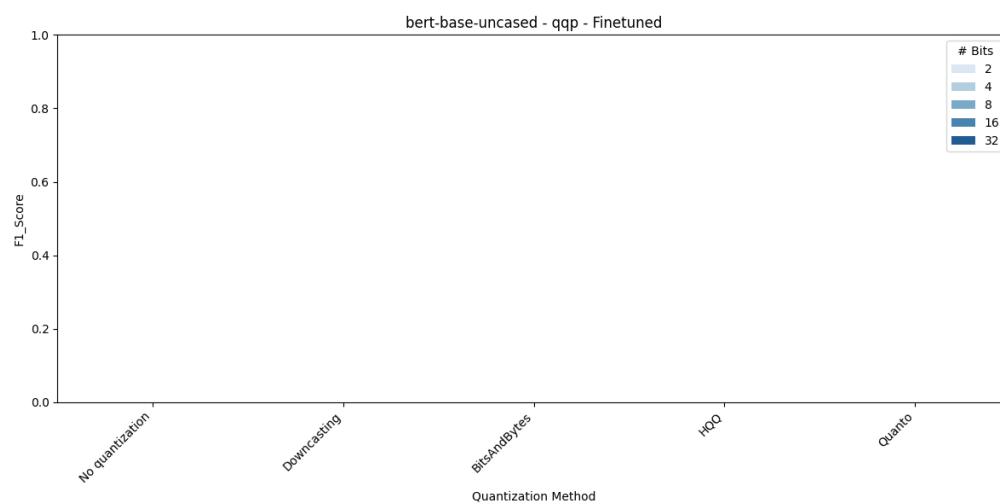


Figure E.22: Results of finetuned bert-base-uncased on qqp using F1_Score

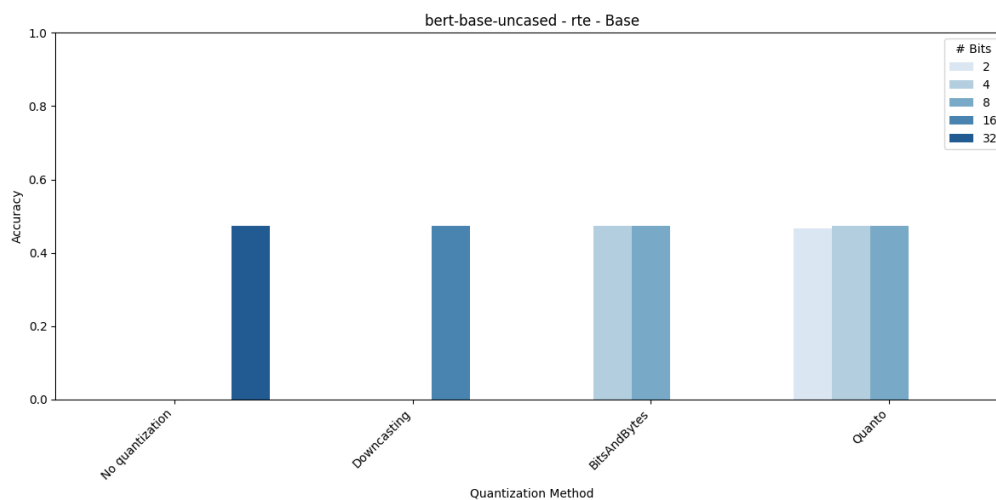


Figure E.23: Results of base bert-base-uncased on rte using Accuracy

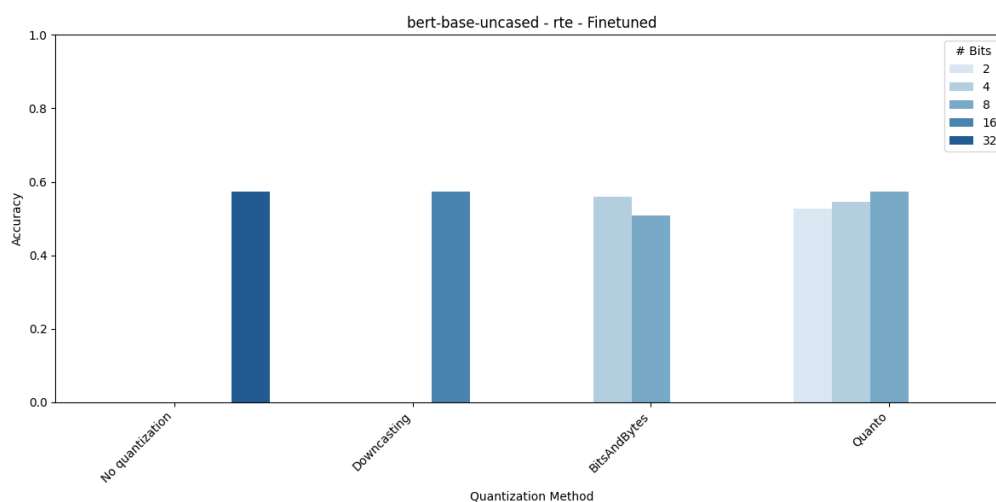


Figure E.24: Results of finetuned bert-base-uncased on rte using Accuracy

E. DETAILED FIGURES

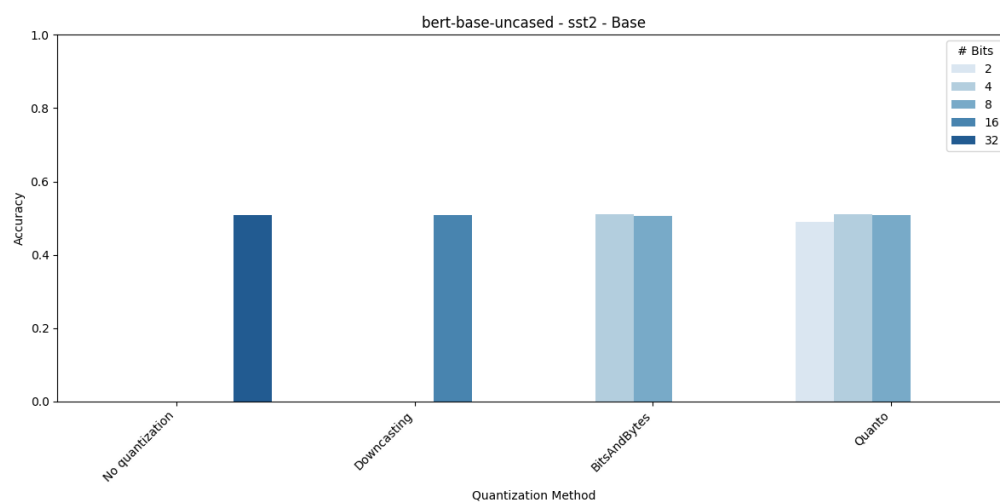


Figure E.25: Results of base bert-base-uncased on sst2 using Accuracy

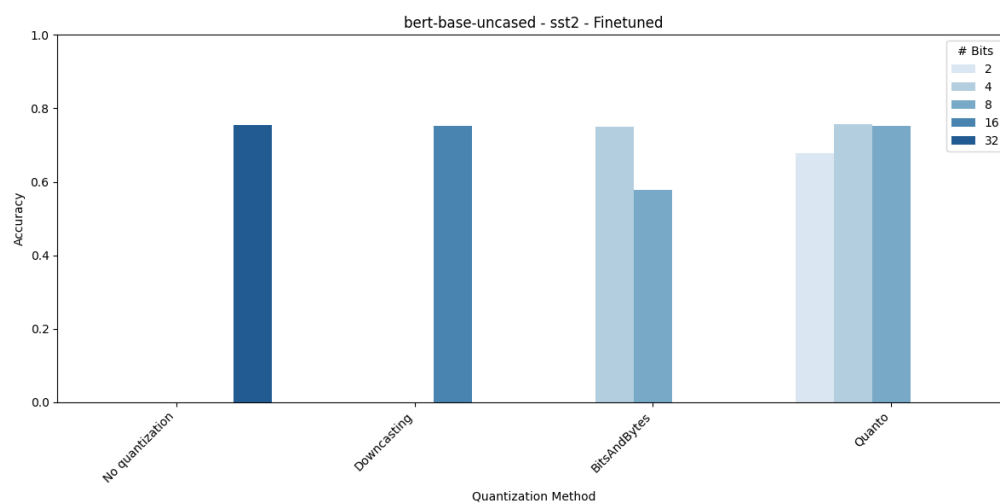


Figure E.26: Results of finetuned bert-base-uncased on sst2 using Accuracy

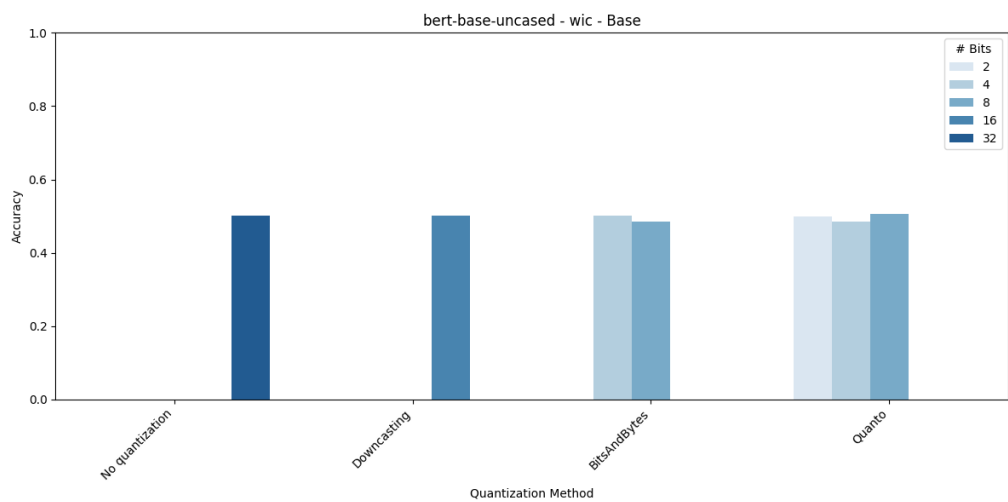


Figure E.27: Results of base bert-base-uncased on wic using Accuracy

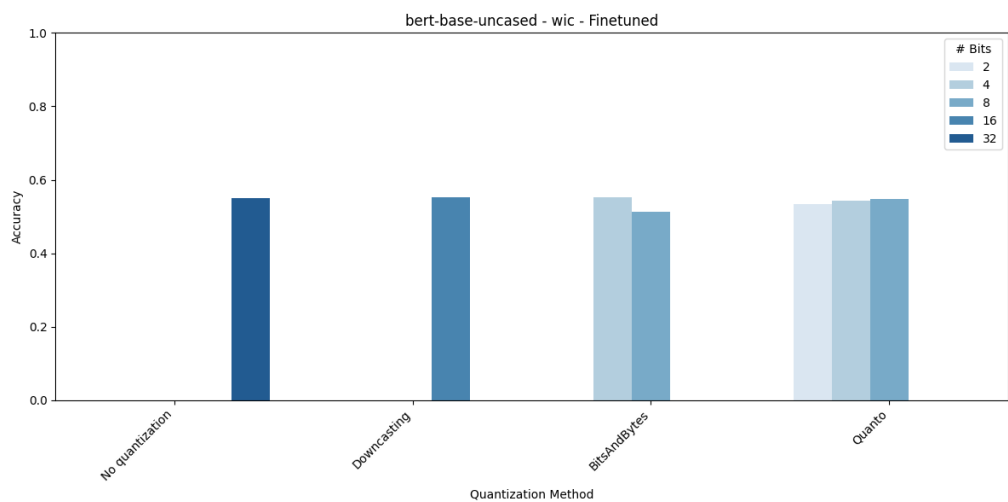


Figure E.28: Results of finetuned bert-base-uncased on wic using Accuracy

E. DETAILED FIGURES

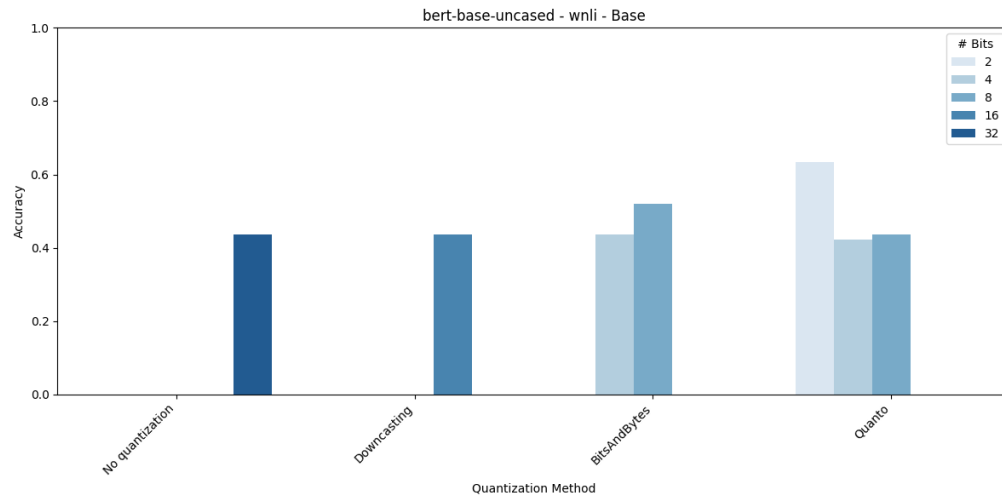


Figure E.29: Results of base bert-base-uncased on wnli using Accuracy

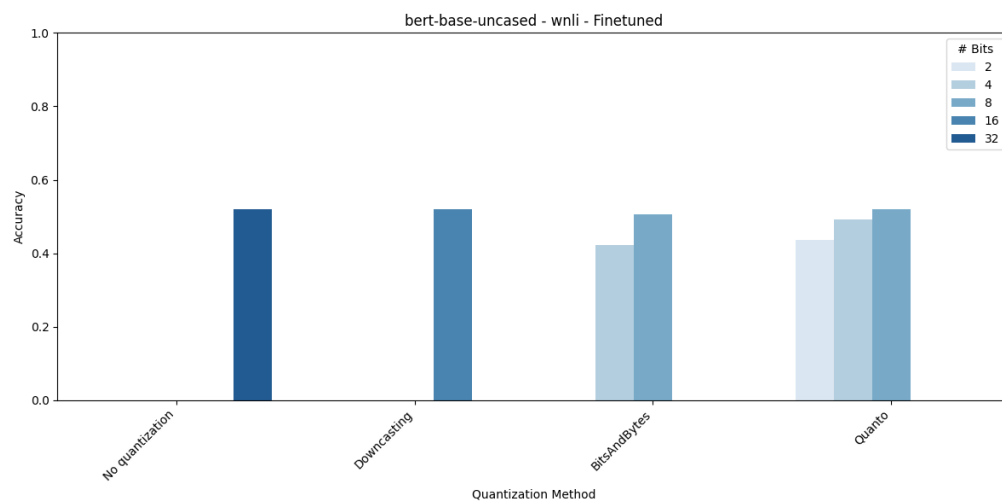


Figure E.30: Results of finetuned bert-base-uncased on wnli using Accuracy

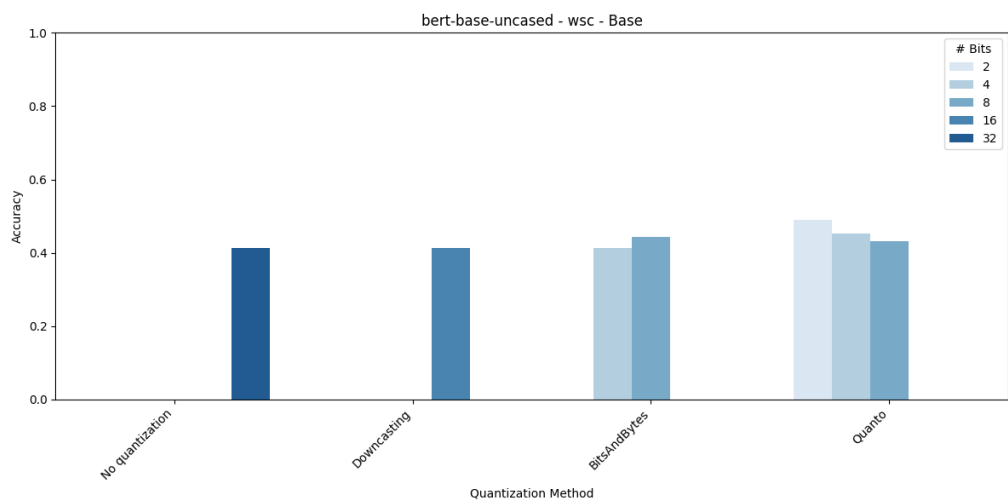


Figure E.31: Results of base bert-base-uncased on wsc using Accuracy

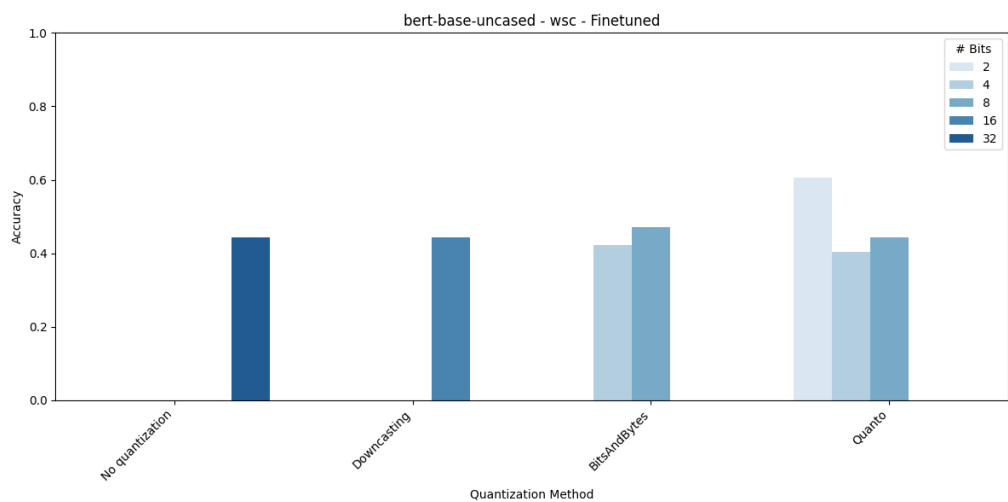


Figure E.32: Results of finetuned bert-base-uncased on wsc using Accuracy

E. DETAILED FIGURES

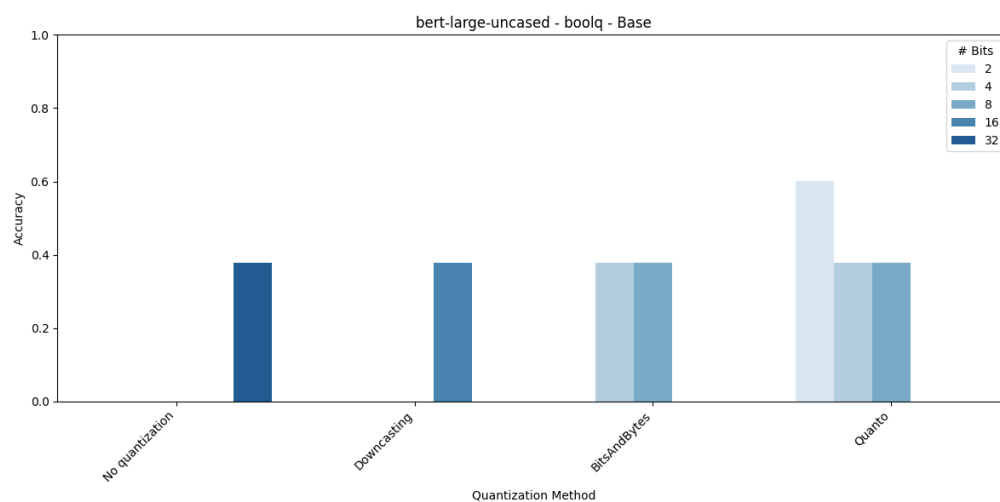


Figure E.33: Results of base bert-large-uncased on boolq using Accuracy

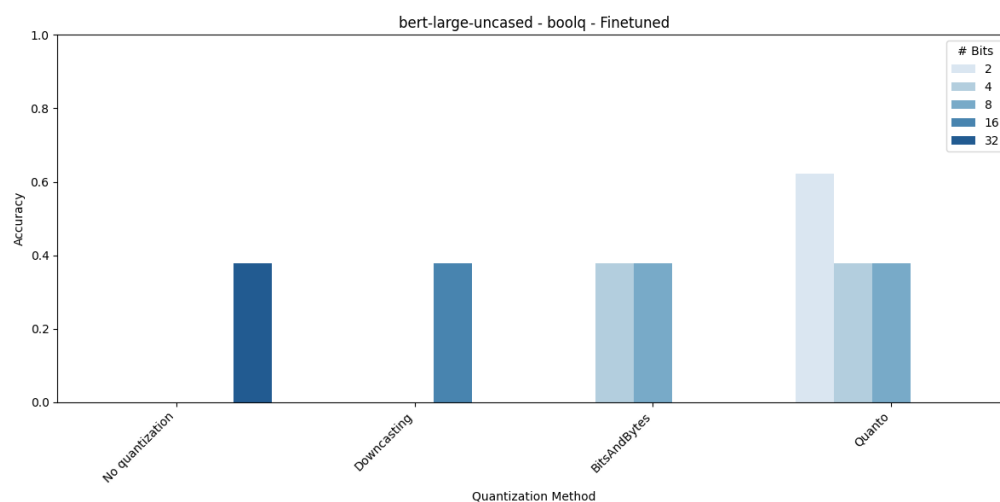


Figure E.34: Results of finetuned bert-large-uncased on boolq using Accuracy

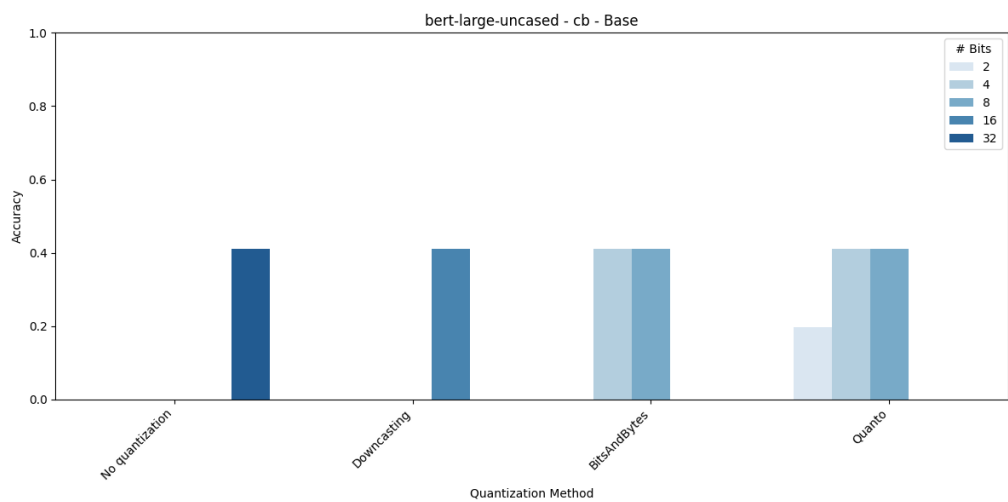


Figure E.35: Results of base bert-large-uncased on cb using Accuracy

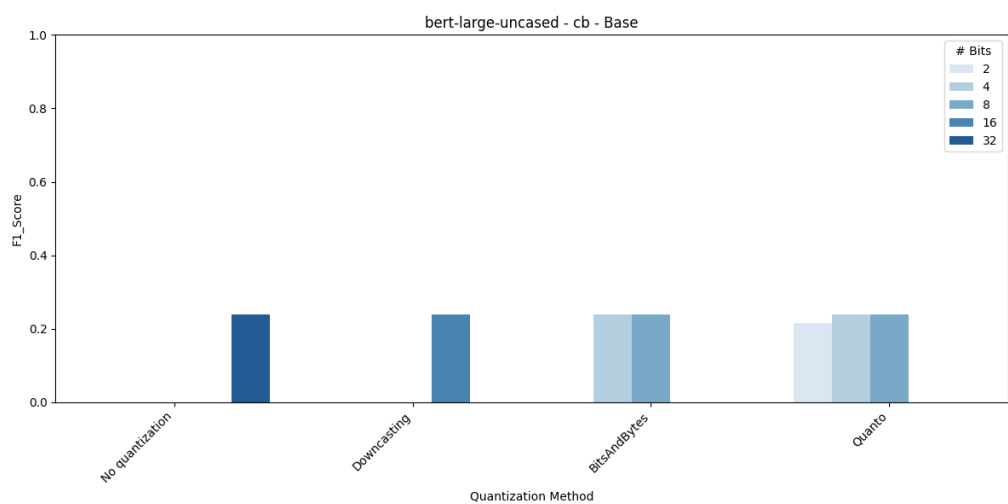


Figure E.36: Results of base bert-large-uncased on cb using F1_Score

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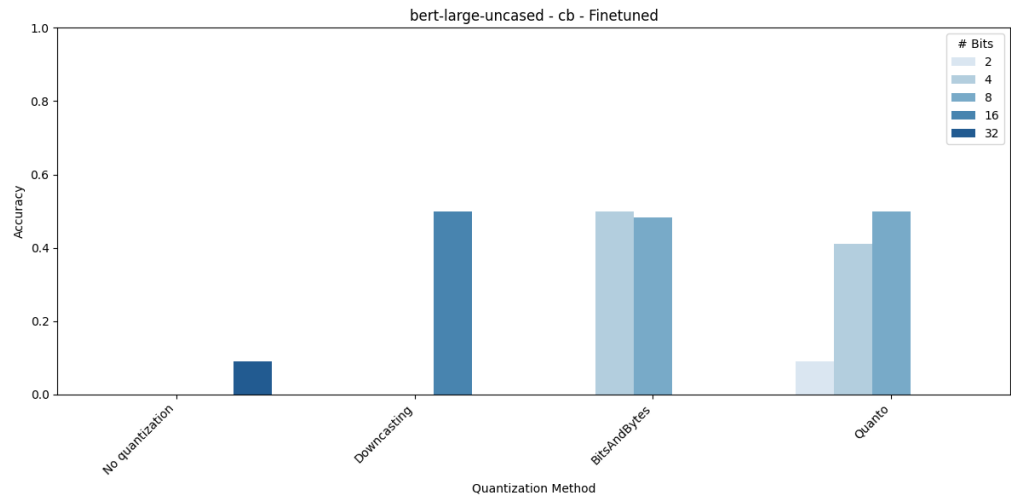


Figure E.37: Results of finetuned bert-large-uncased on cb using Accuracy

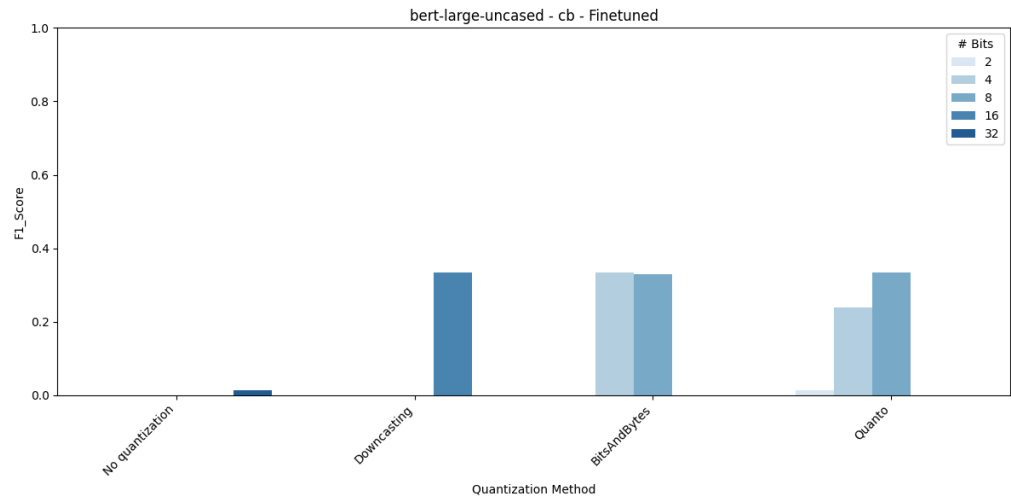


Figure E.38: Results of finetuned bert-large-uncased on cb using F1_Score

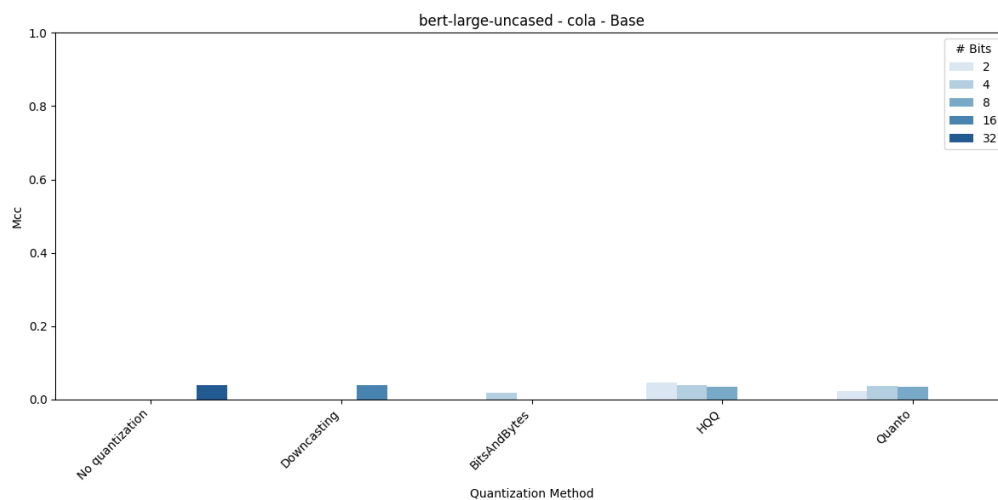


Figure E.39: Results of base bert-large-uncased on cola using Mcc

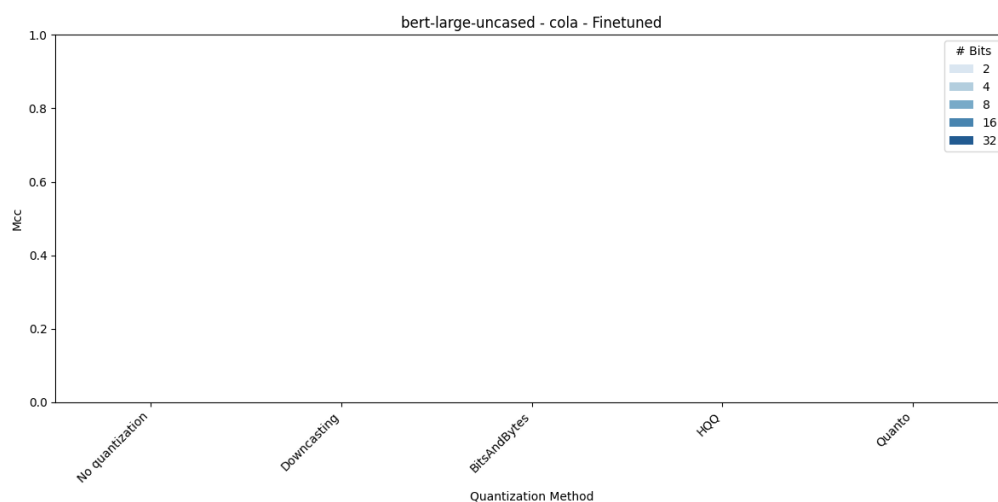


Figure E.40: Results of finetuned bert-large-uncased on cola using Mcc

E. DETAILED FIGURES

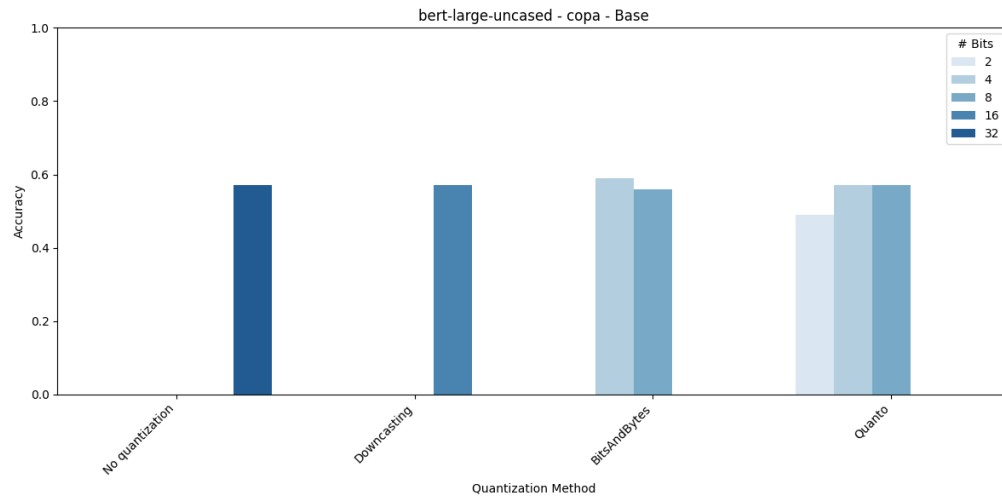


Figure E.41: Results of base bert-large-uncased on copa using Accuracy

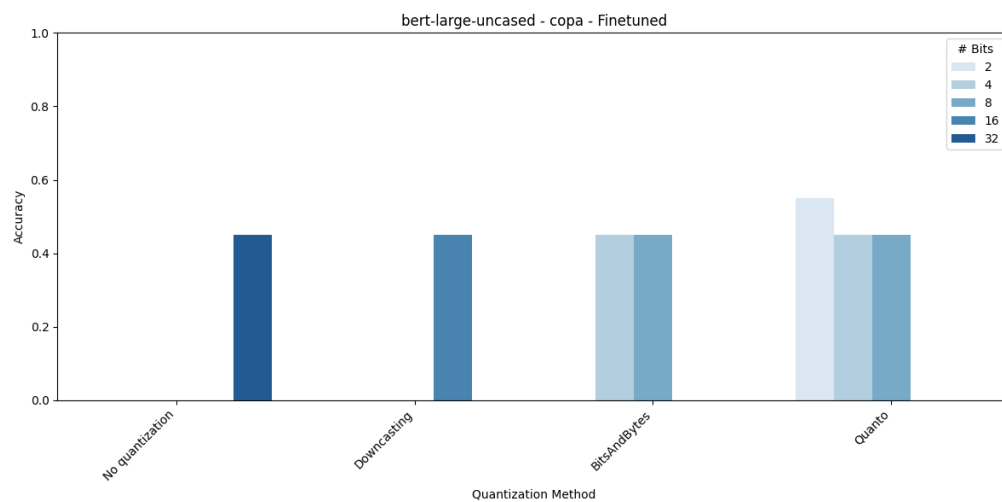


Figure E.42: Results of finetuned bert-large-uncased on copa using Accuracy

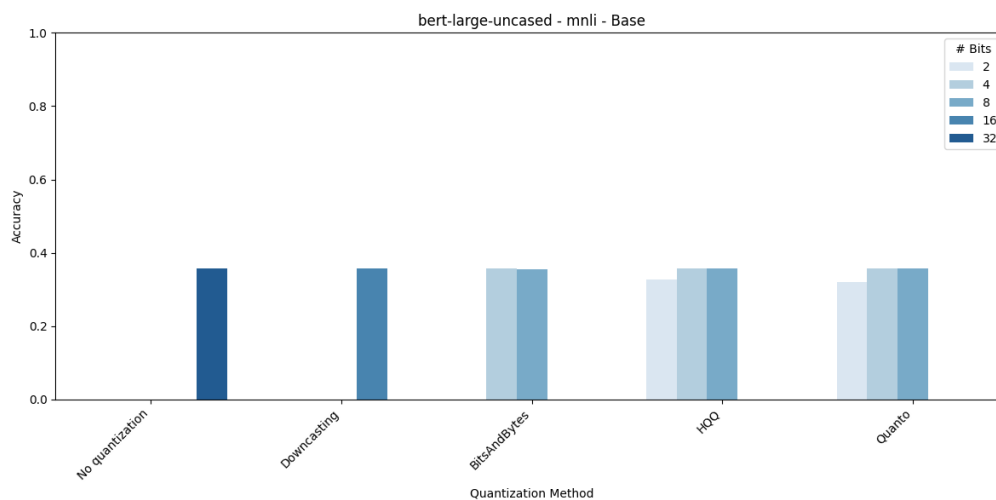


Figure E.43: Results of base bert-large-uncased on mnli using Accuracy

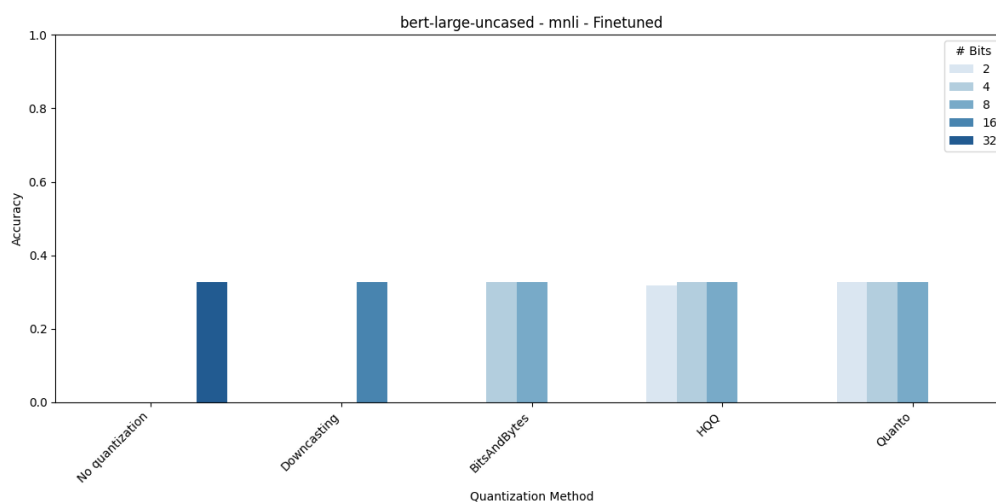


Figure E.44: Results of finetuned bert-large-uncased on mnli using Accuracy

E. DETAILED FIGURES

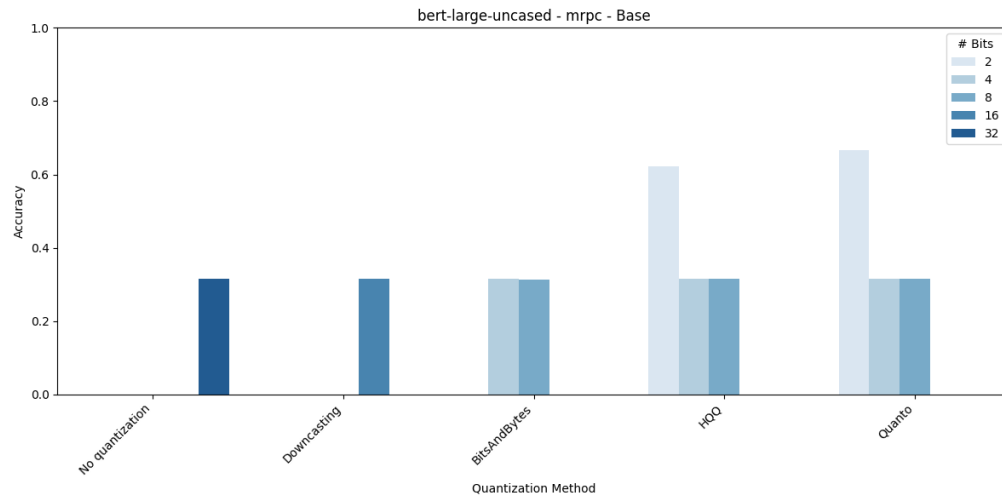


Figure E.45: Results of base bert-large-uncased on mrpc using Accuracy

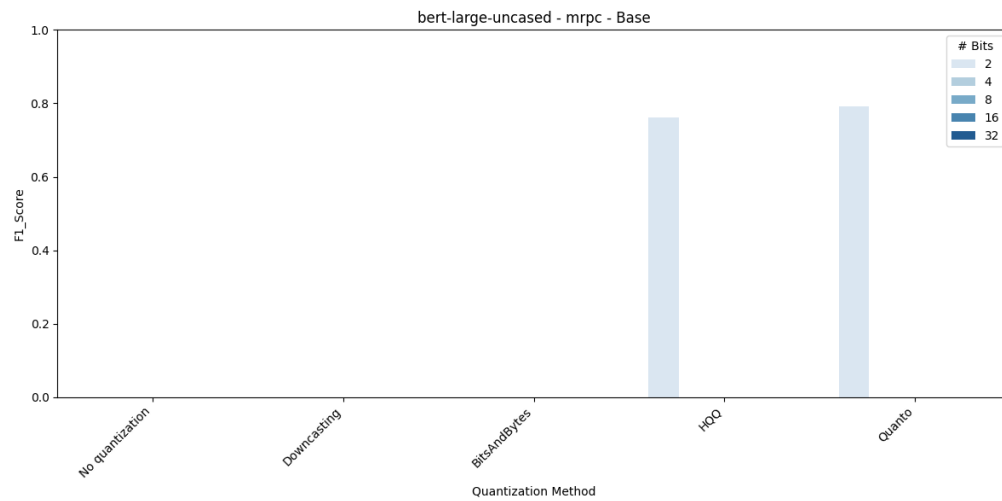


Figure E.46: Results of base bert-large-uncased on mrpc using F1_Score

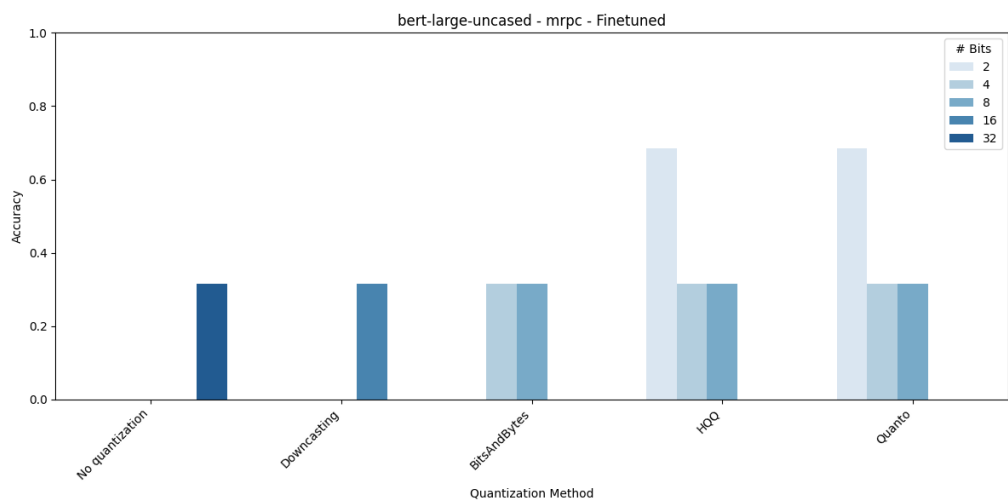


Figure E.47: Results of finetuned bert-large-uncased on mrpc using Accuracy

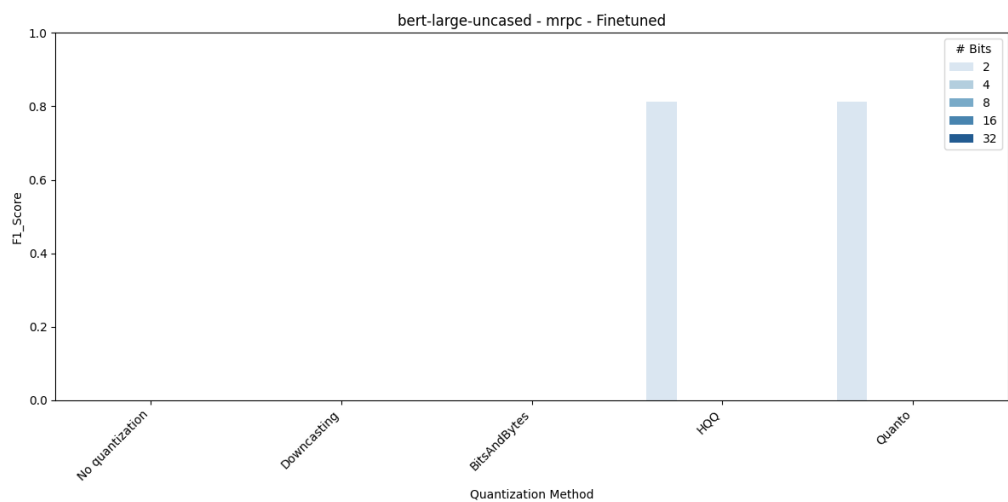


Figure E.48: Results of finetuned bert-large-uncased on mrpc using F1_Score

E. DETAILED FIGURES

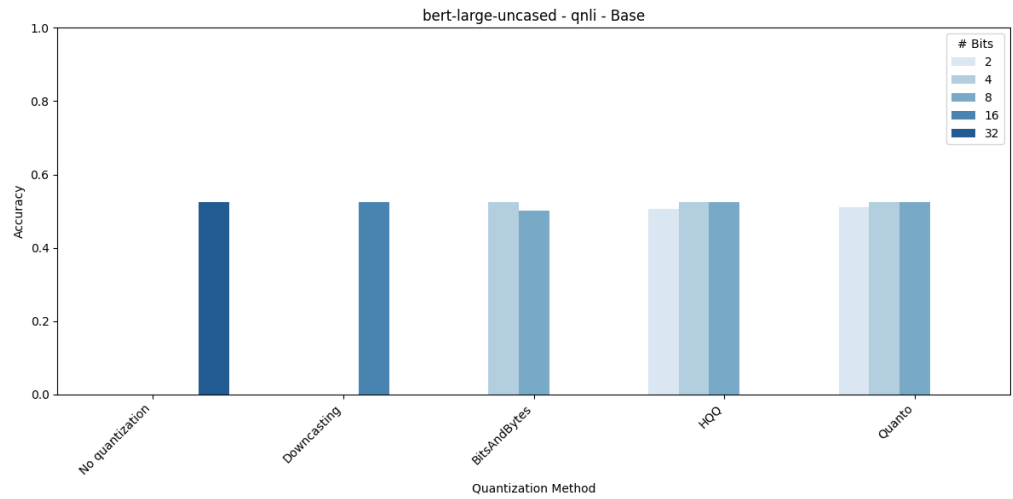


Figure E.49: Results of base bert-large-uncased on qnli using Accuracy

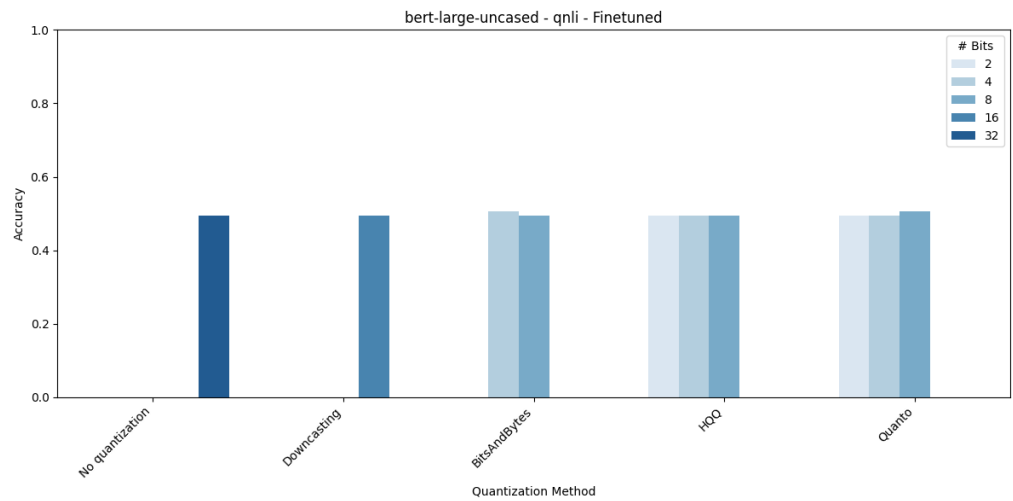


Figure E.50: Results of finetuned bert-large-uncased on qnli using Accuracy

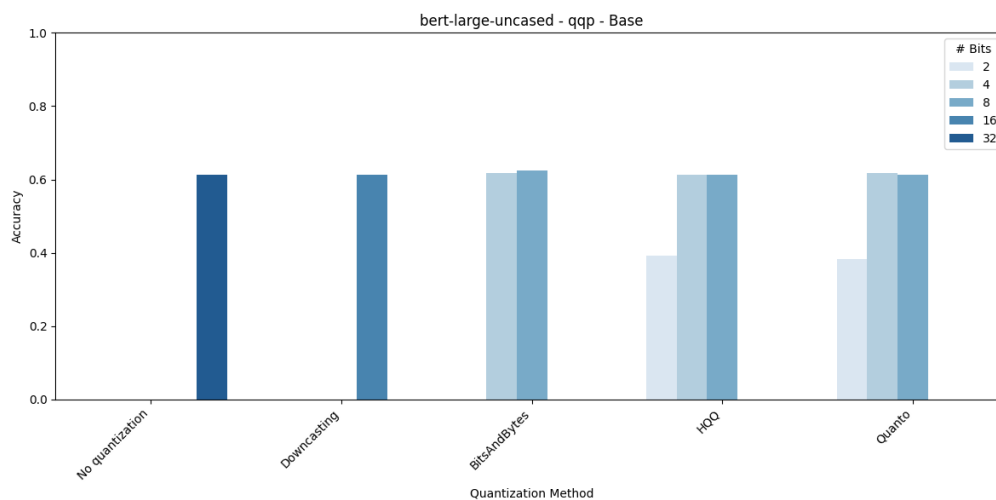


Figure E.51: Results of base bert-large-uncased on qqp using Accuracy

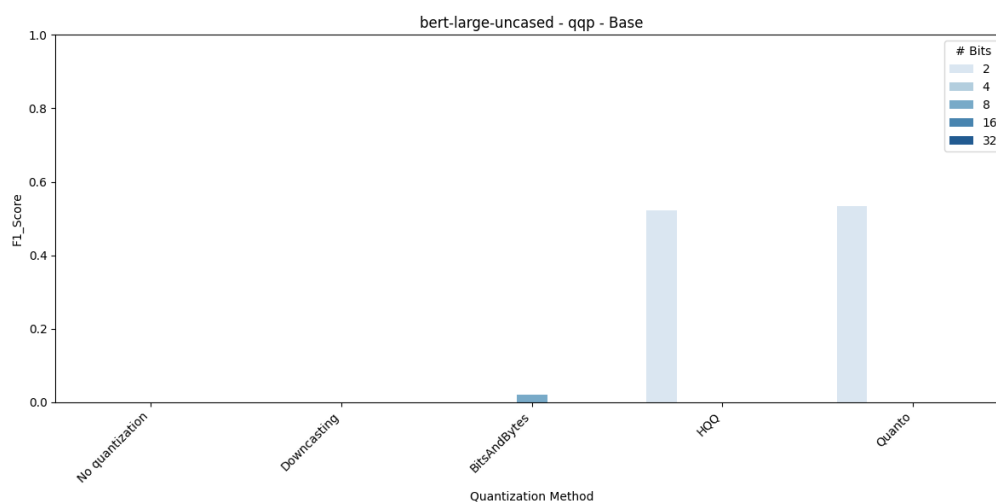


Figure E.52: Results of base bert-large-uncased on qqp using F1_Score

E. DETAILED FIGURES

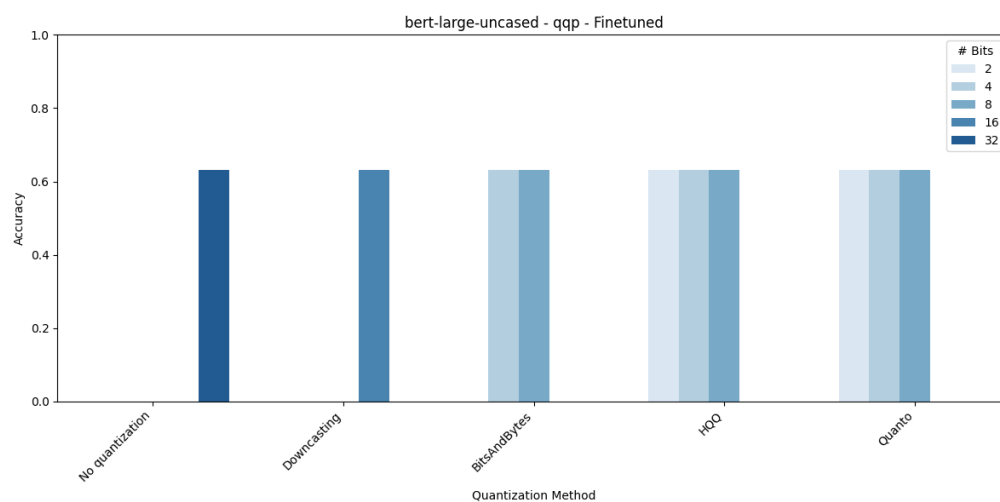


Figure E.53: Results of finetuned bert-large-uncased on qqp using Accuracy

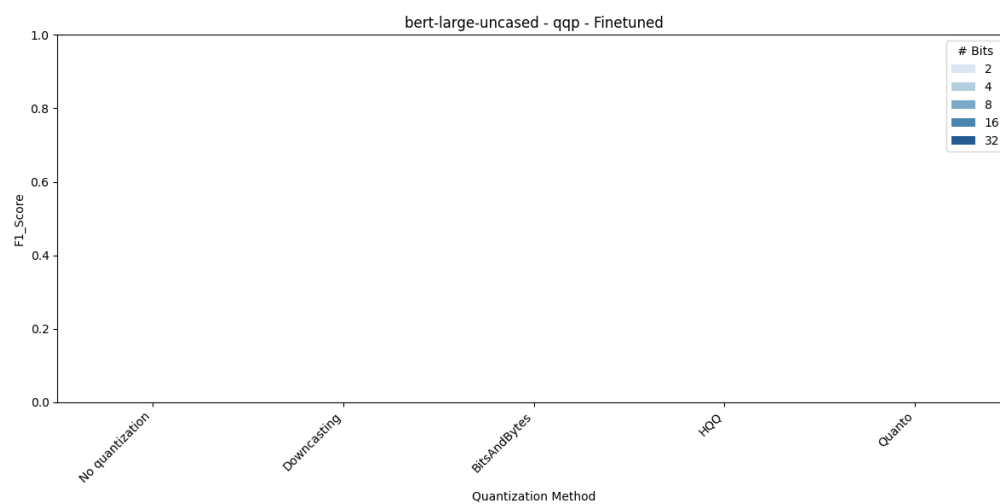


Figure E.54: Results of finetuned bert-large-uncased on qqp using F1_Score

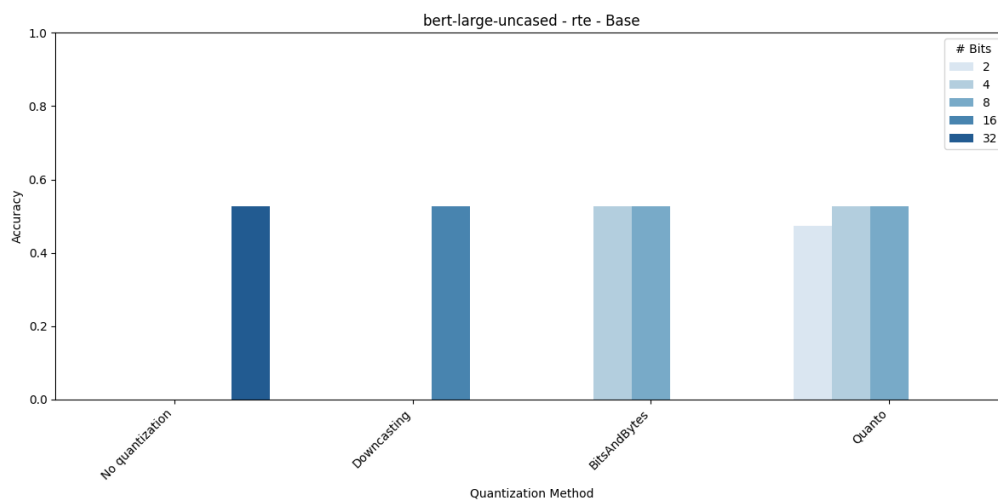


Figure E.55: Results of base bert-large-uncased on rte using Accuracy

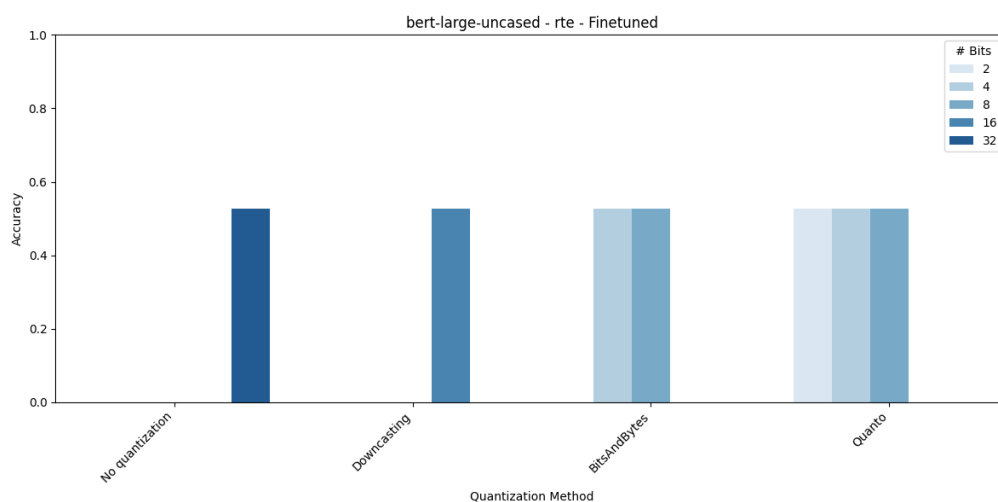


Figure E.56: Results of finetuned bert-large-uncased on rte using Accuracy

E. DETAILED FIGURES

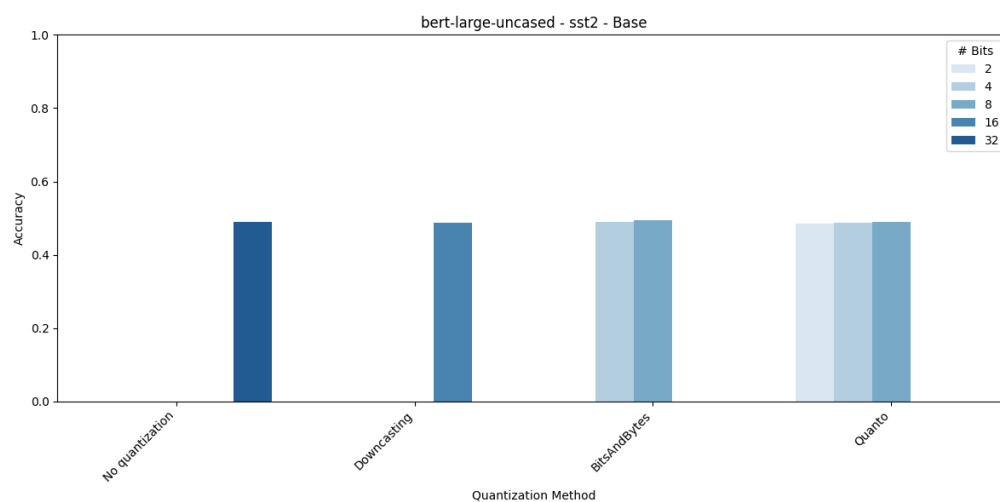


Figure E.57: Results of base bert-large-uncased on sst2 using Accuracy

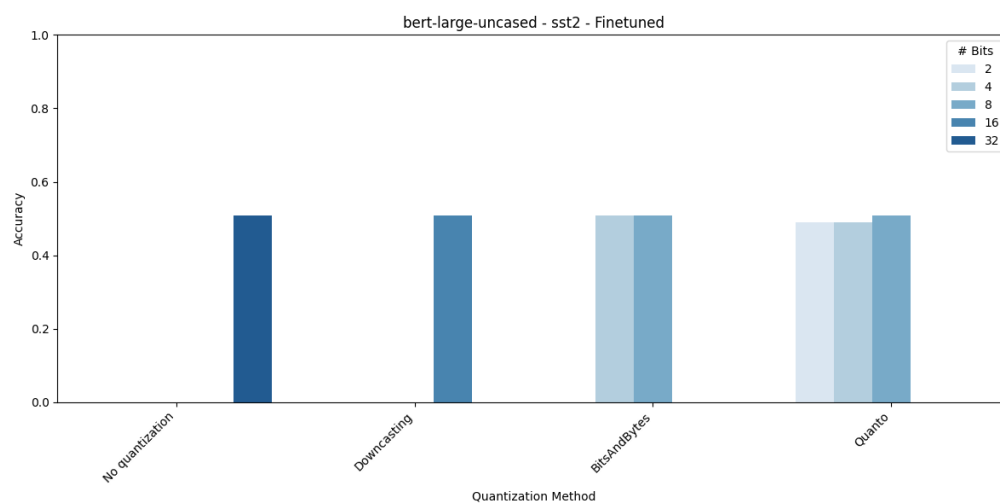


Figure E.58: Results of finetuned bert-large-uncased on sst2 using Accuracy

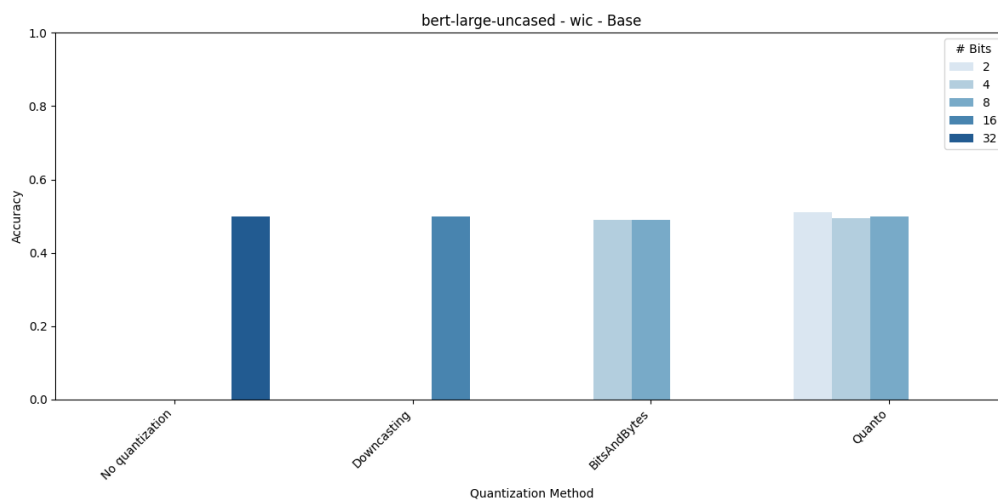


Figure E.59: Results of base bert-large-uncased on wic using Accuracy

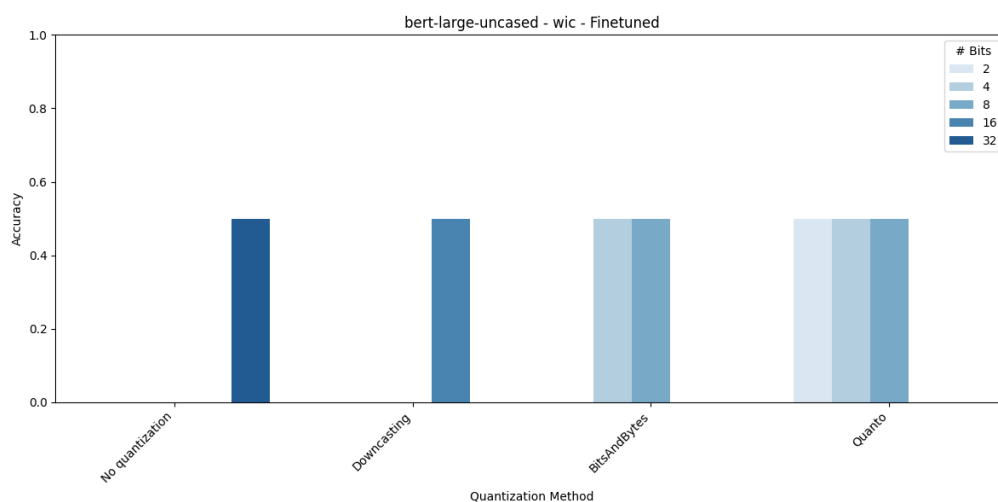


Figure E.60: Results of finetuned bert-large-uncased on wic using Accuracy

E. DETAILED FIGURES

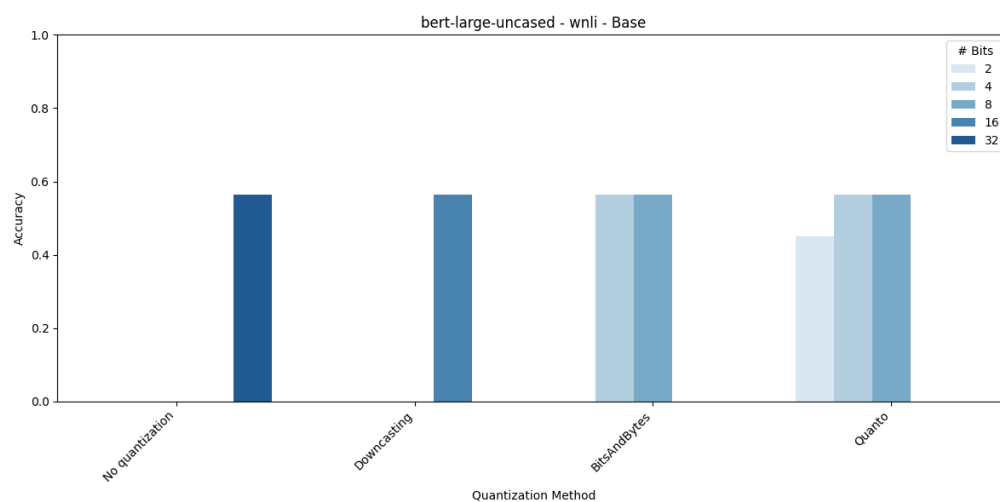


Figure E.61: Results of base bert-large-uncased on wnli using Accuracy

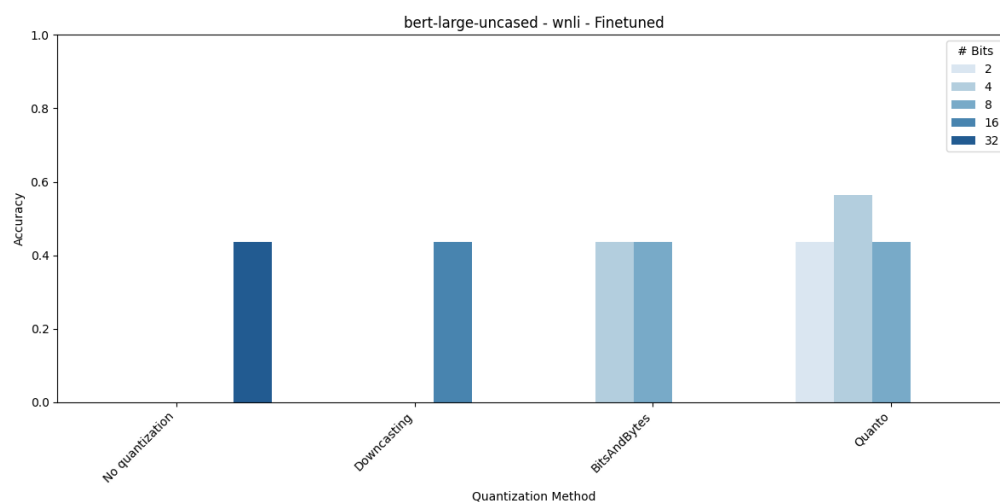


Figure E.62: Results of finetuned bert-large-uncased on wnli using Accuracy

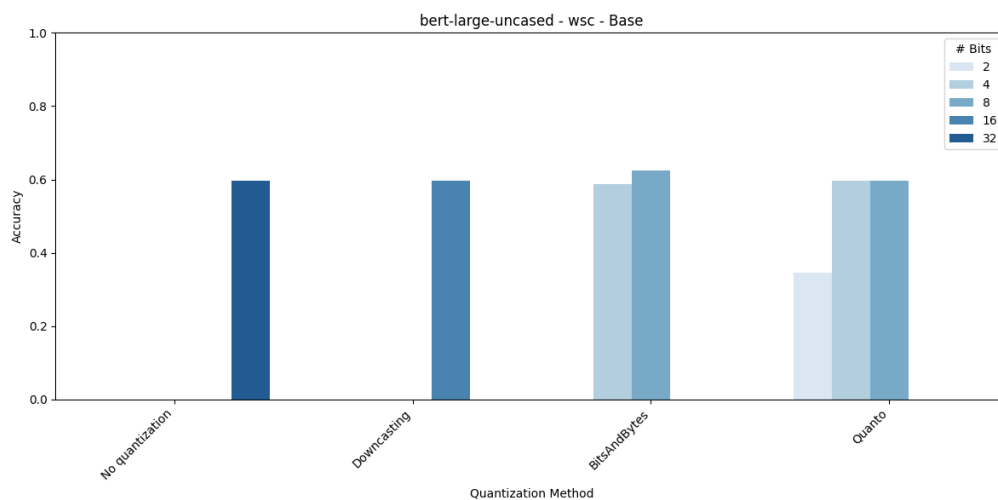


Figure E.63: Results of base bert-large-uncased on wsc using Accuracy

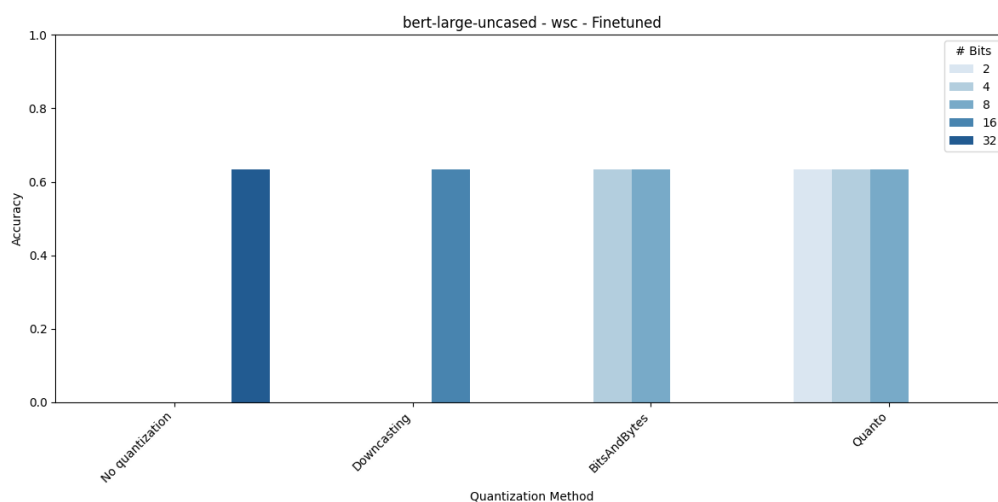


Figure E.64: Results of finetuned bert-large-uncased on wsc using Accuracy

E. DETAILED FIGURES

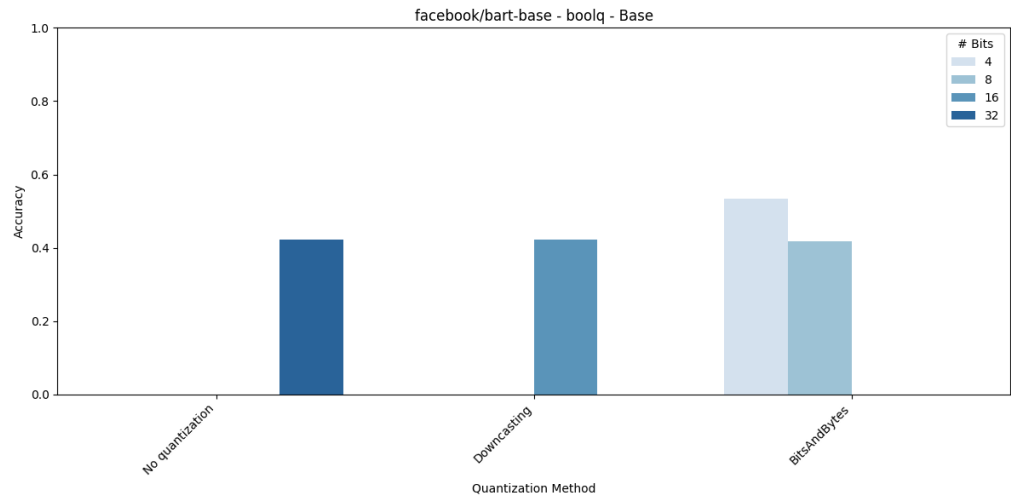


Figure E.65: Results of base facebook-bart-base on boolq using Accuracy

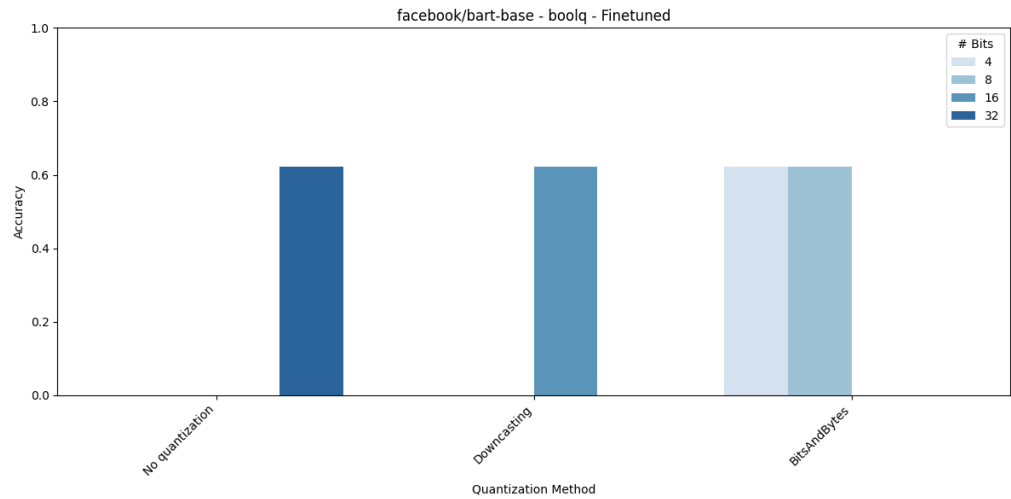


Figure E.66: Results of finetuned facebook-bart-base on boolq using Accuracy

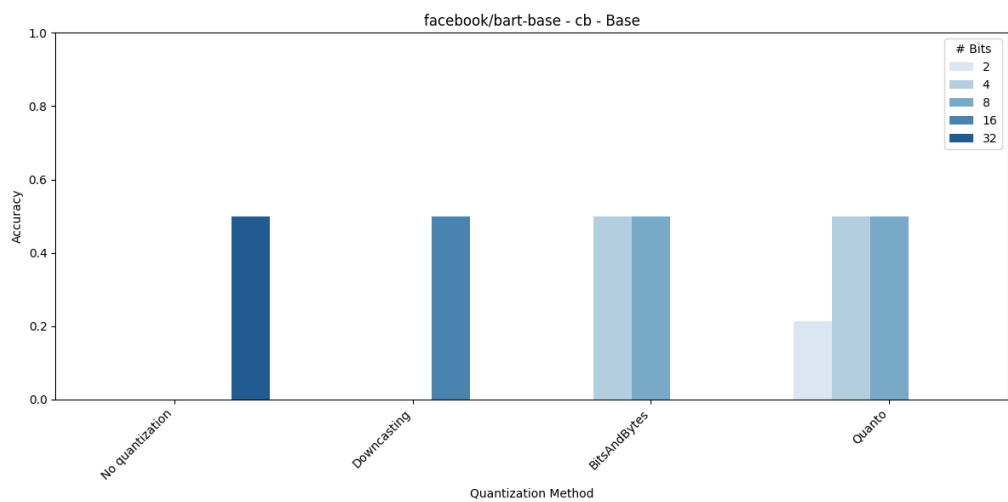


Figure E.67: Results of base facebook-bart-base on cb using Accuracy

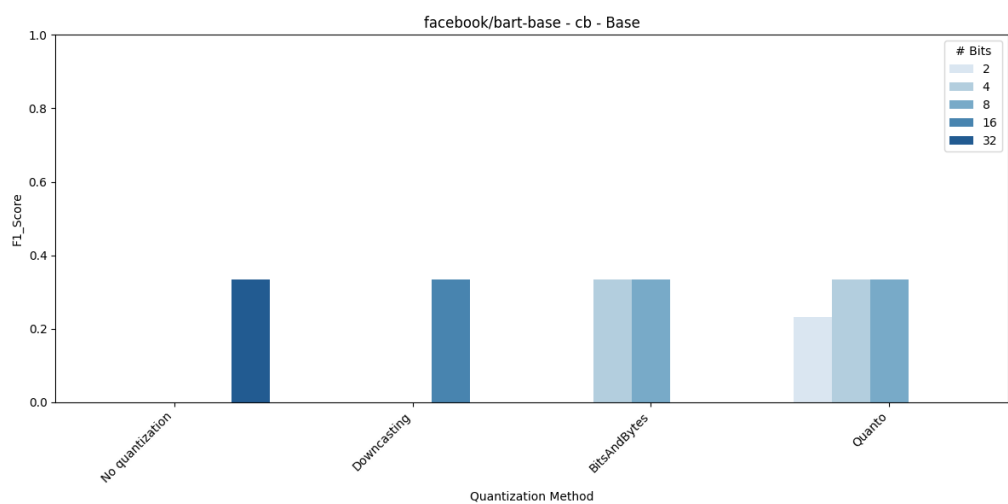


Figure E.68: Results of base facebook-bart-base on cb using F1_Score

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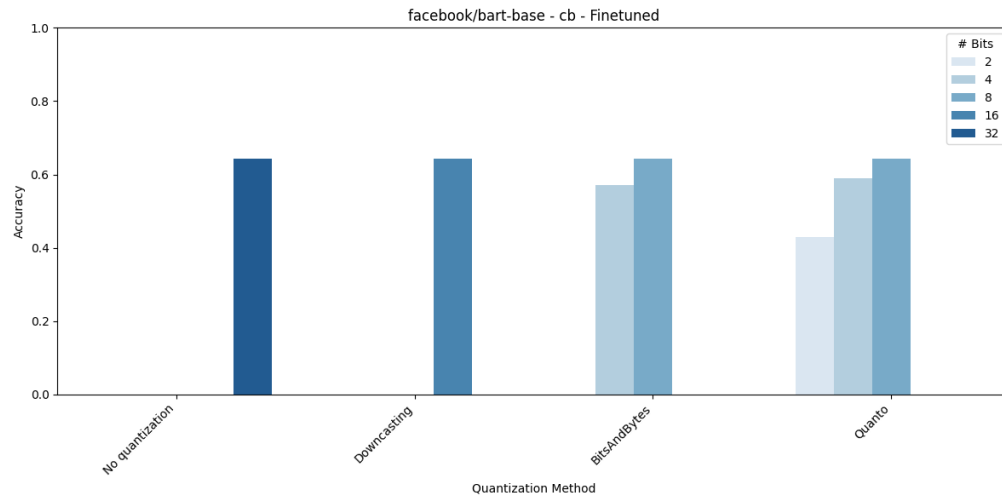


Figure E.69: Results of finetuned facebook-bart-base on cb using Accuracy

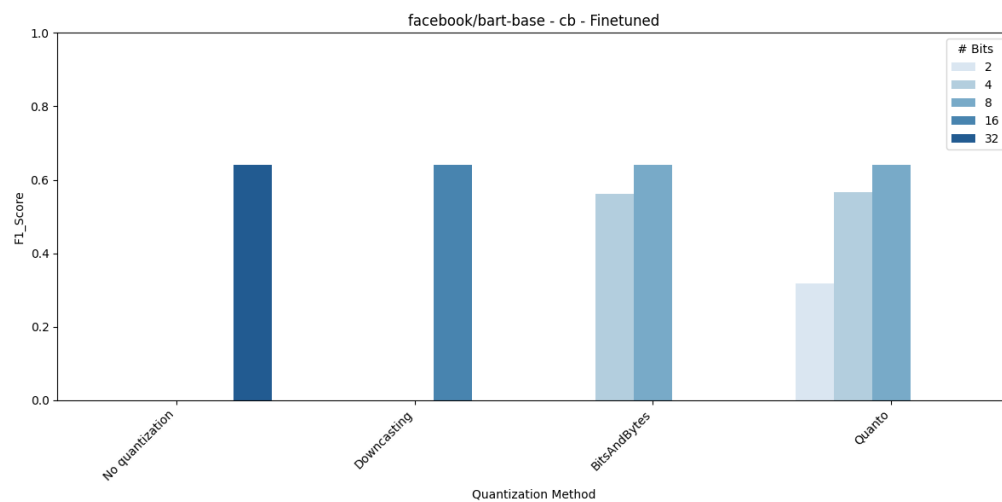


Figure E.70: Results of finetuned facebook-bart-base on cb using F1_Score

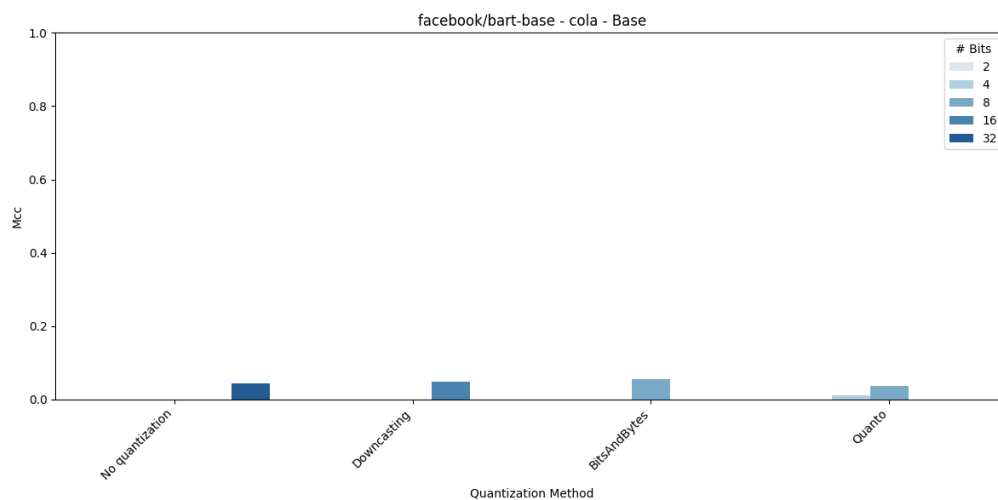


Figure E.71: Results of base facebook-bart-base on cola using Mcc

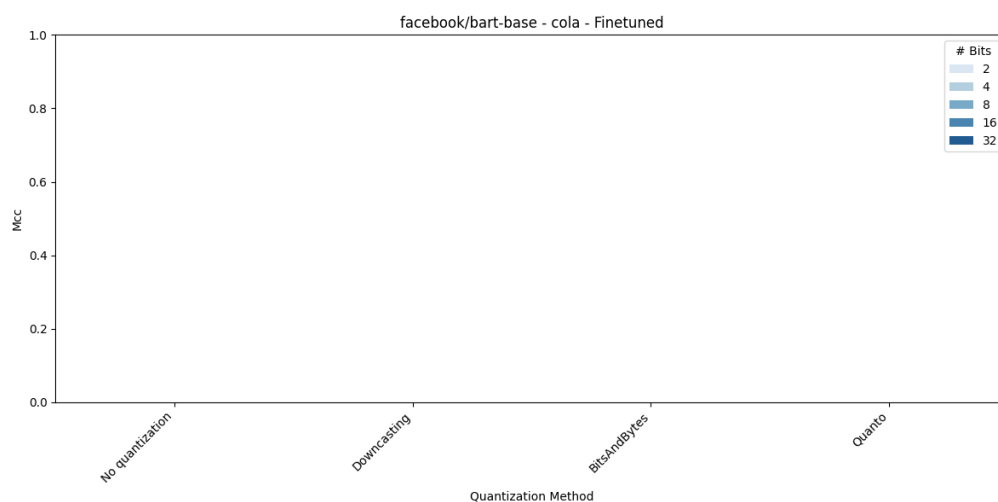


Figure E.72: Results of finetuned facebook-bart-base on cola using Mcc

E. DETAILED FIGURES

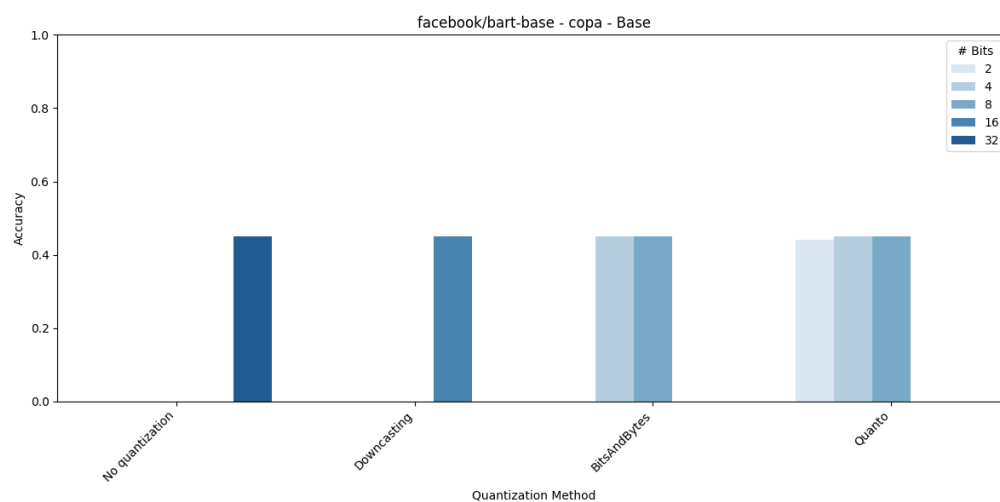


Figure E.73: Results of base facebook-bart-base on copa using Accuracy

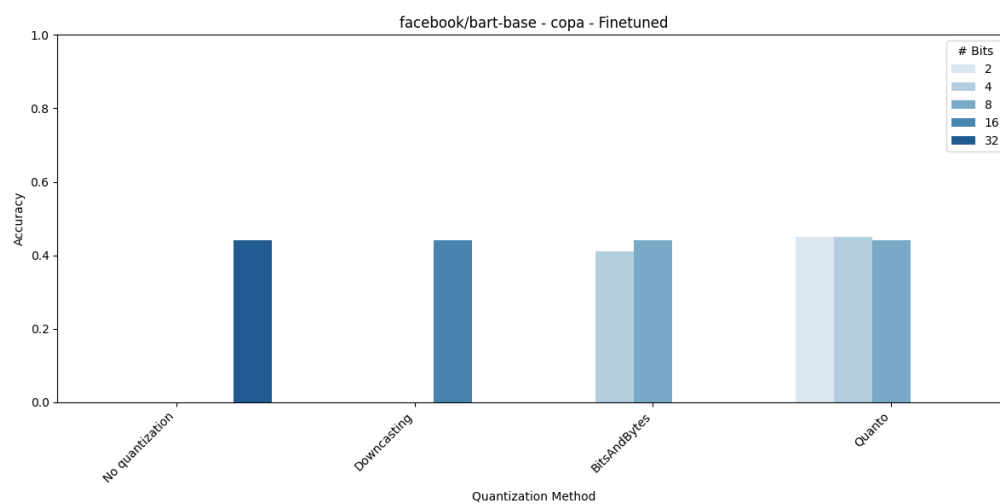


Figure E.74: Results of finetuned facebook-bart-base on copa using Accuracy

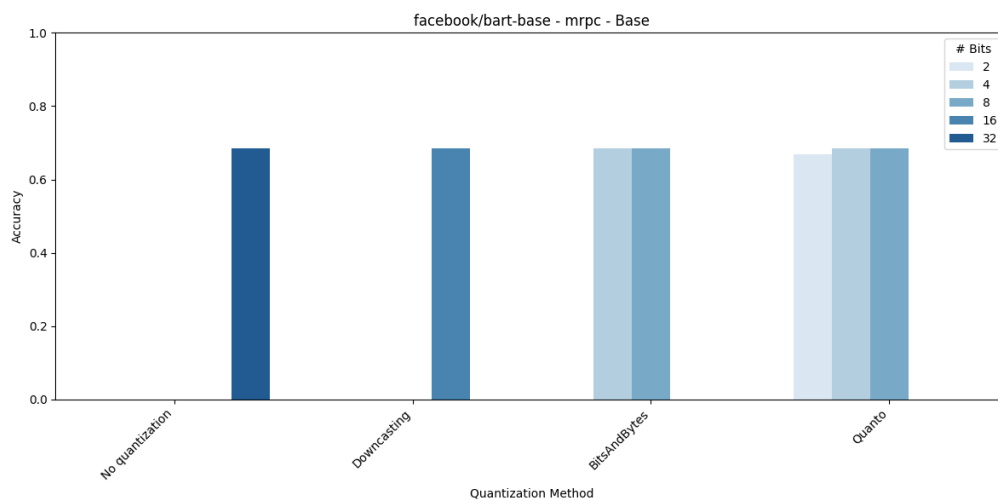


Figure E.75: Results of base facebook-bart-base on mrpc using Accuracy

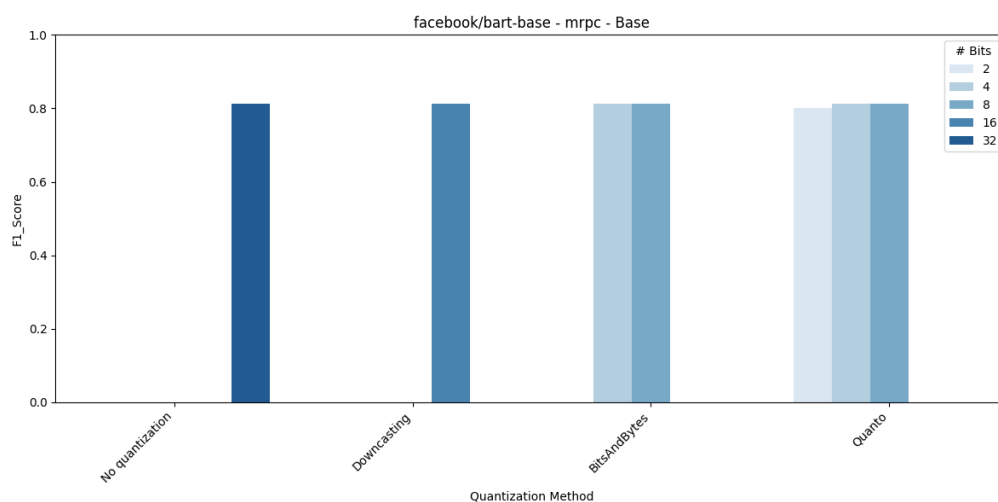


Figure E.76: Results of base facebook-bart-base on mrpc using F1_Score

E. DETAILED FIGURES

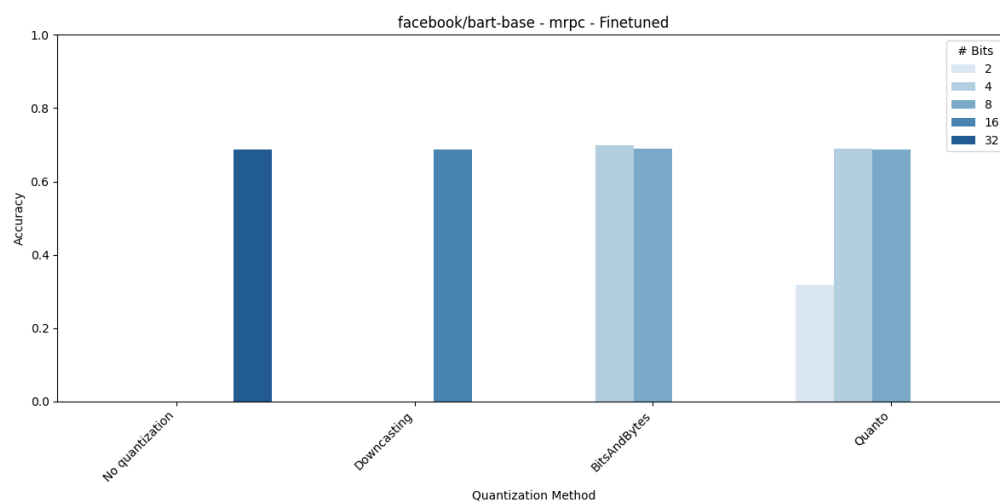


Figure E.77: Results of finetuned facebook-bart-base on mrpc using Accuracy

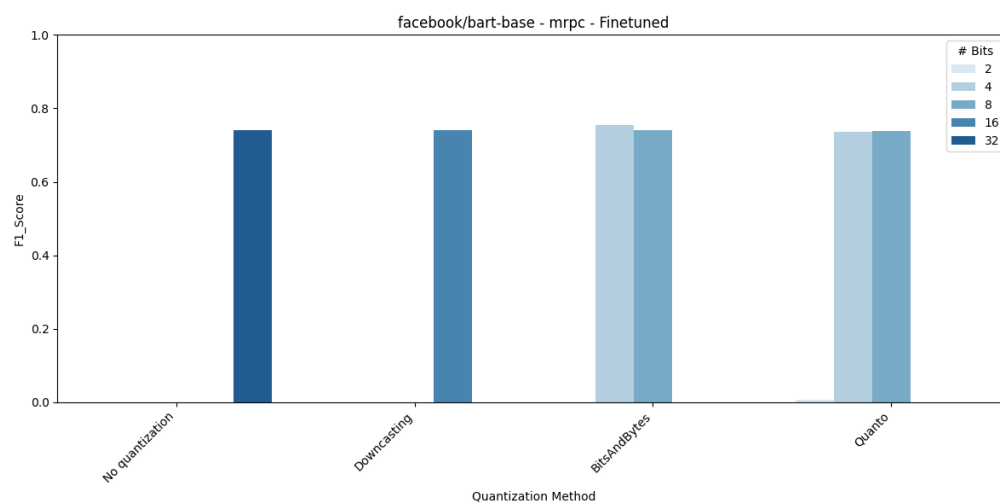


Figure E.78: Results of finetuned facebook-bart-base on mrpc using F1_Score

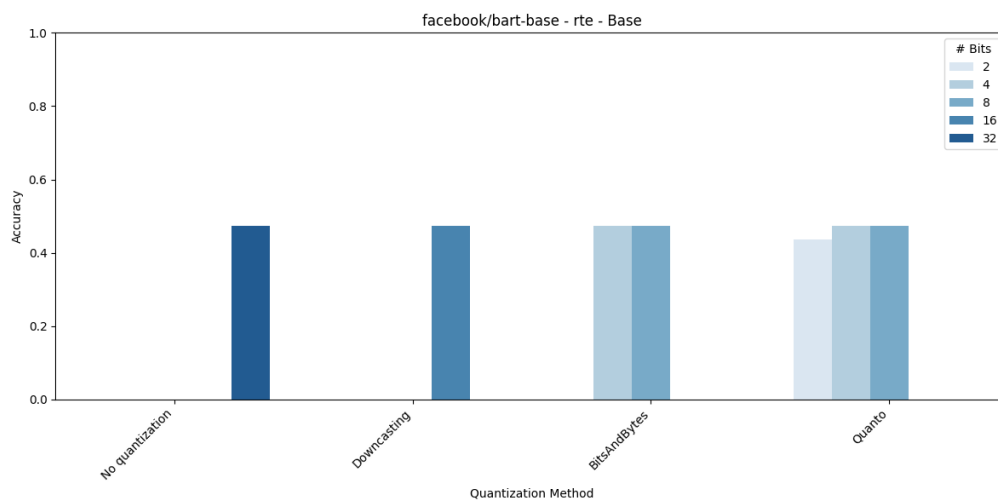


Figure E.79: Results of base facebook-bart-base on rte using Accuracy

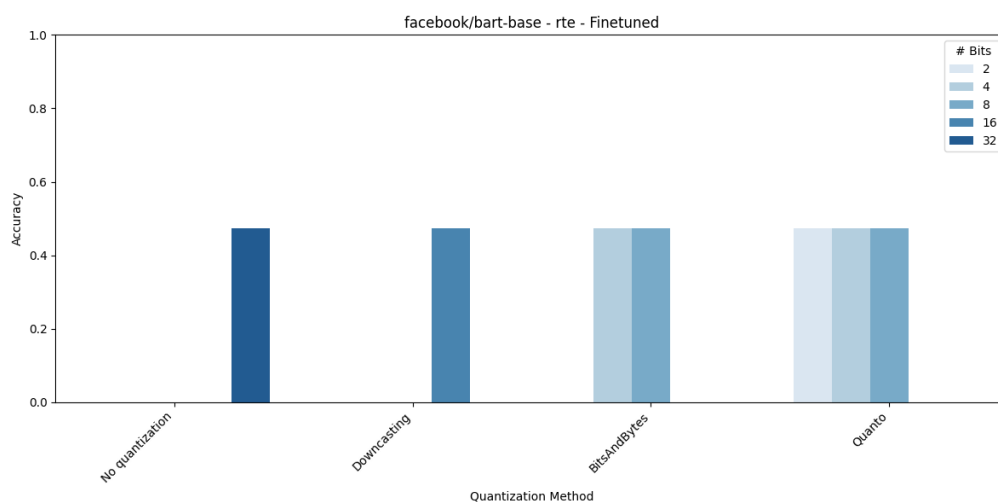


Figure E.80: Results of finetuned facebook-bart-base on rte using Accuracy

E. DETAILED FIGURES

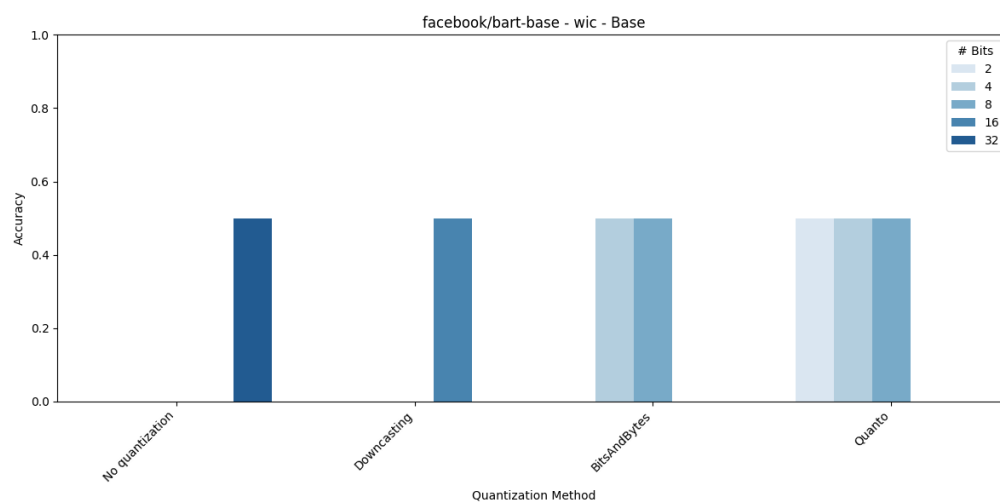


Figure E.81: Results of base facebook-bart-base on wic using Accuracy

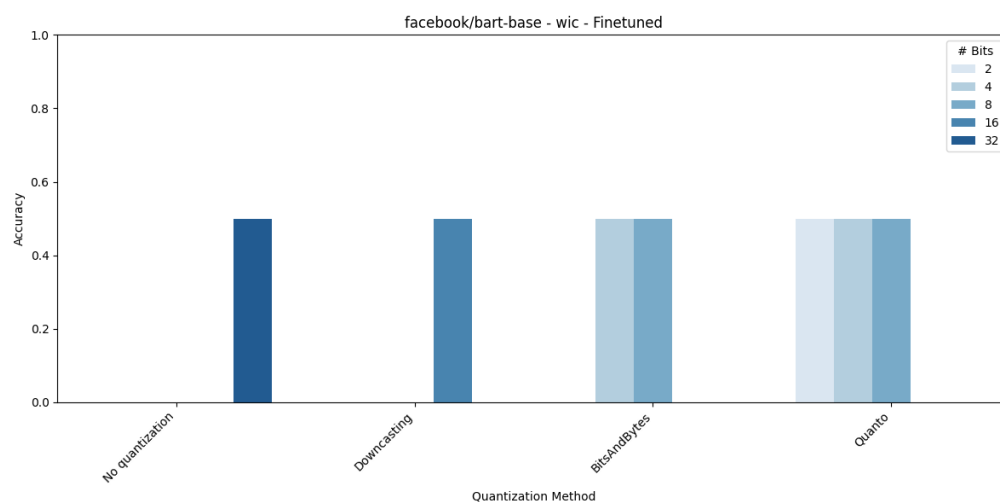


Figure E.82: Results of finetuned facebook-bart-base on wic using Accuracy

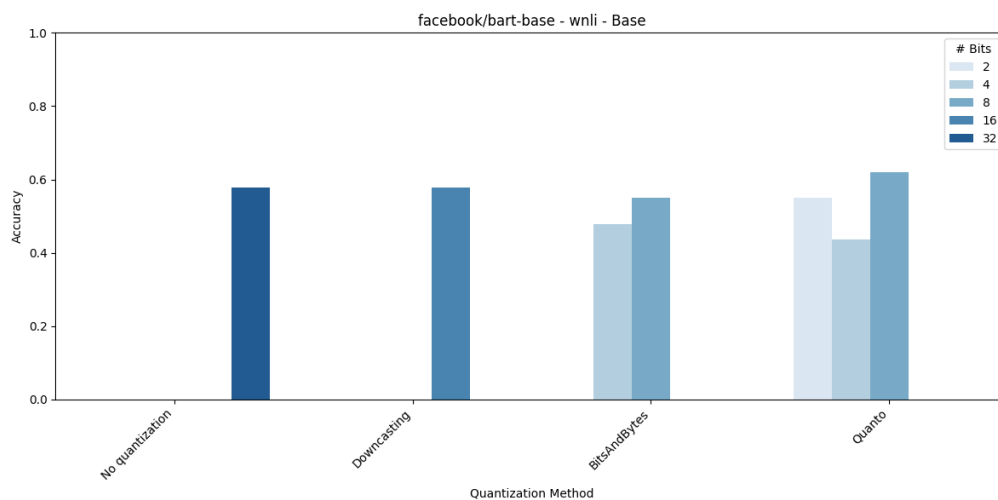


Figure E.83: Results of base facebook-bart-base on wnli using Accuracy

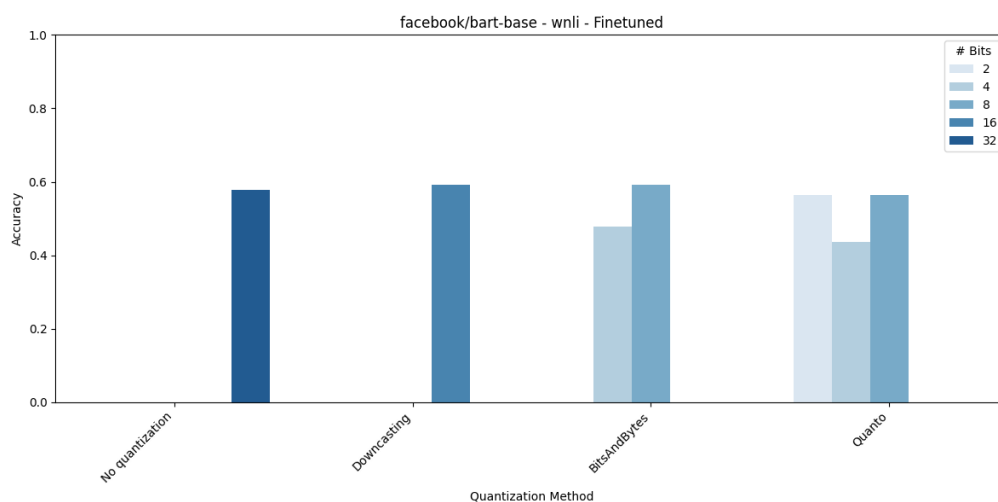


Figure E.84: Results of finetuned facebook-bart-base on wnli using Accuracy

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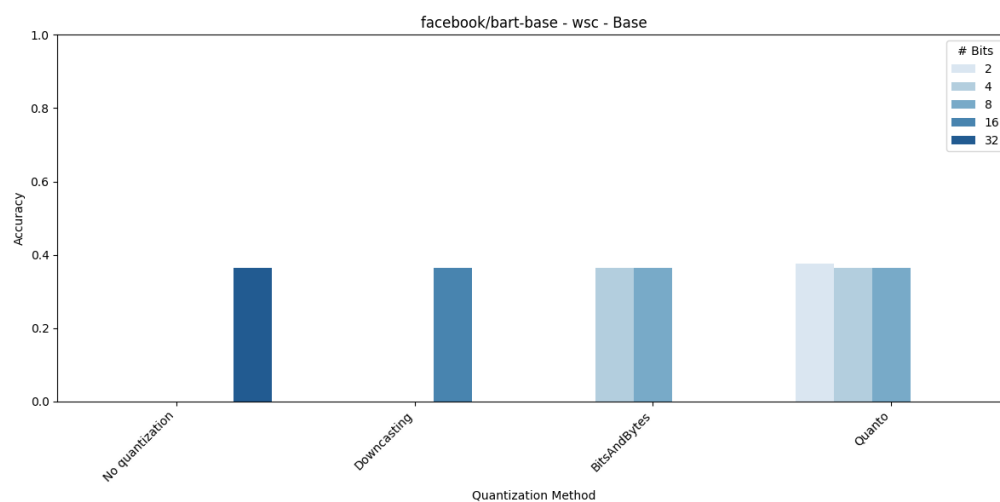


Figure E.85: Results of base facebook-bart-base on wsc using Accuracy

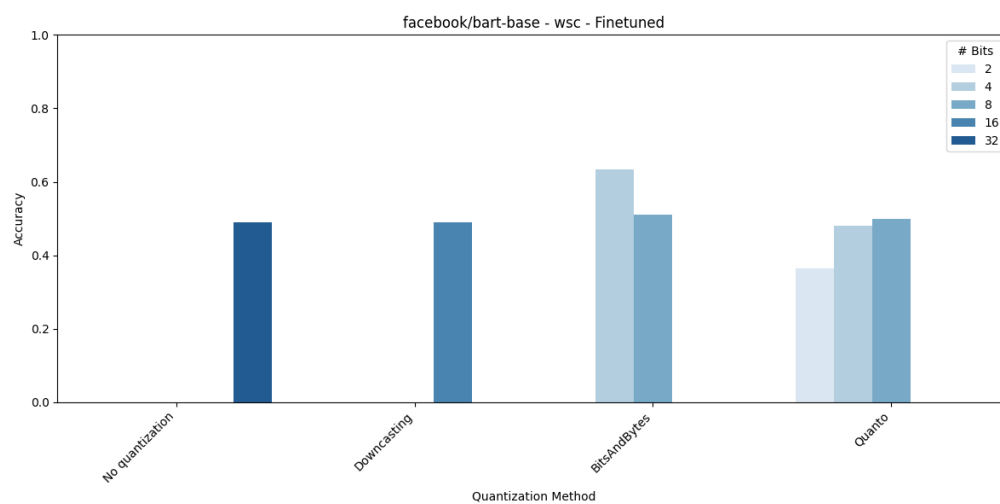


Figure E.86: Results of finetuned facebook-bart-base on wsc using Accuracy

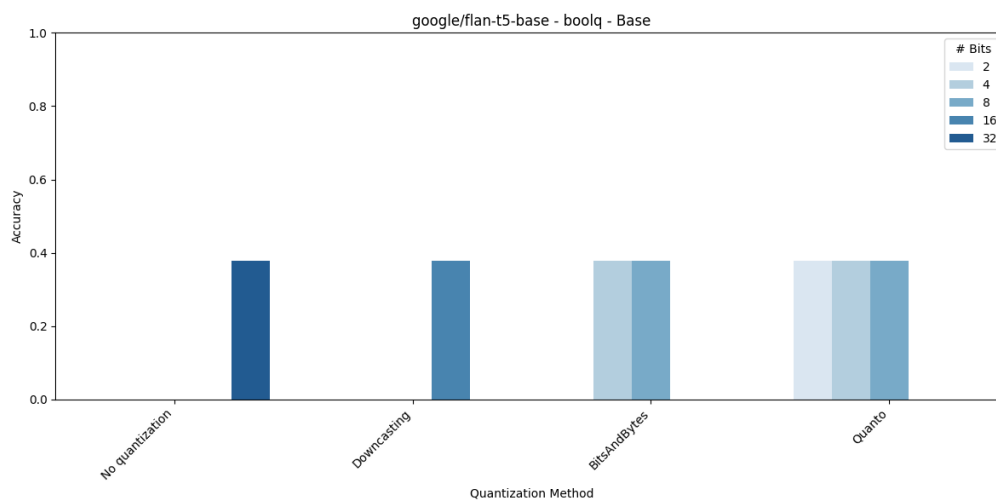


Figure E.87: Results of base google-flan-t5-base on boolq using Accuracy

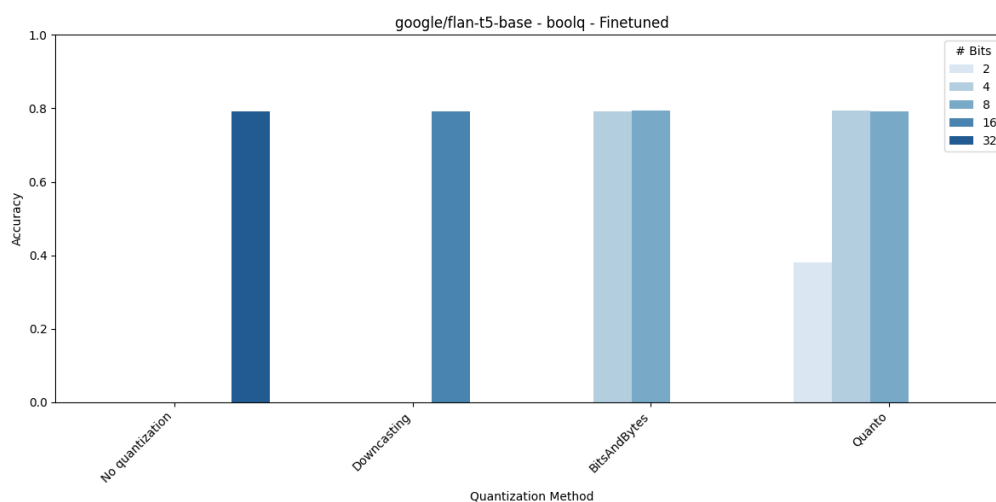


Figure E.88: Results of finetuned google-flan-t5-base on boolq using Accuracy

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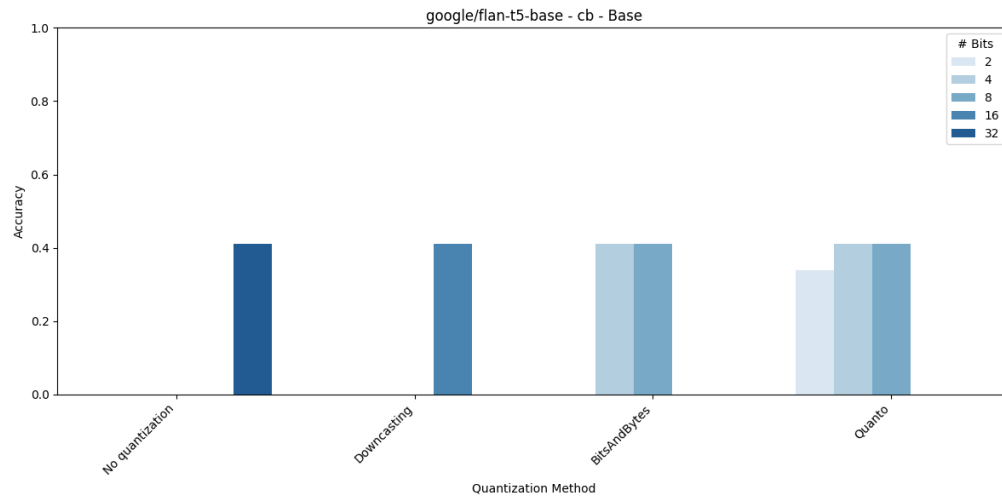


Figure E.89: Results of base google-flan-t5-base on cb using Accuracy

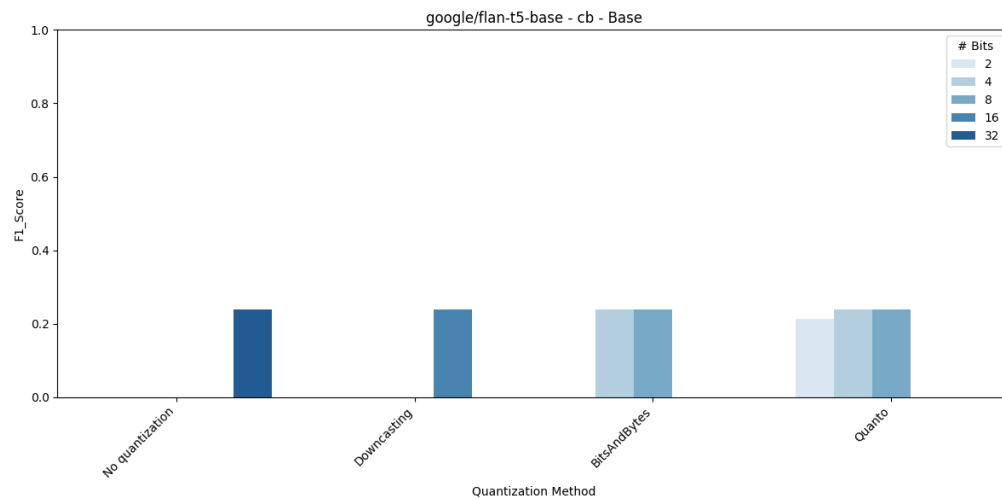


Figure E.90: Results of base google-flan-t5-base on cb using F1_Score

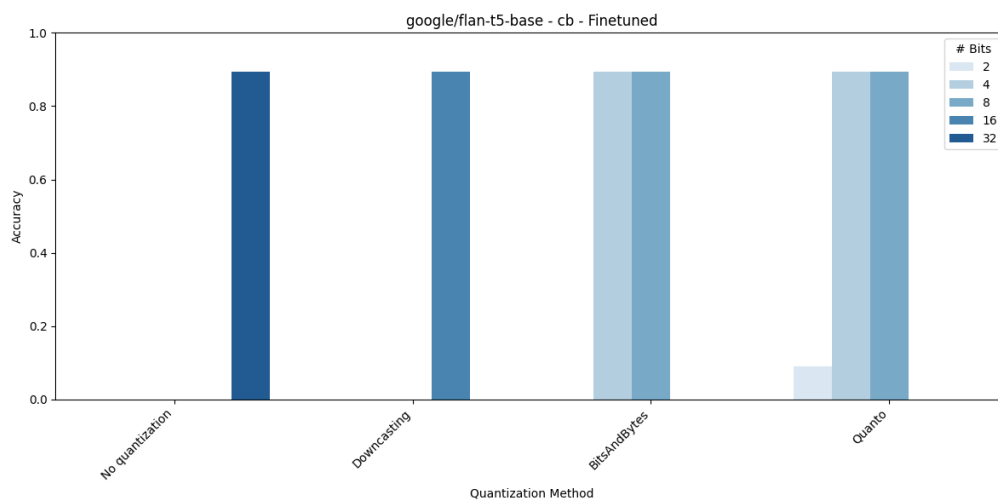


Figure E.91: Results of finetuned google-flan-t5-base on cb using Accuracy

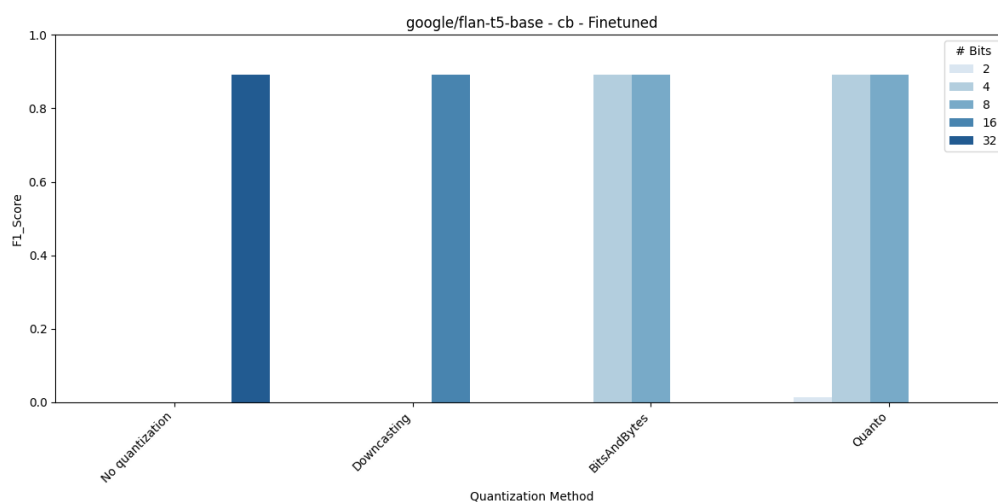


Figure E.92: Results of finetuned google-flan-t5-base on cb using F1_Score

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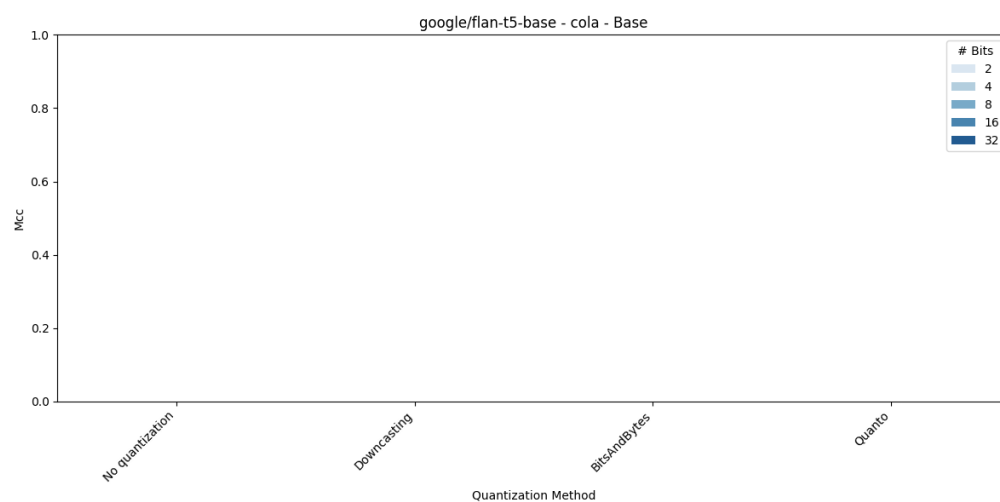


Figure E.93: Results of base google/flan-t5-base on cola using Mcc

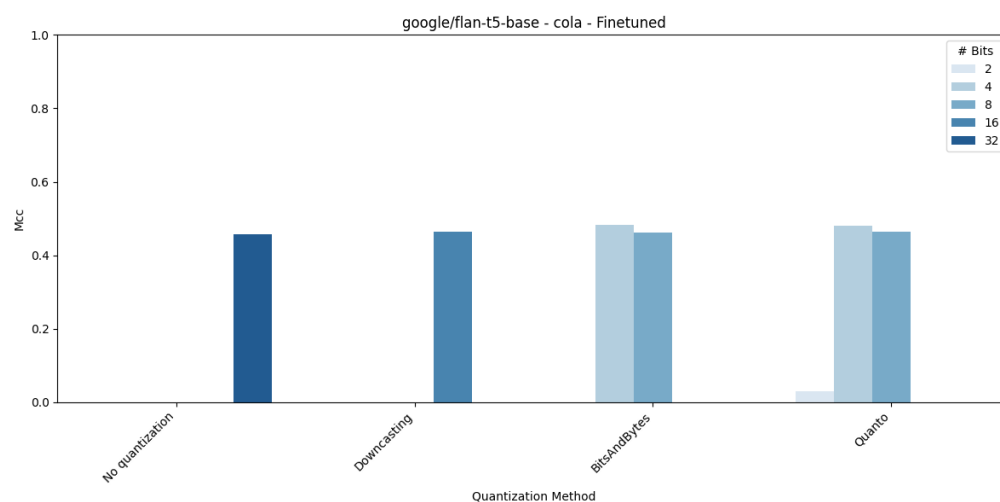


Figure E.94: Results of finetuned google/flan-t5-base on cola using Mcc

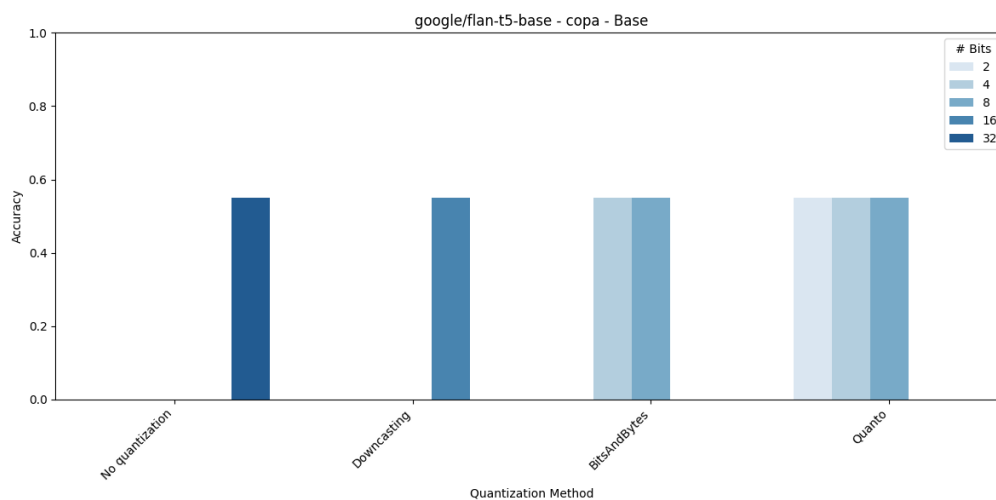


Figure E.95: Results of base google-flan-t5-base on copa using Accuracy

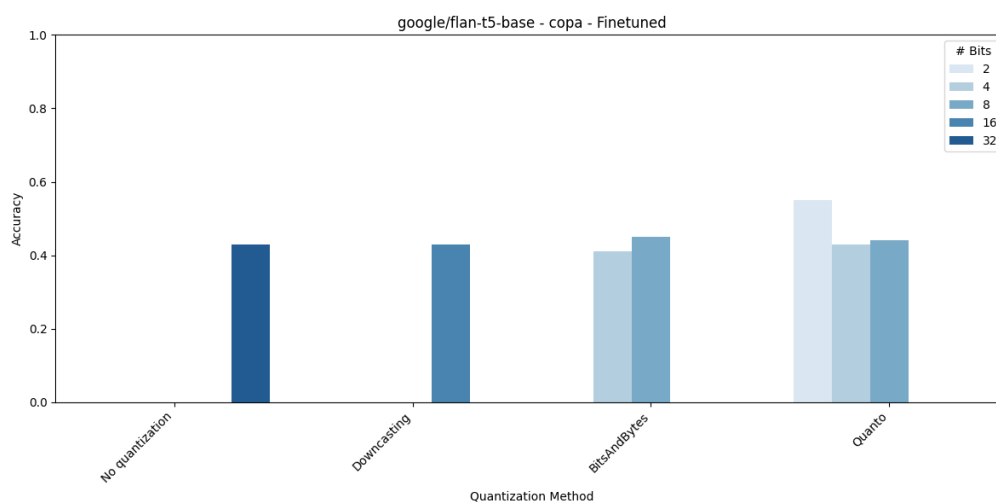


Figure E.96: Results of finetuned google-flan-t5-base on copa using Accuracy

E. DETAILED FIGURES

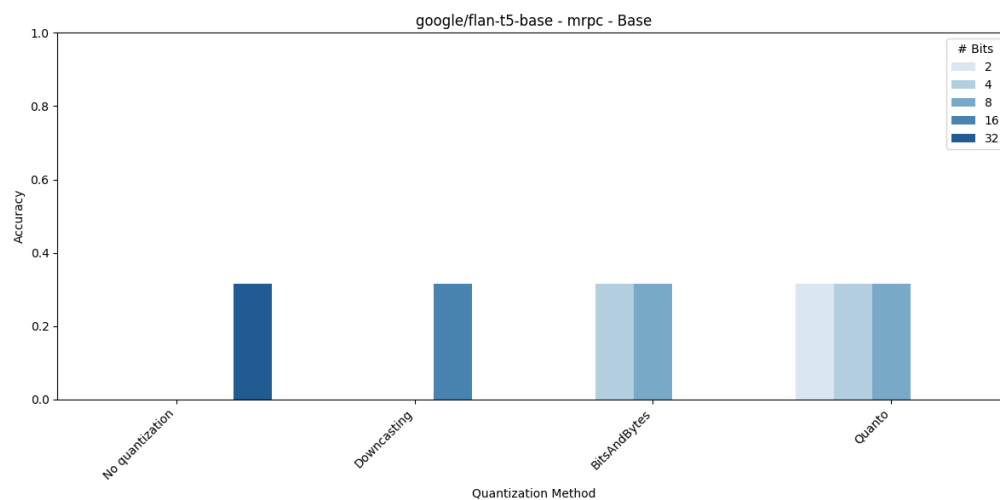


Figure E.97: Results of base google-flan-t5-base on mrpc using Accuracy

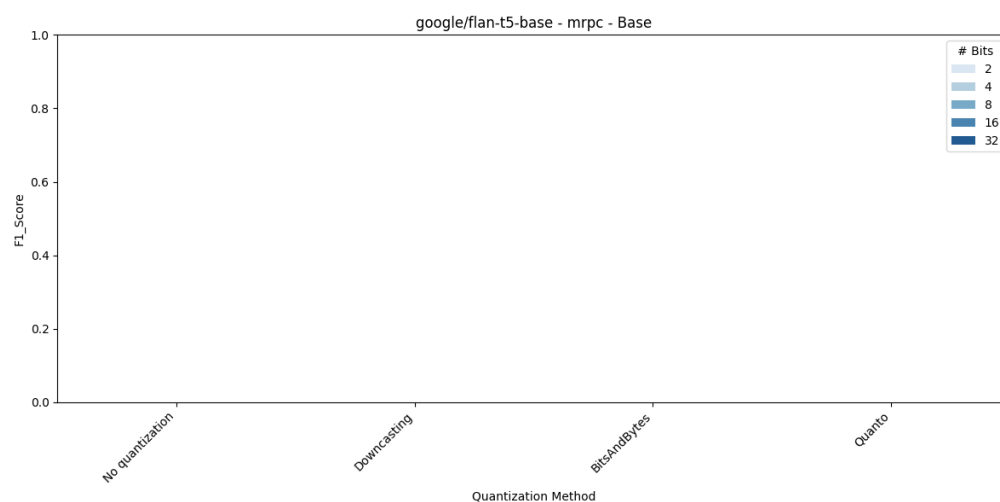


Figure E.98: Results of base google-flan-t5-base on mrpc using F1_Score

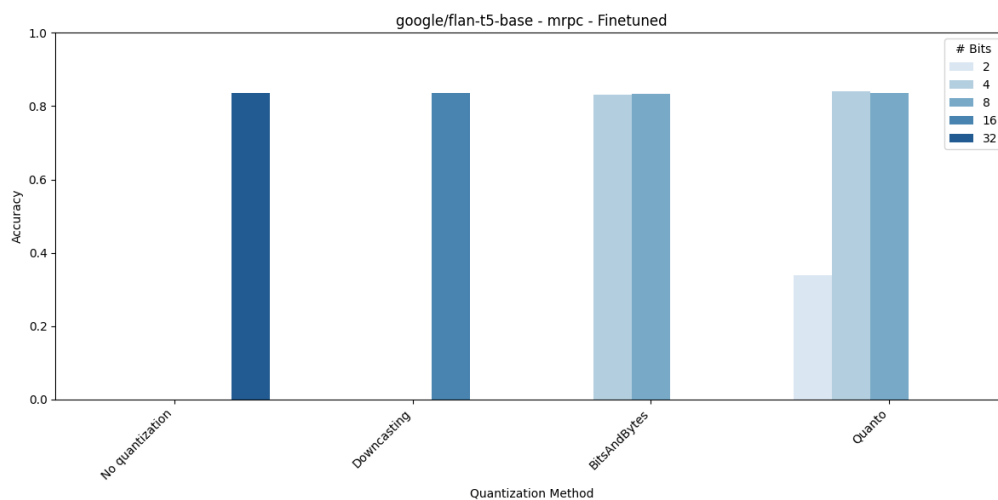


Figure E.99: Results of finetuned google-flan-t5-base on mrpc using Accuracy

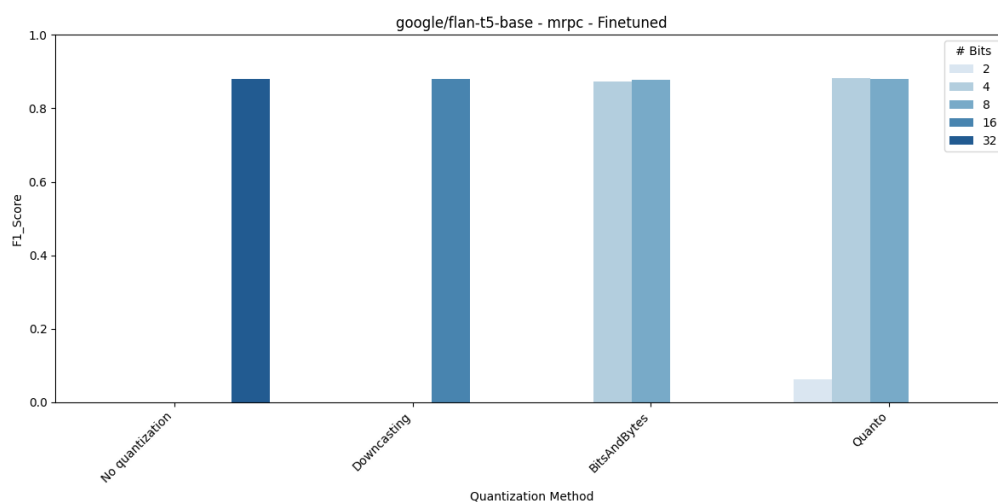


Figure E.100: Results of finetuned google-flan-t5-base on mrpc using F1_Score

E. DETAILED FIGURES

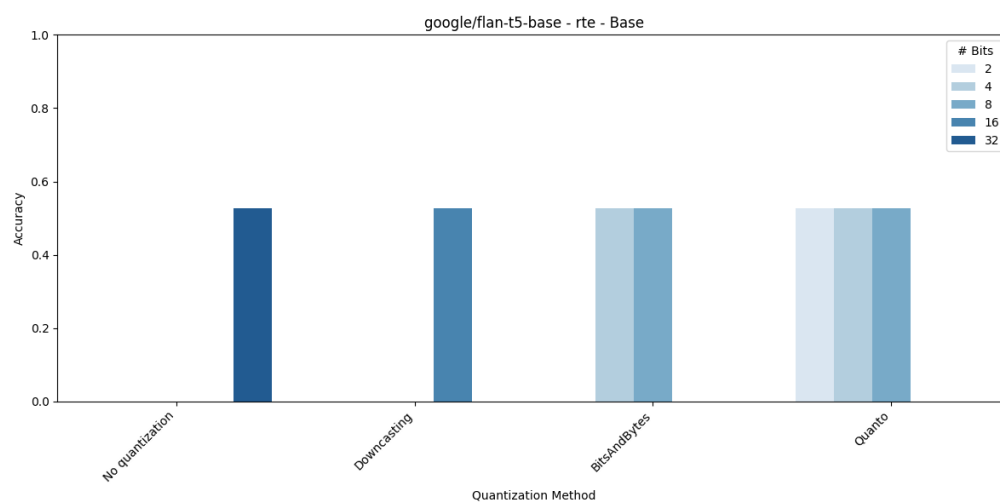


Figure E.101: Results of base google-flan-t5-base on rte using Accuracy

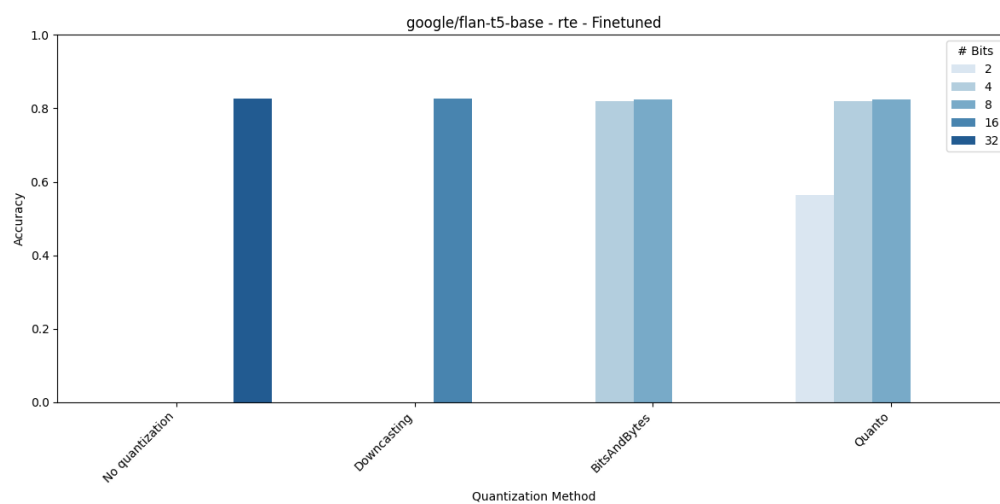


Figure E.102: Results of finetuned google-flan-t5-base on rte using Accuracy

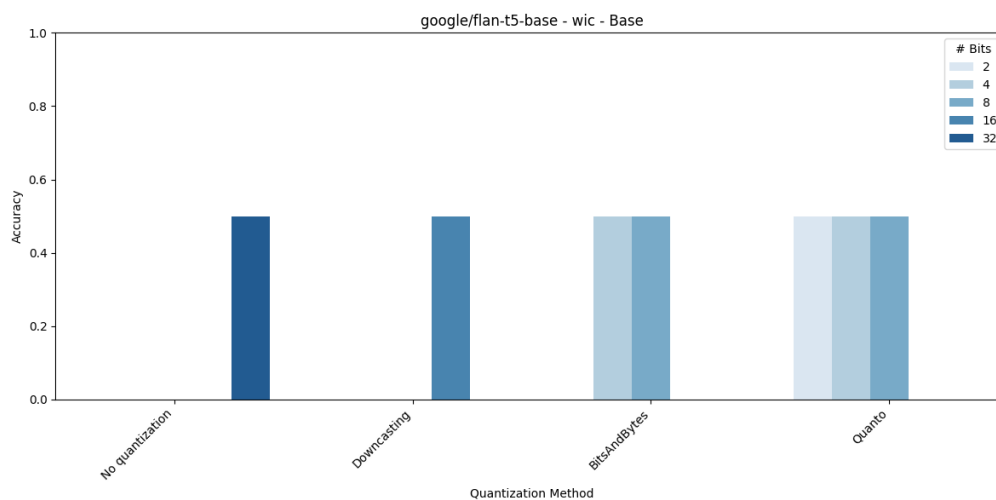


Figure E.103: Results of base google-flan-t5-base on wic using Accuracy

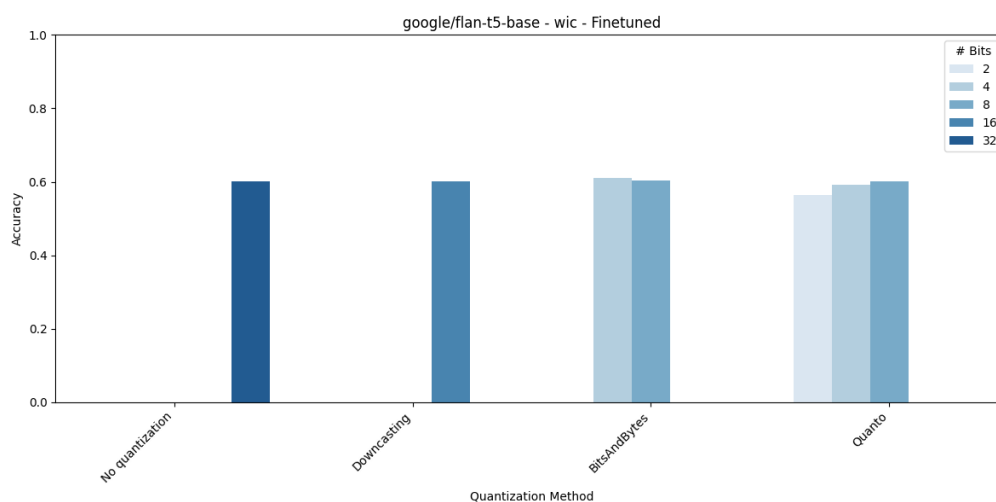


Figure E.104: Results of finetuned google-flan-t5-base on wic using Accuracy

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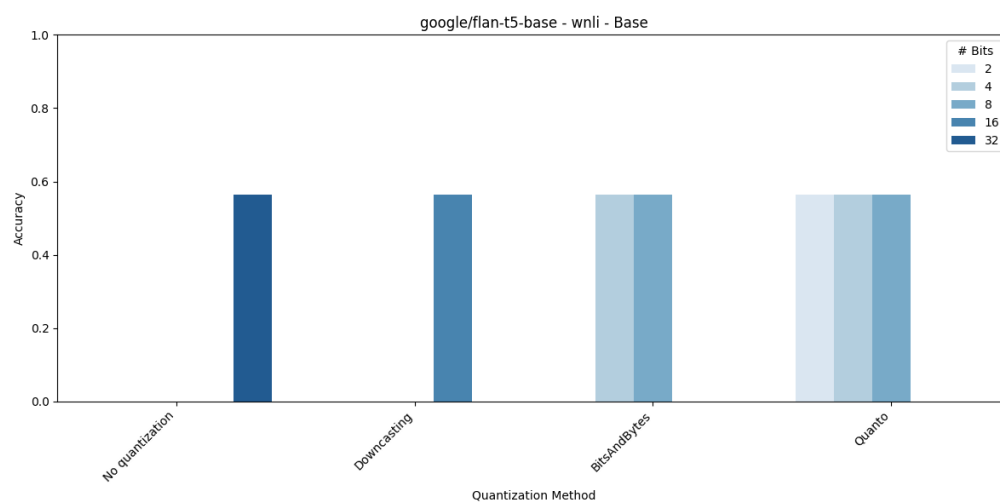


Figure E.105: Results of base google-flan-t5-base on wnli using Accuracy

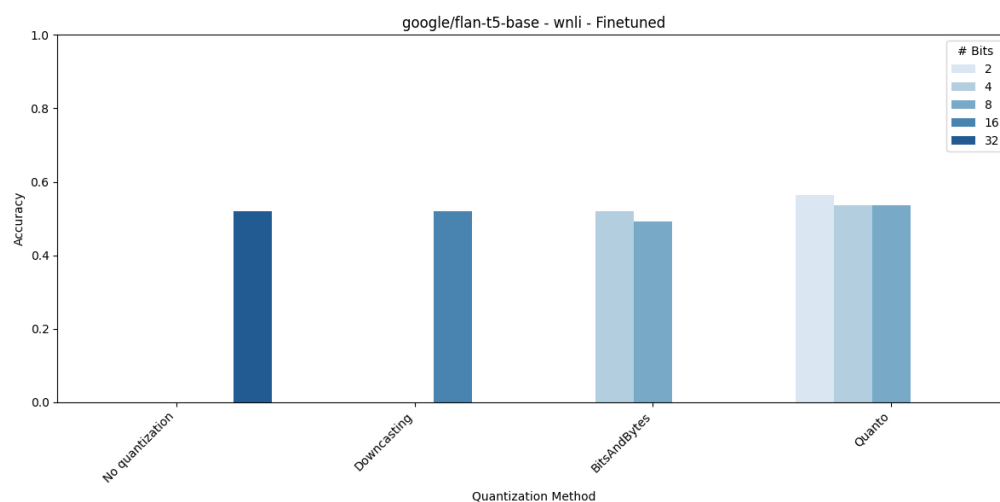


Figure E.106: Results of finetuned google-flan-t5-base on wnli using Accuracy

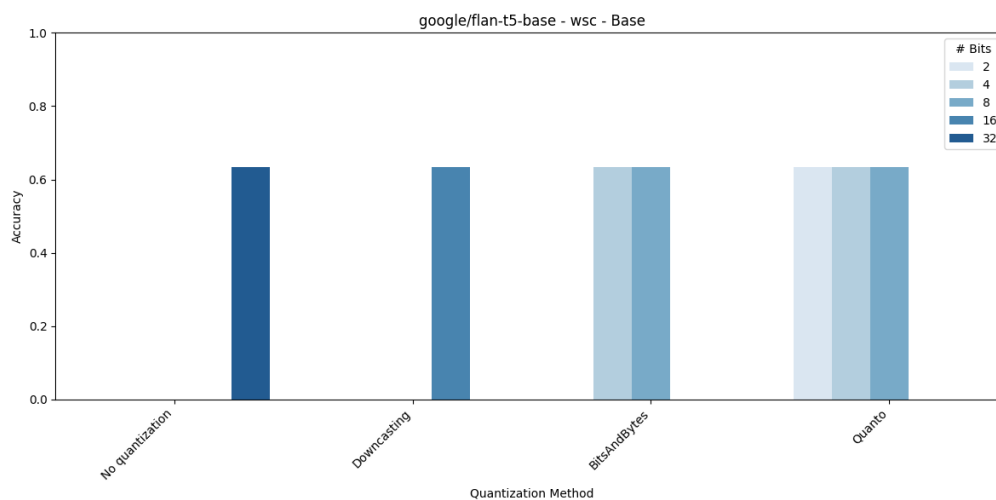


Figure E.107: Results of base google-flan-t5-base on wsc using Accuracy

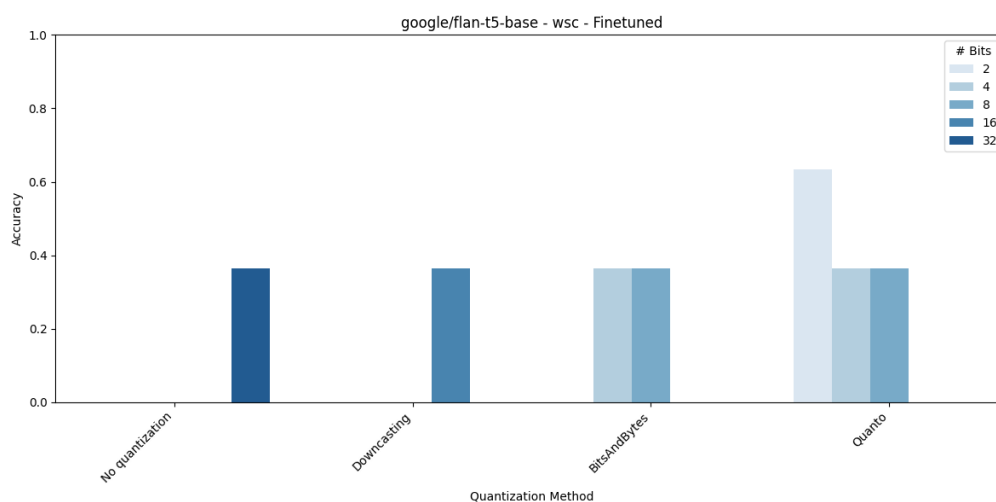


Figure E.108: Results of finetuned google-flan-t5-base on wsc using Accuracy

E. DETAILED FIGURES

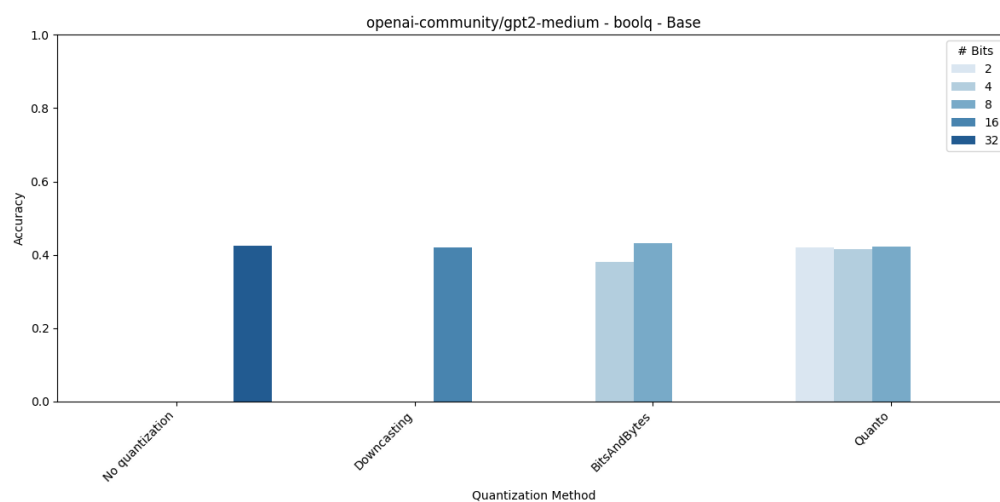


Figure E.109: Results of base openai-community-gpt2-medium on boolq using Accuracy

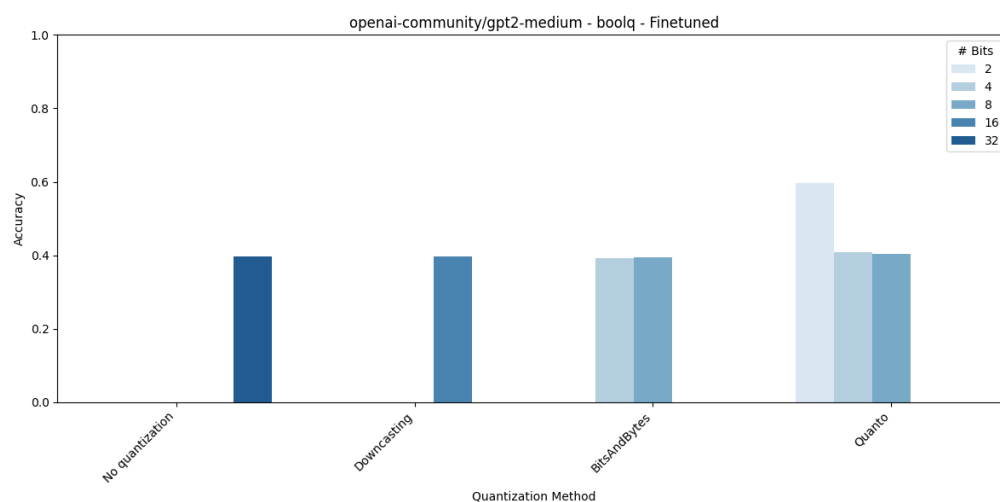


Figure E.110: Results of finetuned openai-community-gpt2-medium on boolq using Accuracy

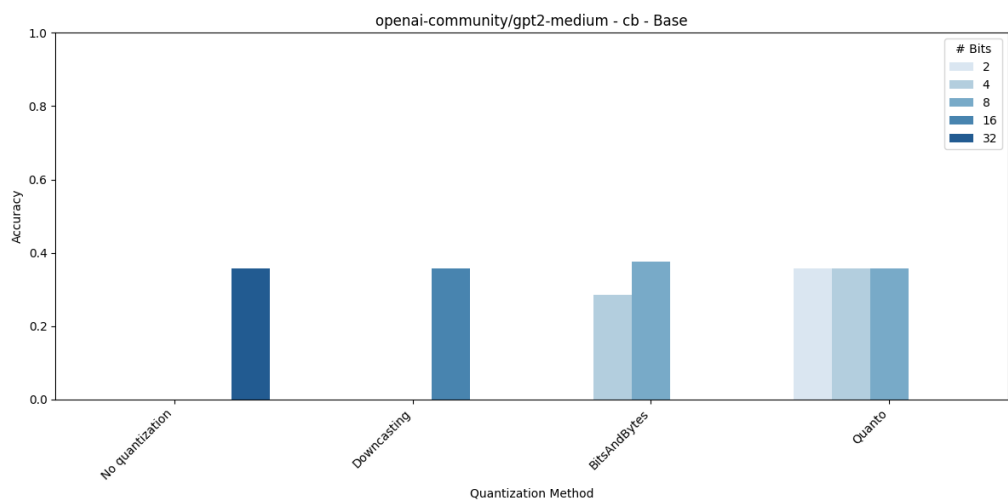


Figure E.111: Results of base openai-community-gpt2-medium on cb using Accuracy

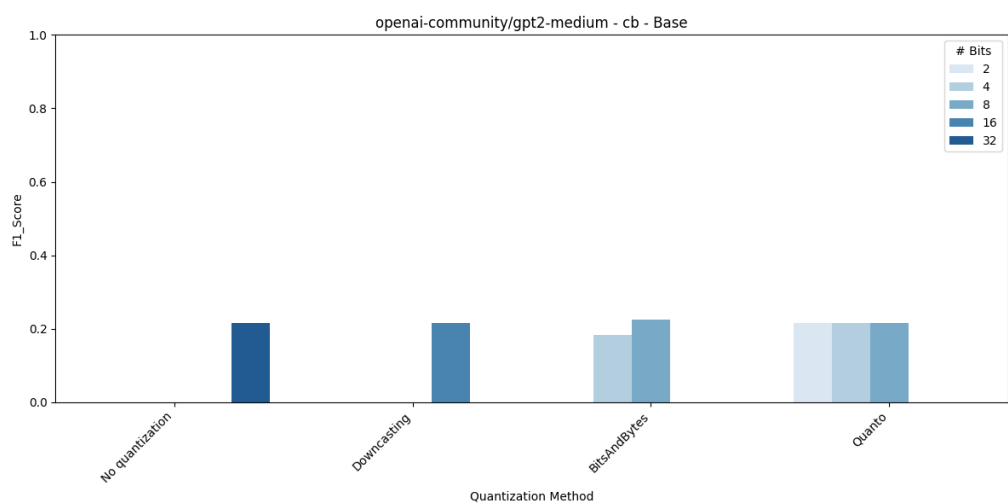


Figure E.112: Results of base openai-community-gpt2-medium on cb using F1_Score

E. DETAILED FIGURES

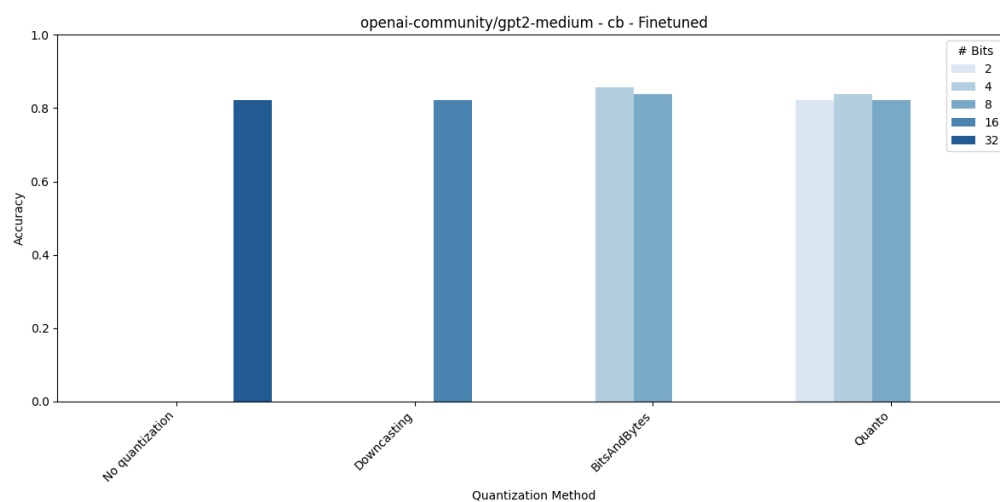


Figure E.113: Results of finetuned openai-community-gpt2-medium on cb using Accuracy

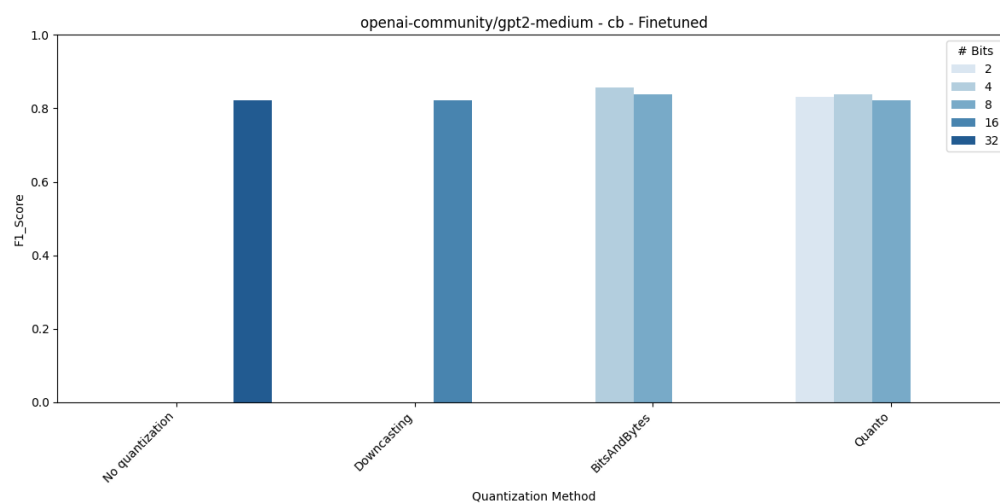


Figure E.114: Results of finetuned openai-community-gpt2-medium on cb using F1_Score

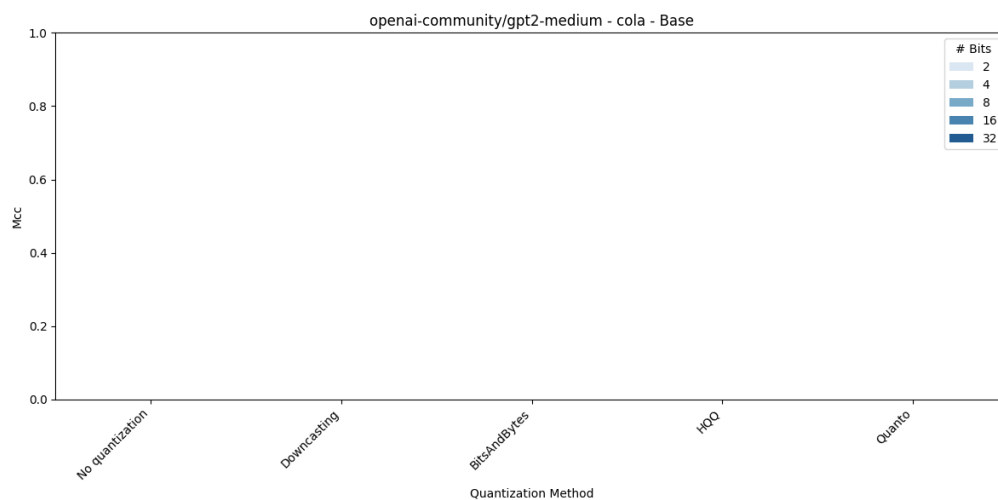


Figure E.115: Results of base openai-community-gpt2-medium on cola using Mcc

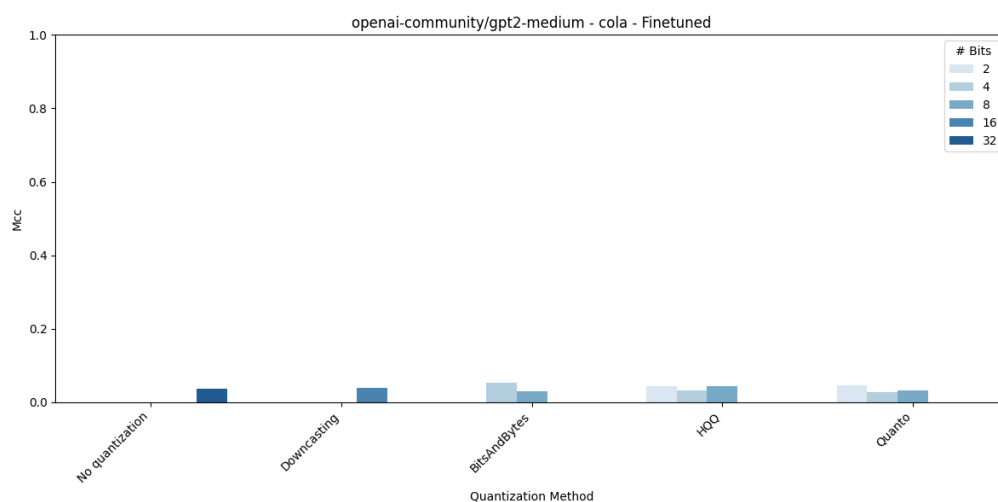


Figure E.116: Results of finetuned openai-community-gpt2-medium on cola using Mcc

E. DETAILED FIGURES

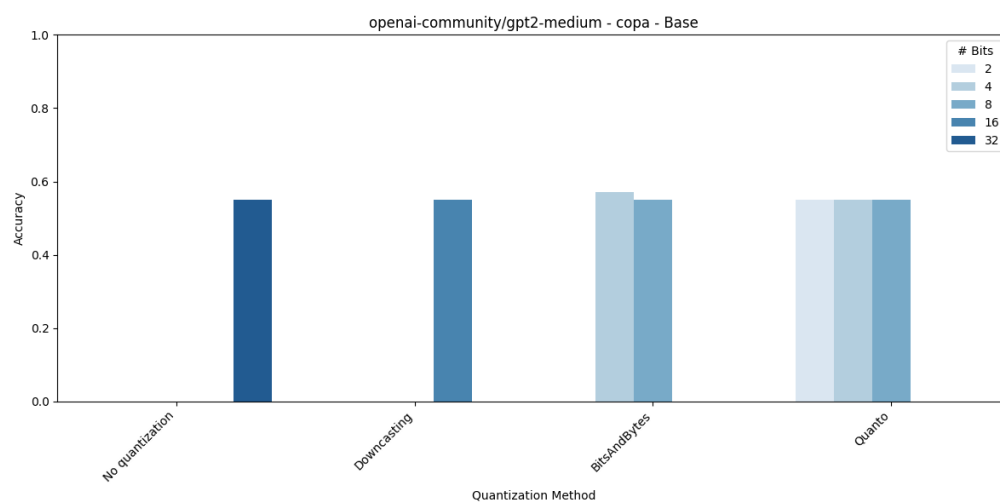


Figure E.117: Results of base openai-community-gpt2-medium on copa using Accuracy

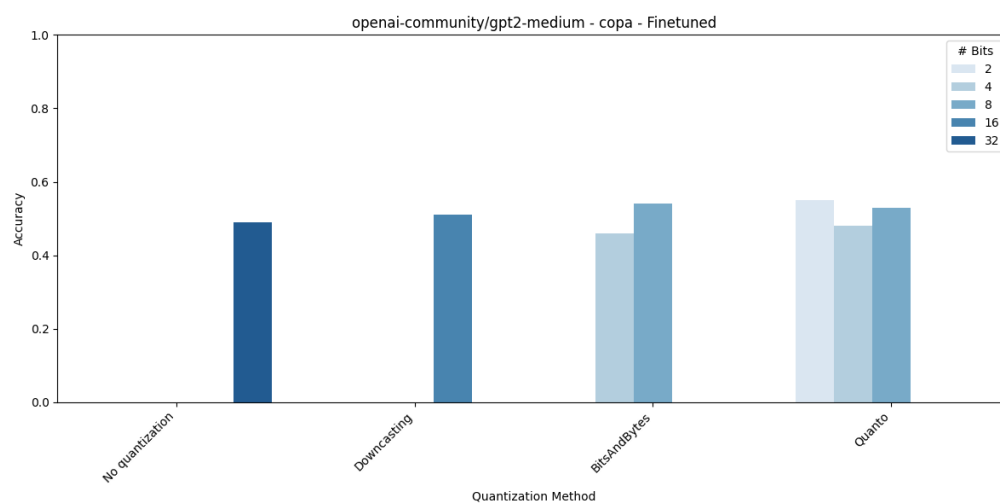


Figure E.118: Results of finetuned openai-community-gpt2-medium on copa using Accuracy

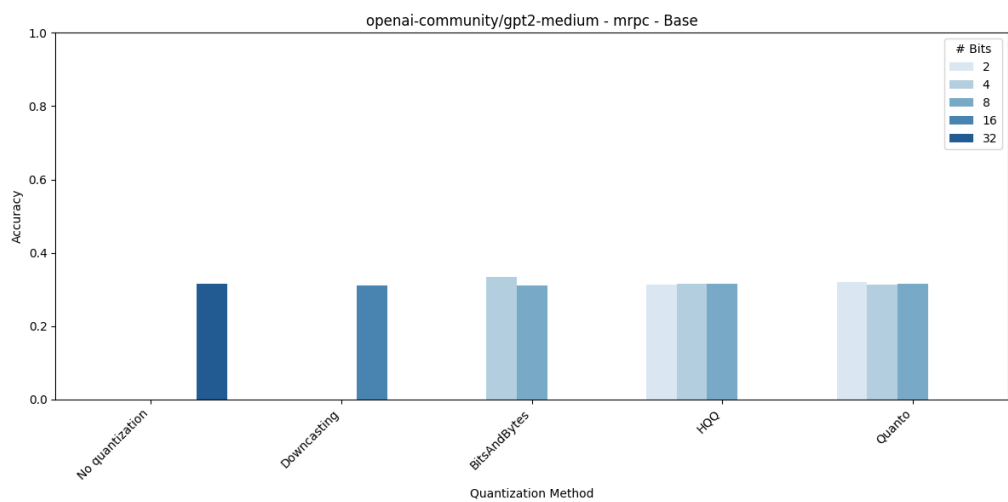


Figure E.119: Results of base openai-community-gpt2-medium on mrpc using Accuracy

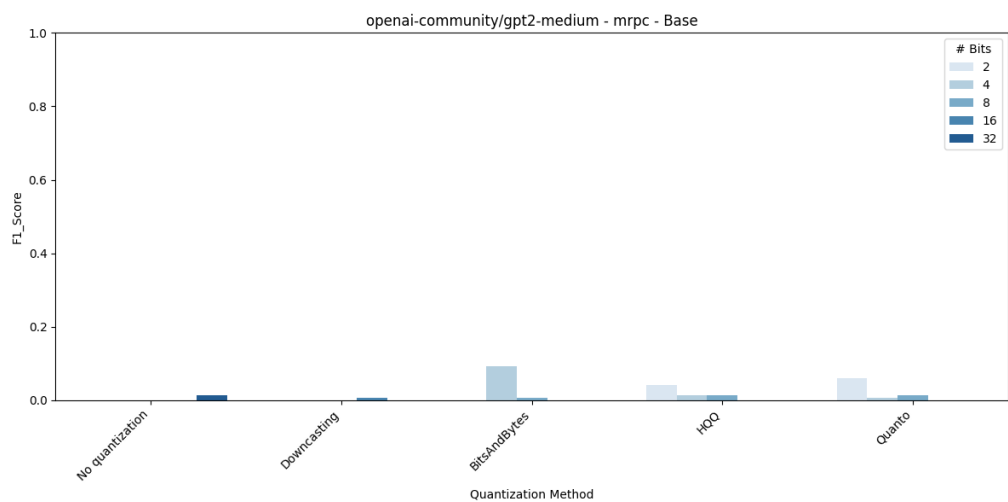


Figure E.120: Results of base openai-community-gpt2-medium on mrpc using F1_Score

E. DETAILED FIGURES

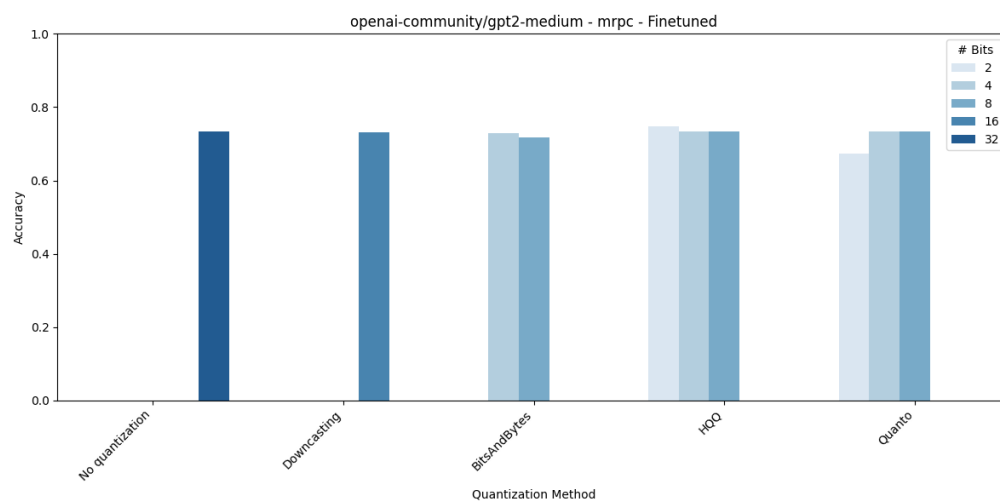


Figure E.121: Results of finetuned openai-community-gpt2-medium on mrpc using Accuracy

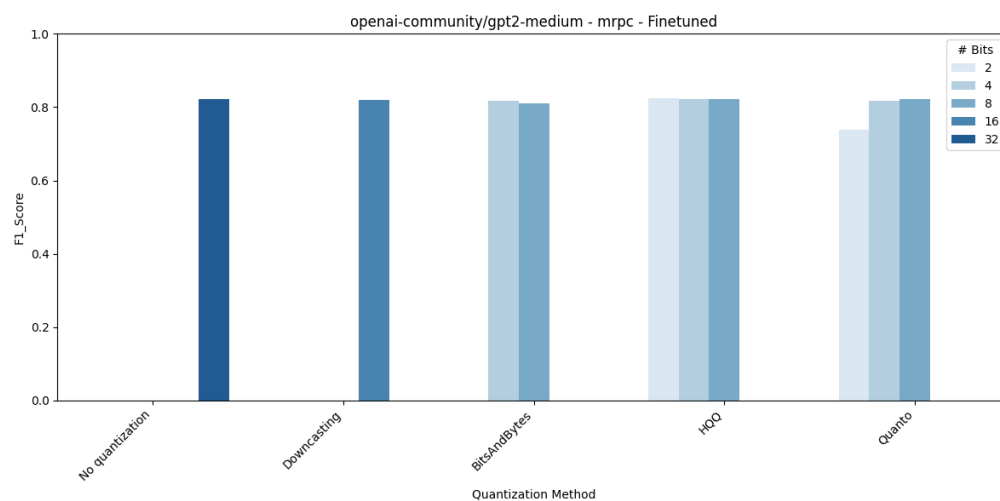


Figure E.122: Results of finetuned openai-community-gpt2-medium on mrpc using F1_Score

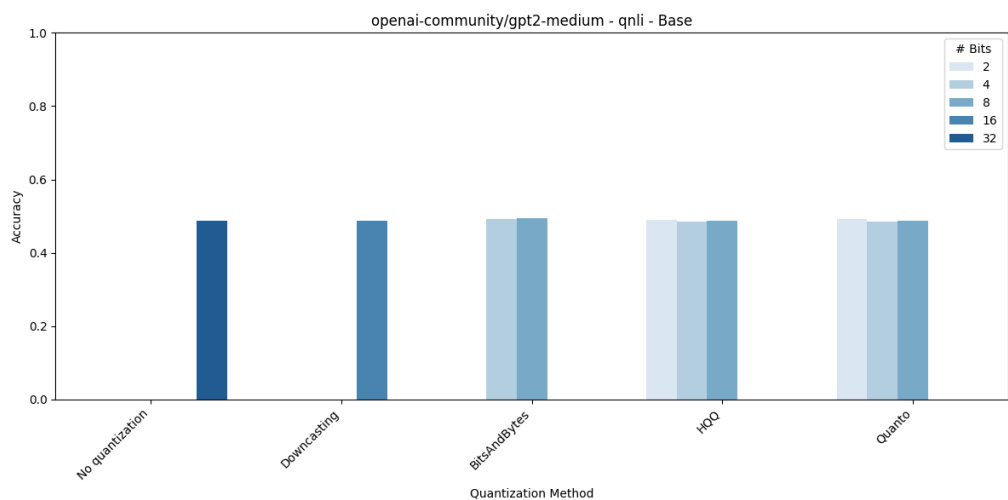


Figure E.123: Results of base openai-community-gpt2-medium on qnli using Accuracy

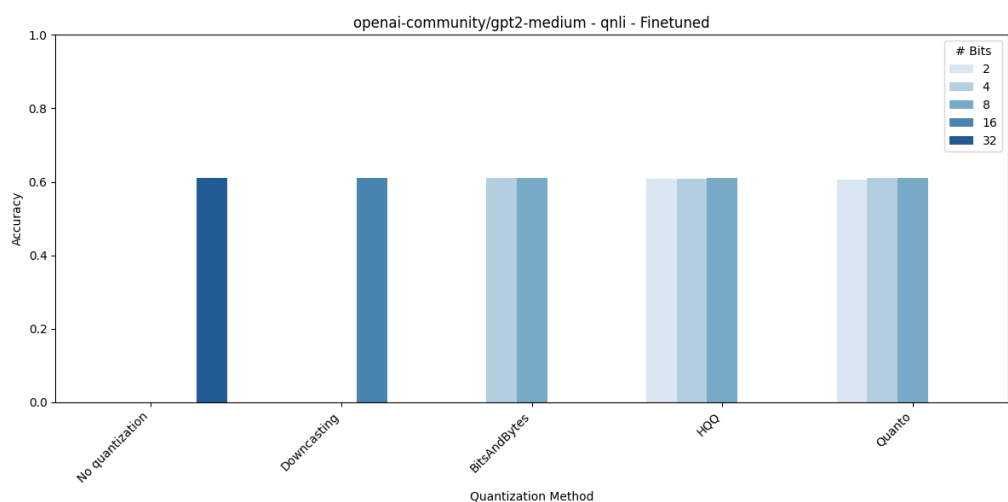


Figure E.124: Results of finetuned openai-community-gpt2-medium on qnli using Accuracy

E. DETAILED FIGURES

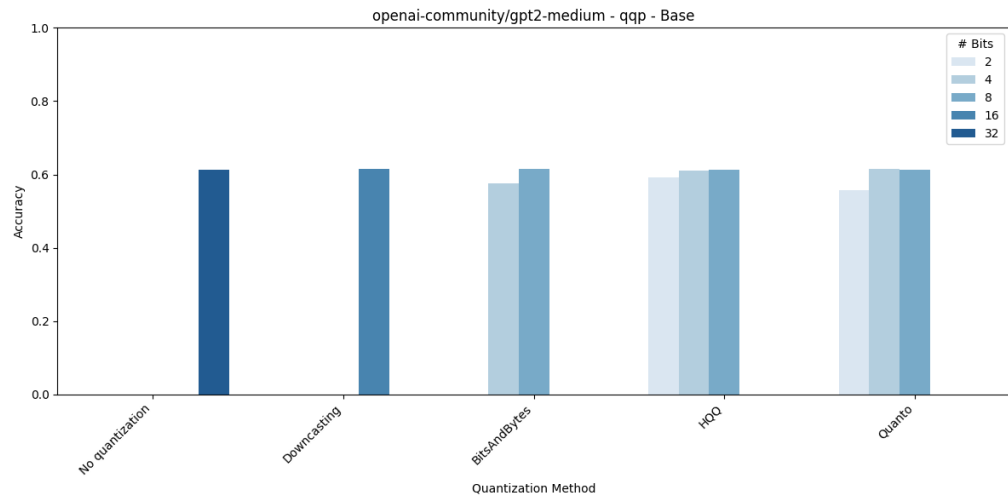


Figure E.125: Results of base openai-community-gpt2-medium on qqp using Accuracy

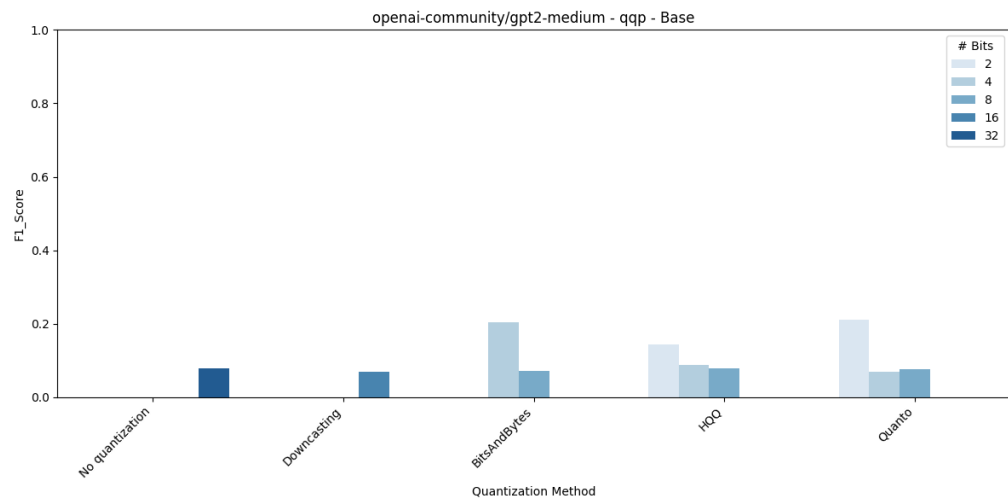


Figure E.126: Results of base openai-community-gpt2-medium on qqp using F1_Score

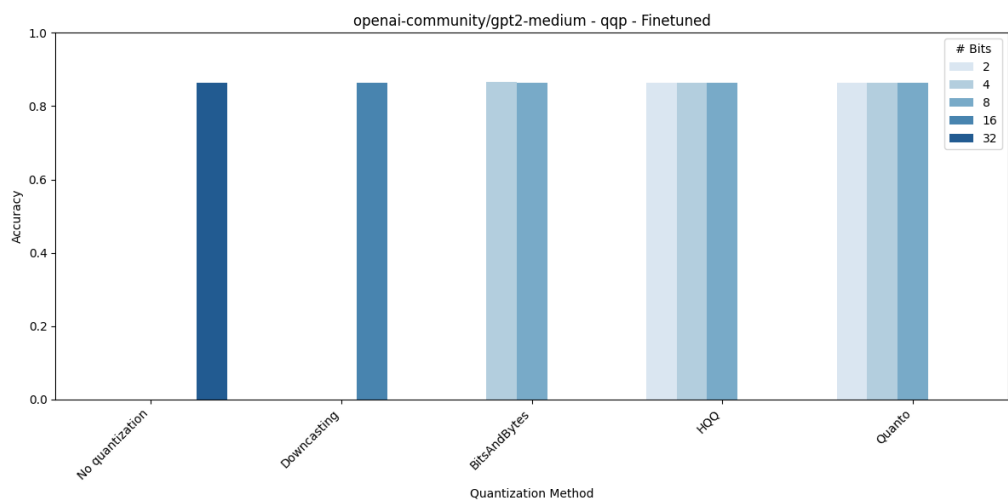


Figure E.127: Results of finetuned openai-community-gpt2-medium on qqp using Accuracy

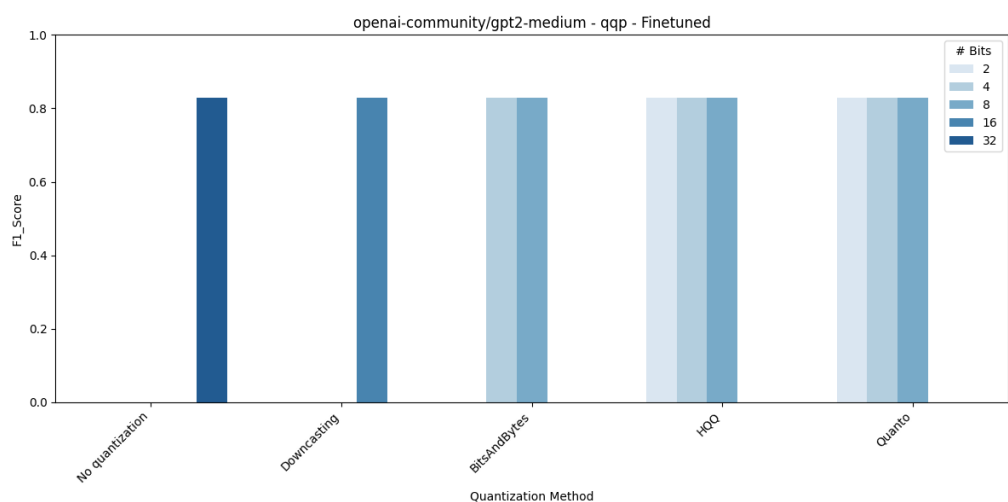


Figure E.128: Results of finetuned openai-community-gpt2-medium on qqp using F1_Score

E. DETAILED FIGURES

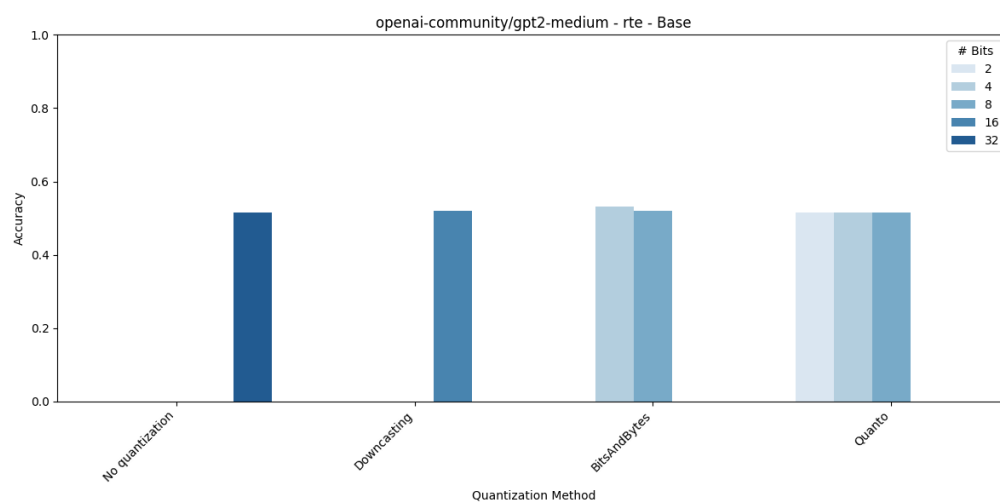


Figure E.129: Results of base openai-community-gpt2-medium on rte using Accuracy

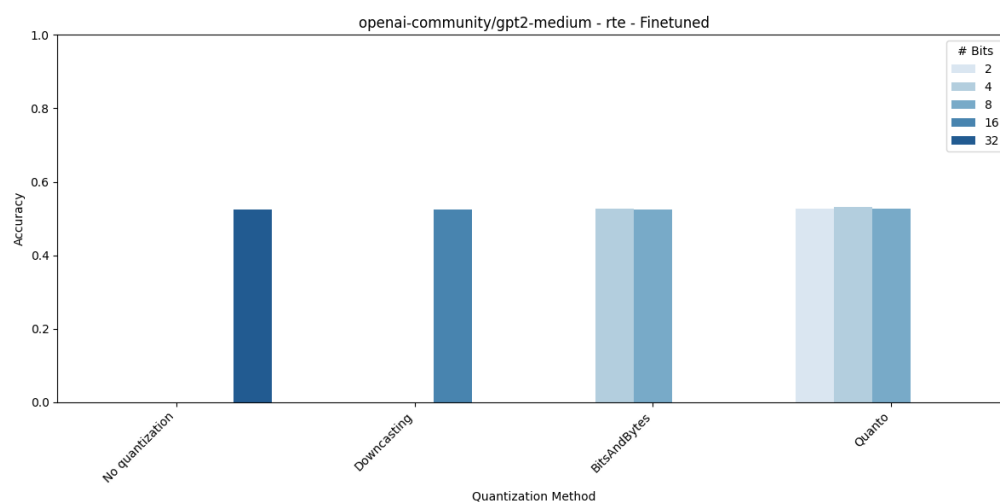


Figure E.130: Results of finetuned openai-community-gpt2-medium on rte using Accuracy

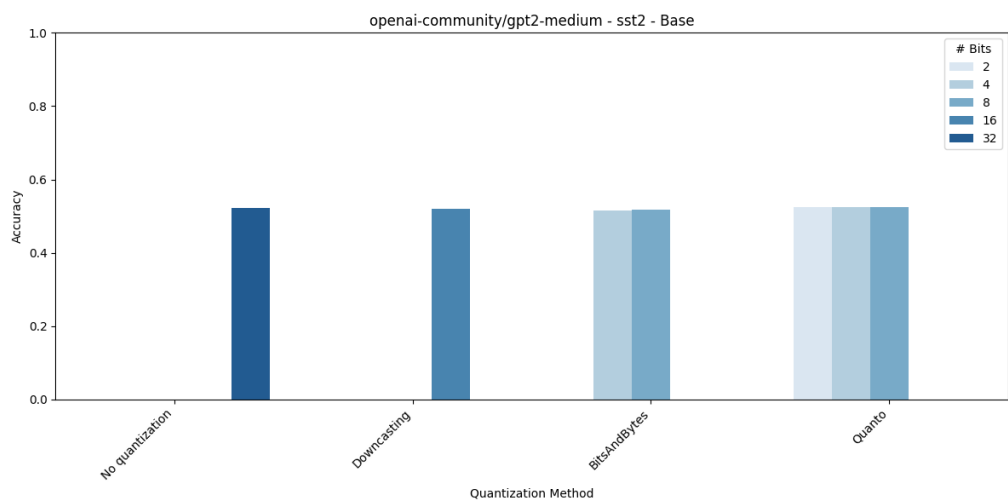


Figure E.131: Results of base openai-community-gpt2-medium on sst2 using Accuracy

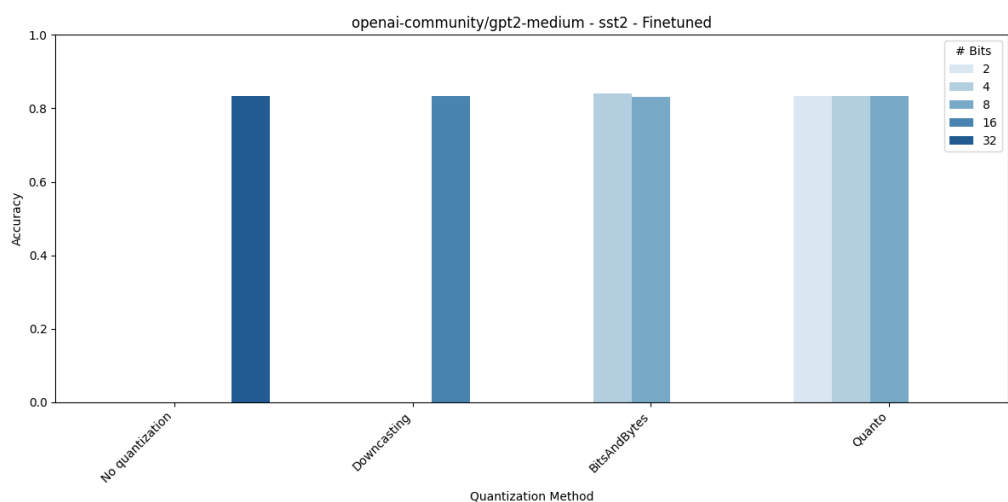


Figure E.132: Results of finetuned openai-community-gpt2-medium on sst2 using Accuracy

E. DETAILED FIGURES

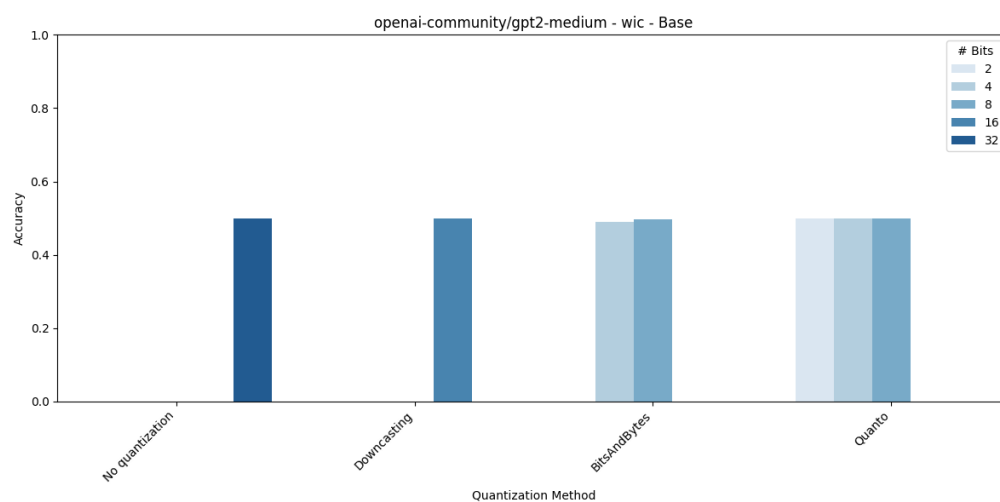


Figure E.133: Results of base openai-community-gpt2-medium on wic using Accuracy

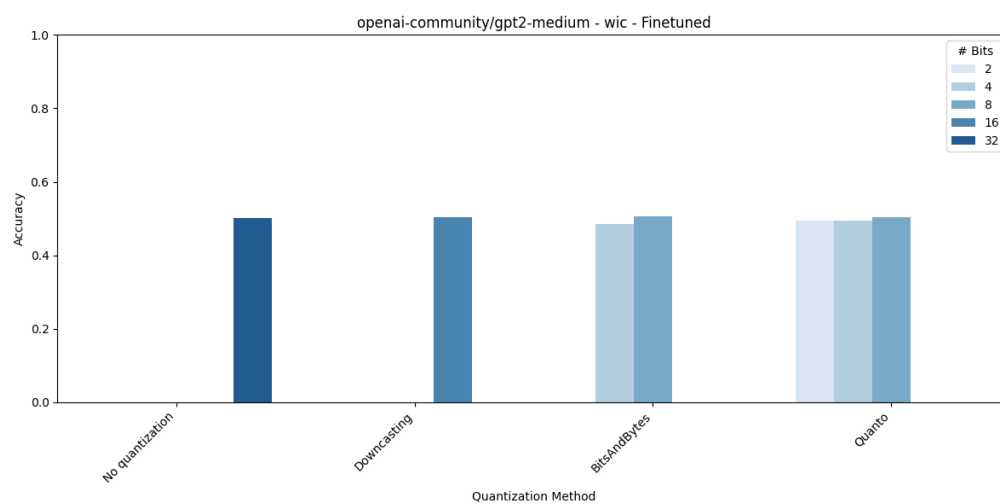


Figure E.134: Results of finetuned openai-community-gpt2-medium on wic using Accuracy

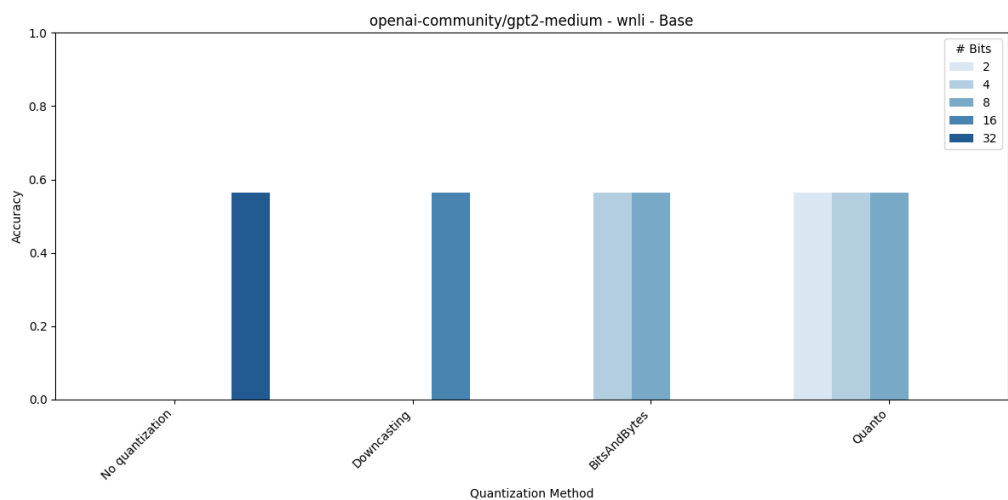


Figure E.135: Results of base openai-community-gpt2-medium on wnli using Accuracy

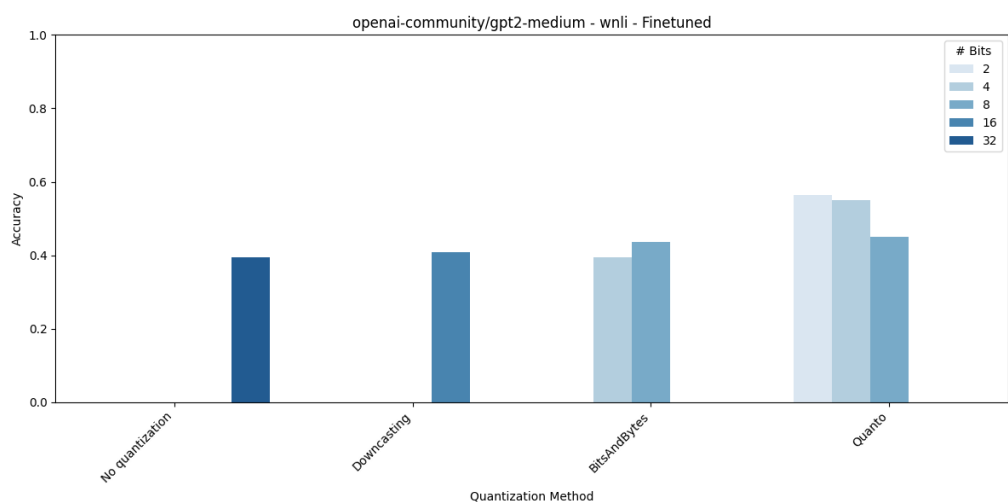


Figure E.136: Results of finetuned openai-community-gpt2-medium on wnli using Accuracy

E. DETAILED FIGURES

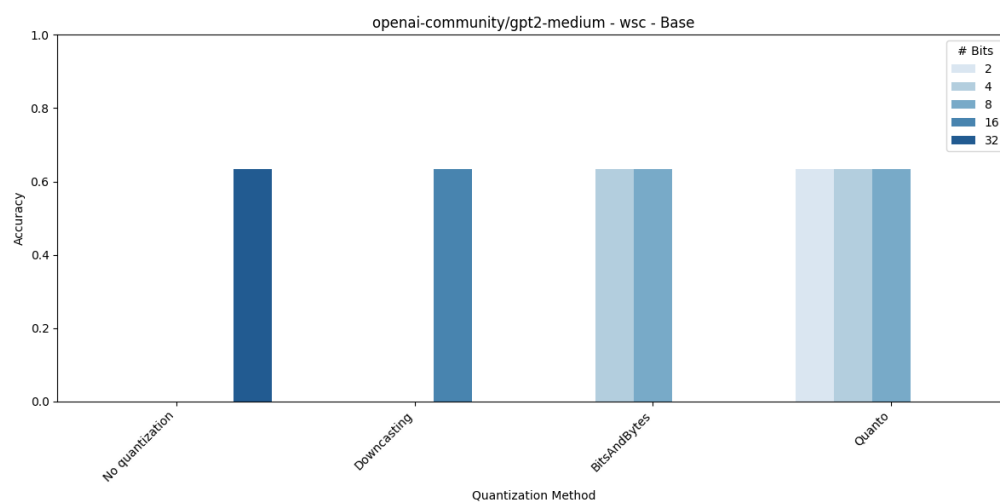


Figure E.137: Results of base openai-community-gpt2-medium on wsc using Accuracy

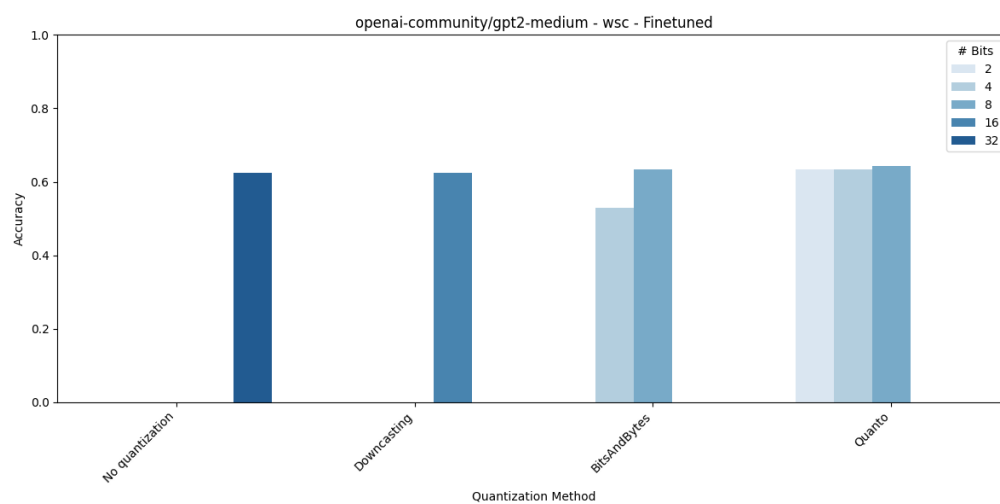


Figure E.138: Results of finetuned openai-community-gpt2-medium on wsc using Accuracy

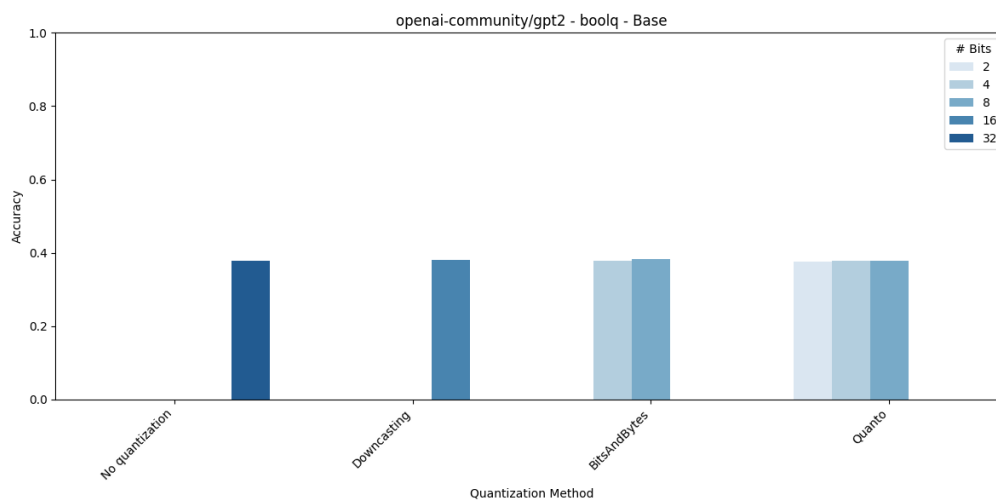


Figure E.139: Results of base openai-community-gpt2 on boolq using Accuracy

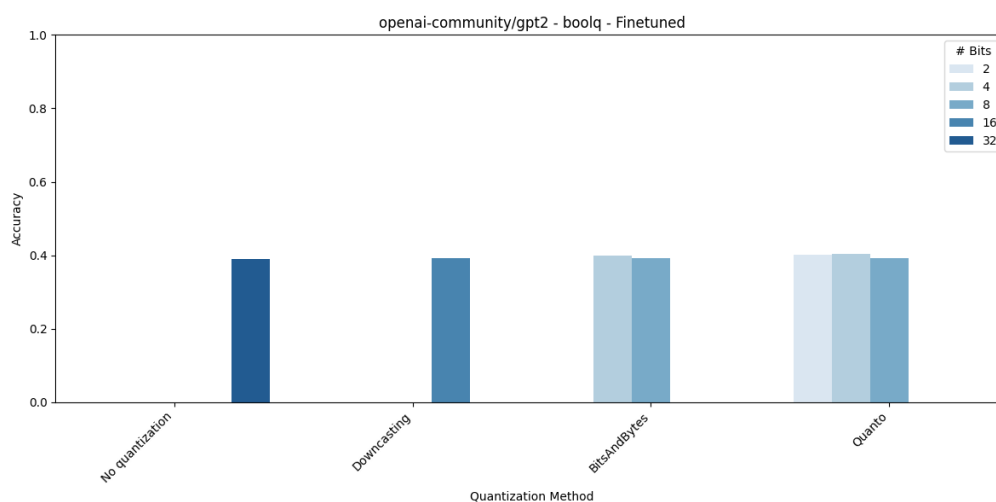


Figure E.140: Results of finetuned openai-community-gpt2 on boolq using Accuracy

E. DETAILED FIGURES

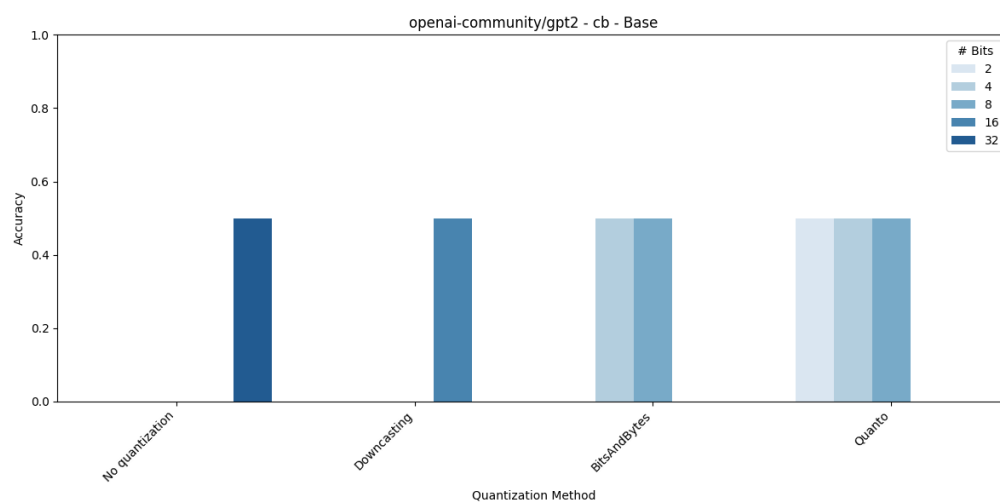


Figure E.141: Results of base openai-community-gpt2 on cb using Accuracy

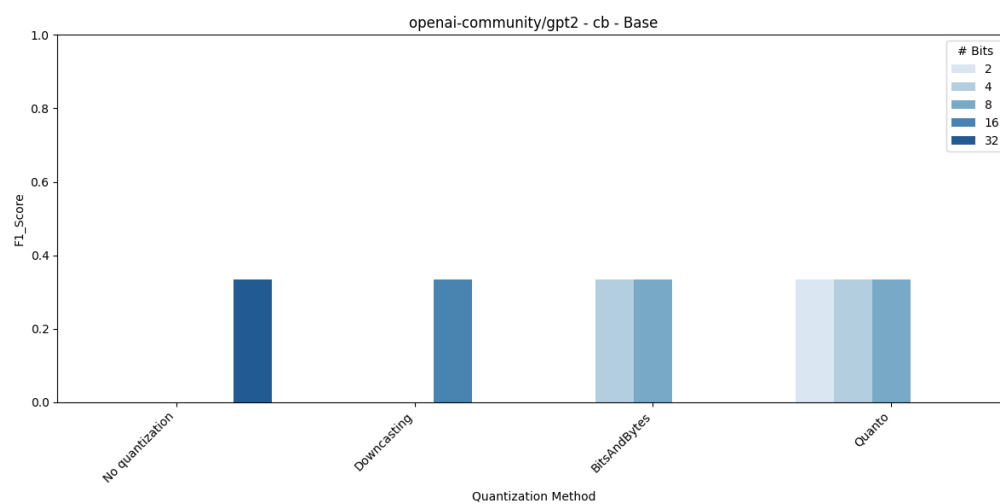


Figure E.142: Results of base openai-community-gpt2 on cb using F1_Score

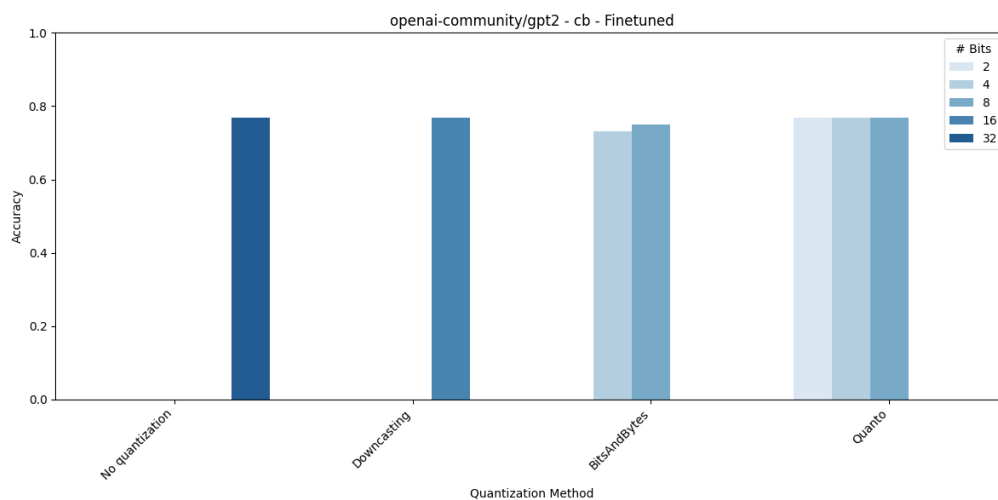


Figure E.143: Results of finetuned openai-community-gpt2 on cb using Accuracy

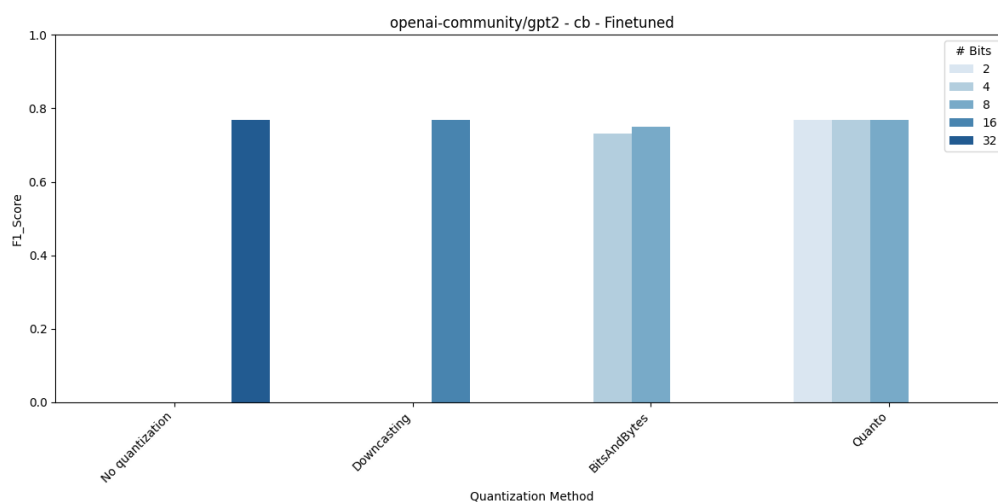


Figure E.144: Results of finetuned openai-community-gpt2 on cb using F1_Score

E. DETAILED FIGURES

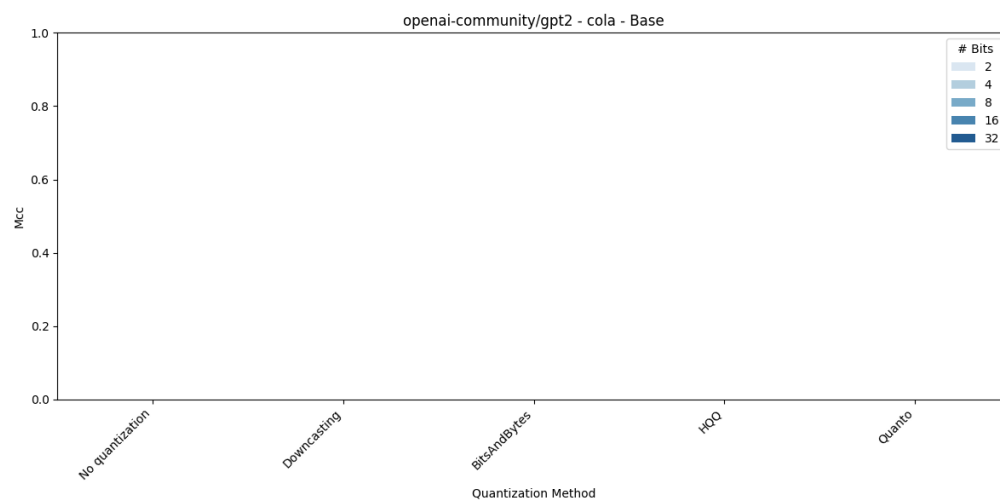


Figure E.145: Results of base openai-community-gpt2 on cola using Mcc

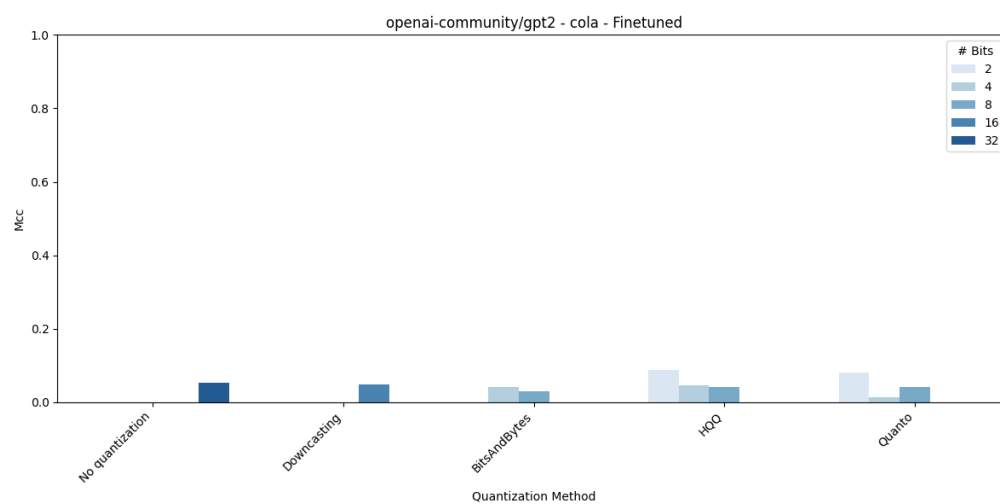


Figure E.146: Results of finetuned openai-community-gpt2 on cola using Mcc

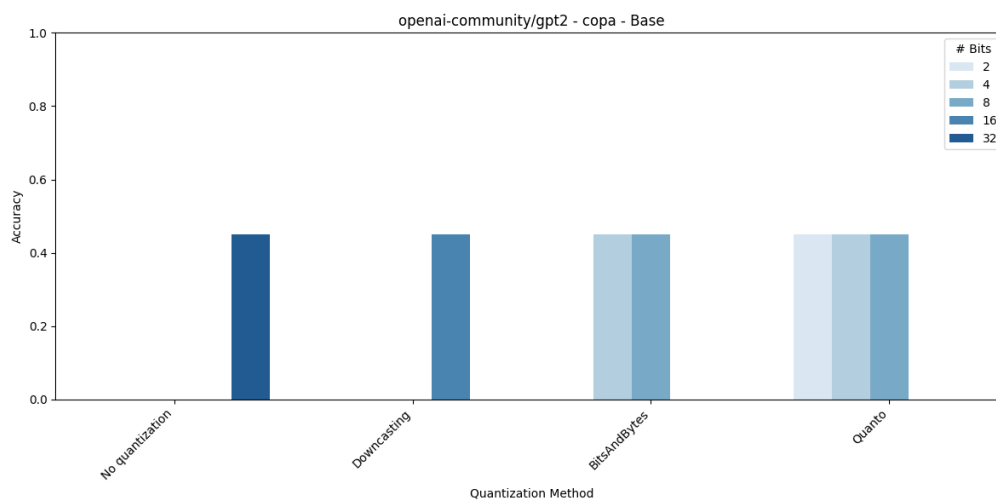


Figure E.147: Results of base openai-community-gpt2 on copa using Accuracy

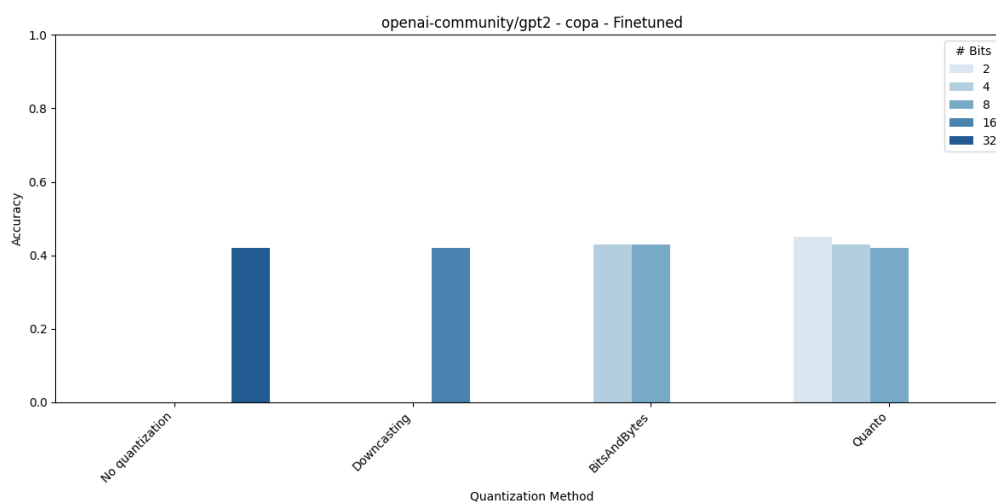


Figure E.148: Results of finetuned openai-community-gpt2 on copa using Accuracy

E. DETAILED FIGURES

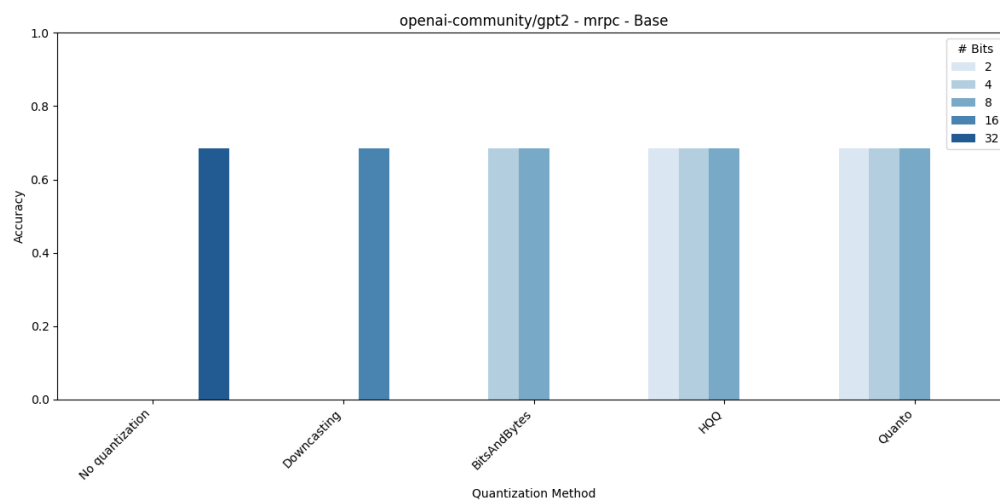


Figure E.149: Results of base openai-community-gpt2 on mrpc using Accuracy

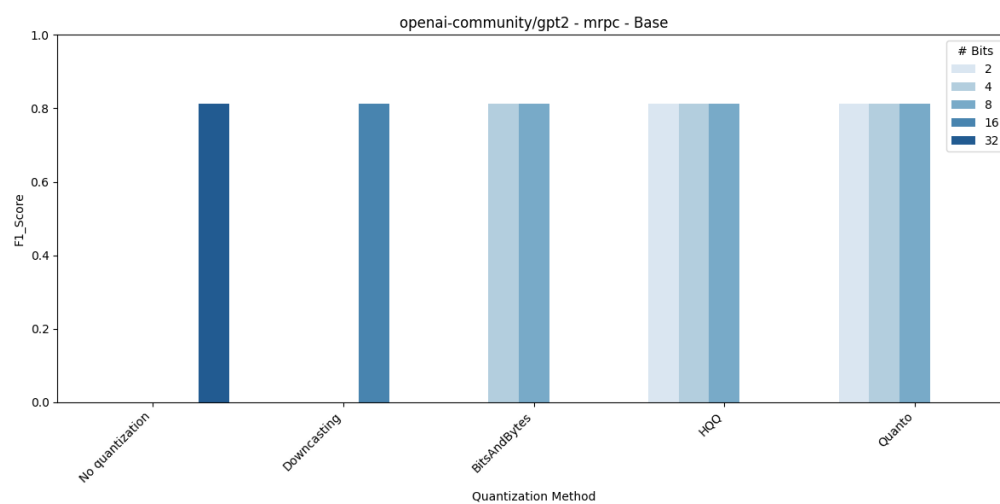


Figure E.150: Results of base openai-community-gpt2 on mrpc using F1_Score

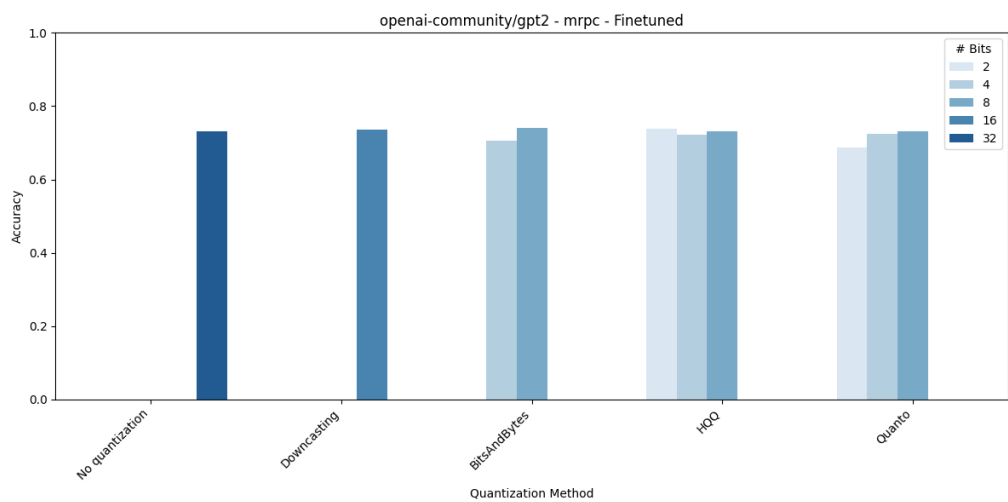


Figure E.151: Results of finetuned openai-community-gpt2 on mrpc using Accuracy

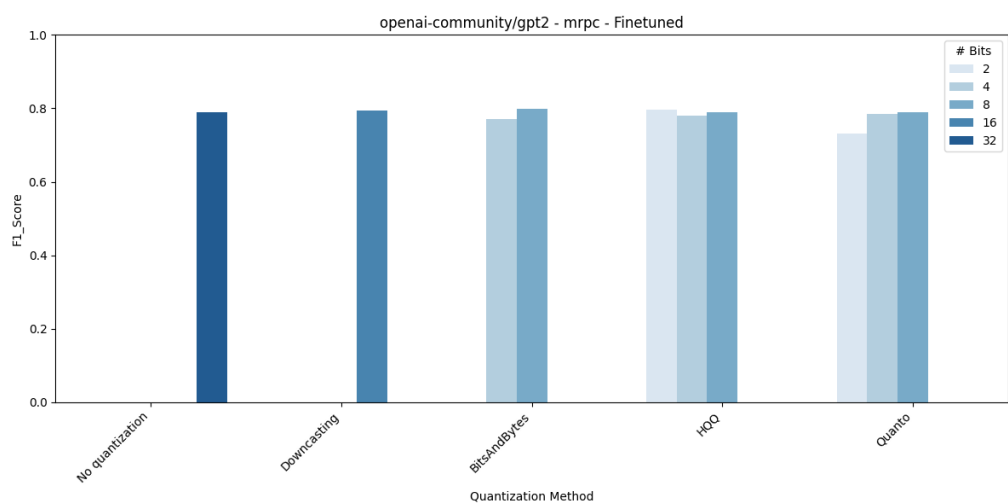


Figure E.152: Results of finetuned openai-community-gpt2 on mrpc using F1_Score

E. DETAILED FIGURES

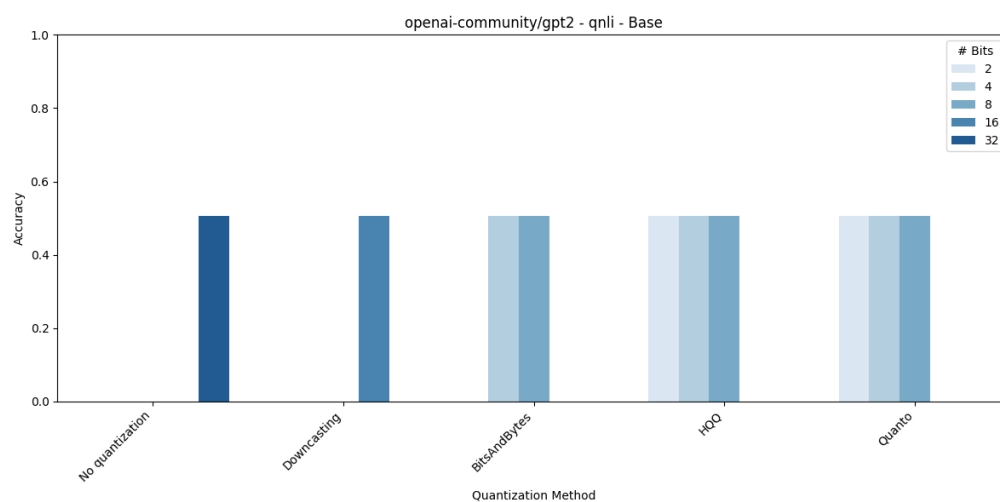


Figure E.153: Results of base openai-community-gpt2 on qnli using Accuracy

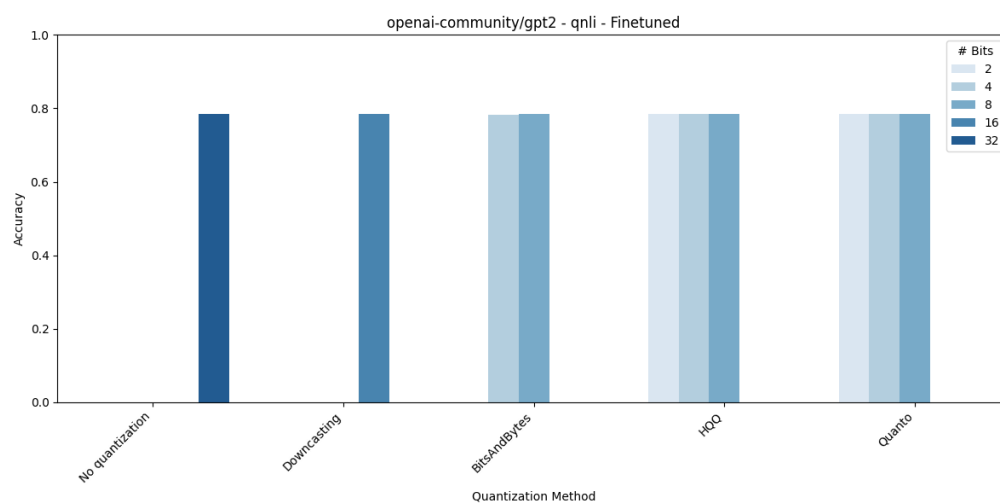


Figure E.154: Results of finetuned openai-community-gpt2 on qnli using Accuracy

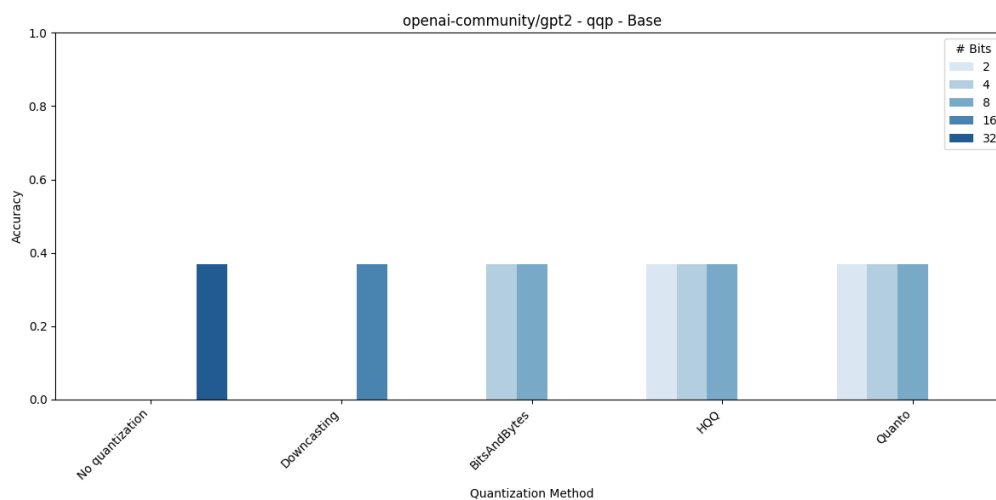


Figure E.155: Results of base openai-community-gpt2 on qqp using Accuracy

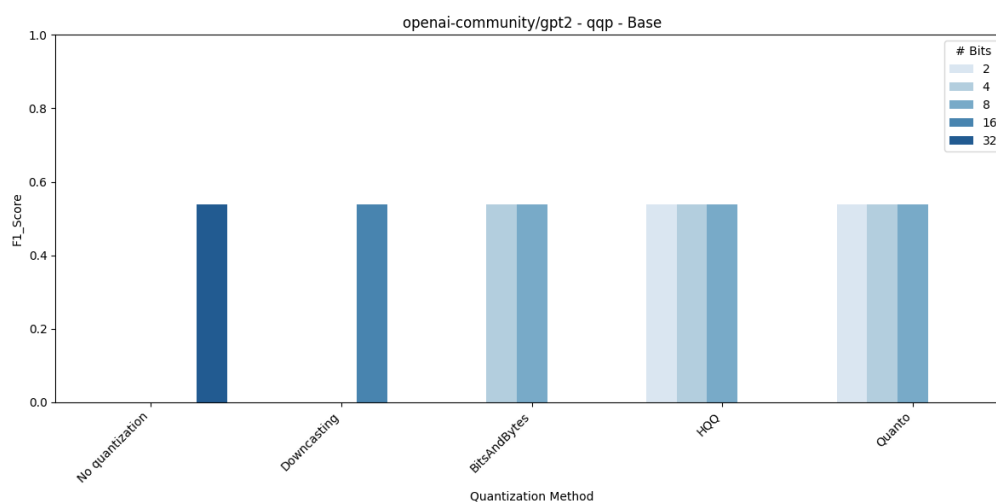


Figure E.156: Results of base openai-community-gpt2 on qqp using F1_Score

E. DETAILED FIGURES

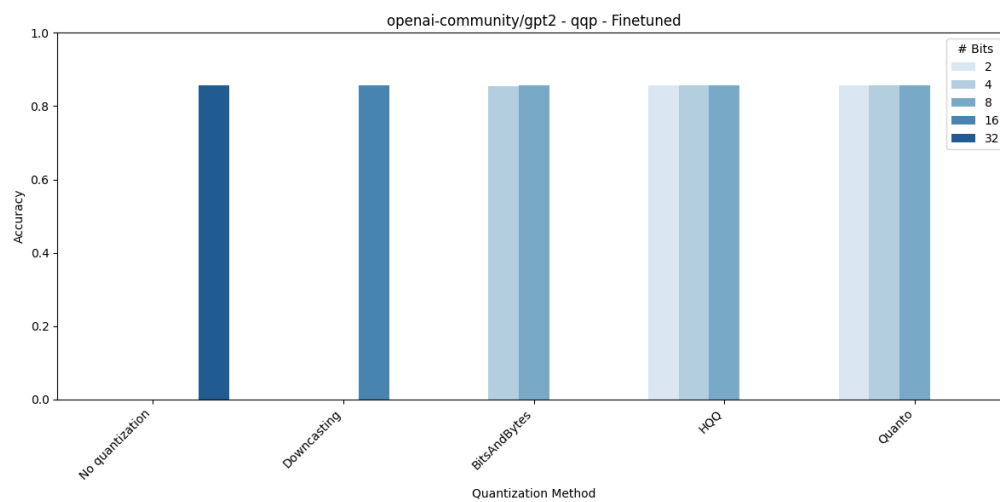


Figure E.157: Results of finetuned openai-community-gpt2 on qqp using Accuracy

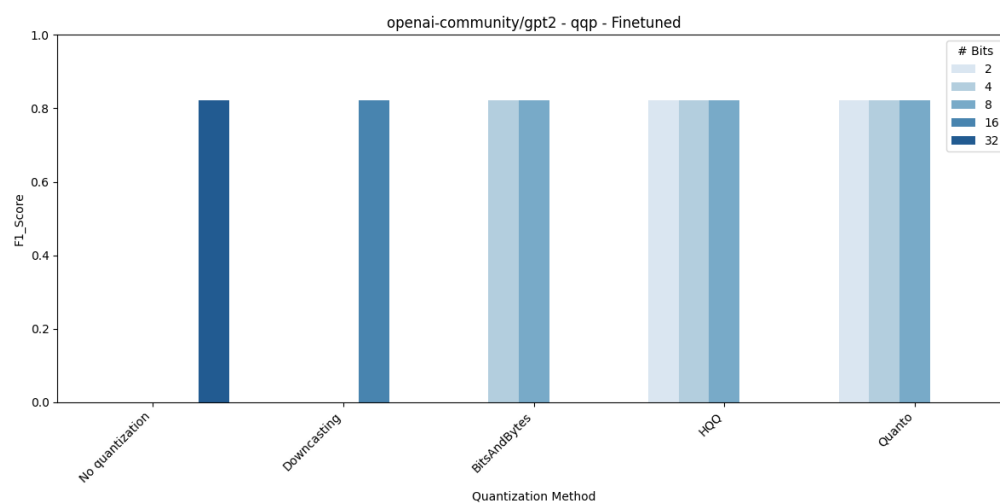


Figure E.158: Results of finetuned openai-community-gpt2 on qqp using F1_Score

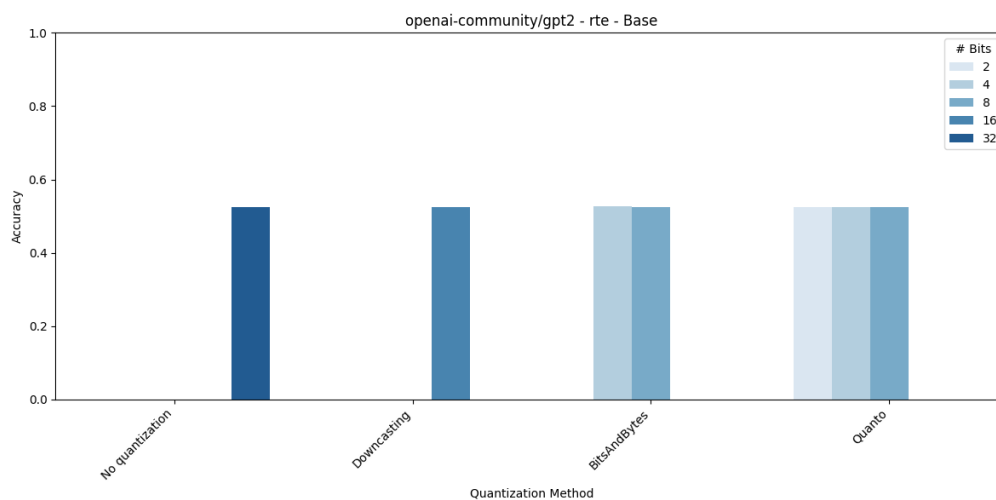


Figure E.159: Results of base openai-community-gpt2 on rte using Accuracy

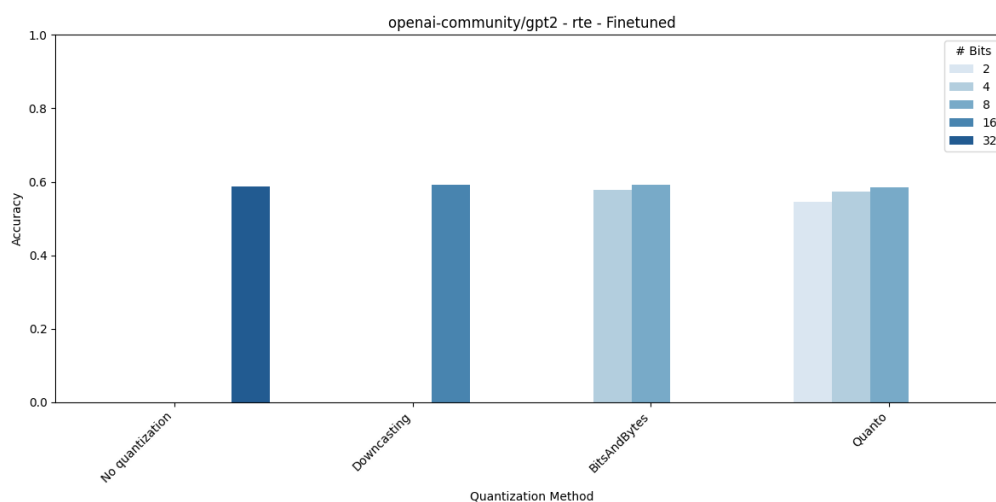


Figure E.160: Results of finetuned openai-community-gpt2 on rte using Accuracy

E. DETAILED FIGURES

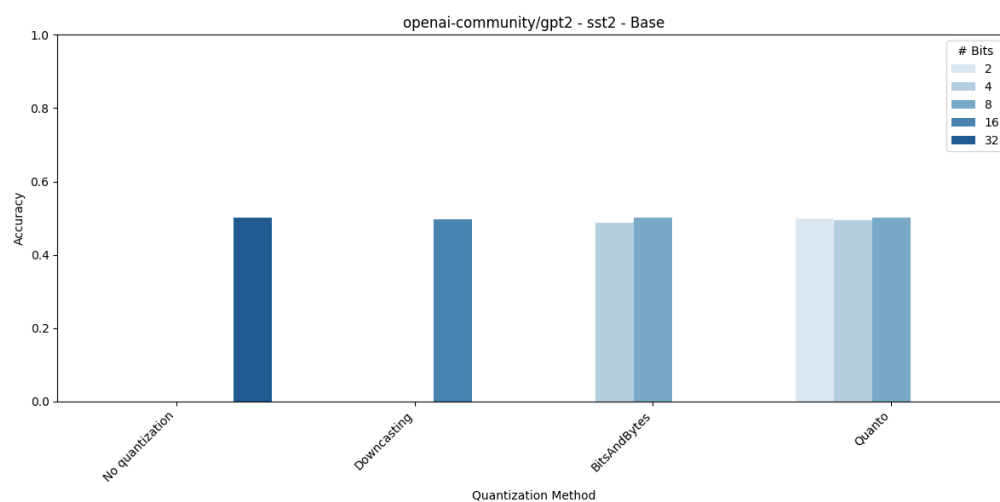


Figure E.161: Results of base openai-community-gpt2 on sst2 using Accuracy

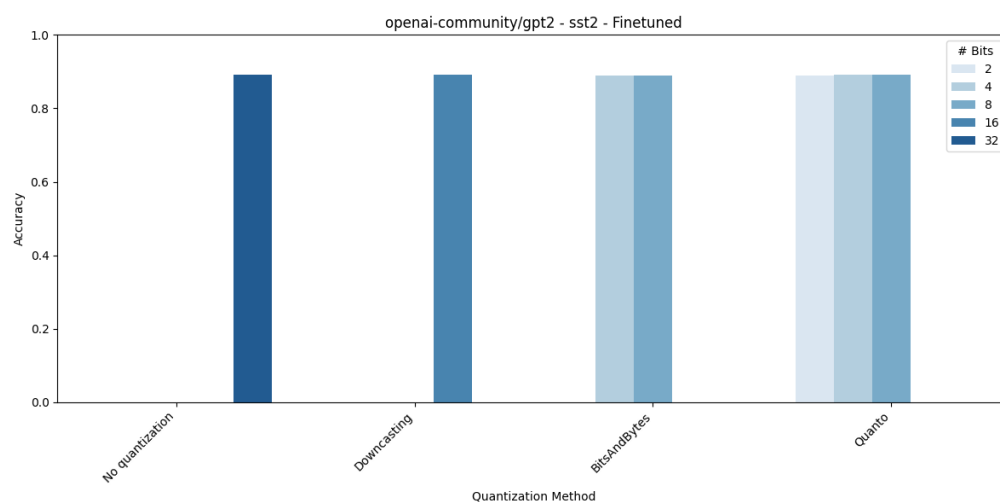


Figure E.162: Results of finetuned openai-community-gpt2 on sst2 using Accuracy

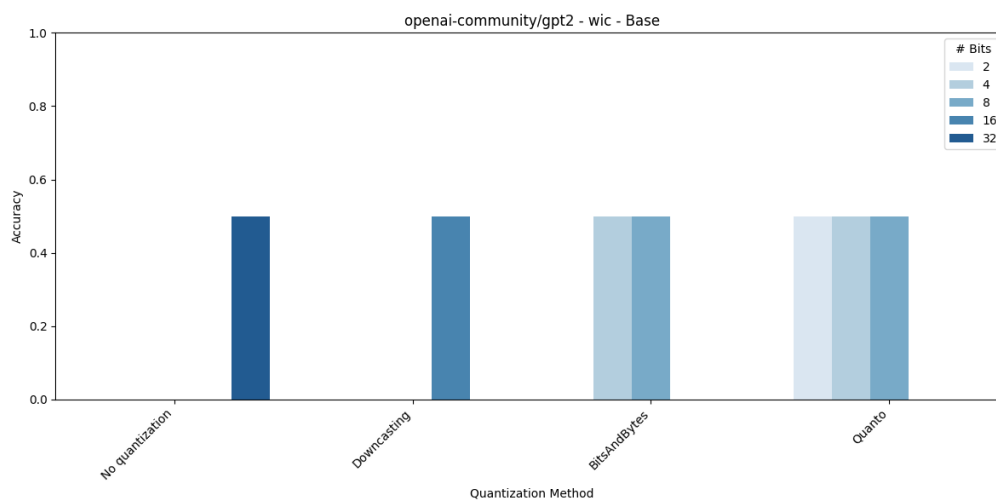


Figure E.163: Results of base openai-community-gpt2 on wic using Accuracy

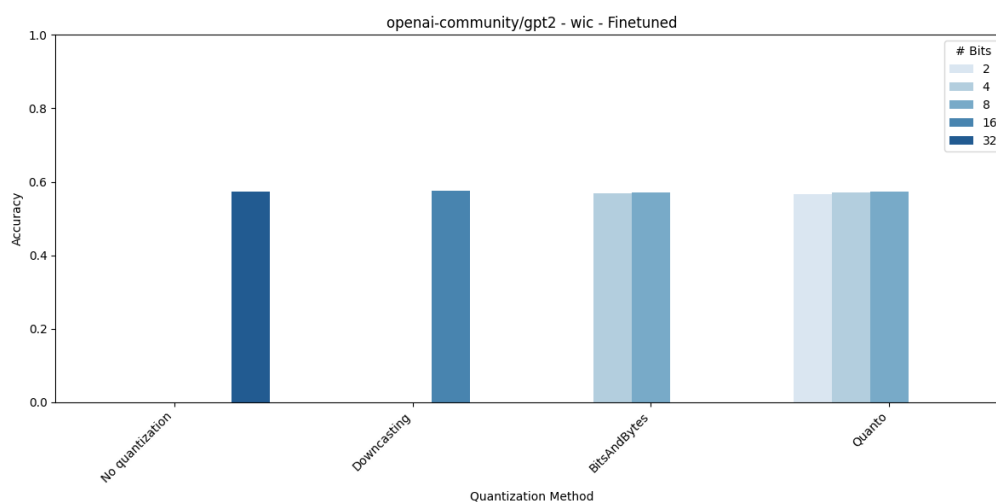


Figure E.164: Results of finetuned openai-community-gpt2 on wic using Accuracy

E. DETAILED FIGURES

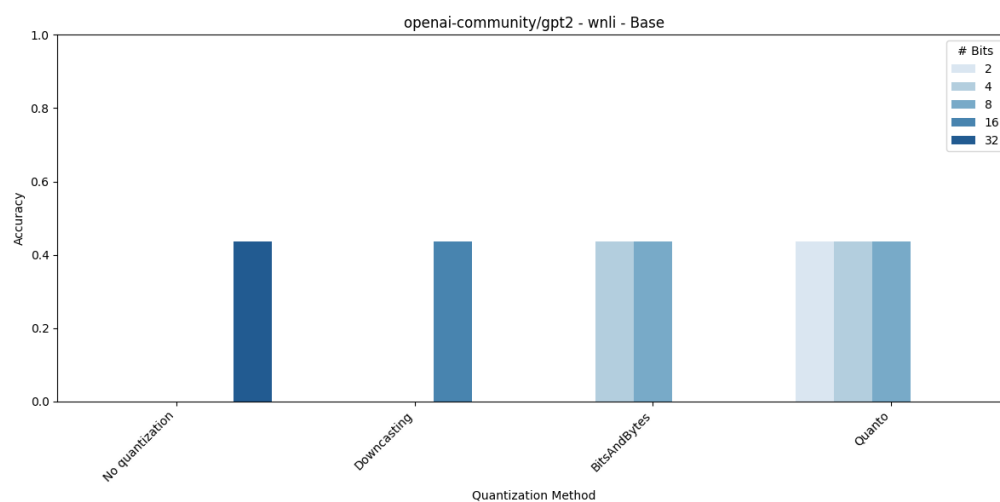


Figure E.165: Results of base openai-community-gpt2 on wnli using Accuracy

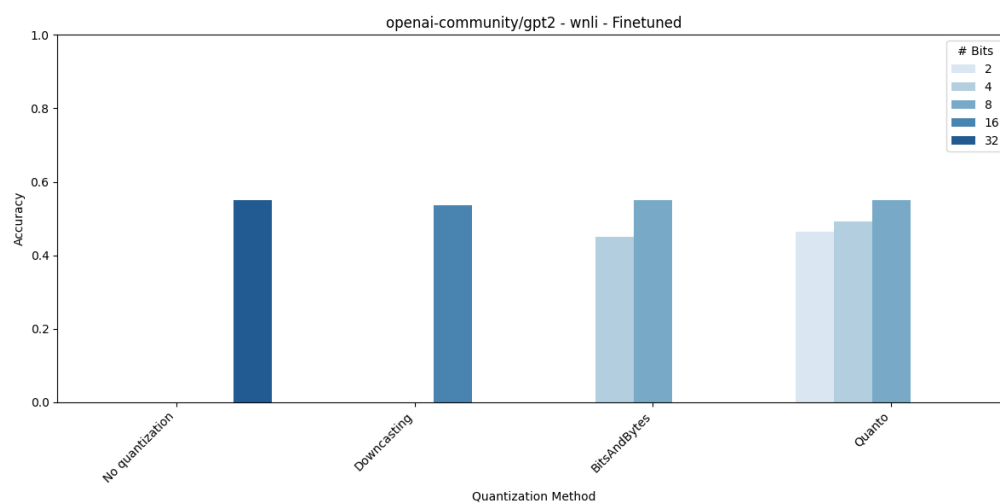


Figure E.166: Results of finetuned openai-community-gpt2 on wnli using Accuracy

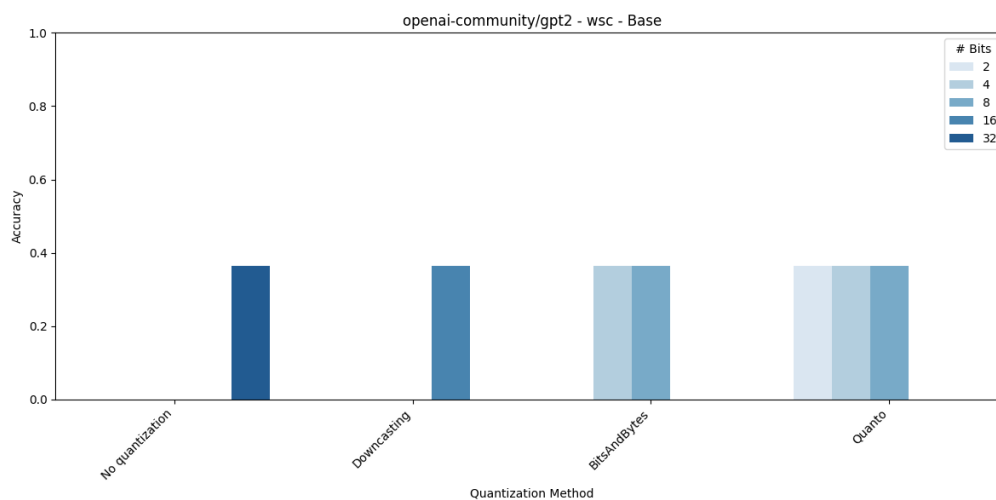


Figure E.167: Results of base openai-community-gpt2 on wsc using Accuracy

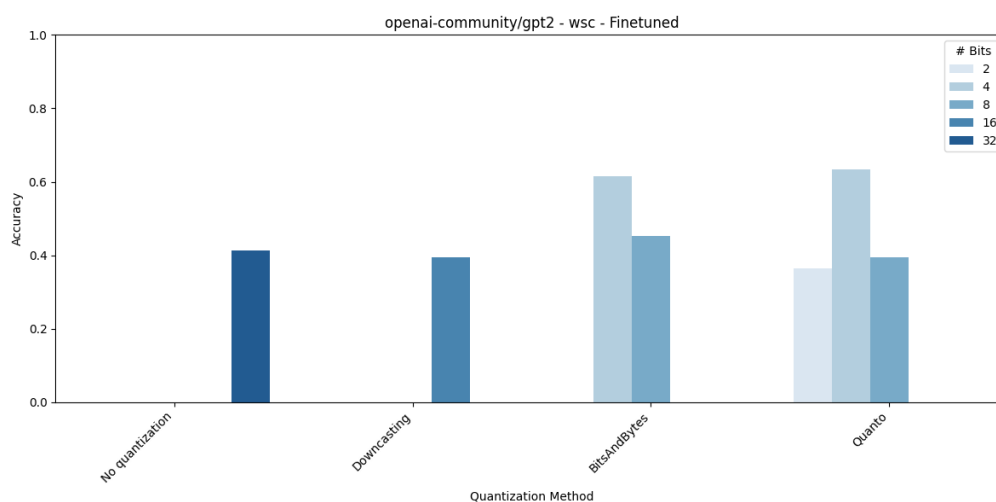


Figure E.168: Results of finetuned openai-community-gpt2 on wsc using Accuracy